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Kelliher et al.

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(54) **SIMULTANEOUS GENE EDITING AND
HAPLOID INDUCTION**

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A01H 1/02 (2006.01)

(Continued)

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CPC **C12N 15/8213** (2013.01); **A01H 1/021**
(2021.01); **A01H 1/045** (2021.01);

(Continued)

(58) **Field of Classification Search**

None

See application file for complete search history.

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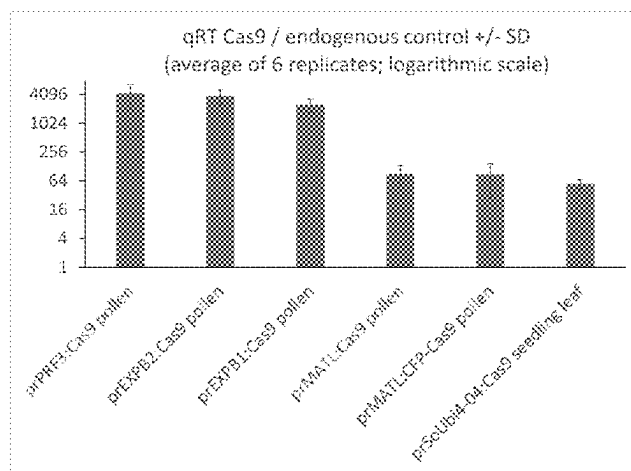
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(57)

ABSTRACT

The presently disclosed subject matter relates to using a
haploid inducing line (whether existing or created) and
transforming the haploid line so that it encodes cellular
machinery capable of editing genes. The transformed hap-
loid inducing line is used as a parent in a cross between two
plants. During pollination, the parental gametes fuse to form
an embryo; and the gene editing machinery is also delivered
to the embryo at this time. During embryonic development,
one set of parental chromosomes are lost, and the gene
editing machinery operates on the remaining set of chromo-
somes. Thus, at least one haploid progeny with edited genes
is produced from the cross. The disclosure is also directed to

(Continued)



methods of testing an edited haploid plant progeny for the presence of a first plant's genomic material.

9 Claims, 24 Drawing Sheets

Specification includes a Sequence Listing.

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A01H 1/04 (2006.01)

C12Q 1/6895 (2018.01)

(52) U.S. Cl.

CPC *C12Q 1/6895* (2013.01); *C12Q 2600/13* (2013.01); *C12Q 2600/158* (2013.01); *C12Q 2600/172* (2013.01)

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Figure 1.

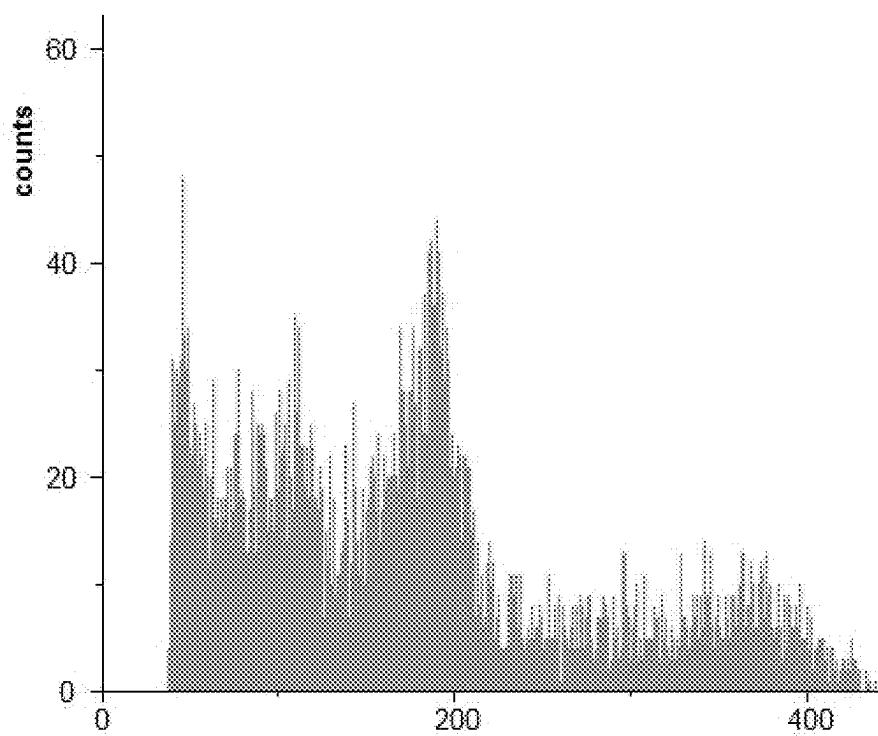


Figure 2.

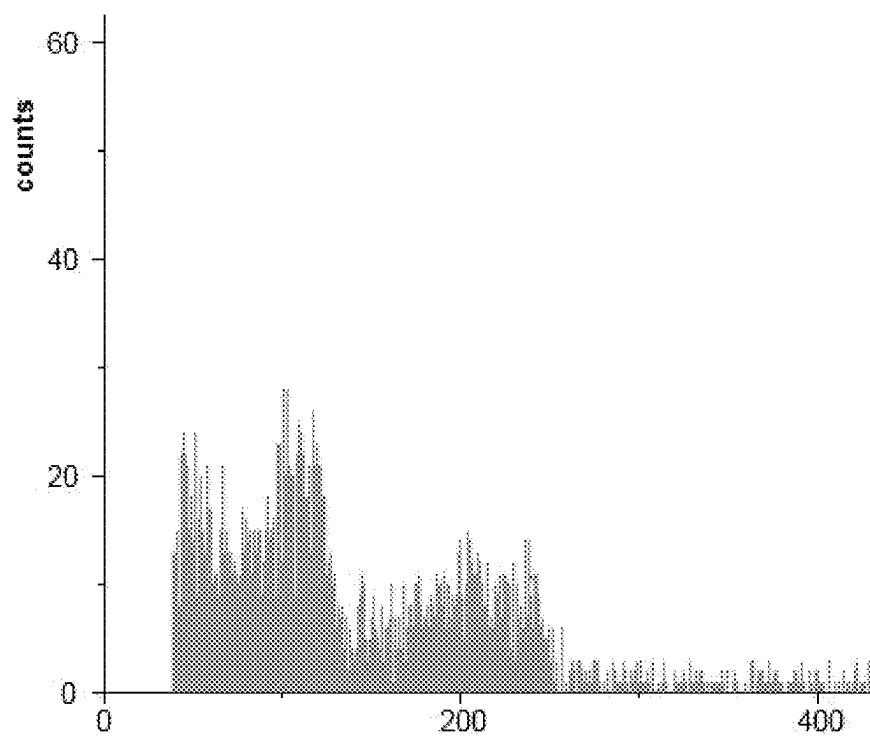


Figure 3.

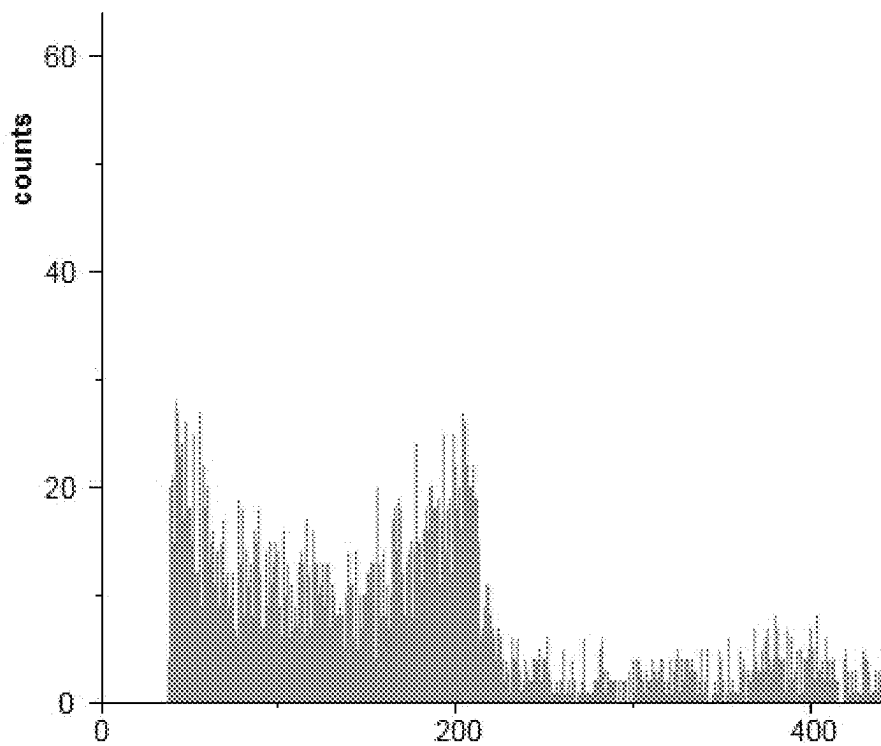


Figure 4.

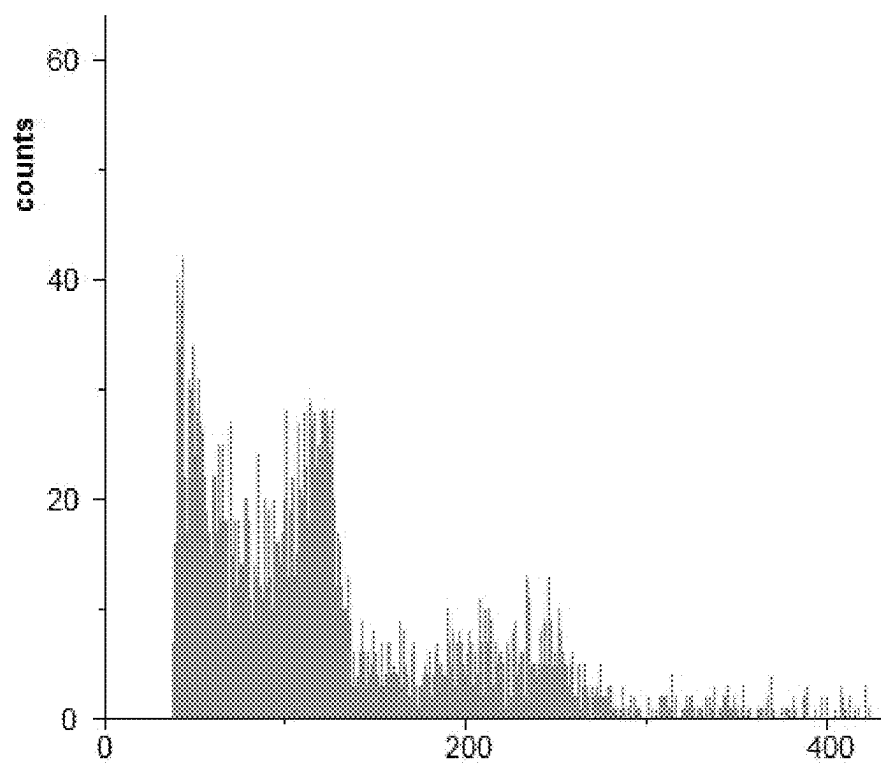


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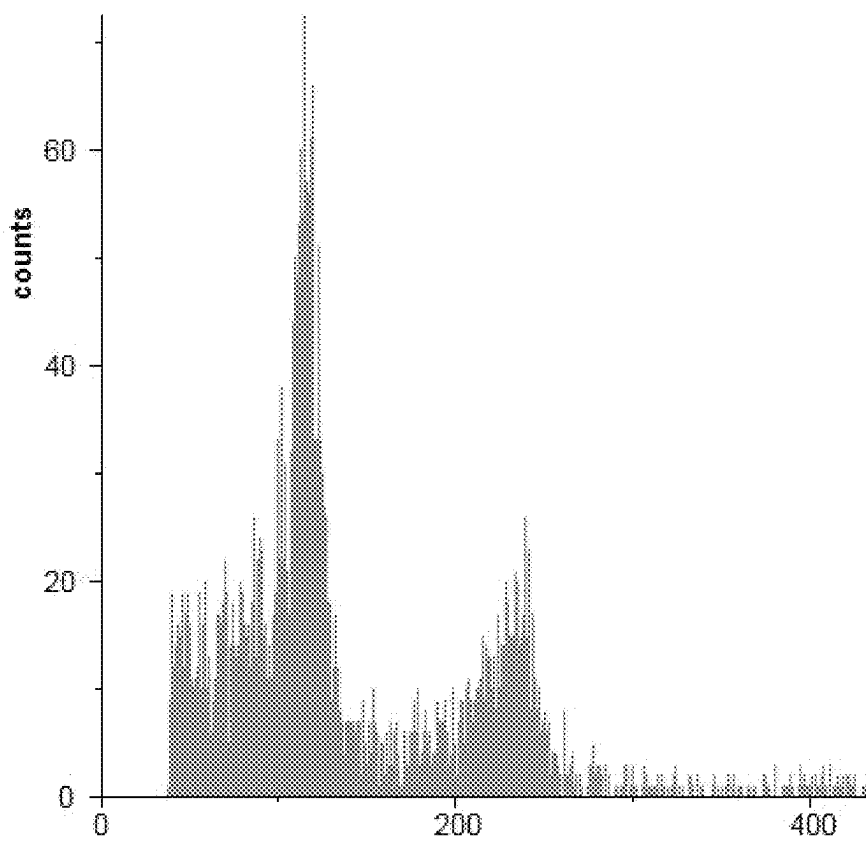


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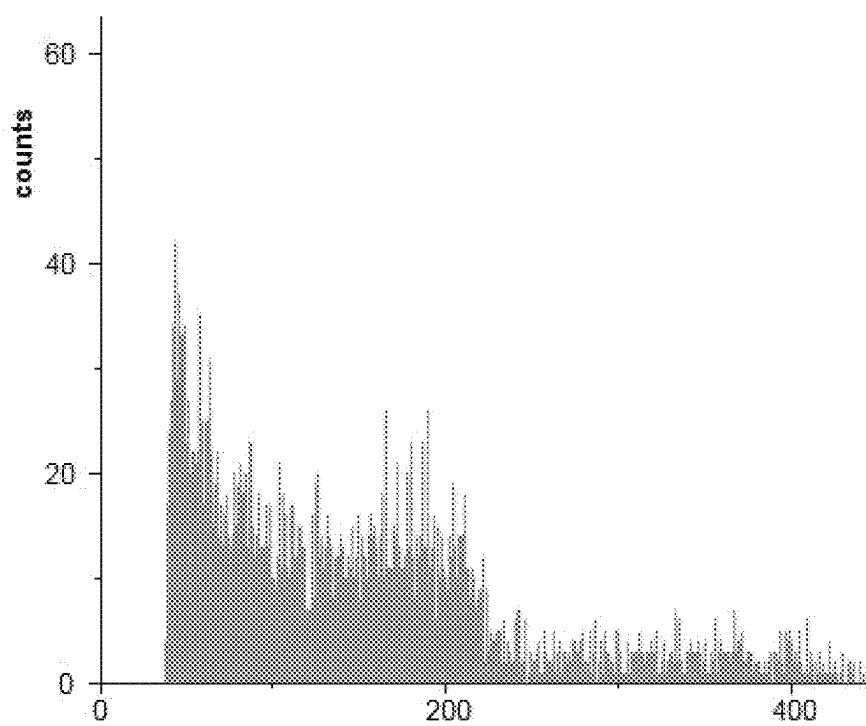


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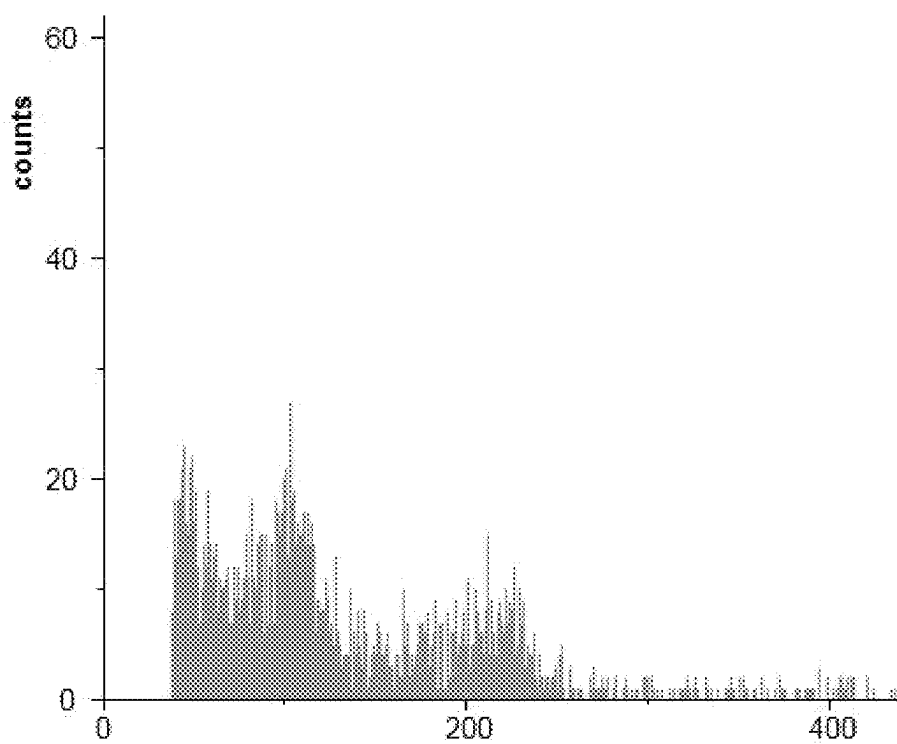


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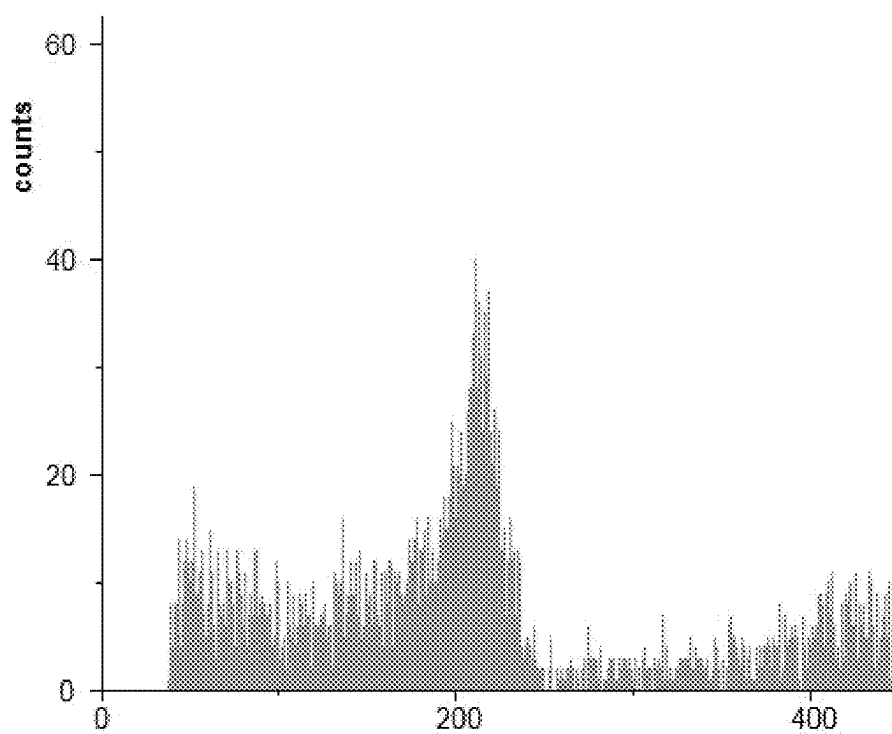


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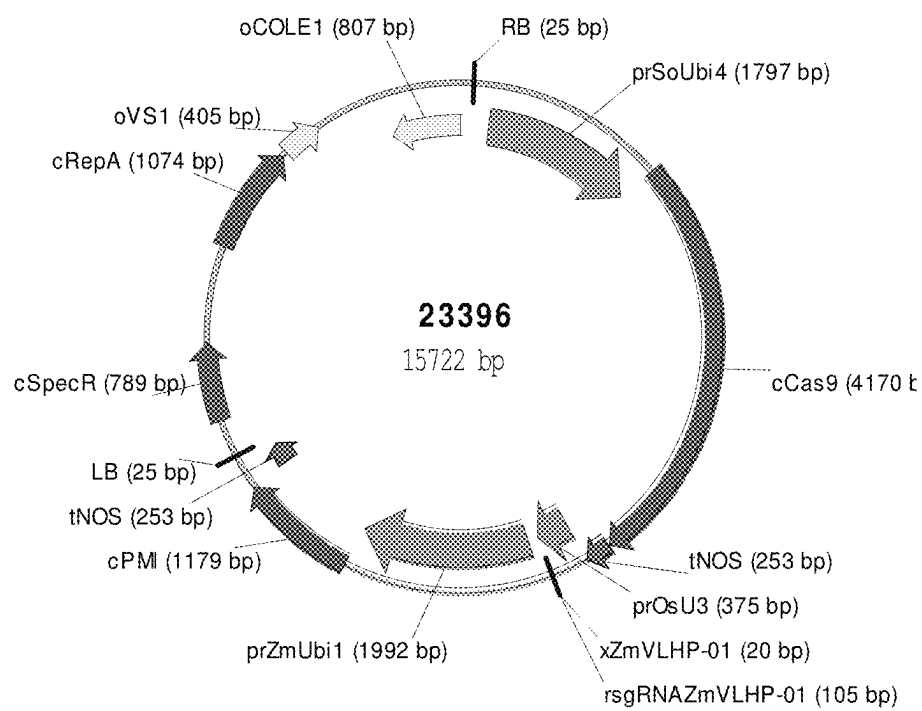


Figure 10.

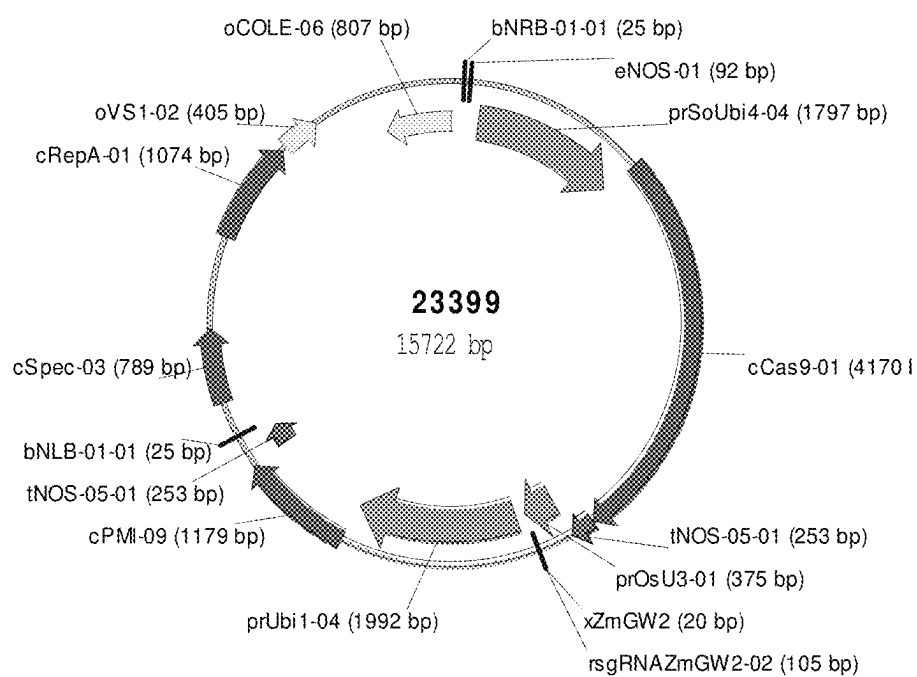


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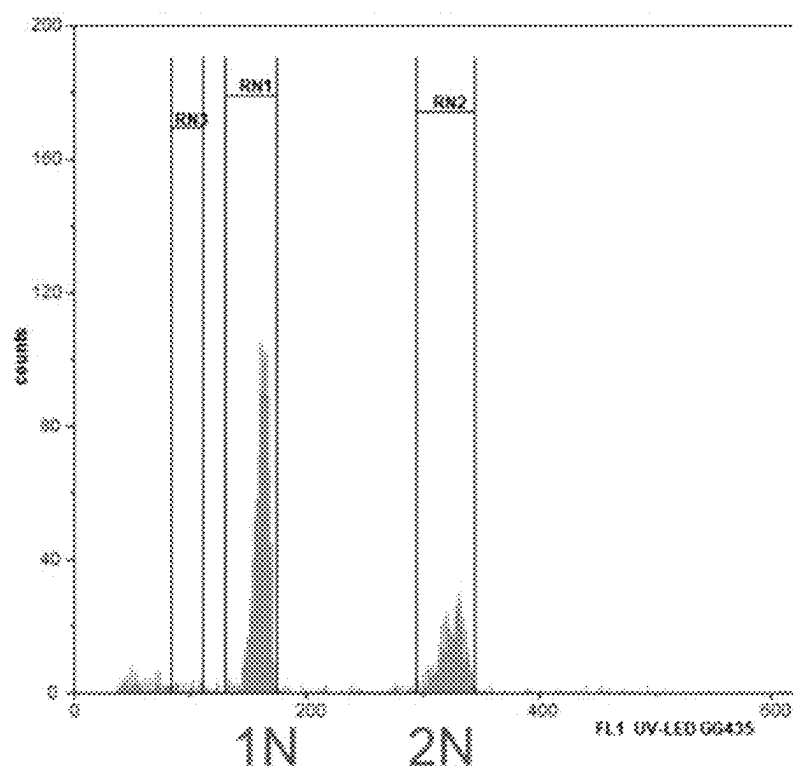


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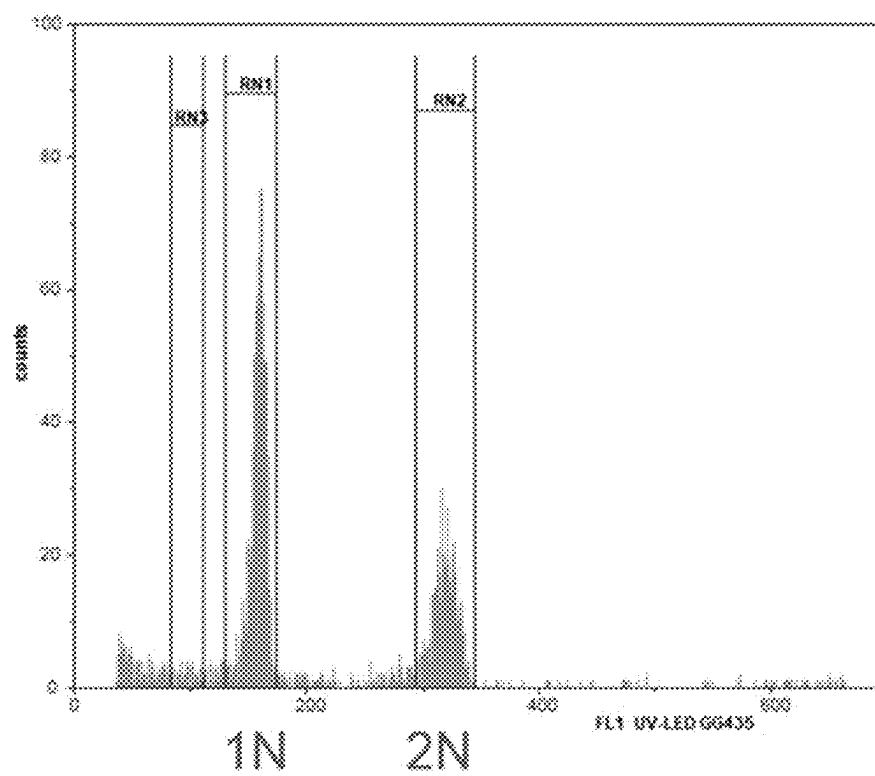


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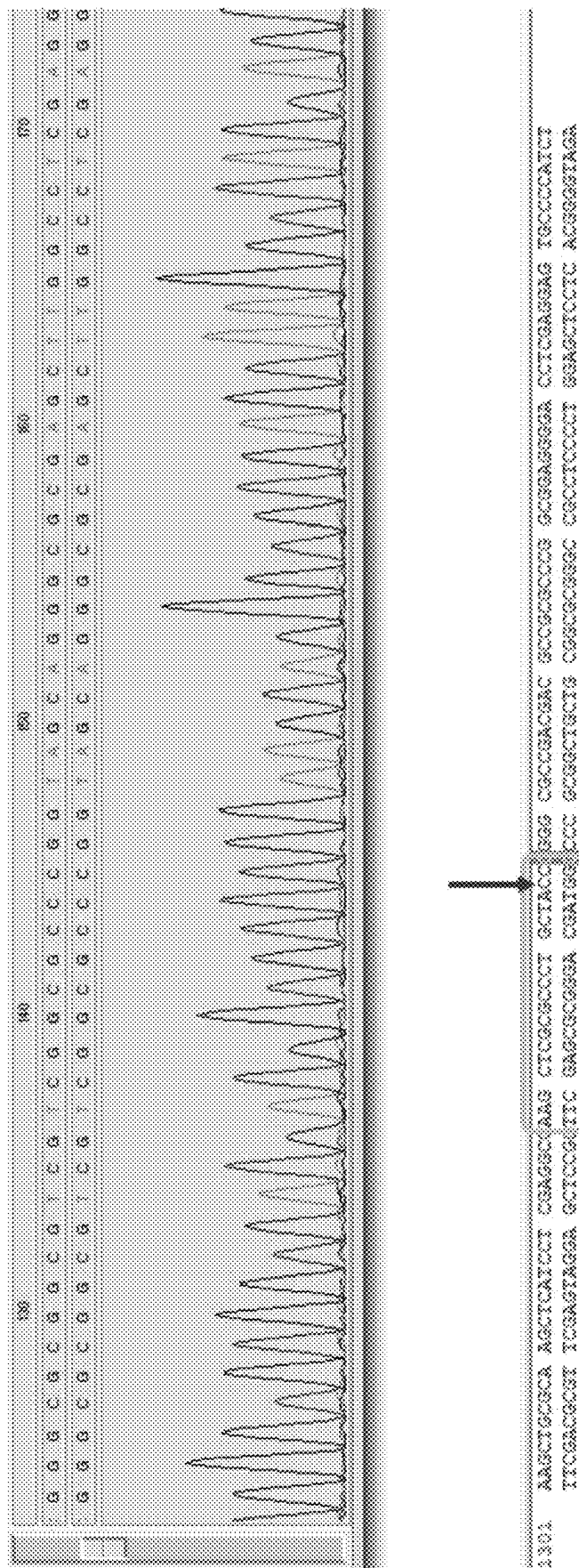


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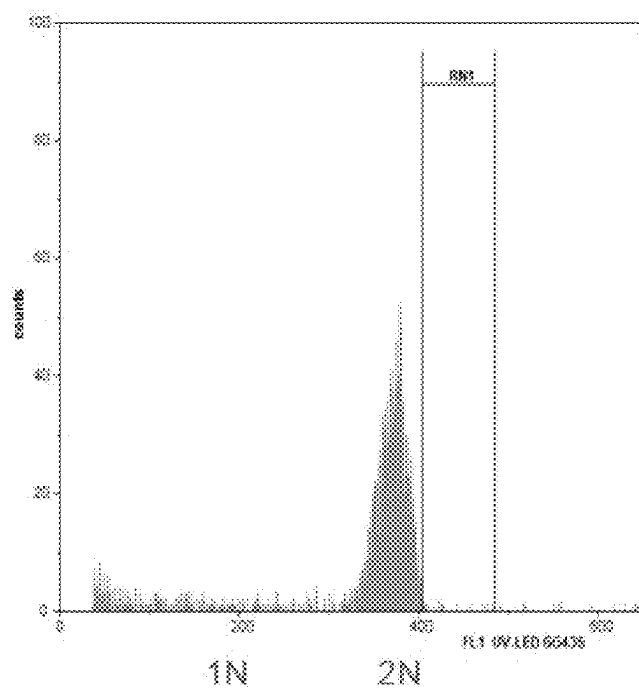


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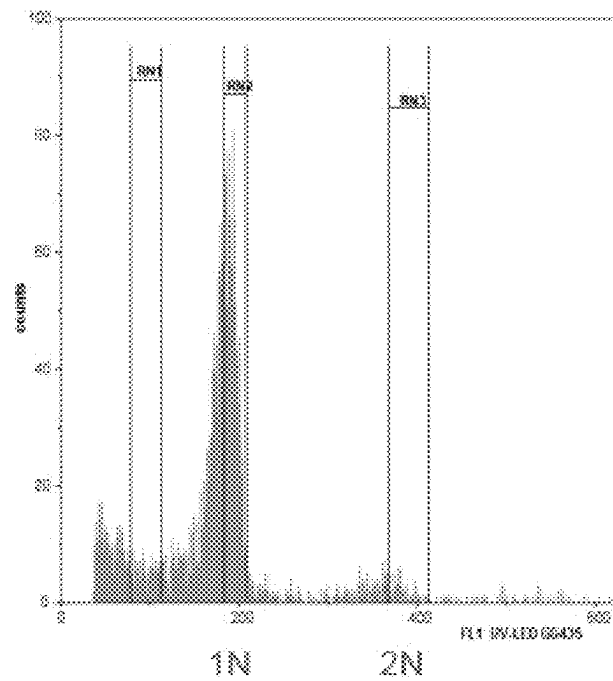


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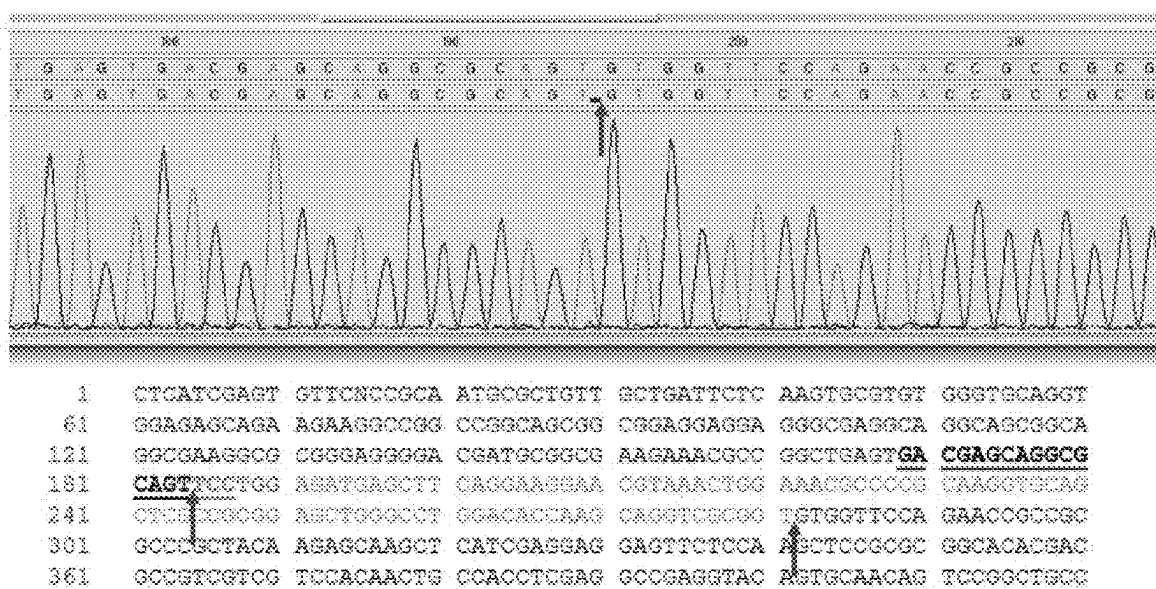


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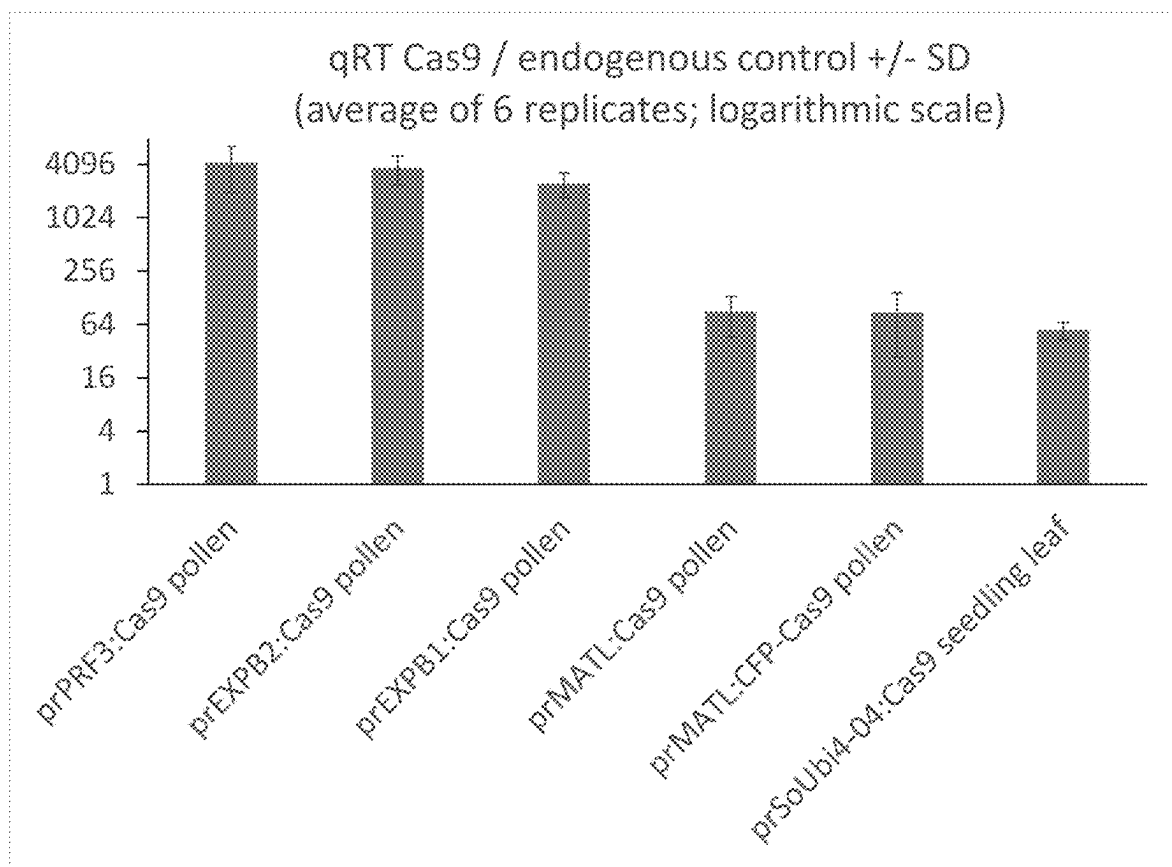


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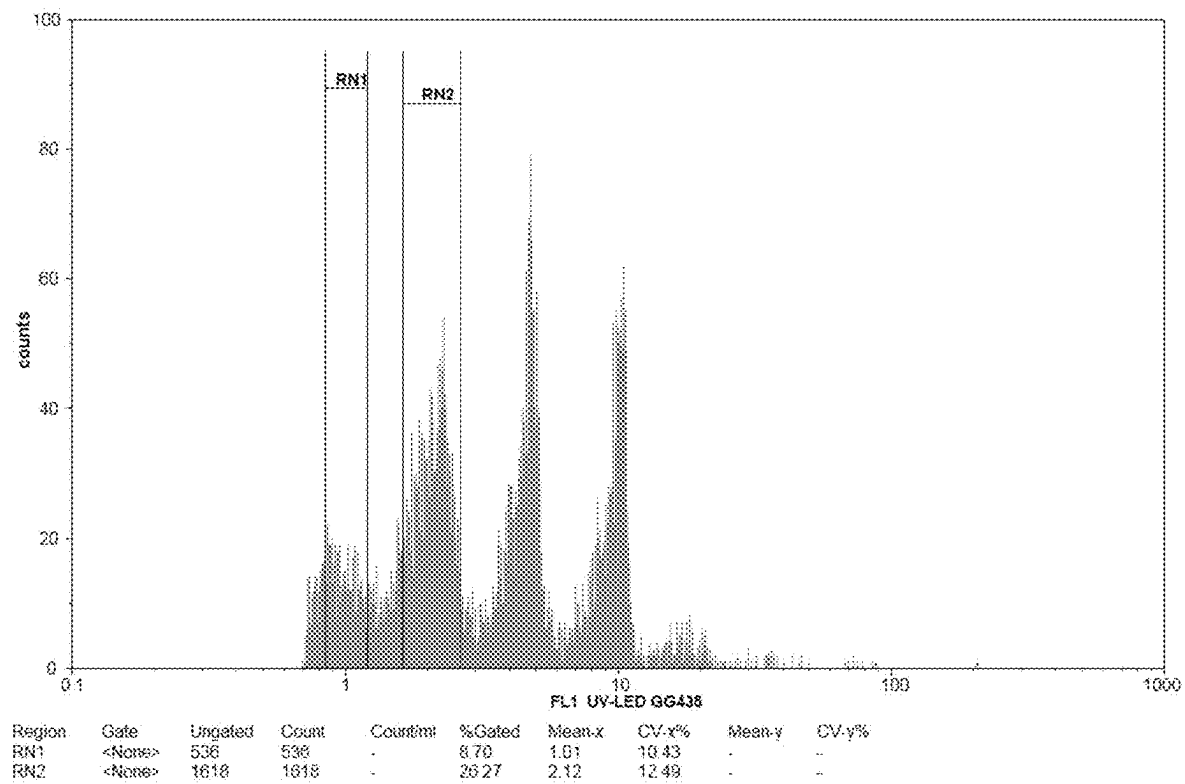


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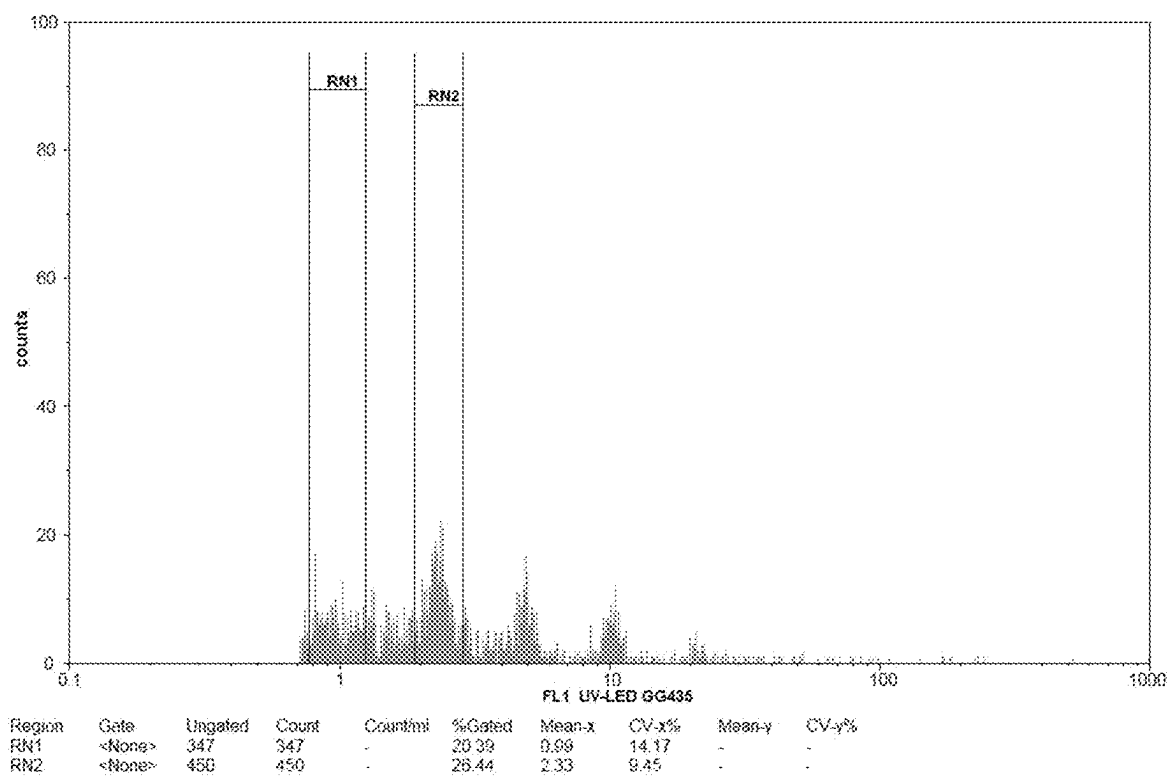


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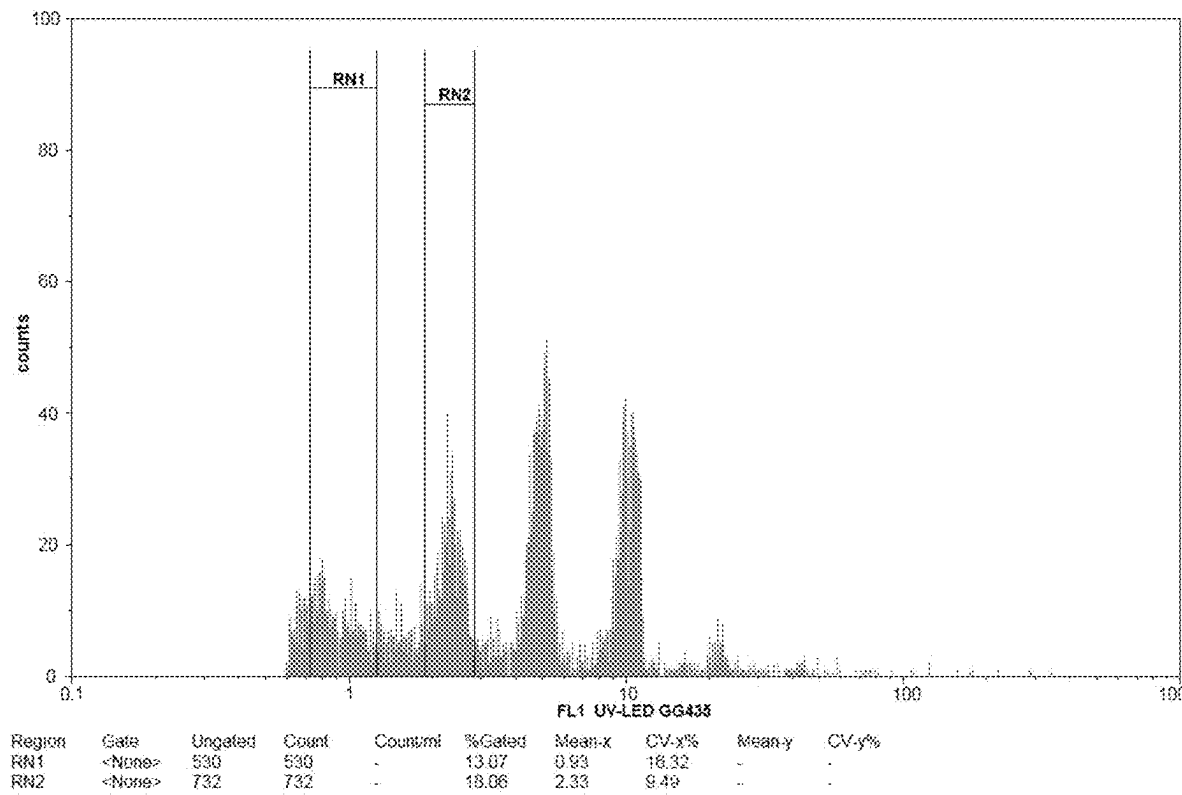


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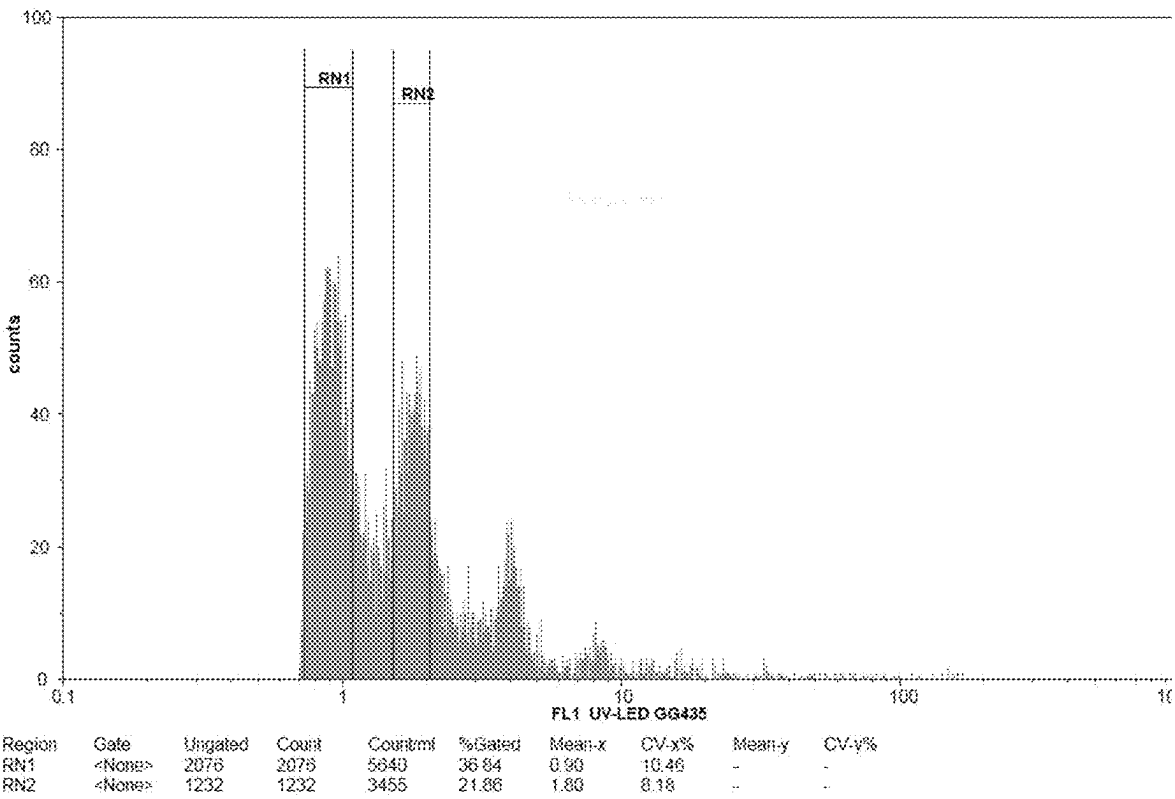


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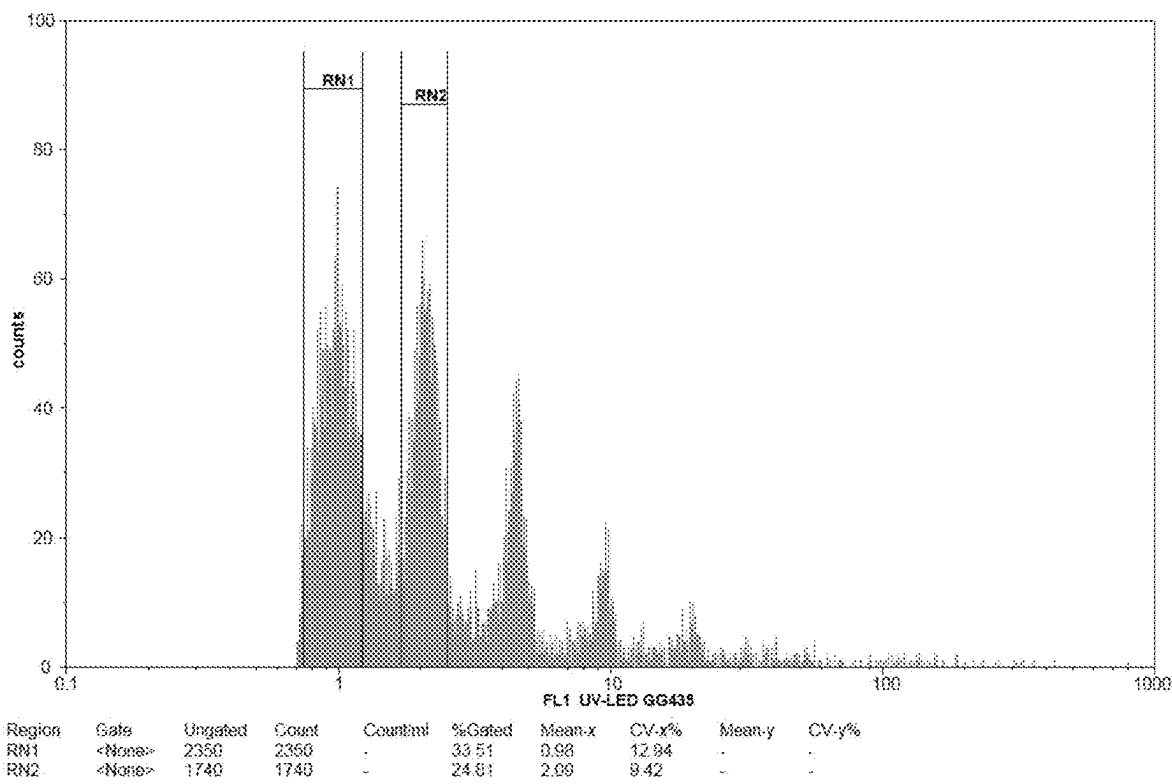
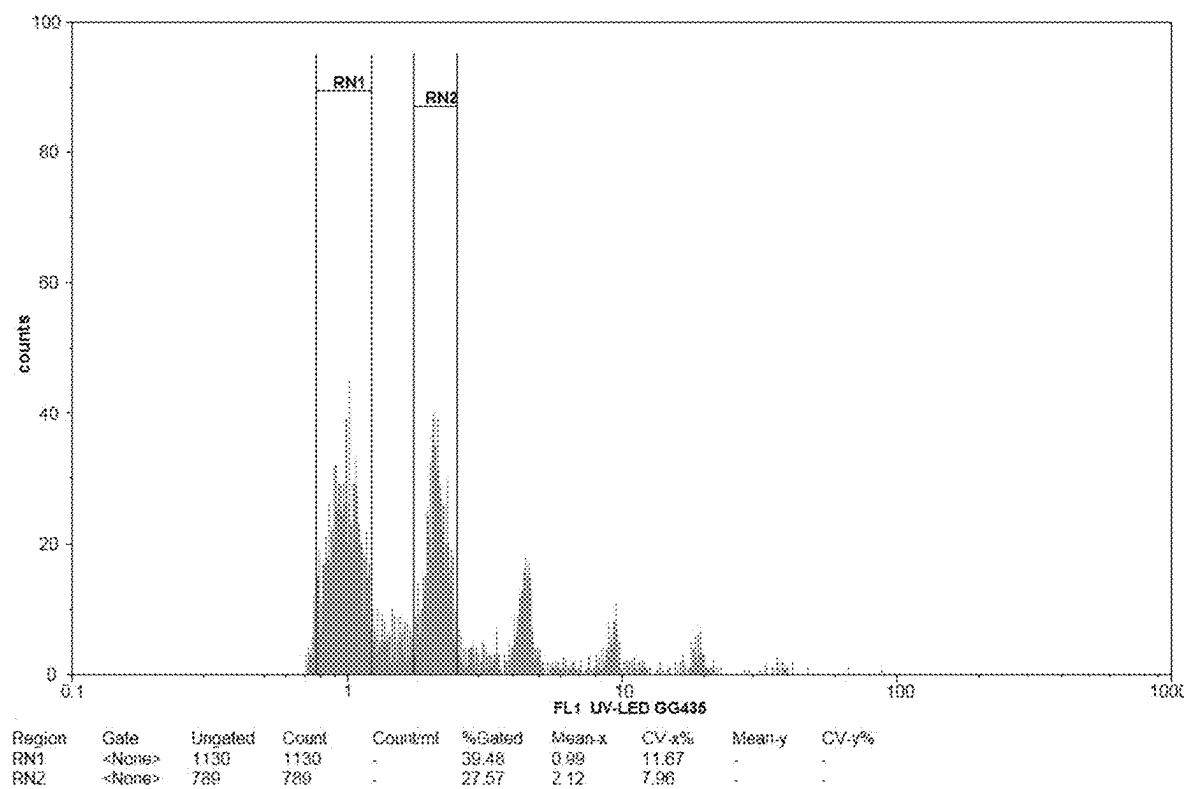


Figure 23.



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SIMULTANEOUS GENE EDITING AND HAPLOID INDUCTION

FIELD OF THE INVENTION

This invention is related to the field of plant biotechnology, specifically agriculture biotechnology and gene editing, as well as plant breeding. The presently disclosed subject matter relates to using a haploid inducing line (whether existing or created) and transforming the haploid inducing line so that it contains DNA coding for cellular machinery capable of editing genes.

STATEMENT REGARDING ELECTRONIC SUBMISSION OF A SEQUENCE LISTING

A Sequence Listing entitled “81771-US-ST26.xml”, 1,001 kilo bytes in size, generated on Oct. 25, 2024 and filed via EFS-Web, is provided in lieu of a paper copy. This Sequence Listing is hereby incorporated by reference into the specification for its disclosures.

BACKGROUND

Targeted mutagenesis (also known as “gene editing”) is a very important technology to crop breeding. There are numerous methods to edit specific gene targets now, including CRISPR, TALEN, meganucleases, and zinc fingers. One method to introduce editing machinery into plants is to use *Agrobacterium* or biolistic transformation of plant tissue. In transformation, DNA coding for the editing machinery (e.g., CAS9 and guide RNA) is introduced into plant callus, seed or embryonic tissue. Stably-transformed plants (“events”) are then recovered, optionally with the help of a selectable marker. But because tissue culture is genotype-dependent, this route will not work for all crops, or even all varieties of the crops for which it does work. These are known as transformation-recalcitrant crops or varieties. These crops or varieties may be valued for their performance but it is a challenge for biotechnology that they cannot be transformed and thus cannot be directly edited via transformation. For recalcitrant varieties, one of two alternative approaches could be used to introduce desirable mutations. First, one could introduce the edits via trait introgression. This route is expensive, laborious, and time-consuming. It also means impurity of the final product because of genetic linkage—that is, there will be a linked block surrounding the introgressed edits, containing genes and alleles from the transformable donor line. This linkage can be an issue if any of those genes or alleles impact the performance of the transformation-recalcitrant line (may also be referred to as an “elite line”). Secondly, one could introduce the editing machinery transiently to the growing plant without tissue culture, such as floral dipping for *Arabidopsis* transformation. The challenge is ensuring edits end up in cells that contribute to the germ-line, so they are passed on to progeny seed. There are few established or routine methods to do this in crops.

We have previously shown new methods to transiently introduce editing machinery during haploid induction in U.S. patent application Ser. No. 15/901,464, which was the national phase of PCT/US2017/064512 (the entire contents of both are incorporated herein by reference). Haploid induction (“HI”) is a class of plant phenomena characterized by loss of one parent’s set of chromosomes (the chromosomes from the haploid inducer parent) from the embryo at some time during or after fertilization, often during early

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embryo development. Haploid induction is also known as gynogenesis if the inducer line is used as the male in the cross, or androgenesis if the inducer line is used as the female in the cross. Haploid induction has been observed in numerous plant species, such as sorghum, barley, wheat, maize, *Arabidopsis*, and many other species.

Commonly, during haploid induction, both parent lines used in the induction cross are both diploids, so their gametes (egg cells and sperm cells) are haploids. Haploid induction is frequently a medium to low penetrance trait of the inducer line, so the resulting progeny, depending on the species or situation, may be either diploid (if no genome loss takes place) or haploids (if genome loss does indeed take place). If the parent line that is crossed to the haploid inducer is not diploid, but rather a tetraploid, hexaploid, or other plant of higher ploidy, the term haploid induction is something of a misnomer, because the “haploid” progeny produced will have a gametic chromosome number, and thus would not really be haploids, but rather diploids (if the parent is tetraploid) or triploids (if the parent is hexaploid) and so on. Therefore, as used herein, “haploids” possess half the number of chromosomes of either parent; thus haploids of diploid organisms (e.g., maize) exhibit monoploidy; haploids of tetraploid organisms (e.g., ryegrasses) exhibit diploidy; haploids of hexaploid organisms (e.g., wheat) exhibit triploidy.

Haploid induction can occur during self-pollination or intercrossing of two lines within the same species, or it can occur during wide crosses, where it can be viewed as a hybridization barrier, preventing the formation of interspecific hybrids. In maize, the most commonly employed method of inducing haploids is through the use of an intraspecific haploid inducer male line, which is primarily triggered by rearrangements of, mutations in, and/or recombinations, insertion, or deletions within a region of chromosome 1, specifically the MATRILINEAL (MATL) gene, also known as NOT LIKE DAD1 (NLD1) and PHOSPHOLIPASE A1 (PLA1) (with the notable exception of the ig type haploid induction, which is a result of a mutation in the INDETERMINATE GAMETOPHYTE1 gene on chromosome 3). In wheat, the most common method of inducing haploids is by wide cross to maize pollen—regardless of parent genotype or lineage, this works with almost any wheat crossed by almost any maize pollen.

HI maize lines contain a quantitative trait locus (“QTL”) on Chromosome 1 responsible for at least 66% of the variation in haploid induction. The QTL causes haploid induction at different rates when it is introgressed into various backgrounds. All maize haploid inducer lines used in the seed industry are derivatives of the founding HI line, known as Stock6, and all have the haploid inducer chromosome 1 QTL mutation.

In maize, haploid seed or embryos are specifically produced by making crosses between a haploid inducer male (i.e., “haploid inducer pollen”) and virtually any ear that one chooses—the ear could be of any inbred, hybrid, or other germplasm. Haploids are produced when the haploid inducer pollen DNA is not fully transmitted and/or maintained through the first cell divisions of the embryos. The resulting phenotype is not fully penetrant, with some ovules containing haploid embryos, and others containing diploid embryos, 2fricana22 embryos, chimeric embryos, or aborted embryos. The haploid kernels have embryos that contain only the maternal DNA plus normal triploid endosperm. After haploid induction, haploid embryos or seed are typically segregated from diploid and 2fricana22 siblings using a phenotypic or genetic marker screen and grown or cultured

into haploid plants. These plants are then converted either naturally or via chemical manipulation (e.g., using an anti-microtubule agent such as colchicine) into doubled haploid (“DH”) plants which then produce inbred seed.

Plant breeding is facilitated by the use of doubled haploid (DH) plants. The production of DH plants enables plant breeders to obtain inbred lines without multi-generational inbreeding, thus decreasing the time required to produce homozygous plants. DH plants provide an invaluable tool to plant breeders, particularly for generating inbred lines, QTL mapping, cytoplasmic conversions, trait introgression, and F2 screening for high throughput trait improvement. A great deal of time is spared as homozygous lines are essentially generated in one generation, negating the need for multi-generational single-seed decent (conventional inbreeding). In particular, because DH plants are entirely homozygous, they are very amenable to quantitative genetics studies. The production of haploid seed is critical for the doubled haploid breeding process. Haploid seed are produced on maternal germplasm when fertilized with pollen from a gynogenetic inducer, such as Stock 6 and Stock 6-derivative lines.

We previously described a novel method in which the in vivo haploid induction process can be co-opted to transiently introduce editing machinery into any germplasm by including it in the haploid inducer parent, either stably integrated as a transgene, or transiently expressed. Simultaneous editing plus haploid induction can be done in almost any crop via wide cross or de novo haploid induction for instance via CENH3 mutation (i.e., CENH3-modified haploid inducer; see, e.g., WO 2017/004375, incorporated herein by reference in its entirety) or via lipid spray (see P.C.T. Patent Application No. PCT/US2016/62548, incorporated herein by reference in its entirety). We showed examples of HI in maize, both field corn and sweet corn, using a haploid inducer male as the editing donor line. Further, we showed examples of HI in *Arabidopsis* using CENH3-modified haploid inducer lines.

We also provided examples of HI in wheat using maize pollen as the editing donor line in a wide cross. In wheat, rice, barley, *Brassica*, and other crops, the route to haploid induction would be to use a pollen donor that induces haploids via wide cross. For example, one could use corn pollen on wheat, millet pollen on wheat, barley pollen on other barley species, or any other wide crossing method. In those cases of gynogenetic haploid induction it would be preferable for the male line to contain the editing machinery, because it is the male (pollen-derived) DNA that is eliminated in the haploid induction process. In cases of androgenic haploid induction, for instance in the *ig1* system in maize or via altered CENH3 in any crop (which can work via either the male or the female), the editing machinery would be optimally present in the female parent, because the female chromosomes are eliminated in the haploid induction process.

In simultaneous editing plus haploid induction, the goal is to rapidly and cost-effectively edit crops and elite lines (“editing destination lines”) without tissue culture. The line that receives the edits could be elite germplasm, and the editing machinery itself would be eliminated during the haploid induction process. At the same time, edited doubled haploid lines are produced.

Here we teach, inter alia, new methods for confirming the presence of haploid progeny, for example, edited haploid progeny, for the presence of haploid inducer genome.

SUMMARY

Tissue culture recalcitrance is a major challenge to rapid elite line editing across crops. Using haploid inducing lines

to deliver the targeted mutagenesis machinery to elite lines and simultaneously induce haploids represents the surmounting of this major obstacle. Next-generation breeding programs may come to depend on this process.

The editing machinery is delivered via the inducer line. The editing machinery is most often DNA-binding proteins combined in some cases with RNA and in some cases also with DNA. The DNA, RNA, and proteins that make up the editing machinery are encoded by and are present in the inducer line because they have been stably inserted in the inducer, for example, via bombardment or agrobacterium mediated transformation. In other examples, the editing machinery is transiently introduced (through exogenous application) or transiently expressed in the gametophyte prior to fertilization. After fertilization, edits are made by the editing machinery in the non-inducer target genes prior to or during elimination of the inducer chromosomes. The result is a haploid embryo or plant or seed that contains the chromosome set only from the non-inducer parent, where that chromosome set contains DNA sequences that have been edited. Haploids may be further tested for evidence of the haploid inducer parent. These edited haploids can be identified, grown, and their chromosomes doubled, preferably by colchicine or other mitotic inhibitor. This line can then be directly used in downstream breeding programs.

In one embodiment, the invention provides a method of editing a plant’s genomic DNA. This is done by taking a first plant—which is a haploid inducing plant and which also has encoded into its DNA the machinery necessary for accomplishing the editing (for example, a Cas9 enzyme and a guide RNA)—and using that first plant’s pollen to pollinate a second plant. The second plant is the plant to be edited. From that pollination event, progeny (e.g., embryos or seeds) are produced; at least one of which will be a haploid seed. This haploid seed will only contain the chromosomes of the second plant; the first plant’s chromosomes have vanished (having been eliminated, lost or degraded), but before doing so, the first plant’s chromosomes permitted the gene-editing machinery to be expressed. Alternately, and without wishing to be bound by theory, the first plant delivers the already-expressed editing machinery upon pollination via the pollen tube. Or, in the case that the haploid inducer line is the female in the cross, the haploid inducing plant’s egg cell contains the editing machinery that is present and perhaps already being expressed, upon fertilization with the “wild type” or non-haploid inducing pollen grain. Through any of these routes, the haploid progeny obtained by the cross will also have had its genome edited. The haploids may be further tested for evidence of the inducer parent genome.

In one aspect, the editing machinery is any DNA modification enzyme, but is preferably a site-directed nuclease. The site-directed nuclease is preferably CRISPR-based, but could also be a meganuclease, a transcription-activator like effector nuclease (TALEN), or a zinc finger nuclease. The nuclease used in this invention could be Cas9, Cpf1, dCas9-FokI, chimeric FEN1-FokI. In one aspect, the DNA modification enzyme is a site-directed base editing enzyme such as Cas9-cytidine deaminase or Cas9-adenine deaminase, wherein the Cas9 can have one or both of its nuclease activity inactivated, i.e. chimeric Cas9 nickase (nCAs9) or deactivated Cas9 (dCas9) fused to cytidine deaminase or adenine deaminase. The optional guide RNA targets the genome at the specific site intended to be edited. In one aspect, the optional guide RNA comprises an 18-21 nucleotide sequence with homology to any of SEQ ID NOs: 2, 4, 8, 21, 23, 25, 29, 32, and 33.

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Once the edited haploid progeny is obtained, it may optionally have its chromosomes doubled by a chromosome doubling agent (for example colchicine, pronamide, dithiopyr, trifluralin, or another known anti-microtubule agent).

In one embodiment, the first plant is a monocot or a dicot. Aspects of the first plant include maize, wheat, rice, barley, oats, triticale, sorghum, pearl millet, teosinte, bamboo, sugar cane, asparagus, onion, and garlic. In one embodiment, the second plant is a monocot or a dicot. Aspects of the second plant include maize, wheat, rice, barley, oats, triticale, sorghum, pearl millet, teosinte, bamboo, sugar cane, asparagus, onion, and garlic. In one embodiment, the first plant is a monocot or a dicot of a different species than the second plant. For example, in one aspect, the first plant is maize and the second plant is wheat. In another aspect, the first plant is wheat and the second plant is maize. In another embodiment, the first plant is a maize plant selected and/or derived from the lines Stock 6, RWK, RWS, UH400, AX5707RS, NP2222-matl, or any of the several other known HI lines. In yet another embodiment, the first plant comprises a mutation in a CENH3 gene, an *igl* gene, or another mutation conferring paternal-haploid inducing systems. In another embodiment, the first plant is a rice plant with the MATL gene modified or knocked out which makes it a haploid inducer line.

In another embodiment, the first plant is not necessarily a haploid inducer, yet the first plant comprises the genes necessary for encoding the gene editing machinery. In this embodiment, haploid induction is produced by administering a compound during, immediately before, or immediately following pollination. In one aspect, the composition comprises a lipid or a phospholipase inhibitor. In another aspect, the composition comprises methyl alpha-linolenoyl fluorophosphate (MALFP), linoleic acid ethyl ester (LLAEE), linoleic acid (LLA), corn oil, distearoyl-phosphatidyl choline (DSPC), methyl arachidonyl fluorophosphonate (MAFP), Palmityl trifluoromethylketone (PACOCF3), Arachidonyl trifluoromethylketone (AACOCF3), Monoalide, Linolenic acid ethyl ester (LNAEE), Linolenic acid ethyl ester (LNAEE), Oleic acid methyl ester (OAME), Oleic acid ethyl ester (OAE), Palmitic acid ethyl ester (PAEE), Palmitoleic acid ethyl ester (PLAEE), Linseed oil, corn oil, alpha-Linolenic acid (aLNA), gamma-Linolenic acid (gLNA), Oleic acid, Arachidonic acid, Stearic Acid, 9(Z)-11E-conjugated Linoleic acid, or 2-oleoyl-1-palmitoyl-sn-glycero-3-phospho-ethanolamine.

The disclosure also includes methods of testing an edited haploid plant progeny for the presence of a first plant's genomic material, wherein the first plant is a haploid inducer line of the plant. In one embodiment, a method comprises isolating a nucleic acid from the edited haploid progeny, and detecting in the nucleic acid the presence of a plurality of codominant markers, wherein the codominant markers have a distinct haplotype for a second plant and wherein the second plant comprises the source of genomic DNA in the edited haploid plant progeny.

The edited haploid may be obtained, for example, by a method comprising: (a) obtaining the first plant, wherein the first plant is capable of expressing a DNA modification enzyme and optionally at least one guide nucleic acid; (b) obtaining the second plant, wherein the second plant comprises the plant genomic DNA which is to be edited; (c) pollinating the second plant with pollen from the first plant; and selecting at least one haploid progeny produced by the pollination of step (c) wherein the haploid progeny comprises the genome of the second plant but not the first plant, and the genome of the haploid progeny has been modified by

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the DNA modification enzyme and optional at least one guide nucleic acid delivered by the first plant. The DNA modification enzyme and guide nucleic acids can be those described herein. Typically, the edited haploid progeny will be treated with a chromosome doubling agent, e.g. colchicine, pronamide, dithiopyr, trifluralin, or another known anti-microtubule agent, after testing, thereby creating an edited doubled haploid progeny.

Plants may vary by embodiment and may include those described herein, e.g. in some embodiments, the first plant is a maize plant selected and/or derived from the lines Stock 6, RWK, RWS, UH400, AX5707RS, NP2222-matl, or any of the several other known HI lines, and the first plant acts as the male and the second plant acts as the female.

The plurality of codominant markers include at least one of at least 5, at least 10, at least 15, at least 20, at least 25, at least 30, at least 35, at least 40, at least 45, at least 50, at least at 55, at least 60, at least 65, and at least 70 markers. Typically, the plurality of codominant markers will include at least two markers selected from the group consisting of the markers in Table 12. Detecting may include the use of at least one assay component in Table 13.

BRIEF DESCRIPTION OF THE SEQUENCES IN THE SEQUENCE LISTING

SEQ ID NO: 1 is a nucleotide sequence for vector 23396.

SEQ ID NO: 2 is the nucleotide sequence encoding the gRNA sequence for editing VLHP1 in maize.

SEQ ID NO: 3 is a nucleotide sequence for vector 23399.

SEQ ID NO: 4 is the gRNA sequence for editing GW2-2 in maize.

SEQ ID NO: 5 is the nucleotide sequence for vector 22808, comprising a TALEN construct.

SEQ ID NO: 6 is the target sequence for the TALEN of 22808.

SEQ ID NO: 7 is the nucleotide sequence for vector 23123 comprising a Cas9 construct.

SEQ ID NO: 8 is the gRNA for editing MATL in maize.

SEQ ID NO: 9 is nucleotide sequence for the relevant portion of MATL in NP2222.

SEQ ID NO: 10 is nucleotide sequence for the relevant portion of MATL in Stock6.

SEQ ID NO: 11 is nucleotide sequence for the relevant portion of MATL in USR01350333-3 Allele

SEQ ID NO: 12 is nucleotide sequence for the relevant portion of MATL in USR01350333-3 Allele 2.

SEQ ID NO: 13 is nucleotide sequence for the relevant portion of MATL in USR01350344-2 Allele 1.

SEQ ID NO: 14 is nucleotide sequence for the relevant portion of MATL in USR01350344-2 Allele 2.

SEQ ID NO: 15 is nucleotide sequence for the relevant portion of MATL in USR01350343-1 Allele 1.

SEQ ID NO: 16 is nucleotide sequence for the relevant portion of MATL in USR01350328-1 Allele 1.

SEQ ID NO: 17 is nucleotide sequence for the relevant portion of MATL in USR01350337-2 Allele 1.

SEQ ID NO: 18 is nucleotide sequence for the relevant portion of MATL in USR01350337-2 Allele 2.

SEQ ID NO: 19 is the nucleotide sequence of cDNA wildtype MATL.

SEQ ID NO: 20 is the nucleotide sequence for vector 23397.

SEQ ID NO: 21 is the gRNA sequence for editing VLHP2 in maize.

SEQ ID NO: 22 is the nucleotide sequence for vector 23398.

SEQ ID NO: 23 is the gRNA sequence for editing GW2-1 in maize.

SEQ ID NO: 24 is the nucleotide sequence for vector 23763.

SEQ ID NO: 25 is the gRNA sequence for VLHP1 in wheat.

SEQ ID NO: 26 is the wheat VLHP target sequence for TaVLHP2.

SEQ ID NO: 27 is the wheat VLHP target sequence for TaVLHP3.

SEQ ID NO: 28 is the target sequence in ZmVLHP2-03 for editing.

SEQ ID NO: 29 is the edited sequence in ZmVLHP2-03.

SEQ ID NO: 30 is the repair donor template sequence for creating E149L mutation in ZmPYL-D.

SEQ ID NO: 31 is the nucleotide sequence for vector 23136.

SEQ ID NO: 32 is the gRNA of vector 23136.

SEQ ID NO: 33 is the nucleotide sequence of rice PLA gene 0503g27610.

SEQ ID NO: 34 is the nucleotide sequence for vector 24038.

SEQ ID NO: 35 is the nucleotide sequence for vector 24039.

SEQ ID NO: 36 is the nucleotide sequence for vector 24079.

SEQ ID NO: 37 is the nucleotide sequence for vector 24091.

SEQ ID NO: 38 is the nucleotide sequence for vector 24094.

SEQ ID NOs: 39 through 97 are primers and probes used in the identified PCR Taqman assays.

SEQ ID NO: 98 is the nucleotide sequence for vector 24075.

SEQ ID NOs: 99-173 are nucleotide sequences containing codominant markers as described in Table 12.

SEQ ID NOs: 174-473 are nucleotide sequences of assay components described in Table 13.

BRIEF DESCRIPTION OF THE FIGURES

FIG. 1 shows the ploidy analysis (flow cytometry) data for USR01350334-3: DIPLOID (major peak at 200, secondary peak at 400).

FIG. 2 shows the ploidy analysis (flow cytometry) data for USR01350333-3: HAPLOID (major peak at 100, secondary peak at 200).

FIG. 3 shows the ploidy analysis (flow cytometry) data for USR01350333-10: DIPLOID (major peak at 200, secondary peak at 400).

FIG. 4 shows the ploidy analysis (flow cytometry) data for USR01350344-2: HAPLOID (major peak at 100, secondary peak at 200).

FIG. 5 shows the ploidy analysis (flow cytometry) data for USR01350343-1: HAPLOID (major peak at 100, secondary peak at 200).

FIG. 6 shows the ploidy analysis (flow cytometry) data for USR01350341-1: DIPLOID (major peak at 200, secondary peak at 400).

FIG. 7 shows the ploidy analysis (flow cytometry) data for USR01350328-1: HAPLOID (major peak at 100, secondary peak at 200).

FIG. 8 shows the ploidy analysis (flow cytometry) data for USR01350321-3: DIPLOID (major peak at 200, secondary peak at 400).

FIG. 9 is a schematic drawing of vector 23396 (SEQ ID NO: 1) used for *Agrobacterium*-mediated transformation of

maize immature embryos to generate targeted mutations in ZmVLHP1 genes. xZmVLHP-01: guide RNA (gRNA) sequence (5'-GCAGGAGGCGTCGAGCAGCG-3', SEQ ID NO: 2); rsgRNAZmVLHP-01: single guide RNA (sgRNA) comprising of gRNA, tracrRNA and PolIII termination sequences. cPMI: PMI selectable marker gene; cCas9: Cas9 nuclease gene; RB: T-DNA right border; LB: T-DNA left border; tNOS: Nopaline synthetase terminator. cSpec: Spectinomycin resistance gene.

FIG. 10 is a schematic drawing of vector 23399 (SEQ ID NO: 3) used for *Agrobacterium*-mediated transformation of maize immature embryos to generate targeted mutations in ZmGW2 genes. xZmGW2-02: guide RNA (gRNA) sequence (5'-AAGCTCGCGCCCTGCTACCC-3', SEQ ID NO: 4); rsgRNAZmGW2-02: single guide RNA (sgRNA) comprising of gRNA, tracrRNA and PolIII termination sequences. cPMI-09: PMI selectable marker gene; cCas9-01: Cas9 nuclease gene; RB: T-DNA right border; LB: T-DNA left border; tNOS: Nopaline synthetase terminator. cSpec: Spectinomycin resistance gene.

FIG. 11 shows ploidy assay of edited haploid sweet corn line JSER82A056 and

FIG. 12 shows the same for edited haploid sweet corn line JSER82A063. These lines were obtained through crossing with RWKS haploid induction line carrying transgene locus of CRISPR-Cas9 expression vector 23399.

FIG. 13 shows sequencing confirmation of GW2-02 target site editing in haploid sweet corn line JSER82A063. A single base C next to the predicted Cas9 cleavage site was deleted.

FIG. 14 shows ploidy analysis of wild type control, and

FIG. 15 shows ploidy analysis of edited haploid wheat line JSWER30A22.

FIG. 16 shows sequencing confirmation of TaVLHP1-4B target site editing in haploid wheat line JSWER30A22. Lower panel showing 97 bp of TaVLHP1-4B sequence was deleted immediately downstream of the predicted Cas9 cleavage site. The 97 bp deleted sequences were marked by 2 arrows.

FIG. 17 shows pollen expression as measured by pollen collected from transgenic maize T0 plants carrying T-DNA of vector 24038, 24039, 24079, 24091, and 24094, which were used to pollinate emasculated spring wheat line AC-Nanda. The expression was high in the pollen, averaging about 100 fold higher in plants carrying T-DNA vectors 24038, 24039, and 24079 compared to the sugar cane ubiquitin promoter used in many of the corn and wheat examples. The expression was also higher in pollen from plants containing vectors 24038, 24039, and 24079 when compared to the pollen carrying the MATRILINEAL promoter constructs (24094 and 24091).

FIG. 18 shows the ploidy analysis histogram of a diploid control (parent USR01424135). The first peak is located at the "2" position on the logarithmic X-axis. Because this is leaf tissue from *Arabidopsis*, we see multiple peaks indicating the normal level of endoreduplication.

FIG. 19 shows the ploidy analysis histogram of a diploid control (parent USR01431603). The first peak is located at the "2" position on the logarithmic X-axis. Because this is leaf tissue from *Arabidopsis*, we see multiple peaks indicating the normal level of endoreduplication.

FIG. 20 shows the ploidy analysis histogram of a diploid control (parent USR01431609). The first peak is located at the "2" position on the logarithmic X-axis. Because this is leaf tissue from *Arabidopsis*, we see multiple peaks indicating the normal level of endoreduplication.

FIG. 21 shows the ploidy analysis histogram of an edited haploid from plate 1033, well C3 (USR01424135 X Ler-

427). The first peak is located at the “1” position on the logarithmic X-axis, indicating it is a haploid. Because this is leaf tissue from *Arabidopsis*, we see multiple peaks indicating the normal level of endoreduplication.

FIG. 22 shows the ploidy analysis histogram of an edited haploid from plate 1033, well E4 (USR01424135 X Ler-437). The first peak is located at the “1” position on the logarithmic X-axis, indicating it is a haploid. Because this is leaf tissue from *Arabidopsis*, we see multiple peaks indicating the normal level of endoreduplication.

FIG. 23 shows the ploidy analysis histogram of an edited haploid from plate 1046, well H12 (USR01431609 X Ler-123). The first peak is located at the “1” position on the logarithmic X-axis, indicating it is a haploid. Because this is leaf tissue from *Arabidopsis*, we see multiple peaks indicating the normal level of endoreduplication.

FIG. 24 shows the GL1 target site sequence mutations in the parent #USR01424135 and all of the sequenced edited haploids from outcrosses by *Landsberg erecta* pollen. It is clear that the precise edit made is different in the different haploids.

DEFINITIONS

While the following terms are believed to be well understood by one of ordinary skill in the art, the following definitions are set forth to facilitate explanation of the presently disclosed subject matter.

All technical and scientific terms used herein, unless otherwise defined below, are intended to have the same meaning as commonly understood by one of ordinary skill in the art. References to techniques employed herein are intended to refer to the techniques as commonly understood in the art, including variations on those techniques and/or substitutions of equivalent techniques that would be apparent to one of skill in the art. While the following terms are believed to be well understood by one of ordinary skill in the art, the following definitions are set forth to facilitate explanation of the presently disclosed subject matter.

Following long-standing patent law convention, the terms “a,” “an,” and “the” refer to “one or more” when used in this application, including the claims. For example, the phrase “a cell” refers to one or more cells, and in some embodiments can refer to a tissue and/or an organ. Similarly, the phrase “at least one”, when employed herein to refer to an entity, refers to, for example, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20, 25, 30, 35, 40, 45, 50, 75, 100, or more of that entity, including but not limited to all whole number values between 1 and 100 as well as whole numbers greater than 100.

Unless otherwise indicated, all numbers expressing quantities of ingredients, reaction conditions, and so forth used in the specification and claims are to be understood as being modified in all instances by the term “about.” The term “about,” as used herein when referring to a measurable value such as an amount of mass, weight, time, volume, concentration or percentage is meant to encompass variations of in some embodiments $\pm 20\%$, in some embodiments $\pm 10\%$, in some embodiments $\pm 5\%$, in some embodiments $\pm 1\%$, in some embodiments $\pm 0.5\%$, and in some embodiments $\pm 0.1\%$ from the specified amount, as such variations are appropriate to perform the disclosed methods and/or employ the disclosed compositions, nucleic acids, polypeptides, etc. Accordingly, unless indicated to the contrary, the numerical parameters set forth in this specification and attached claims are approximations that can vary depending upon the desired properties sought to be obtained by the presently disclosed subject matter.

As used herein, the term “allele” refers to a variant or an alternative sequence form at a genetic locus. In diploids, a single allele is inherited by a progeny individual separately from each parent at each locus. The two alleles of a given locus present in a diploid organism occupy corresponding places on a pair of homologous chromosomes, although one of ordinary skill in the art understands that the alleles in any particular individual do not necessarily represent all of the alleles that are present in the species.

As used herein, the term “and/or” when used in the context of a list of entities, refers to the entities being present singly or in combination. Thus, for example, the phrase “A, B, C, and/or D” includes A, B, C, and D individually, but also includes any and all combinations and subcombinations of A, B, C, and D (e.g., AB, AC, AD, BC, BD, CD, ABC, ABD, and BCD). In some embodiments, one of more of the elements to which the “and/or” refers can also individually be present in single or multiple occurrences in the combinations(s) and/or subcombination(s).

As used herein, the phrase “associated with” refers to a recognizable and/or assayable relationship between two entities. For example, the phrase “associated with HI” refers to a trait, locus, gene, allele, marker, phenotype, etc., or the expression thereof, the presence or absence of which can influence an extent and/or degree at which a plant or its progeny exhibits HI. As such, a marker is “associated with” a trait when it is linked to it and when the presence of the marker is an indicator of whether and/or to what extent the desired trait or trait form will occur in a plant/germplasm comprising the marker. Similarly, a marker is “associated with” an allele when it is linked to it and when the presence of the marker is an indicator of whether the allele is present in a plant/germplasm comprising the marker. For example, “a marker associated with HI” refers to a marker whose presence or absence can be used to predict whether and/or to what extent a plant will display haploid induction.

The term “comprising,” which is synonymous with “including,” “containing,” and “characterized by,” is inclusive or open-ended and does not exclude additional, unrecited elements and/or method steps. “Comprising” is a term of art that means that the named elements and/or steps are present, but that other elements and/or steps can be added and still fall within the scope of the relevant subject matter.

As used herein, the phrase “consisting of” excludes any element, step, or ingredient not specifically recited. When the phrase “consists of” appears in a clause of the body of a claim, rather than immediately following the preamble, it limits only the element set forth in that clause; other elements are not excluded from the claim as a whole.

As used herein, the phrase “consisting essentially of” limits the scope of the related disclosure or claim to the specified materials and/or steps, plus those that do not materially affect the basic and novel characteristic(s) of the disclosed and/or claimed subject matter.

With respect to the terms “comprising,” “consisting essentially of,” and “consisting of,” where one of these three terms is used herein, the presently disclosed and claimed subject matter can include in some embodiments the use of either of the other two terms. For example, if a subject matter relates in some embodiments to nucleic acids that encode polypeptides comprising amino acid sequences that are at least 95% identical to a SEQ ID NO: 9 or 19. It is understood that the disclosed subject matter thus also encompasses nucleic acids that encode polypeptides that in some embodiments consist essentially of amino acid sequences that are at least 95% identical to that SEQ ID NO: 9 or 19 as well as nucleic acids that encode polypeptides that

in some embodiments consist of amino acid sequences that are at least 95% identical to that SEQ ID NO: 9 or 19. Similarly, it is also understood that in some embodiments the methods for the disclosed subject matter comprise the steps that are disclosed herein, in some embodiments the methods for the presently disclosed subject matter consist essentially of the steps that are disclosed, and in some embodiments the methods for the presently disclosed subject matter consist of the steps that are disclosed herein.

As used herein, the term “de novo haploid induction” refers to the triggering of haploid induction by the introduction of a spontaneous haploid inducing agent. Such introduction can be achieved by topical spray, hand-pollination, mutagenesis, or transgenic methods. The terms “de novo haploid induction,” “de novo HI,” and “haploid induction de novo” are used interchangeably throughout this specification.

As used herein, the term “gene” refers to a hereditary unit including a sequence of DNA that occupies a specific location on a chromosome and that contains the genetic instruction for a particular characteristic or trait in an organism.

A “genetic map” is a description of genetic linkage relationships among loci on one or more chromosomes within a given species, generally depicted in a diagrammatic or tabular form.

As used herein, a plant referred to as “haploid” has a reduced number of chromosomes (n) in the haploid plant, and its chromosome set is equal to that of the gamete. In a haploid organism, only half of the normal number of chromosomes are present. Thus haploids of diploid organisms (e.g., maize) exhibit monoploidy; haploids of tetraploid organisms (e.g., ryegrasses) exhibit diploidy; haploids of hexaploid organisms (e.g., wheat) exhibit triploidy; etc. As used herein, a plant referred to as “doubled haploid” is developed by doubling the haploid set of chromosomes. A plant or seed that is obtained from a doubled haploid plant that is selfed to any number of generations may still be identified as a doubled haploid plant. A doubled haploid plant is considered a homozygous plant. A plant is considered to be doubled haploid if it is fertile, even if the entire vegetative part of the plant does not consist of the cells with the doubled set of chromosomes; that is, a plant will be considered doubled haploid if it contains viable gametes, even if it is chimeric in vegetative tissues.

As used herein, the term “human-induced mutation” refers to any mutation that occurs as a result of either direct or indirect human action. This term includes, but is not limited to, mutations obtained by any method of targeted mutagenesis.

As used herein, “introduced” means delivered, expressed, applied, transported, transferred, permeated, or other like term to indicate the delivery, whether of nucleic acid or protein or combination thereof, of a desired object to an object. For example, nucleic acids encoding a site directed nuclease and optionally at least one guide RNA may be introduced into a haploid embryo upon haploid induction. Likewise, extant editing machinery (comprising a site directed nuclease protein and optionally at least one guide RNA) may be introduced to a haploid embryo upon application of appropriate cell-penetrating peptides.

As used herein, the terms “marker probe” and “probe” refer to a nucleotide sequence or nucleic acid molecule that can be used to detect the presence or absence of a sequence within a larger sequence, e.g., a nucleic acid probe that is complementary to all of or a portion of the marker or marker locus, through nucleic acid hybridization. Marker probes

comprising about 8, 10, 15, 20, 30, 40, 50, 60, 70, 80, 90, 100 or more contiguous nucleotides can be used for nucleic acid hybridization.

As used herein, the term “molecular marker” can be used to refer to a genetic marker, as defined above, or an encoded product thereof (e.g., a protein) used as a point of reference when identifying the presence/absence of a HI-associated locus. A molecular marker can be derived from genomic nucleotide sequences or from expressed nucleotide sequences (e.g., from an RNA, a cDNA, etc.). The term also refers to nucleotide sequences complementary to or flanking the marker sequences, such as nucleotide sequences used as probes and/or primers capable of amplifying the marker sequence. Nucleotide sequences are “complementary” when they specifically hybridize in solution (e.g., according to Watson-Crick base pairing rules). This term also refers to the genetic markers that indicate a trait by the absence of the nucleotide sequences complementary to or flanking the marker sequences, such as nucleotide sequences used as probes and/or primers capable of amplifying the marker sequence.

As used herein, the terms “nucleotide sequence,” “polynucleotide,” “nucleic acid sequence,” “nucleic acid molecule,” and “nucleic acid fragment” refer to a polymer of RNA or DNA that is single- or double-stranded, optionally containing synthetic, non-natural, and/or altered nucleotide bases. A “nucleotide” is a monomeric unit from which DNA or RNA polymers are constructed and consists of a purine or pyrimidine base, a pentose, and a phosphoric acid group. Nucleotides (usually found in their 5'-monophosphate form) are referred to by their single letter designation as follows: “A” for adenylate or deoxyadenylate (for RNA or DNA, respectively), “C” for cytidylate or deoxycytidylate, “G” for guanylate or deoxyguanylate, “U” for uridylylate, “T” for deoxythymidylate, “R” for purines (A or G), “Y” for pyrimidines (C or T), “K” for G or T, “H” for A or C or T, “I” for inosine, and “N” for any nucleotide.

As used herein, the term “nucleotide sequence identity” refers to the presence of identical nucleotides at corresponding positions of two polynucleotides. Polynucleotides have “identical” sequences if the sequence of nucleotides in the two polynucleotides is the same when aligned for maximum correspondence (e.g., in a comparison window). Sequence comparison between two or more polynucleotides is generally performed by comparing portions of the two sequences over a comparison window to identify and compare local regions of sequence similarity. The comparison window is generally from about 20 to 200 contiguous nucleotides. The “percentage of sequence identity” for polynucleotides, such as about 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 98, 99 or 100 percent sequence identity, can be determined by comparing two optimally aligned sequences over a comparison window, wherein the portion of the polynucleotide sequence in the comparison window can include additions or deletions (i.e., gaps) as compared to the reference sequence for optimal alignment of the two sequences. In some embodiments, the percentage is calculated by: (a) determining the number of positions at which the identical nucleic acid base occurs in both sequences; (b) dividing the number of matched positions by the total number of positions in the window of comparison; and (c) multiplying the result by 100. Optimal alignment of sequences for comparison can also be conducted by computerized implementations of known algorithms, or by visual inspection. Readily available sequence comparison and multiple sequence alignment algorithms are, respectively, the Basic Local Alignment Search Tool (BLAST) and ClustalW/ClustalW2/Clustal

Omega programs available on the Internet (e.g., the website of the EMBL-EBI). Other suitable programs include, but are not limited to, GAP, BestFit, Plot Similarity, and FASTA, which are part of the Accelrys GCG Package available from Accelrys, Inc. of San Diego, California, U.S.A. See also Smith & Waterman, 1981; Needleman & Wunsch, 1970; Pearson & Lipman, 1988; Ausubel et al., 1988; and Sambrook & Russell, 2001.

One example of an algorithm that is suitable for determining percent sequence identity and sequence similarity is the BLAST algorithm, which is described in Altschul et al., 1990. In some embodiments, a percentage of sequence identity refers to sequence identity over the full length of one of the gDNA, cDNA, or the predicted protein sequences in the largest ORF of SEQ ID No: 1 being compared. In some embodiments, a calculation to determine a percentage of nucleic acid sequence identity does not include in the calculation any nucleotide positions in which either of the compared nucleic acids includes an "N" (i.e., where any nucleotide could be present at that position).

The term "open reading frame" (ORF) refers to a nucleic acid sequence that encodes a polypeptide. In some embodiments, an ORF comprises a translation initiation codon (i.e., start codon), a translation termination (i.e., stop codon), and the nucleic acid sequence there between that encodes the amino acids present in the polypeptide. The terms "initiation codon" and "termination codon" refer to a unit of three adjacent nucleotides (i.e., a codon) in a coding sequence that specifies initiation and chain termination, respectively, of protein synthesis (mRNA translation).

Patatin-like phospholipase A2a may also be known as PLA, pPLA, pPLAIIA pPLAII α , PLA2 α , or PLA2, or other similar variation. Patatin-like phospholipase AII α is also referred to as MATRILINEAL (MATL). These terms are used interchangeably throughout. A MATRILINEAL gene comprising a four basepair frameshift mutation is referred to as matrilineal (matl).

As used herein, the terms "phenotype," "phenotypic trait" or "trait" refer to one or more traits of a plant or plant cell. The phenotype can be observable to the naked eye, or by any other means of evaluation known in the art, e.g., microscopy, biochemical analysis, or an electromechanical assay. In some cases, a phenotype is directly controlled by a single gene or genetic locus (i.e., corresponds to a "single gene trait"). In the case of haploid induction use of color markers, such as R Navajo, and other markers including transgenes visualized by the presences or absences of color within the seed evidence if the seed is an induced haploid seed. The use of R Navajo as a color marker and the use of transgenes is well known in the art as means to detect induction of haploid seed on the female plant. In other cases, a phenotype is the result of interactions among several genes, which in some embodiments also results from an interaction of the plant and/or plant cell with its environment.

As used herein, the term "plant" can refer to a whole plant, any part thereof, or a cell or tissue culture derived from a plant. Thus, the term "plant" can refer to any of: whole plants, plant components or organs (e.g., leaves, stems, roots, etc.), plant tissues, seeds and/or plant cells.

A plant cell is a cell of a plant, taken from a plant, or derived through culture from a cell taken from a plant. Thus, the term "plant cell" includes without limitation cells within seeds, suspension cultures, embryos, meristematic regions, callus tissue, leaves, shoots, gametophytes, sporophytes, pollen, and microspores. The phrase "plant part" refers to a part of a plant, including single cells and cell tissues such as plant cells that are intact in plants, cell clumps, and tissue

cultures from which plants can be regenerated. Examples of plant parts include, but are not limited to, single cells and tissues from pollen, ovules, leaves, embryos, roots, root tips, anthers, flowers, fruits, stems, shoots, and seeds; as well as scions, rootstocks, protoplasts, calli, and the like.

As used herein, the term "primer" refers to an oligonucleotide which is capable of annealing to a nucleic acid target (in some embodiments, annealing specifically to a nucleic acid target) allowing a DNA polymerase and/or reverse transcriptase to attach thereto, thereby serving as a point of initiation of DNA synthesis when placed under conditions in which synthesis of a primer extension product is induced (e.g., in the presence of nucleotides and an agent for polymerization such as DNA polymerase and at a suitable temperature and pH). In some embodiments, one or more pluralities of primers are employed to amplify plant nucleic acids (e.g., using the polymerase chain reaction; PCR).

As used herein, the term "probe" refers to a nucleic acid (e.g., a single stranded nucleic acid or a strand of a double stranded or higher order nucleic acid, or a subsequence thereof) that can form a hydrogen-bonded duplex with a complementary sequence in a target nucleic acid sequence. Typically, a probe is of sufficient length to form a stable and sequence-specific duplex molecule with its complement, and as such can be employed in some embodiments to detect a sequence of interest present in a plurality of nucleic acids.

As used herein, the terms "progeny" and "progeny plant" refer to a plant generated from vegetative or sexual reproduction from one or more parent plants. In gynogenesis-mediated haploid induction, the haploid embryo on the female parent comprises female chromosomes to the exclusion of male chromosomes—thus it is not a progeny of the male haploid-inducing line. The haploid corn seed typically still has normal triploid endosperm that contains the male genome. The edited haploid progeny and subsequent edited doubled haploid plants and subsequent seed is not the only desired progeny. There is also the seed from the haploid inducer line itself, often carrying the Cas9 transgene, and subsequent plant and seed progeny of the haploid inducing plant. Both the haploid seed and the haploid inducer (self-pollination-derived) seed can be progeny. A progeny plant can be obtained by cloning or selfing a single parent plant, or by crossing two or more parental plants. For instance, a progeny plant can be obtained by cloning or selfing of a parent plant or by crossing two parental plants and include selfings as well as the F1 or F2 or still further generations. An F1 is a first-generation progeny produced from parents at least one of which is used for the first time as donor of a trait, while progeny of second generation (F2) or subsequent generations (F3, F4, and the like) are specimens produced from selfings, intercrosses, backcrosses, and/or other crosses of F1s, F2s, and the like. An F1 can thus be (and in some embodiments is) a hybrid resulting from a cross between two true breeding parents (i.e., parents that are true-breeding are each homozygous for a trait of interest or an allele thereof), while an F2 can be (and in some embodiments is) a progeny resulting from self-pollination of the F1 hybrids.

As used herein, the phrase "recombination" refers to an exchange of DNA fragments between two DNA molecules or chromatids of paired chromosomes (a "crossover") over in a region of similar or identical nucleotide sequences. A "recombination event" is herein understood to refer in some embodiments to a meiotic crossover.

As used herein, the term "reference sequence" refers to a defined nucleotide sequence used as a basis for nucleotide sequence comparison. In some embodiments, any of SEQ ID

NOs: 2, 4, 8, 21, 23, 25, 29, 32, and 33 can serve as a reference sequence for comparing to other sequences obtained from plants.

As used herein, the term “regenerate,” and grammatical variants thereof, refers to the production of a plant from tissue culture.

As used herein, the phrase “stringent hybridization conditions” refers to conditions under which a polynucleotide hybridizes to its target subsequence, typically in a complex mixture of nucleic acids, but to essentially no other sequences. Stringent conditions are sequence-dependent and can be different under different circumstances.

Longer sequences typically hybridize specifically at higher temperatures. An extensive guide to the hybridization of nucleic acids is found in Sambrook & Russell, 2001. Generally, stringent conditions are selected to be about 5-10° C. lower than the thermal melting point T_m for the specific sequence at a defined ionic strength, pH, and nucleic acid concentration) at which 50% of the probes complementary to the target hybridize to the target sequence at equilibrium (as the target sequences are present in excess, at T_m , 50% of the probes are occupied at equilibrium). Exemplary stringent conditions are those in which the salt concentration is less than about 1.0 M sodium ion, typically about 0.01 to 1.0 M sodium ion concentration (or other salts) at pH 7.0 to 8.3 and the temperature is at least about 30° C. for short probes (e.g., 10 to 50 nucleotides) and at least about 60° C. for long probes (e.g., greater than 50 nucleotides).

Stringent conditions can also be achieved with the addition of destabilizing agents such as formamide. Additional exemplary stringent hybridization conditions include 50% formamide, 5×SSC, and 1% SDS incubating at 42° C.; or SSC, 1% SDS, incubating at 65° C.; with one or more washes in 0.2×SSC and 0.1% SDS at 65° C. For PCR, a temperature of about 36° C. is typical for low stringency amplification, although annealing temperatures can vary between about 32° C. and 48° C. (or higher) depending on primer length. Additional guidelines for determining hybridization parameters are provided in numerous references (see e.g., Ausubel et al., 1999).

As used herein, the term “trait” refers to a phenotype of interest, a gene that contributes to a phenotype of interest, as well as a nucleic acid sequence associated with a gene that contributes to a phenotype of interest. For example, a “HI trait” refers to a haploid induction phenotype as well as a gene (e.g., *mat1* in maize or Os03g27610 in rice) that contributes to a haploid induction and a nucleic acid sequence (e.g., a HI-associated gene product) that is associated with the presence or absence of the haploid induction phenotype.

As used herein, the term “transgene” refers to a nucleic acid molecule introduced into an organism or one or more of its ancestors by some form of artificial transfer technique. The artificial transfer technique thus creates a “transgenic organism” or a “transgenic cell.” It is understood that the artificial transfer technique can occur in an ancestor organism (or a cell therein and/or that can develop into the ancestor organism) and yet any progeny individual that has the artificially transferred nucleic acid molecule or a fragment thereof is still considered transgenic even if one or more natural and/or assisted breedings result in the artificially transferred nucleic acid molecule being present in the progeny individual.

As used herein, the term “targeted mutagenesis” or “mutagenesis strategy” refers to any method of mutagenesis that results in the intentional mutagenesis of a chosen gene.

Targeted mutagenesis includes the methods CRISPR, TILLING, TALEN, and other methods not yet discovered but which may be used to achieve the same outcome.

As used herein, haploid induction rate (“HIR”) means the number of surviving haploid kernels over the total number of kernels after an ear is pollinated with haploid inducer pollen.

Particular problems plague that haploid induction: increased embryo abortion rates and increased fertilization failure rates (reduced seed set rates). For these reasons, there exists a need to successfully determine the cause of HI, and to use that knowledge to determine methods of stably or increasingly creating haploid plants while simultaneously reducing fertilization failure and embryo abortions.

It is specifically contemplated that one could mutagenize a promoter to potentially improve the utility of the elements for the expression of transgenes in plants. The mutagenesis of these elements can be carried out at random and the mutagenized promoter sequences screened for activity in a trial-by-error procedure. Alternatively, particular sequences which provide the promoter with desirable expression characteristics, or the promoter with expression enhancement activity, could be identified and these or similar sequences introduced into the promoter via mutation. It is further contemplated that one could mutagenize these sequences in order to enhance their expression of transgenes in a particular species. The means for mutagenizing a DNA segment encoding a promoter sequence of the current invention are well-known to those of skill in the art. As indicated, modifications to promoter or other regulatory element may be made by random, or site-specific mutagenesis procedures. The promoter and other regulatory element may be modified by altering their structure through the addition or deletion of one or more nucleotides from the sequence which encodes the corresponding unmodified sequences.

Mutagenesis may be performed in accordance with any of the techniques known in the art, such as, and not limited to, synthesizing an oligonucleotide having one or more mutations within the sequence of a particular regulatory sequence. In particular, site-specific mutagenesis is a technique useful in the preparation of promoter mutants, through specific mutagenesis of the underlying DNA. RNA-guided endonucleases (“RGEN,” e.g., CRISPR/Cas9) may also be used. The technique further provides a ready ability to prepare and test sequence variants, for example, incorporating one or more of the foregoing considerations, by introducing one or more nucleotide sequence changes into the DNA. Site-specific mutagenesis allows the production of mutants through the use of specific oligonucleotide sequences which encode the DNA sequence of the desired mutation, as well as a sufficient number of adjacent nucleotides, to provide a primer sequence of sufficient size and sequence complexity to form a stable duplex on both sides of the deletion junction being traversed. Typically, a primer of about 17 to about 75 nucleotides or more in length is preferred, with about 10 to about 25 or more residues on both sides of the junction of the sequence being altered.

Where a clone comprising a promoter has been isolated in accordance with the instant invention, one may wish to delimit the essential promoter regions within the clone. One efficient, targeted means for preparing mutagenized promoters relies upon the identification of putative regulatory elements within the promoter sequence. This can be initiated by comparison with promoter sequences known to be expressed in similar tissue specific or developmentally unique patterns. Sequences which are shared among promoters with similar expression patterns are likely candidates

for the binding of transcription factors and are thus likely elements which confer expression patterns. Confirmation of these putative regulatory elements can be achieved by deletion analysis of each putative regulatory sequence followed by functional analysis of each deletion construct by assay of a reporter gene which is functionally attached to each construct. As such, once a starting promoter sequence is provided, any of a number of different deletion mutants of the starting promoter could be readily prepared.

The invention disclosed herein provides polynucleotide molecules comprising regulatory element fragments that may be used in constructing novel chimeric regulatory elements. Novel combinations comprising fragments of these polynucleotide molecules and at least one other regulatory element or fragment can be constructed and tested in plants and are considered to be within the scope of this invention. Thus the design, construction, and use of chimeric regulatory elements is one embodiment of this invention. Promoters of the present invention include homologues of cis elements known to affect gene regulation that show homology with the promoter sequences of the present invention.

Functional equivalent fragments of one of the transcription regulating nucleic acids described herein comprise at least 50, 100, 150, 200, 250, 300, 350, 400, 450, 500, 550, 600, 650, 700, 750, 800, 850, 900, 950, or 1000 base pairs of a transcription regulating nucleic acid. Equivalent fragments of transcription regulating nucleic acids, which are obtained by deleting the region encoding the 5'-untranslated region of the mRNA, would then only provide the (untranscribed) promoter region. The 5'-untranslated region can be easily determined by methods known in the art (such as 5'-RACE analysis). Accordingly, some of the transcription regulating nucleic acids, described herein, are equivalent fragments of other sequences.

As indicated above, deletion mutants of the promoter of the invention also could be randomly prepared and then assayed. Following this strategy, a series of constructs are prepared, each containing a different portion of the promoter (a subclone), and these constructs are then screened for activity. A suitable means for screening for activity is to attach a deleted promoter or intron construct which contains a deleted segment to a selectable or screenable marker, and to isolate only those cells expressing the marker gene. In this way, a number of different, deleted promoter constructs are identified which still retain the desired, or even enhanced, activity. The smallest segment which is required for activity is thereby identified through comparison of the selected constructs. This segment may then be used for the construction of vectors for the expression of exogenous genes.

An expression cassette as described herein may comprise further regulatory elements. The term in this context is to be understood in the broad meaning comprising all sequences which may influence construction or function of the expression cassette. Regulatory elements may, for example, modify transcription and/or translation in prokaryotic or eukaryotic organisms. The expression cassette described herein may be downstream (in 3' direction) of the nucleic acid sequence to be expressed and optionally contain additional regulatory elements, such as transcriptional or translational enhancers. Each additional regulatory element may be operably linked to the nucleic acid sequence to be expressed (or the transcription regulating nucleotide sequence). Additional regulatory elements may comprise additional promoters, minimal promoters, promoter elements, or transposon elements which may modify or enhance the expression regulating properties. The expres-

sion cassette may also contain one or more introns, one or more exons and one or more terminators.

Furthermore, it is contemplated that promoters combining elements from more than one promoter may be useful. For example, U.S. Pat. No. 5,491,288 discloses combining a Cauliflower Mosaic Virus promoter with a histone promoter. Thus, the elements from the promoters disclosed herein may be combined with elements from other promoters. Promoters which are useful for plant transgene expression include those that are inducible, viral, synthetic, constitutive (Odell Nature 313: 810-812 (1985)), temporally regulated, spatially regulated, tissue specific, and spatially temporally regulated. Using the regulatory elements described herein, numerous agronomic genes can be expressed in transformed plants. More particularly, plants can be genetically engineered to express various phenotypes of agronomic interest.

DETAILED DESCRIPTION

One embodiment of the invention provides a method of editing plant genomic DNA, comprising: (i) obtaining a first plant, wherein the first plant is a haploid inducer line of the plant, and wherein said first plant is capable of expressing a DNA modification enzyme and optionally a guide nucleic acid; (ii) obtaining a second plant, wherein the second plant comprises the plant genomic DNA which is to be edited; (iii) pollinating the second plant with pollen from the first plant; and (iv) selecting at least one haploid progeny produced by the pollination of step (i) wherein the haploid progeny comprises the genome of the second plant but not the first plant, and the genome of the haploid progeny has been modified by the DNA modification enzyme and optional guide nucleic acid delivered by the first plant.

In one aspect of the method, the DNA modification enzyme is a site-directed nuclease selected from the group consisting of meganucleases (MNs), zinc-finger nucleases (ZFNs), transcription-activator like effector nucleases (TALENs), Cas9 nuclease, Cpf1 nuclease, dCas9-FokI, dCpf1-FokI, chimeric Cas9-cytidine deaminase, chimeric Cas9-adenine deaminase, chimeric FEN1-FokI, and Mega-TALs, a nickase Cas9 (nCas9), chimeric dCas9 non-FokI nuclease and dCpf1 non-FokI nuclease; and further wherein the guide nucleic acid is a guide RNA.

In another aspect of the method, the edited haploid progeny is treated with a chromosome doubling agent, thereby creating an edited doubled haploid progeny. For example, the chromosome doubling agent is colchicine, pronamide, dithiopyr, trifluralin, or another known anti-microtubule agent.

In another aspect of the method, the first plant is a monocot or a dicot. For example, the first plant is a monocot selected from the group consisting of maize, wheat, rice, barley, oats, triticale, sorghum, pearl millet, teosinte, bamboo, sugar cane, asparagus, onion, and garlic. In another aspect, the second plant is a monocot or a dicot. For example the second plant is a monocot selected from the group consisting of maize, wheat, rice, barley, oats, triticale, sorghum, pearl millet, teosinte, bamboo, sugar cane, asparagus, onion, and garlic.

In another aspect of the method, the optional guide RNA is an 18-21 nucleotide sequence and is homologous to a sequence selected from the group consisting of SEQ ID NOs: 2, 4, 8, 21, 23, 25, 29, 32, and 33. In another aspect, the first plant expresses a marker gene. For example, the marker gene is selected from the group consisting of GUS, PMI, PAT, GFP, RFP, CFP, B1, C1, R-nj, anthocyanin pigments, and any other marker gene.

In another aspect of the method, the first plant is a maize plant selected and/or derived from the lines Stock 6, RWK, RWS, UH400, AX5707RS, NP2222-matl, or any of the several other known HI lines.

In one embodiment, the first plant and the second plant are different species. In one aspect, first plant is a wheat plant and the second plant is a maize plant. In another aspect, the first plant is a maize plant and the second plant is a wheat plant.

One object of the invention is a gene-edited plant produced by the method provided.

In another embodiment, the invention provides a method of editing plant genomic DNA, comprising: (i) obtaining a first plant, wherein said first plant is capable of expressing a DNA modification enzyme and optionally a guide nucleic acid; (ii) obtaining a second plant, wherein the second plant comprises the plant genomic DNA which is to be edited; (iii) pollinating the second plant with pollen from the first plant; (iv) applying a composition comprising a lipid or a phospholipase inhibitor immediately preceding, during, or following the pollination of step (iii); and (v) selecting at least one haploid progeny produced by the pollination of step (iii) wherein the haploid progeny comprises the genome of the second plant but not the first plant, and the genome of the haploid progeny has been modified by the DNA modification enzyme and optional guide nucleic acid delivered by the first plant. In one aspect, the composition comprises methyl alpha-linolenoyl fluorophosphonate (MALFP), linoleic acid ethyl ester (LLAEE), linoleic acid (LLA), corn oil, distearoyl-phosphatidyl choline (DSPC), methyl arachidonyl fluorophosphonate (MAFP), Palmityl trifluoromethylketone (PACOCF₃), Arachidonyl trifluoromethylketone (AA-COCF₃), Manoalide, Linolenic acid ethyl ester (LNAEE), Linolenic acid ethyl ester (LNAEE), Oleic acid methyl ester (OAME), Oleic acid ethyl ester (OAE), Palmitic acid ethyl ester (PAEE), Palmitoleic acid ethyl ester (PLAEE), Linseed oil, corn oil, alpha-Linolenic acid (aLNA), gamma-Linolenic acid (gLNA), Oleic acid, Arachidonic acid, Stearic Acid, 9(Z)-11E-conjugated Linoleic acid, or 2-oleoyl-1-palmitoyl-sn-glycero-3-phospho-ethanolamine.

In another embodiment, the invention provides a method of editing plant genomic DNA, comprising: (i) obtaining a first plant, wherein said first plant is capable of expressing a DNA modification enzyme and optionally a guide nucleic acid; (ii) obtaining a second plant, wherein the second plant comprises the plant genomic DNA which is to be edited; (iii) crossing the first plant with the second plant; and (iv) selecting at least one haploid progeny produced by the crossing of step (iii) wherein the haploid progeny comprises the genome of the second plant but not the first plant, and the genome of the haploid progeny has been modified by the DNA modification enzyme and optional guide nucleic acid delivered by the first plant. In one aspect, the first plant acts as the female parent in the cross of step (iii). In another aspect, the first plant comprises a mutation in a CENH3 gene, an ig1 gene, or another mutation conferring paternal-haploid inducing systems.

EXAMPLES

I. Producing New Haploid Inducer Lines Comprising the Editing Machinery.

We transformed a transformable line of maize called NP2222 with a TALEN construct, and separately transformed this line with a Cas9 and guide RNA construct. The TALEN construct (pBSC22808 (SEQ ID NO: 5), with TALENs targeting cleavage within target sequence,

5'-TCCAGGGTCAACGTGGAGACAGGGAGGTACG-AACCGGTGACTGGCGAAGGAAGCA-3', SEQ ID NO: 6; TALEN recognition sequence underlined) and the Cas9 construct (pBSC23123 (SEQ ID NO: 7) with guide RNA sequence of xZmPLAIIA, 5'-GGGTCAACGTGGA-GACAGGG-3', SEQ ID NO: 8) were designed to target mutations into the fourth exon of maize gene called MATRILINEAL (MATL; GRAMENE ID: GRMZM2G471240). This gene, when mutated at the target site by the TALEN or by the Cas9 and guide RNA, is knocked out, resulting in a loss of function of the protein product. We previously established that lines that are homozygous for loss of function mutations in MATL are haploid inducer lines, meaning that when they are used as pollen donors in crosses, they induce the formation of haploids on the resulting ears (see P.C.T. Patent Application No. PCT/US2016/62548, filed Nov. 17, 2016, incorporated herein by reference in its entirety).

We produced several events and self-pollinated them to make T1 seed. We grew up T1 individuals from event MZET152408A042A. We recovered five T1 progeny that retained two copies of the Cas9 and guide RNA editing machinery stably transformed, and were also homozygous mutant for the MATL gene. See Table 1.

TABLE 1

New HI lines comprising the genome editing machinery.			
New HI Line Individual ID	wt MATL Presence	Cas9 Presence	Mutation in MATL
USR01283349	-	+	13 bp deletion, homozygous
USR01283378	-	+	13 bp deletion, homozygous
USR01283388	-	+	8 bp deletion, homozygous
USR01283391	-	+	8 bp deletion, homozygous
USR01283398	-	+	13 bp deletion, homozygous

The MATL mutations are detected using a TaqMan assay, which amplifies the wildtype copy of MATL (referred to herein as MATL or wt-MATL; these terms are used interchangeably throughout). When both copies of MATL are mutated, this assays reads negative (i.e., "-"). The Cas9 and guide RNA editing machinery were stably inserted via Construct 23123 (SEQ ID NO: 7). We sequenced the mutations in MATL via PCR and subcloning. Four colonies of each PCR product was sequenced, and all of the colonies for a given individual had the same sequence, indicating these plants are all homozygous mutant for the MATL allele (also referred to herein as matl when referencing the 4 basepair insertion in MATRILINEAL found in Stock6 and other Stock6-derived lines, or pMATL when referencing any other human-induced mutation in MATRILINEAL). There were two plants that had 8 bp deletions, and three plants that had 13 bp deletions.

II. Using the New HI Lines as Male Parents and Progeny Analysis.

We crossed the above new HI plants as male pollen donors to a female tester line, which contained a recessive color marker but were wild type for the MATL gene. The male haploid inducer line is homozygous wild type for the same color marker. This female line was thus a non-haploid inducer and were homozygous wild-type for the MATL gene but homozygous mutant for the color marker. We recovered seeds from the crosses, and germinated seedlings therefrom.

Progeny seedlings were subjected to several assays. Progeny seedlings were scored as diploids if they do not exhibit the color marker (because the recessive marker is comple-

mented by the male inducer DNA). Progeny seedlings were scored as putative haploids if they do exhibit the color marker because the recessive marker is not complemented. Of the 2656 seeds planted, we used the color assay and identified 90 seedlings as putative haploids.

We further analyzed the 90 putative haploids for presence of the wildtype MATL gene using a Taqman marker assay. Of these, 82 were positive for MATL, meaning they were not edited by the editing machinery provided by the male parent.

true haploid, putative edited individuals, specifically focusing on the sequence flanking the guide RNA target mutagenesis site. This was to determine the nature of the edits that may or may not have occurred there. We sub-cloned the PCR fragment using commercially-available TOPO Blunt IV kit, and sequenced at least four colonies each (forward and reverse sequencing). See Table 2, below, for comparisons of the edited alleles and the reference wt-MATL allele.

TABLE 2

Comparing the Edited Alleles against wt-MATL.			
Individual ID	Allele Type	Sequence (corresponds to 1126-1166 of SEQ ID NO: 19)	SEQ ID NO:
NP2222	vit-MATL	AGGGTCAACGTGGAGACAGGGAGGTACGAACCGGTGACTGG	9
Stock6	matl	AGGGTCAACGTGGAGACAGGGAGGTACGAACCGGTGACTGG	10
USR01350333-3 Allele 1	edited	AGGGTCAACGTGGAGACA <u>AGGG</u> GAGGTACGAACCGGTGACTGG	11
USR01350333-3 Allele 2	PCR contamination	AGGGTCAACGTGGA::::::::::::GAACCGGTGACTGG	12
USR01350344-2 Allele 1	edited	AGGGTCAACGTGGAGAC:GGGAGGTACGAACCGGTGACTGG	13
USR01350344-2 Allele 2	PCR contamination	AGGGTCAACGTGGA::::::::::::GAACCGGTGACTGG	14
USR01350343-1 Allele 1	edited	AGGGTCAACGTGGAGACA <u>AGGG</u> GAGGTACGAACCGGTGACTGG	15
USR01350328-1 Allele 1	edited	AGGGTCAACGTGGAGAC:GGGAGGTACGAACCGGTGACTGG	16
USR01350337-2 Allele 1	not edited	AGGGTCAACGTGGAGACAGGGAGGTACGAACCGGTGACTGG	17
USR01350337-2 Allele 2	PCR contamination	AGGGTCAACGTGGA::::::::::::GAACCGGTGACTGG	18

The remaining 8 putative haploid seedlings were negative for wildtype MATL using the Taqman marker, indicating that they may have been edited by the editing machinery provided by the male parent.

We performed ploidy analysis via Flow Cytometry on these 8 putative, edited haploid seedlings using leaf tissue in a ploidy analyzer. See FIGS. 1-8. We found that four of them were true haploids, while the others were actually diploids. As we discuss below, we ran PCR and sequenced the mutations in the MATL gene in these four true haploids as well as for plant USR01350337-2 which, according to the MATL Taqman assay, was not edited by the genome editing machinery.

The finding that there were four diploids among the 90 putative haploids was not unexpected—the seedling assay is not perfect and there are occasional false positives. We tested the 90 haploids for the presence of the Cas9 construct (Construct 23123), and found it was missing in 86 out of 90, including the four true haploids above. In contrast, the four edited diploids that we found during the ploidy analysis all had the Cas9 construct present, confirming their status as hybrid diploids that were falsely identified by the haploid seedling assay as being haploids.

We then used the leaf tissue to isolate genomic DNA and ran a PCR reaction to sequence the MATL gene in those four

Individual USR01350333-3 produced an edited MATL allele with an insertion of alanine at basepair 1143 of the cDNA sequence (underlined in Table 2). This would be sufficient to cause a frameshift in the coding sequence, which would produce a premature STOP codon. What we previously thought was Edited Allele #2 of USR01350333-3 (a 13 basepair deletion of GACAAGGGAGGTAC) was actually the result of PCR contamination. After resequencing, we confirmed that this plant only has one edited allele, and it was found in 6 out of 6 colonies.

This allele is novel in that it is not in either the male or the female parent plant of this individual. The male parent ID for this individual was USR01283391, and that plant was found to be homozygous for an 8 bp deletion.

Individual USR01350344-2 provides a deletion of A (a deletion of basepair 1143 of the wild type cDNA sequence). This mutation would be sufficient to cause a frame-shift in the coding sequence, and produce a premature STOP codon. After resequencing and discovering the PCR contamination, we confirmed this was found in 6 out of 6 colonies. Previously identified as Edited Allele #2 of USR01350344-2, this was identified as PCR contamination.

Individual USR01350343-1 provides an insertion of A at basepair 1143 of the cDNA sequence. This would be sufficient to cause a frame-shift in the coding sequence, and produce a premature STOP codon. This was found in 4 out of 4 colonies.

Individual USR01350328-1 provides a deletion of A (a deletion of basepair 1143 from the wild type cDNA sequence). This mutation would be sufficient to cause a frame-shift in the coding sequence, and produce a premature STOP codon. It was found in 4 out of 4 colonies.

Individual USR01350337-2 had no change: its sequence was 100% identical to that of wt-MATL.

In summary, we found that 4 out of 86 confirmed haploids had mutations in the MATL gene. We have confirmed that these plants are haploids and do not contain any Cas9 DNA. It is clear that the Cas9 transgene has been eliminated along with the rest of the male-derived DNA during embryogenesis, and that edits have occurred to the female (egg cell-derived) genome in the process of embryogenesis.

We know that the edits are novel and occurred in the female genome in the process of embryogenesis because the haploid inducer line typically makes maternal haploids and we have confirmed that these are indeed haploids. One might try to argue that there is a chance that these are actually paternal haploids, and that the edits we are seeing are actually edits that were already present in the paternal DNA. However, we can prove that this is not the case. First, the mutations do not match those of the paternal parent. This can clearly be seen in Table 3 and 4 (shown below). The edited haploid plant USR01350343-1 was homozygous for an insertion of a single nucleotide (an "A"), but the male parent plant had a deletion of 13 nucleotides. Similarly, plant USR01350328-1 was homozygous for a deletion of an A, but the male parent had a deletion of 13 nucleotides. These examples, taken together, prove that during the haploid induction process, it is possible to have editing of the maternal genome occur, resulting in the formation of edited maternal haploids. According to these and based on the

embryo sac, the egg cell, in the pollen, or in sperm cells might be more effective, particularly in the case of wide crosses, in which the male DNA is eliminated in a much more robust and rapid fashion than in intraspecific haploid inducer systems like the maize haploid inducer system or CENH3 type haploid inducer systems. In other words, during a wide cross, for instance when crossing maize pollen on to wheat ears, which is done in order to induce wheat maternal haploids, it might work best to have the editing machinery in the maize pollen driven by a promoter that has strong pollen or sperm cell expression, perhaps in addition to zygote expression, so that abundant editing machinery (RNA and protein) is delivered and present in the zygote cell and during the subsequent two, four, or eight cell embryo stage, even if the male DNA is eliminated or lost very quickly.

TABLE 3

Haploid Progeny Produced				
Individual Progeny ID code	wt MATL Presence	Ploidy Analysis	Cas9 Presence	Allele 1
USR01350333-3	-	Haploid	-	insertion of A
USR01350344-2	-	Haploid	-	deletion of A
USR01350343-1	-	Haploid	-	insertion of an A
USR01350328-1	-	Haploid	-	deletion of A
USR01350337-2	+	Haploid	-	no mutation
USR01350334-3	-	Diploid	+	
USR01350333-10	-	Diploid	+	
USR01350341-1	-	Diploid	+	
USR01350321-3	-	Diploid	+	

TABLE 4

Male Parent Information and Their Progeny				
Male Parent ID	wt MATL Presence	Sequencing (# colonies)	Cas9 Presence	Progeny ID
USR01283391	-	deletion of 8 nt (4)	+	USR01350333-3 and USR01350333-10
USR01283349	-	deletion of 13 nt (4)	+	USR01350344-2, USR01350328-1 and USR01350321-3
USR01283378	-	deletion of 13 nt (4)	+	USR01350343-1 and USR01350341-1
USR01283398	-	deletion of 13 nt (4)	+	USR01350337-2
USR01283388	-	deletion of 8 nt (4)	+	USR01350334-3

assay detecting MATL presence and the confirmation via ploidy analysis, and using the Cas9 transgene on the male side under control of the maize ubiquitin promoter, the rate of editing during the haploid induction process is about 4/86, or 4.65%.

Furthermore, the rate of editing during haploid induction may be very different when using different haploid inducer lines or using wide crosses. It appears that both haploid induction in maize using MATL mutant lines and wide crosses in barley, wheat, or other crops all work via similar mechanisms: fertilization is followed by genome elimination. It also appears that the time period between fertilization and genome elimination is long enough for the editing machinery to edit the target gene in the genome of the line to which the inducer line has been hybridized (the target germplasm). It is noted that the choice of promoter driving expression of the stably transformed editing proteins system may have a large impact on the rate of editing in haploids. We used a constitutive sugarcane promoter (prSoUbi4) but other promoters driving high or specific expression in the

III. Simultaneous Haploid Induction and Editing in Elite Maize Inbred Lines.

A transformable haploid inducer line, NP2222-HI, RWK, RWS, or UH400 or Stock6 or any other haploid inducer line, all of which already have the mutant versions of MATL, is stably transformed with construct expressing genome modification system such as Cas9+guide RNA (Cong, L. et al. 2013. Multiplex genome engineering using CRISPR/Cas systems. *Science* 339, 819-823), dCas9-FokI+ guide RNA (Tsai, S. Q. et al. 2014, Dimeric CRISPR RNA-guided FokI nucleases for highly specific genome editing. *Nature Biotechnol.* 32, 569-576), TALEN (Li et al., 2012, High-efficiency TALEN-based gene editing produces disease-resistant rice. *Nature Biotech.* 30, 390-392), engineered meganuclease (Gao et al., 2010, Heritable targeted mutagenesis in maize using a designed endonuclease. *Plant Journal.* 61:176-187), zinc finger nuclease (Shukla et al. 2009. Precise genome modification in the crop species *Zea mays* using zinc-finger nucleases. *Nature* 459, 437-441), dCas9-cytidine deaminase (Komor et al. 2016, Program-

mable editing of a target base in genomic DNA without double-stranded DNA cleavage. Nature doi:10.1038/nature17946) or any other genome modification system. The transgenic haploid inducer line also expressing the editing machinery is then used as pollen donor to create mutations and haploids in target lines via outcrossing. Haploid embryos or seeds are then recovered, identified as haploids, and tested for the edits at the target site (whatever target site is chosen by virtue of the TALEN construct design or the Cas9 guide RNA design). Haploids containing the desired edits is chromosomally doubled using standard procedures using standard means such as colchicine, trifluralin or other chromosome doubling agent. Identification of the induced haploids can be simplified by using a color marker as is typically done in corn doubled haploid production—this color marker can display in the resulting embryos, seeds, seedlings, or adult plant. Presence of mutations at the target site can be checked by sequence analysis (DNA sequencing), by marker analysis, or by phenotype. Because there is only one copy of the DNA to mutate in haploid plants, recessive phenotypes should display so that could be another way to identify the haploids that were edited.

A. Mutagenesis of VLHP Targets in Elite Maize Inbred Line with Transgenic Editing Locus Generated Directly in a Haploid Inducer Line.

VLHP1 and VLHP2 are homeodomain-leucine zipper I-class homeobox genes and members of a class of proteins that is unique to plants. The HD domain is involved in DNA binding whereas the Zip domain is involved in protein homo- and hetero-dimerization. HD-Zip I proteins are generally involved in responses related to abiotic stress, abscisic acid (ABA), blue light, de-etiolation and embryogenesis (Elhiti and Stasolla, 2009. Structure and function of homodomain-leucine zipper (HD-Zip) proteins. Plant Signal Behav. 4: 86-88). VLHP1 and VLHP2 are in the same gene family as Grassy Tillers1 (GT1). GT1 promotes lateral bud dormancy and suppresses elongation of lateral ear branches in maize.

In this example, vector 23396 (SEQ ID NO: 1; see also FIG. 9) for expressing Cas9 and single guide RNA (sgRNA) was made to target maize VLHP1 (GRMZM2G104204) and its homolog VLHP2 (GRMZM2G062244) genes. Vector 23396 expresses a sgRNA with 20-nucleotide targeting sequence xZmVLHP-01 (5'-GCAGGAGGCGTCGAG-CAGCG-3', SEQ ID NO: 2). xZmVLHP-01 targets both VLHP1 and VLHP2 genes at the second exon. Vector 23396 was introduced into a transformable haploid inducer line NP2222-HI using *Agrobacterium*-mediated transformation with mannose selection. NP2222-HI was derived from crossing of transformable maize inbred line NP2222 with Stock 6 derivative line RWKS to introgress the haploid induction (HI) locus. NP2222-HI has an average haploid induction rate of about 9.2%.

NP2222-HI transformants from vector 23396 were assayed for modification of genomic VLHP target sequences (5'-GCAGGAGGCGTCGAGCA/GCG-3'; SEQ ID NO: 2). The slash ("/") represents the Cas9 cleavage position. Target locus editing activity was determined using quantitative PCR Taqman method as described before (WO2016106121, incorporated herein by reference). Transgenic lines with high target site modification activities—i.e., both VLHP1 and VLHP2 genes were modified, and preferably containing single copy transgene—were selected for further studies and used for crossing or progeny production.

Pollen from T0 transformants of 23396 is used directly to pollinate ears of elite inbred line ID5829 or other maize lines including sweet corn lines to induce production of haploid

embryos. Alternatively, T0 transformants of 23396 in NP2222-HI background are selfed to produce progeny lines carrying homozygous transgene and pollen from the progeny plants are used to pollinate other corn lines to induce haploid embryo formation. The induced haploid embryos are extracted from kernels and placed on embryo rescue media for direct germination or allowed to mature to form seeds. Tissues from the induced haploid embryos and the resulting plants are assayed to determine if editing has occurred in the VLHP target sequences. If the induced haploid embryos or plants contain desired mutations, chromosome doubling treatment is applied to produce doubled haploid lines from them. For example, using embryo rescue method, embryos are extracted from elite line ID5829 ears pollinated with transgenic haploid inducer line carrying 23396 editing locus at 18-22 days after pollination ("DAP," extraction between 10-25 DAP is theoretically possible). DNA is isolated from germinated haploid seedlings and used for assay. Colchicine treatment is applied to seedling for chromosome doubling. Alternatively, chromosome doubling agent can be applied to the isolated embryos during germination. DNA is extracted from germinated seedlings and used for determining that mutation has occurred at the xZmVLHP-01 target sequence.

Alternate methods are available. One could allow the seed to mature and select haploids later by another phenotype. One could let the seed dry down and at a later date germinate the seeds to determine haploids without a marker (e.g., using plant size rather than a gene conferring a color marker), at which point one would test for edits and apply chromosome doubling agents where appropriate. This method may have its advantages in that embryo screening and/or rescue is avoided.

B. Mutagenesis of GW2 Targets in Elite Maize Inbred Line with Transgenic Editing Locus Introduced Directly in a Haploid Inducer Line.

A mutation in DA2, an E3-ubiquitin ligase gene, in rice resulted in larger seeds (Song et al., 2007). Rice DA2 has 2 maize homologs, GW2-1 (GRMZM2G170088) and GW2-2 (GRMZM2G007288). The maize genes are 94% identical at the protein level and 90% identical at the DNA level. GRMZM2G170088 has a large 177 bp insert (59 aa) in comparison with GRMZM2G007288.

In this example, vector 23399 (SEQ ID NO: 3, see also FIG. 10) was made for expression of Cas9 and sgRNA to target both maize GW2-1 (GRMZM2G170088) and its homolog GW2-2 (GRMZM2G007288) genes. Both GW2-1 and GW2-2 genes contain target sequence xZmGW2-02 (5'-AAGCTCGCGCCCTGCTACCC-3', SEQ ID NO: 4) in exon 1 and this sequence was used to design sgRNA expressed from vector 23399. Binary vector 23399 expresses single guide RNA (sgRNA) with 20-nucleotide targeting sequence xZmGW2-02 fused to single guide RNA scaffold comprising of both crRNA and tracrRNA. Vector 23399 was introduced into a transformable haploid inducer line NP2222-HI using *Agrobacterium*-mediated transformation with mannose selection. NP2222-HI was derived from crossing of transformable maize inbred line NP2222 with Stock 6 derivative line RWKS to introgress the haploid induction (HI) locus.

NP2222-HI transformants of vector 23399 were assayed for modification of genomic GW2-2 target sequences (5'-AAGCTCGCGCCCTGCTA/CCC-3', SEQ ID NO: 4; the slash ("/") indicates the Cas9 cleavage position). Target sequence editing activity was determined using quantitative PCR Taqman method as described before (WO2016106121). Transgenic lines with high target site

modification activities—i.e. both GW2-1 and GW2-2 genes were modified, and preferably containing single copy transgene—were selected for further studies and used for crossing or progeny production.

Pollen from T0 transformants of 23399 is used directly to pollinate ears of elite inbred line ID5829 or other maize line including sweet corn lines to induce production of haploid embryos. Alternatively, T0 transformants of 23399 in NP2222-HI background are selfed to produce progeny lines carrying homozygous transgene and pollen from the progeny plants are used to pollinate other corn lines to induce haploid embryo formation. The induced haploid embryos are extracted from kernels and placed on embryo rescue media for direct germination or allowed to mature to form seeds. Tissues from the induced haploid embryos and the resulting plants are assayed to determine if editing has occurred in the maize GW2 target sequences. If the induced haploid embryos or plants contain desired mutations, chromosome doubling treatment is applied to produce doubled haploid lines from them. For example, using embryo rescue method, embryos are extracted from elite line ID5829 ears pollinated with transgenic haploid inducer line carrying 23396 editing locus at 18-22 days after pollination. DNA is isolated from germinated haploid seedlings and used for assay. Colchicine treatment is applied to seedling for chromosome doubling. Alternatively, chromosome doubling agent can be applied to the isolated embryos during germination. DNA is extracted from germinated seedlings and used for determining if mutation has occurred at the xZmGW2-02 target sequence. Alternately, one could allow the seed to mature and select haploids later by another phenotype. One could even let the seed dry down and at a later date germinate the seeds to determine haploids without a marker (e.g., using plant size rather than a gene conferring a color marker), at which point one would test for edits and apply chromosome doubling agents where appropriate. This method may have its advantages in that embryo screening and/or rescue is avoided.

IV. Simultaneous Haploid Induction and Editing in Corn, Rice, Sunflower, or Any Other Crop Via Chemical-Based Haploid Induction

Any line of corn, rice, wheat, tomato, sunflower, barley, or any other crop is transformable with the editing construct (Cas9 plus guide RNAs designed to mutate a particular target site) and then optionally make the editing construct either heterozygous or homozygous (via self-pollination of the transformed event), and then using lipid or oil applications during outcrossing (pollination onto target lines) in order to induce de novo haploids and simultaneously edit the target sites in the target genomes. These lipid applications have the ability to induce haploids when applied to pollen, silks, flowers, or tassels of any plant—regardless of male parent. In particular, the male parent is not required to have any mutations in the MATL gene (i.e., it can be homozygous wild type for the MATRILINEAL gene). These lipid applications induce haploids de novo, without any genetic requirement on behalf of either parent. See P.C.T. Patent Application No. PCT/US2016/62548, incorporated herein by reference in its entirety. The mechanism of de novo haploid induction via lipid spray apparently works the same way as it does in matl mutant (genetic haploid inducer) lines: via chromosome elimination post-fertilization. Haploid progeny are isolated and checked for the induced mutations (caused via the editing process) and then doubled to make edited, doubled haploid plants.

V. Mutagenesis of Target Sequences in Elite Field Corn and Sweet Corn Inbred Lines with Transgenic Editing Locus Introgressed into a Haploid Inducer Line.

Transgenic locus expressing genome editing machinery can also be generated in conventional transformable maize line without haploid inducing activity such as A188, Hi-II or NP2222 and then introgressed into haploid inducer line such as NP2222-HI, RWK, RWKS, RWS, or UH400 or Stock6 or any other haploid inducer line.

In this example, maize inbred line NP2222 is transformed with VLHP Cas9-sgRNA vectors (23396 and 23397) and GW2 Cas9-sgRNA vectors (23398 and 23399). Vectors 23396 and 23399 have been described in previous examples (Example IIIA and Example IIIB). Vector 23397 (SEQ ID NO: 20) is identical to 23396 except the gRNA-coding sequence xZmVLHP-01 (5'-GCAGGAGGCGTCG-AGCAGCG-3', SEQ ID NO: 2) is replaced with xZmVLHP-02 (5'-GCTGGAGCTGAGCTTCCGGG-3', SEQ ID NO: 21). Vector 23398 (SEQ ID NO: 23) is identical to 23399 except the gRNA-coding sequence xZmGW2-02 (5'-AAGCTCGCGCCCTGCTACCC-3', SEQ ID NO: 4) in 23399 is replaced by xZmGW2-01 (5'-GAGCGGTT-CACGCGGCCGCA-3', SEQ ID NO: 23). These vectors were introduced into *Agrobacterium* strain LBA4404 (pVGW7). The resulting *Agrobacterium* strain containing vector 23396, 23397, 23398, or 23399 was used to transform immature embryos of transformable elite inbred line NP2222. Calli were induced from infected immature embryos and selected on mannose media to recover transgenic calli. Transgenic calli were placed on regeneration and rooting media to recover transgenic plants expressing the CRISPR-Cas9 editing machinery. Transgenic plants were assayed for transgene copy number and moved to greenhouse for seed production.

Single copy transformants of vector 23396 (MZET154902A004A, MZET154902B006A), 23397 (MZET154903B009A, MZET154903B012A), 23398 (MZET154904B005A, MZET154904B014A) and 23399 (MZET154905A002A, MZET154905A010A) were identified and backcrossed with non-transgenic NP2222. Ears of transgenic progeny plants containing T-DNA insert of each of the above vectors were pollinated with pollen of haploid inducer line RWKS to produce F1 progeny. F1 progeny containing transgenic locus and haploid induction locus were identified by genotyping assays and self-pollinated to produce F2 progeny seeds. F2 progeny seeds were planted and seedling plants assayed to identify plants homozygous for transgenic Cas9-sgRNA locus (assay #2540) and haploid induction locus (assay #2827) with qPCR Taqman assays.

Lines homozygous for the haploid induction locus and preferably homozygous transgenic 23396, 23397, 23398, and 23399 Cas9-sgRNA editing locus were used to pollinate ears from target elite field corn line ID5829 and sweet corn lines (SWC726 or SWC412F) for haploid induction. Induced haploid embryos were isolated from pollinated ID5829, SWC412F, SWC726 ears and germinated on embryo rescue media. Alternatively, pollinated ears were allowed to mature and kernels with haploid embryos were germinated. Leaf samples were collected and analyzed with Taqman assay to identify plants containing mutations in VLHP and GW2 genes but absence of genetic components from induction line such as transgenic Cas9-sgRNA or other non-transgenic marker gene sequences. Identified haploid plants with targeted GW2 or VLHP gene mutations were treated with colchicine for chromosome doubling to recover doubled haploid plants for seed production. Alternatively, extracted haploid embryos can be treated with chromosome

doubling agent such as colchicine and the resulting plants are analyzed for ploidy level and presence of targeted mutations in GW2 or VLHP genes. Plants with targeted GW2 and VLHP gene mutations are grown to maturity for seed production and further progeny evaluation.

For example, edited haploid lines (JSER82A056 and JSER82A063) were identified from crosses between sweet corn line SWC412F ears pollinated with haploid inducer containing 23399 Cas9-sgRNA transgene. Line JSER82A056 has both GW2-01 and GW2-02 target genes mutated, whereas line JSER82A063 only has GW2-02 gene mutated (See Table 5). Neither of these lines contain Cas9 transgene (assay #2540 for Cas9 or #1750 for PMI selectable marker gene) or haploid inducer gene (assay #2827) as the male genome has been eliminated from the haploids. Ploidy level analysis confirmed that both lines are haploids (FIGS. 11 and 12). Note that wildtype (“VVT”) genes in the haploids have a copy number of “2” and mutant will be “0” since the copy call is relative to the endogenous ADH gene copy number. Therefore, haploid lines carrying VVT unedited GW2-01 or GW2-02 genes will have a copy call of “2.” VVT haploid inducer locus will have copy call of “2” for assay #2826 and “0” for assay #2827 (haploid inducer variant). If a corn plant line is a diploid between sweet corn and transgenic inducer, it will be heterozygous for the haploid inducer gene and thus have copy call of “1” for both assay #2826 and assay #2827.

TABLE 5

Progeny zygosity analysis from crosses. Taqman analysis results showing the lines do not contain transgene or haploid inducer locus from pollen donor, but have edits in GW2-01 and/or GW2-02 targets.							
Allele:							
		cCas9-01	cPMI-09	CRISPR target in GW2-01 (23399)	CRISPR target in GW2-02 (23399)	pPLAIIa WT allele	RWK (Haploid Inducer) allele of pPLAIIa
		Assay ID:					
Plant ID	Construct ID	2540 Copy# level	1750 Copy# level	3065 Copy# level	3095 Copy# level	2826 Copy# level	2827 Copy# level
1-copy control		+	1	ND	ND	1	1
wild type control		0	0	2	2	2	0
JSER82A056	23399	0	0	0	0	2	0
JSER82A063	23399	0	0	1 or 2	0	2	0
JSER85A021	23399	0	0	0	>2	2	0
JSER85A022	23399	0	0	0	>2	2	0
JSER85A024	23399	0	0	0	>2	2	0
JSER85A027	23399	0	0	0	>2	2	0
JSER85A037	23399	0	0	0	>2	2	0
JSER85A039	23399	0	0	0	>2	2	0
JSER85A044	23399	0	0	0	>2	2	0
JSER85A055	23399	0	0	0	>2	2	0

To further confirm target-specific editing in these haploid lines, GW2-02 target region was amplified from JSER82A063 by PCR and the PCR product was sequenced. A single base C was deleted in JSER82A063 in comparison with the VVT sequence precisely at the Cas9 cleavage site (FIG. 13). These results clearly demonstrated that editing machinery brought into the egg cell from the male gametophyte can edit the female genome before the male genome is eliminated after double fertilization to form haploid embryo. Candidate edited haploid lines without transgene were treated with injection of 0.125% colchicine in 0.5% DMSO or seedling drenching in 0.06% colchicine solution

(Eder and Chalyk, 2002, In vivo haploid induction in maize. Theor. Appl. Genetics 104:703-708). Treated lines were planted in soil and grown in greenhouse for progeny seed production.

5 VI. Simultaneous Haploid Induction and Editing in Wheat and Other Monocots Via Wide Cross.

Haploid induction is also achieved using interspecific or intergeneric wide crosses (Kasha and Kao, 1970, High frequency haploid production in barley (*Hordeum vulgare* L.). Nature 225:874-886). For example, wheat haploids can be obtained by pollination with various intergeneric crosses with maize (Suenaga and Nakajima 1989), pearl millet (Inagaki and Mujeeb-Kazi 1995), teosinte (Ushiyama et al. 1991), *H. bulbosum* (Barclay 1975), and sorghum (Ohkawa et al. 1992). Barley haploids are obtained by pollination with *Hordeum bulbosum* pollen. Tobacco haploids can be obtained by crossing with *N. glauca* pollen. Many other examples exist in other crops.

Similar to examples above in introducing transgenic editing locus into Stock6 induction line, transgenic editing locus can be introduced into these lines used for wide crosses to induce haploid induction and targeted sequence mutation. Transgenic lines expressing editing machinery can be generated in any line of corn, wheat, barley, rye, pearl millet, rice, *Brassica*, lettuce, tomato, or any other crop by direct transformation or out-crossing. Preferably the transgenic locus is made homozygous and then the line is used as pollen

donor in a wide cross with other compatible recipient crops to induce haploids to produce desired edits. The process of post-fertilization genome elimination in wide crosses is basically the same as the process in the maize MATL mutant system, although in some cases the foreign pollen-derived DNA and editing machinery may be eliminated slightly earlier in embryo development, which is why this method is preferably practiced using a promoter that drives expression of the editing machinery in the pollen, sperm cells, and/or zygote cell, so that the editing RNA and protein is present and able to edit the target genome even though the male DNA is eliminated rather quickly after fertilization.

To demonstrate feasibility of simultaneous haploid induction and editing via wide crosses, maize transgenic lines expressing Cas9 and sgRNA targeting wheat VLHP gene sequences were generated. Vector 23763 (SEQ ID NO: 24) contains expression cassettes for Cas9 and sgRNA containing protospacer sequence xTaVLHP1 (5'-GACGAGCAGGCCGAGTTCC-3', SEQ ID NO: 25) for guiding Cas9-mediated cleavage of TaVLHP1 target sites in wheat. The wheat genome has three xTaVLHP1 targets in total (TaVLHP1-4A, TaVLHP1-4B and TaVLHP1-4D), with each one in its three sub-genomes. The guide sequence in 23397 (SEQ ID NO: 20), xZmVLHP (5'-GCTGGAGCTGAGCTTCCGGG-3', SEQ ID NO: 21) will also direct cleavage of wheat VLHP target sequences, xTaVLHP2-1A (5'-GCTGGAGCTGAGCTTCCGGG-3', SEQ ID NO: 26) or xTaVLHP2-1B (5'-TCTGGAGCTGAGCTTCCGGG-3', SEQ ID NO: 27). There are three VLHP2A genes containing xTaVLHP2-1A and 3 VLHP2B genes containing xTaVLHP2-1B sequences in the Chinese Spring wheat genome. Vectors 23397 and 23763 were transformed into maize inbred line NP2222 using *Agrobacterium*-mediated transformation to generate transgenic lines expressing Cas9 and sgRNA. Transgenic maize lines were grown in greenhouse and selfed to produce T1 plants.

Pollen collected from transgenic maize T0 or progeny T1 plants carrying T-DNA of vector 23397 or 23763 were used to pollinate emasculated spring wheat line AC-Nanda. At one to two days before anthesis, wheat florets were emasculated and two days later are pollinated with fresh maize pollen carrying the editing machinery. For convenience, spikelets from a Syngenta elite cytoplasmic male sterile ("CMS") wheat line (16A300292) were also directly used as female donors to induce haploid embryo formation with transgenic maize pollen expressing 23397 or 23763 Cas9-sgRNA. Embryos were extracted from pollinated florets at 14-20 days after pollination for embryo rescue to recover haploid plantlets from the wheat x maize haploid induction system. Excised embryos were cultured on either full strength MS (Murashige and Skoog 1962) or 1/2 MS or B5

basal medium containing various modifications of organic supplements and grown in vitro for 3-5 weeks at 20-25° C. and 16-hour day length.

For example, pollen of T1 progeny from transgenic maize line MZET164902A044A containing vector 23763 was used to pollinate spikes of CMS line 16A300292 to induce wheat haploids. Haploid embryos were rescued and the resulting wheat haploid seedling were sampled for qPCR analysis to determine the copy number of VLHP target sequences (See Table 6). One of the haploid lines (JSWER30A22) was found to contain mutation in TaVLHP1-4B gene, but not in its orthologs TaVLHP1-4A and TaVLHP1-4D in the A and D sub-genomes. Ploidy level analysis confirmed that JSWER30A22 is a true haploid (See FIGS. 14 and 15). The mutation within the TaVLHP1-4B target region was further characterized by sequencing and was found to contain 97 bp deletion starting from the predicted Cas9 cleavage site (FIG. 16). We also identified another line JSW16A07 with "0" copy in TaVLHP1-4A gene (assay #3252), suggesting targeted editing in the target sequence. However, the deletion in this target gene is probably quite large in deleting the primer binding site(s) since we were not able to recover PCR product for sequencing. Haploid seedlings with an edited target site were transplanted to soil after 3-5 weeks in vitro culture. The transplanted seedlings were hardened for one week in a growth chamber under the same environmental regime. Colchicine was added after shoots had formed. However, the chromosome doubling treatment can be done earlier at embryo rescue in vitro culture stage or later after transplanting. When whole wheat seedlings are treated for doubling, roots of the haploid seedling are trimmed leaving a zone of 2-3 cm and then submerged in a 0.1% colchicine solution with 2% dimethyl sulfoxide (DMSO) and ca. 0.05% Tween-20 at 20° C. for 5 hours. After this treatment, the roots are washed to remove residual colchicine and potted in peat soil. Plant tissue samples can be removed from haploid seedlings for mutation detection to identify plants containing mutations in TaVLHP target gene sequences but with the maize chromosomes including sequences encoding the transgenic editing machinery completely eliminated. Since JSWER30A22 is from a CMS line, the plant is pollinated with a restorer to produce progeny seeds.

TABLE 6

Taqlman analysis for wheat progeny from wide crosses. Line JSW30A22 is edited.						
Plant ID	Construct ID	Allele:				
		TAV_4A	TAV_4B	TAV_4D	PMI	CAS9
		Assay ID:				
		3252 Copy# level	3253 Copy# level	3254 Copy# level	1750 Copy# level	2540 Copy# level
WT, AC-Nanda	N/A	>2	2	>2	0	0
WT, AC-Nanda	N/A	2	2	2	0	0
WT, CMS	N/A	2	2	2	0	0
WT, CMS	N/A	2	2	2	0	0
JSW29A01	23763	2	2	2	0	0
JSW29A02	23763	2	2	2	0	0
JSW29A03	23763	2	2	2	0	0
JSW29A04	23763	2	2	2	0	0
JSW29A05	23763	1 or 2	2	2	0	0
JSW29A06	23763	2	2	2	0	0
JSW29A07	23763	2	2	2	0	0
JSW29A08	23763	2	2	2	0	0
JSW29A09	23763	2	2	2	0	0
JSW29A10	23763	2	2	2	0	0
JSW29A11	23763	2	2	2	0	0

TABLE 6-continued

Taqman analysis for wheat progeny from wide crosses. Line JSW30A22 is edited.						
Plant ID	Construct ID	Allele:				
		TAV_4A	TAV_4B	TAV_4D	PMI	CAS9
		Assay ID:				
		3252 Copy# level	3253 Copy# level	3254 Copy# level	1750 Copy# level	2540 Copy# level
JSW29A12	23763	2	2	2	0	0
JSW29A13	23763	2	2	2	0	0
JSW29A14	23763	2	2	2	0	0
JSW29A15	23763	2	2	2	0	0
JSW29A16	23763	2	2	2	0	0
JSW30A01	23763	2	2	2	0	0
JSW30A02	23763	2	1 or 2	2	0	0
JSW30A03	23763	2	2	2	0	0
JSW30A04	23763	2	2	2	0	0
JSW30A05	23763	2	2	2	0	0
JSW30A06	23763	2	2	2	0	0
JSW30A07	23763	2	2	2	0	0
JSW30A08	23763	2	2	2	0	0
JSW30A09	23763	2	2	2	0	0
JSW30A10	23763	2	2	2	0	0
JSW30A11	23763	2	2	2	0	0
JSW30A12	23763	2	2	2	0	0
JSW30A13	23763	2	2	2	0	0
JSW30A14	23763	2	2	2	0	0
JSW30A15	23763	2	2	2	0	0
JSW30A16	23763	2	2	2	0	0
JSW30A17	23763	2	2	2	0	0
JSW30A18	23763	2	2	2	0	0
JSW30A19	23763	>2	2	2	0	0
JSW30A20	23763	2	2	2	0	0
JSW30A21	23763	2	2	2	0	0
JSW30A22	23763	2	0	2	0	0
JSW30A23	23763	2	2	1 or 2	0	0
JSW30A24	23763	2	2	2	0	0
JSW30A25	23763	2	2	2	0	0
JSW30A26	23763	2	2	2	0	0
JSW30A27	23763	2	2	2	0	0
JSW30A28	23763	2	2	2	0	0
JSW30A29	23763	2	2	2	0	0
JSW30A30	23763	2	1 or 2	1 or 2	0	0
JSW30A31	23763	2	2	2	0	0

To further demonstrate feasibility of simultaneous haploid induction and editing via wide crosses, maize transgenic lines expressing Cas9 from five promoters that have high and/or specific expression in pollen, along with sgRNA targeting wheat VLHP gene sequences, were generated. These five vectors were 24038 (SEQ ID NO: 34), 24039 (SEQ ID NO: 35), 24079 (SEQ ID NO: 36), 24091 (SEQ ID NO: 37), and 24094 (SEQ ID NO: 38). All five of these vectors utilized the same sgRNA containing protospacer sequence xTaVLHP2 (5'-GCTGGAGCTGAGCTTCCGGG-3', SEQ ID NO: 21) for guiding Cas9-mediated cleavage of TaVLHP2 target sites in wheat. The wheat genome has three xTaVLHP2 targets in total (TaVLHP2-2A, TaVLHP2-2B and TaVLHP2-20), with each one in its three sub-genomes. The guide sequence in these five constructs also directs cleavage of wheat VLHP target sequences, xTaVLHP2 (5'-GCTGGAGCTGAGCTTCCGGG-3', SEQ ID NO: 26) or xTaVLHP3 (5'-TCTGGAGCTGAGCTTCCGGG-3', SEQ ID NO: 27). There are three TaVLHP2 genes containing xTaVLHP2 and 3 TaVLHP3 genes containing xTaVLHP2-1 B sequences in the Chinese Spring wheat genome.

Vector 24038 (SEQ ID NO: 34) contains expression cassettes for Cas9 under control of a pollen-preferred high expression promoter prZmGRMZM5G876285 and termina-

tor tZmGRMZM5G876285 from the maize prf3 (34fric-ana34 homolog3) gene, which has an extremely high native expression at the RNA and protein level in pollen and which has evidence of high sperm cell expression.

Vector 24039 (SEQ ID NO: 35) contains expression cassettes for Cas9 under control of a pollen-preferred high expression promoter prZmGRMZM2G020852 and terminator tZmGRMZM2G020852 from the maize EXPB2 (BETA EXPANSIN2) gene, which has an extremely high native expression at the RNA and protein level in pollen and which has evidence of sperm cell expression.

Vector 24079 (SEQ ID NO: 36) contains expression cassettes for Cas9 under control of a pollen-preferred high expression promoter prZmGRMZM2G146551 and terminator tZmGRMZM2G146551 from the maize EXPB1 (BETA EXPANSIN1) gene, which has an extremely high native expression at the RNA and protein level in pollen and which has evidence of sperm cell expression.

Vector 24091 (SEQ ID NO: 37) contains expression cassettes for Cas9 under control of a pollen-preferred promoter prZmGRMZM2G471240 and terminator tZmGRMZM2G471240 from the maize MATL (MATRI-LINEAL) gene, which shows evidence of pollen and possibly sperm cell expression at the RNA and protein level.

Vector 24094 (SEQ ID NO: 38) contains expression cassettes for Cas9 under control of a pollen-preferred pro-

moter prZmGRMZM2G471240 and terminator tZmGMRMZM2G471240 from the maize MATL (MATRILINEAL) gene, which shows evidence of pollen and possibly sperm cell expression at the RNA and protein level. This construct additionally has an N-terminal fusion of

AmCyan fluorescent protein on the Cas9 molecule for imaging and visualization of the Cas9 localization in pollen. These five vectors (24038, 24039, 24079, 24091, and 24094) were transformed into maize inbred line NP2222 using *Agrobacterium*-mediated transformation to generate

transgenic lines expressing Cas9 and sgRNA. Transgenic maize lines were grown in greenhouse and single and two-copy transgenic plants were outcrossed onto spring wheat and a CMS wheat line. Pollen collected from transgenic maize T0 plants carrying T-DNAs of one of the vectors 24038, 24039, 24079, 24091, and 24094 were used to pollinate emasculated spring wheat line AC-Nanda. Pollen was also used for a qRT experiment, in which the expression of the Cas9 was measured at the RNA level and compared to Cas9 expression in leaf samples when the Cas9 was driven by a sugar cane ubiquitin promoter used in many of the corn and wheat examples given above. As you can see in FIG. 17, the expression was high in the pollen, averaging about 100 fold higher in plants carrying the T-DNA vectors 24038, 24039, and 24079 compared to the Ubiquitin promoter. The expression was also higher in pollen from plants containing vectors 24038, 24039, and 24079 when compared to the pollen carrying the MATRILINEAL promoter constructs (24094 and 24091), which is known to have lower native gene expression. All five of these promoters have expression patterns that are restricted to pollen. As an indication that the promoters were working properly, we observed no T0 expression of Cas9 in callus seedling leaves, and there was no editing of the VLHP target sites in the T0 maize leaves (without wishing to be bound by theory, editing may happen at the maize target sites, in all likelihood, at the mature pollen stage, when the Cas9 is expressed for the first time).

At one to two days before anthesis, wheat florets were emasculated from the CMS line and the AC Nanda line. Two days later the florets were pollinated with fresh maize pollen carrying the editing machinery, Cas9-sgRNA, from either construct 24038, 24039, 24091, or 24094 (T0 plants transformed with construct 24079 were delayed, and not crossed to wheat in this manner). Wheat embryos were extracted from pollinated florets at 14-20 days after pollination for embryo rescue to recover haploid plantlets from the wheat x maize haploid induction system. Excised embryos were cultured on either full strength MS (Murashige and Skoog 1962) or 1/2 MS or B5 basal medium containing various modifications of organic supplements and grown in vitro for 1-5 weeks at 20-25° C. and 16-hour day length. For example, pollen of T0 progeny from transgenic maize line MZKE172601A100A containing vector 24039 was used to pollinate spikes of CMS line 16A300292 to induce wheat haploids. Haploid embryos were rescued and the resulting wheat haploid seedlings were sampled for qPCR analysis to determine the copy number of VLHP target sequences (Table 7). In this analysis, we tested for the Cas9 transgene using assay #2540. All wheat embryos rescued and tested lacked this transgene and gave scores of "0" for Cas9, because they do not have any corn DNA in the developing embryo and therefore do not have the transgene. The corn DNA is totally eliminated, kicked out or fails to be fully delivered in the first place during the haploid induction process, taking place during and/or after fertilization). In addition to Cas9, we test for assays #3332 and #3333, which

give non-specific amplification of both VLHP2-2A and -2D alleles. These assays typically read as "2" or ">2" in haploid wheat, and the majority of the haploids we produced using the transgenic maize pollen scored 2 or >2 for these assays. We used these assays to look for putative edited haploids, by looking for scores of 0 or 1. A call of "1" might indicate that one of the two alleles, either VLHP2-2A, or -2D, was edited. Finally, we tested for assay 3255 in AC Nanda haploids, which detects VLHP2-2B specifically. The CMS line does not amplify this assay, even when it is wild-type, so we did not use it for the CMS haploids. The unedited haploids give a score of a "2," while putative edited haploids are found because they have a score of "0." A score of "1" might indicate a faulty reading or a chimeric, partially-edited sample.

As an example, one of the AC Nanda haploid plants 440-A5 was found to contain mutation in TaVLHP2-2B gene, but not in its orthologs TaVLHP2-2A and TaVLHP2-2D in the A and D sub-genomes (Table 7). The Taqman data also showed that it lacked the Cas9 transgene. The mutation within the TaVLHP2-2B target region was further characterized by sequencing, but although we were able to amplify the A and D alleles, we could no longer amplify the B allele, suggesting that there is a larger edit present, likely a large deletion, that results in the PCR product no longer amplifying.

As another example, one of the CMS haploid plants 450-D11 was found to contain mutation in either the TaVLHP2-2D or -2A homologues, according to the score of "1" for both assays 3332 and 3333. (Table 7). The Taqman data showed that it lacked the Cas9 transgene. The TaVLHP2-2A, 2B and 2D target regions were further characterized by sequencing, but although we were able to amplify the A and B alleles, we could no longer amplify the D allele, suggesting that there is a larger edit present that led to PCR failure.

Considering the 2295 wheat haploids produced from crosses to maize pollen carrying one of the following five preferred-pollen expression constructs (24038, 24039, 24091, and 24094), we found 15 haploids that gave Taqman assay data that indicated possible editing at either the VLHP2-2A, VLHP2-2D, or VLHP2-2B target sites. After sequencing, seven of those haploids were found to have wild-type sequences at the target sites, and were called false positives due to Taqman error. These errors are thought to be either due to the fact that assays #3332 and #3333 gave non-specific amplification of both VLHP-2A and -2D alleles, leading to some missed calls, or due to low DNA quantity.

Of the remaining 8 putative edited haploids, six were AC Nanda (440-B3, 440-D3, 440-A5, 447-G8, 456-G9, 459-A2) where the editing transgene was from construct 24038. Four of those (440-B3, 440-D3, 440-A5, and 456-G9) contained edits in VLHP2-2B. These were found because they had a Taqman score of "0" for assay 3255. These plants lacked Cas9 (score of "0") but had wild-type "2" scores for VLHP2-2A or VLHP2-2D (assays #3332 and #3333) indicating they were not edited at those sites. These six plants were confirmed to be haploids by ploidy analysis. We attempted to sequence the edited alleles, but while the PCR and sequencing reactions worked well for 2A and 2D, we were not able to obtain a PCR product for 2B. We repeated the PCR several times using a range of reaction conditions but could not amplify the 2B homeologs from these haploid plants. This may indicate that the editing caused a large change in the 2B gene in these plants that may end up deleting the primer annealing site. We expect that many of

the CMS plants also have edits at the VLHP2-2B target site, but we did not have an assay to detect the VLHP2-2B allele from the CMS line.

Considering AC Nanda alone, we calculate an overall editing rate at that allele of 0.7% for all constructs, but a particularly high editing rate of 1.4% for construct 24038.

In addition to these four edited haploids with scores of “0” for 3255, several other plants gave scores of “0 or 1” or “1” for 3255, which indicates possible chimerism (partial editing in certain cell lineages of the embryo or plantlet), but we did not follow up on those plants. For AC Nanda homolog VLHP2-2A, plant 447-G8 contained an edit which we were also not able to sequence because the PCR reaction failed, even though 2B and 2D did amplify and contained wild-type sequence. We repeated the PCR several times using a range of reaction conditions but could not amplify the 2A homolog. Similarly, for VLHP2-20, plant 459-A2 contained an edit which we were not able to sequence because the PCR reaction failed. We repeated the PCR several times using a range of reaction conditions but could not amplify the 2D homolog. We also found putative edits in 447-H12 and 440-G6, but upon sequencing we found that these were false positives.

For the CMS haploids, plant 450-011 gave scores of “1” for both assay #3332 and 3333 (Table 7). Upon sequencing, we found that the 2A homolog had wild-type sequence, but

we could not PCR-amplify the 2D homolog, suggesting that a large edit had occurred. We repeated the PCR several times using a range of reaction conditions but could not amplify the 2D homolog. For plant 452-B11, the Taqman score was “0” for #3332 (VLHP2-2A), and we could not amplify that allele for sequencing, even though the 2D and 2B PCR products and sequences were normal. We repeated the PCR several times using a range of reaction conditions but could not amplify the 2A homolog. We also found five plants that had putative edits according to the Taqman data for assays 3332 and 3333, but PCR-sequencing showed these to be false positives; the sequence was wild-type (unedited).

In total, we found two edited CMS haploids and six edited AC Nanda haploids. There may be many more edited haploids that we were not able to detect because we did not have assays for the 2B gene for the CMS plants, nor for the VLHP3 gene target sites of the guide RNA in these five constructs.

The sequencing data from these edited haploids are consistent with the concept of a large deletion, inversion or rearrangement around the guide RNA target site, and extending far enough away to possibly include removal of one of the primer binding sites. This type of large change is not uncommon during editing by Cas9, especially in tissues where DNA repair via non-homologous end-joining is slower or inhibited—which may be the case in the just-fertilized zygote or early haploid wheat embryo.

TABLE 7

Sequencing data from edited wheat haploids.													
			TAV 2A 3332		TAV 2D 3333		TAV 2B 3255		PMI 1750		Cas9 2540		
Plant ID	Con-struct ID	copy #	Raw Copy #	Copy # level	Raw Copy #	Copy # level	Raw Copy #	Copy # level	Raw Copy #	Copy # level	Raw Copy #	Copy # level	Sequencing result
CMS													
427-A2	WT	N/A	2.44	>2	2.38	2	Not tested	0.00	0	0.00	0	not sequenced	
427-B2	WT	N/A	1.99	2	1.99	2	Not tested	0.00	0	0.00	0	not sequenced	
427-C2	WT	N/A	2.02	2	2.07	2	Not tested	0.00	0	0.00	0	not sequenced	
427-D2	WT	N/A	2.31	2	2.16	2	Not tested	0.00	0	0.00	0	not sequenced	
427-A10	24091	2	2.07	2	1.66	2	Not tested	0.00	0	0.00	0	not sequenced	
427-B10	24091	2	1.95	2	1.84	2	Not tested	0.00	0	0.00	0	not sequenced	
427-C10	24091	2	1.93	2	2.28	2	Not tested	0.00	0	0.00	0	not sequenced	
427-D10	24091	2	2.59	>2	2.48	>2	Not tested	0.00	0	0.00	0	not sequenced	
427-E10	24091	2	1.90	2	1.78	2	Not tested	0.00	0	0.00	0	not sequenced	
427-F10	24091	2	2.03	2	1.96	2	Not tested	0.00	0	0.00	0	not sequenced	
427-G10	24091	2	2.08	2	2.25	2	Not tested	0.00	0	0.00	0	not sequenced	
427-H10	24091	2	0.58	1	0.81	1	Not tested	0.00	0	0.00	0	A and D were both WT	
427-A11	24091	2	1.57	1 or 2	1.93	2	Not tested	0.00	0	0.00	0	not sequenced	
427-B11	24091	2	1.41	1 or 2	1.63	2	Not tested	0.00	0	0.00	0	not sequenced	
427-C11	24091	2	1.06	1	1.21	1	Not tested	0.01	0	0.01	0	not sequenced	
427-D11	24091	2	1.98	2	2.03	2	Not tested	0.00	0	0.00	0	not sequenced	
427-E11	24091	2	1.94	2	1.94	2	Not tested	0.00	0	0.00	0	not sequenced	
427-F11	24091	2	1.84	2	1.84	2	Not tested	0.00	0	0.00	0	not sequenced	
427-G11	24091	2	1.54	1 or 2	1.76	2	Not tested	0.00	0	0.00	0	not sequenced	
427-H11	24091	2	1.75	2	1.76	2	Not tested	0.00	0	0.00	0	not sequenced	
427-A12	24091	2	1.99	2	2.15	2	Not tested	0.00	0	0.00	0	not sequenced	
427-B12	24091	2	0.72	1	1.26	1	Not tested	0.00	0	0.00	0	A and D were both WT	
427-C12	24091	2	1.69	2	1.50	1 or 2	Not tested	0.00	0	0.01	0	not sequenced	
427-D12	24091	1	2.34	2	2.03	2	Not tested	0.00	0	0.00	0	not sequenced	
427-E12	24091	1	1.98	2	2.04	2	Not tested	0.00	0	0.00	0	not sequenced	
427-F12	24091	1	1.89	2	1.97	2	Not tested	0.00	0	0.00	0	not sequenced	
427-G12	24091	1	1.56	1 or 2	1.77	2	Not tested	0.00	0	0.00	0	not sequenced	
427-H12	24091	1	1.57	1 or 2	2.36	2	Not tested	0.00	0	0.00	0	not sequenced	
428-A3	24091	1	2.12	2	1.75	2	Not tested	0.00	0	0.00	0	not sequenced	
428-B3	24091	1	2.69	>2	1.89	2	Not tested	0.00	0	0.00	0	not sequenced	
428-C3	24091	1	2.09	2	2.44	>2	Not tested	0.00	0	0.00	0	not sequenced	
428-D3	24091	1	2.05	2	2.39	2	Not tested	0.00	0	0.00	0	not sequenced	
428-E3	24091	1	2.48	>2	2.87	>2	Not tested	0.00	0	0.00	0	not sequenced	
428-F3	24091	1	2.33	2	2.76	>2	Not tested	0.00	0	0.00	0	not sequenced	
428-G3	24091	1	2.84	>2	0.22	0	Not tested	0.00	0	0.00	0	A and D were both WT	

TABLE 7-continued

Sequencing data from edited wheat haploids.													
Plant ID	Con-struct ID	copy #	TAV 2A		TAV_2D		TAV_2B		PMI		Cas9		Sequencing result
			Raw Copy #	Copy level	Raw Copy #	Copy level	Raw Copy #	Copy level	Raw Copy #	Copy level	Raw Copy #	Copy level	
428-H3	24091	1	2.83	>2	2.60	>2	Not tested		0.00	0	0.00	0	not sequenced
450-A11	24094	1	1.97	2	2.24	2	Not tested		0.00	0	0.00	0	not sequenced
450-B11	24094	1	2.13	2	2.04	2	Not tested		0.00	0	0.00	0	not sequenced
450-C11	24094	1	2.15	2	2.18	2	Not tested		0.00	0	0.00	0	not sequenced
450-D11	24094	1	1.04	1	0.99	1	Not tested		0.00	0	0.00	0	A & B were WT; D failed
450-E11	24094	1	2.35	2	2.01	2	Not tested		0.00	0	0.00	0	not sequenced
450-F11	24094	1	2.02	2	1.90	2	Not tested		0.00	0	0.00	0	not sequenced
450-G11	24039	1	1.76	2	1.72	2	Not tested		0.00	0	0.00	0	not sequenced
450-H11	24039	1	2.07	2	2.04	2	Not tested		0.00	0	0.00	0	not sequenced
452-H4	24038	2	2.62	>2	0.01	0	Not tested		0.00	0	0.00	0	A and D were both WT
452-A11	24038	2	2.24	2	2.28	2	Not tested		0.00	0	0.00	0	not sequenced
452-B11	24038	2	0.00	0	2.22	2	Not tested		0.00	0	0.00	0	B & D were WT; A failed
452-C11	24038	2	2.55	>2	2.22	2	Not tested		0.00	0	0.00	0	not sequenced
452-D11	24038	2	0.82	1	1.26	1	Not tested		0.00	0	0.00	0	A and D were both WT
452-E11	24038	2	2.43	>2	2.36	2	Not tested		0.00	0	0.00	0	not sequenced
452-F11	24038	2	2.12	2	2.21	2	Not tested		0.00	0	0.00	0	not sequenced
452-G11	24038	2	2.38	2	1.99	2	Not tested		0.00	0	0.00	0	not sequenced
452-H11	24038	2	1.82	2	1.83	2	Not tested		0.00	0	0.00	0	not sequenced
NANDA													
425-A2	WT	N/A	2.30	2	2.62	>2	1.908	2	0.00	0	0.00	0	not sequenced
425-B2	WT	N/A	2.28	2	2.41	>2	2.274	2	0.00	0	0.00	0	not sequenced
425-C2	WT	N/A	2.47	>2	1.92	2	1.962	2	0.00	0	0.00	0	not sequenced
425-D2	WT	N/A	2.10	2	2.11	2	1.772	2	0.00	0	0.00	0	not sequenced
447-A12	24038	2	1.72	2	1.90	2	2.02	2	0.00	0	0.00	0	not sequenced
447-B12	24039	2	2.18	2	1.62	2	1.47	1 or 2	0.00	0	0.00	0	not sequenced
447-C12	24039	2	1.78	2	2.40	2	1.90	2	0.00	0	0.00	0	not sequenced
447-D12	24039	2	1.58	1 or 2	1.70	2	2.18	2	0.00	0	0.00	0	not sequenced
447-E12	24039	2	2.13	2	1.82	2	2.14	2	0.00	0	0.00	0	not sequenced
447-F12	24039	2	2.25	2	1.78	2	2.15	2	0.00	0	0.00	0	not sequenced
447-G12	24039	2	1.90	2	2.30	2	2.23	2	0.00	0	0.00	0	not sequenced
447-H12	24039	1	2.34	2	1.95	2	0.89	1	0.00	0	0.00	0	WT
440-A2	24039	1	1.72	2	1.71	2	1.24	1	0.00	0	0.00	0	not sequenced
440-B2	24039	1	2.30	2	2.56	>2	1.77	2	0.00	0	0.00	0	not sequenced
440-C2	24039	1	3.05	>2	1.85	2	2.01	2	0.00	0	0.00	0	not sequenced
440-D2	24039	1	1.66	2	1.70	2	1.44	1 or 2	0.00	0	0.00	0	not sequenced
440-E2	24039	1	2.23	2	1.91	2	2.04	2	0.00	0	0.00	0	not sequenced
440-F2	24039	1	2.63	>2	2.07	2	2.28	2	0.00	0	0.00	0	not sequenced
440-G2	24038	11	1.91	2	1.87	2	2.10	2	0.00	0	0.00	0	not sequenced
440-H2	24038	1	1.85	2	1.80	2	1.97	2	0.00	0	0.00	0	not sequenced
440-A3	24038	1	2.52	>2	2.05	2	1.99	2	0.00	0	0.00	0	not sequenced
440-B3	24038	1	2.16	2	2.19	2	0.00	0	0.00	0	0.00	0	A & D were WT; B failed
440-C3	24038	1	2.58	>2	2.02	2	2.78	>2	0.00	0	0.00	0	not sequenced
440-D3	24038	1	2.34	2	2.32	2	0.00	0	0.00	0	0.00	0	A & D were WT; B failed
440-E3	24038	1	2.49	>2	2.23	2	2.47	>2	0.00	0	0.00	0	not sequenced
440-F3	24038	1	2.08	2	2.10	2	2.17	2	0.00	0	0.00	0	not sequenced
440-F4	24038	1	1.73	2	1.47	1 or 2	1.41	1 or 2	0.00	0	0.00	0	not sequenced
440-G4	24038	1	1.53	1 or 2	2.02	2	1.99	2	0.00	0	0.00	0	not sequenced
440-H4	24038	1	2.22	2	1.90	2	0.00	0	0.00	0	0.00	0	not sequenced
440-A5	24038	1	2.22	2	1.90	2	0.00	0	0.00	0	0.00	0	A & D were WT; B failed
440-A6	24039	2	2.49	>2	2.32	2	1.84	2	0.00	0	0.00	0	not sequenced
440-B6	24039	2	2.12	2	2.03	2	2.21	2	0.00	0	0.00	0	not sequenced
440-C6	24039	2	2.63	>2	2.07	2	2.28	2	0.00	0	0.00	0	not sequenced
440-D6	24039	2	2.49	>2	2.23	2	2.47	>2	0.00	0	0.00	0	not sequenced
440-E6	24039	2	2.45	>2	2.20	2	2.32	2	0.00	0	0.00	0	not sequenced
440-F6	24039	2	2.10	2	1.92	2	1.91	2	0.00	0	0.00	0	not sequenced
440-G6	24039	2	0.57	1	0.66	1	0.53	1	0.00	0	0.00	0	A, B & D were all WT
440-H6	24039	2	1.81	2	1.96	2	2.51	>2	0.00	0	0.00	0	not sequenced
447-A8	24038	1	2.42	>2	2.21	2	2.10	2	0.00	0	0.00	0	not sequenced
447-B8	24038	1	2.46	>2	2.32	2	2.09	2	0.00	0	0.00	0	not sequenced
447-C8	24038	1	2.09	2	2.08	2	2.29	2	0.00	0	0.00	0	not sequenced
447-D8	24038	1	2.13	2	2.14	2	2.34	2	0.00	0	0.00	0	not sequenced
447-E8	24038	11	2.36	2	2.31	2	2.44	>2	0.00	0	0.00	0	not sequenced
447-F8	24038	1	2.72	>2	2.28	2	2.00	2	0.00	0	0.00	0	not sequenced

TABLE 7-continued

Sequencing data from edited wheat haploids.													
Plant ID	Con-struct ID	copy #	TAV 2A		TAV_2D		TAV_2B		PMI		Cas9		Sequencing result
			Raw Copy #	Copy # level	Raw Copy #	Copy # level	Raw Copy #	Copy # level	Raw Copy #	Copy # level	Raw Copy #	Copy # level	
447-G8	24038	1	0.71	1	1.34	1 or 2	2.33	2	0.00	0	0.00	0	B & D were WT; A failed
447-H8	24038	1	2.25	2	2.29	2	2.01	2	0.00	0	0.00	0	not sequenced
456-A9	24038	2	2.19	2	1.59	1 or 2	2.03	2	0.00	0	0.00	0	not sequenced
456-B9	24038	2	2.13	2	2.11	2	2.02	2	0.00	0	0.00	0	not sequenced
456-C9	24038	2	2.16	2	1.85	2	1.45	1 or 2	0.00	0	0.00	0	not sequenced
456-D9	24038	2	2.56	>2	2.18	2	1.76	2	0.00	0	0.00	0	not sequenced
456-E9	24038	2	2.29	2	2.03	2	1.65	2	0.00	0	0.00	0	not sequenced
456-F9	24038	2	2.24	2	2.02	2	2.05	2	0.00	0	0.00	0	not sequenced
456-G9	24038	2	2.49	>2	2.03	2	0.00	0	0.00	0	0.00	0	A & D were WT; B failed
456-H9	24038	2	1.78	2	1.62	2	1.38	1 or 2	0.00	0	0.00	0	not sequenced
459-A2	24038	2	1.38	1 or 2	1.11	1	0.94	1	0.00	0	0.00	0	A & B were WT; D failed
459-B2	24038	2	1.86	2	1.91	2	2.12	2	0.00	0	0.00	0	not sequenced
459-C2	24038	2	1.94	2	2.09	2	1.42	1 or 2	0.00	0	0.00	0	not sequenced
459-D2	24038	2	2.09	2	2.05	2	1.91	2	0.00	0	0.00	0	not sequenced
459-E2	24038	2	2.18	2	2.12	2	2.12	2	0.00	0	0.00	0	not sequenced

Overall, we found that the editing frequency (number of edited haploids identified divided by the total number of haploids) for construct 24038 was 0.79%; for construct 24039 it was 0%; for construct 24091 it was 0%, and for construct 24094 it was 0.75%. However, this editing rate is certainly an under-estimate because we did not have assays to detect edits at many of the guide RNA target sites. Additionally, because we used T0 pollen that was either 1 or 2 copy, we know that with the 1-copy pollen, only 50% of the fertilizing pollen grains will contain the Cas9, and so only half of the embryos have the opportunity to be edited; similarly, for 2 copy parents, assuming random segregation of the transgenes in the male meiosis, we would expect about 75% of the pollen to contain Cas9, so 25% of the embryos cannot be edited. It is reasonable to conclude that, when one is trying to use this simultaneous editing plus haploid induction technology with the editing machinery carried by the pollen, it may in some cases be more optimal to use a promoter that express specifically or highly in pollen and in sperm cells, so that the Cas9 can be expressed at a higher level. In cases where the gene target might impact development of the haploid inducer plant, having a pollen or sperm-preferred promoter that does not express in leaves might be useful because it would avoid editing the target gene in the haploid inducer plant during development—perhaps editing it for the first time in pollen.

Because the sperm cells fertilize the egg, they have the potential to deliver Cas9 RNA and protein (as well as the transgene DNA itself, integrated into one of the male chromosomes that will be eliminated). As we demonstrated in the wide-cross work in this example, it may work well to have the Cas9 and/or guide RNA under the control of a promoter that specifically or highly expresses in pollen, and in particular in sperm cells, when using a haploid inducer as the male to edit elite lines. We do not know exactly whether MATRILINEAL, EXPB1, EXPB2, and PRF3 express in the vegetative nucleus, the sperm cells, or both, and whether there might be any expression in a zygote cell type, but these were chosen because they are supposedly highly and/or

specifically expressed in pollen. The PRF3 promoter has a DUO1 binding motif in the promoter, which may indicate it expresses in sperm cells. This is consistent with that promoter having higher editing frequency. The fact that we found many edited wheat haploids after the wide cross makes it clear that when there is high expression of Cas9 in pollen, using these or any other promoter, that expression can lead to editing in the wheat embryos after the wide cross. There is a strong possibility that these promoters, as well as other promoters that drive expression in pollen, or in particular in the sperm cells, might increase the efficiency of the editing process during corn haploid induction, or rice haploid induction.

Similarly, in the next example below, we show haploid editing in a dicot using a CENH3-modified-haploid inducer line, and we use constitutive promoter to drive the Cas9. But in an attempt to increase the efficiency of the haploid editing, we could opt to use a promoter that drives high and/or specific expression in egg cells, such as the EGG APPARATUS1 gene's promoter ("prEA1") (see, e.g., Gray-Mitsumune, M. and Matton, D. P., The Egg apparatus 1 gene from maize is a member of a large gene family found in both monocots and dicots, *Planta* 223(3):618-625 (February 2006)) or EGG CELL1 (EC1) (see, e.g., Sprunck S, et al., Egg cell-secreted EC1 triggers sperm cell activation during double fertilization. *Science* 2012; 338:1093-97; PMID: 23180860; <http://dx.doi.org/10.1126/science.1223944>).

As an example of this, one could use a sperm-cell expressed promoter, such as the *Arabidopsis* sperm-specific DUO1 promoter (see, e.g., Engel, et al., Green Sperm. Identification of Male Gamete Promoters in *Arabidopsis*, *Plant Physiology* August 2005, 138 (4) 2124-2133; DOI: 10.1104/pp.104.054213), or homologs of DUO1 from other species (for instance, the maize genes GRMZM2G105137 and GRMZM2G046443 are both DUO1 homologs that share a similar pollen-specific expression pattern). If one used any of these to drive Cas9 expression in the sperm cells of a haploid inducer line like RWK, NP2222-II, or an *matl* mutant, it might make a highly efficient haploid editor line

for use in editing diverse elite maize or wheat germplasm, via intraspecific or wide cross, respectively.

Other suitable sperm-expressed promoters for this concept of driving high Cas9 expression in sperm cells would include the DUO1 homologs in wheat, rice, barley, tomato, sunflower, or other monocots or dicots. Other suitable promoters for this concept are shown in Table 8 below. These promoters, or their homologs in crop species—might be very useful for this concept. The principal at work is that gamete cell expression of the editing machinery can increase the rate or efficiency of this invention because it means that there will be abundant editing protein or RNA present or delivered to the embryo during fertilization so that editing can happen rapidly.

TABLE 8

Promoters List: promoters one can use in a transgene to drive high sperm cell expression of editing machinery to boost the efficiency of simultaneous editing and doubled-haploid induction ("SEDHI").			
Gene Name	Gene ID	Maize Ortholog	Rice Ortholog
DUO1	At3G60460	GRMZM2G105137, GRMZM2G046443	LOC_Os04g46384
MGH3	At1G19890	NA	NA
GEX1	At5G55490	GRMZM2G388045	LOC_Os09g27040, LOC_Os07g47194
GEX2	At5G49150	GRMZM2G036832	LOC_Os09g25650
GEX3	At5G16020	GRMZM2G458159	LOC_Os01g42060
HAP2/GSC1	At4G11720	GRMZM2G412911	LOC_Os05g18730
CycB1	At4G37490	NA	NA
DAZ1	At2G17180	GRMZM2G132057	NA
DAZ2	At4G35280	NA	LOC_Os02g19180
DAZ3	At4G35700	NA	NA
PCR11	At1G68610	NA	NA
DAN1	At3G04620	NA	NA
TIP1	At3G47440	NA	LOC_Os04g46490
MKKK20	At3G50310	NA	NA
DAF1	At3G62230	NA	NA
DAW1	At4G35560	GRMZM2G176647	NA
DAU2/DMP9	At5G39650	NA	NA

VII. Simultaneous Haploid Induction and Editing in Dicots Via Wide Cross or Via Crosses to CENH3-Altered Lines or Other Haploid Inducing Lines.

In vivo haploid induction can also be achieved using interspecific or intergeneric wide crosses on dicot plant species, for example, in cotton (Turcotte et al. 1969, Semigametic production of haploids in pima cotton. *Crop Sci.* 9:653-655) and tobacco (Burke et al, 1979, Maternal haploids of *Nicotiana tabacum* L. *Science* 206:585; Wernsman et al. 1989, Androgenetic vs. gynogenetic doubled haploids of tobacco. *Crop Sci.* 29:1151-1155). Haploid *Arabidopsis* plants can be obtained by crossing with pollen from mutant CENH3 plant, or by crossing said plants as females to wild type pollen (Ravi and Chan, 2010, Haploid plants produced by centromere-mediated genome elimination. *Nature* 464: 615-618). Other candidate genes which may be modified to generate a haploid inducer and SEDHI editing line include KNL2 and CENPC (both of which may operate via centromere-mediated uniparental genome elimination) as well as MSI2 and sunflower PLA2. In this case, the haploid-inducing genome (be it the male or female in the cross) also contains the editing machinery, so that the editing can be achieved during the haploid induction process, with the result being an edited maternal or paternal haploid progeny plant without altered CENH3 or editing transgenes. See, e.g., WO 2017/004375, incorporated herein by reference in its entirety. Transgenic locus expressing editing machinery can be introduced into any dicot crops or their wild relatives

of *Brassica*, tomato, pepper, lettuce, eggplant, soybean, sunflower, sugar beet, cotton, alfalfa, tobacco, and others. The transgenic lines expressing editing machinery are then used as pollen donors, or in the case of CENH3, either pollen donors or acceptors, in interspecific or intergeneric wide crosses for haploid induction and simultaneous genome editing. For example, *N. glauca* transgenic CRISPR-Cas9 lines expressing sgRNA targeting tobacco gibberellin 20-oxidase are created through *Agrobacterium*-mediated transformation and used to pollinate emasculated tobacco flowers to induce haploid plants with their genome edited at the gibberellin 20-oxidase locus. Preferably, an easily transformable line with large number of pollen is used as pollen donor for haploid induction and to provide the editing machinery transiently. The recipient plant for haploid production has flowers that are easy to emasculate or is male sterile. More preferably, a color or other visual marker is present in the induction line or is included in the editing locus to easily differentiate haploid embryos or plants from diploids resulted from normal zygote development.

We exemplified this by utilizing an *Arabidopsis* haploid inducer line in the Columbia ecotype, and transforming it with a construct encoding expression of Cas9 and a single guide RNA targeting the GLABROUS1 gene (GL1) which, when knocked out, gives a trichome-less phenotype. We crossed the T0s as females by *Landsberg erecta* (Ler) ecotype pollen, and recovered gl1 edited haploid progeny.

The haploid inducer materials were obtained from the Comai lab at UC Davis. These materials are typically utilized as paternal haploid inducer lines (causing androgenesis, when crossed as females to wild-type males) but can also act as maternal haploid inducers (causing gynogenesis, when crossed as males to wild-type females). These lines have been altered to become haploid inducers by replacing the native CENH3 gene with a *Zea mays* CENH3 transgene as reported in (Maheshwari, et al, 2017, Centromere location in *Arabidopsis* is unaltered by extreme divergence in CENH3 protein sequence. *Genome Research* 27(3)).

In particular, both copies of the native AtCENH3 gene was knocked out and complemented with the stably inserted ZmCENH3 transgene, which did not impact normal plant development, and did not produce haploids upon self-pollination, but did produce about 10% haploids upon outcross. This is a modification to the original concept of CENH3-tailswap described in detail in (Ravi and Chan, 2010, Haploid plants produced by centromere-mediated genome elimination. *Nature* 464: 615-618) and many subsequent publications.

After we obtained the CENH3* lines from UC Davis, we grew them up, confirmed that they had the ZmCENH3 transgene and were homozygous "null" for the native AtCENH3 gene. We did this by designing a taqman qPCR assay for ZmCENH3 (assay #2298) and by using PCR and gel electrophoresis to test 183 seedlings for the zygosity of the AtCENH3 genotype by running PCR using the XbaI forward and reverse primers (SEQ NO TKX and TKY) and Reddy mix at 60° C. annealing temperature and cutting with the XbaI restriction enzyme overnight at 37° C. The wild-type allele would be cut by this enzyme and produce two bands (189 bp, 25 bp) while the mutant would remain at 215 bp. These tests showed that all of the seed that UC Davis sent were homozygous for the mutant allele Atcenh3-1, and that there were multiple copies of the ZmCENH3 transgene present.

Confident that these acquired seeds were indeed haploid inducers, we kept 100 plants and initiated floral dip transformation with binary vector 24075 (SEQ ID NO: 98)

containing a sgRNA cassette targeting the *Arabidopsis* (GL1) gene (AT3G27920) at two target sites. The target sequences are 5'-GGAAAAGTTGTAGACTGAGA-3', and 5'-GCAGTGATGAACAATGACGG-3' (complementary strand). The disruption of the GL1 gene produces visible phenotypes of partially or completely glabrous plants (glabrous plants lack trichomes). The Cas9 gene (cCas9-05) in this vector was driven by the *Arabidopsis thaliana* elongation factor promoter. The vector also contains two selectable marker cassettes conferring Kan resistance and AmCyan fluorescence driven by the CMP-02 promoter and Glycine max UBI-01 promoter respectively. The vector was moved into the agrobacterium strain EHA101 and then floral dip transformed into the haploid inducer *Arabidopsis* plants.

The transformation protocol was as follows: In the morning we spread 24075 EHA101 RecA *Agrobacterium* obtained from plates to YPSpec100Kan50 plates. We cultured these in 28° C. dark for 24 hours. We prepared infiltration medium (4 L): ½ XMS salts (8.66 g), 1×gamborg's B5 vitamins (4 ml), 5% (W/V) sucrose (200 g), 0.044 µM BAP (12.5 mg-12.5 mL DMSO) 40 µL, followed by filter sterilization. We then added 250 µl 40 mg/ml AS (20 mg/L) and 25 µl Silwet L-77(50 µl/L) to 500 ml Infiltration media. Using a loop to collect the *Agrobacterium* and put in 50 ml tube with ~10 ml of the filter sterilization, we suspended the *Agrobacterium* until it produced 1 L with an optical density 600 of 0.54. We dipped the inflorescence shoot in to the suspension medium for 20-30 seconds and used the lid to cover the tray. We repeated this for a second time with another suspension of OD600 of 0.552.

About 4 weeks after transformation, approximately 100,000 self-pollinated seeds were harvested and incubated at 4° C. for two days vernalization, and then the seeds were sterilized by soaking in 70% ethanol for 1 minute and then soaking in 50% (V/V) bleach with 0.05% (v/v) Triton X-100 for a further 10 minutes, then washing the seeds in four changes of sterile water. The seeds were then placed on kanamycin (50 µg/ml) plates for germination-screening/selection in a plant tissue culture room (23° C. day, 24° C. night, 16 hours lighting). 38 positive transformants were identified because they were resistant to the kanamycin selection, and they were grown into seedlings before being transferred onto soil and sampled to test for the presence of the Cas9 T-DNA (assay #3049) as well as the status of the two guide RNA cut sites (assays #3321 and #3322). 10 single copy and 15 2-copy events were identified that had both alleles of GL1 mutated and that had a trichomeless phenotype. These plants were prioritized because they had shown evidence of Cas9 activity (by virtue of the mutated GL1 and the glabrous phenotype), they had the Cas9 transgene and they had the ZmCENH3 transgene by qPCR assay. These plants were induced to flower for a long period of time by keeping them in the following growth conditions: 16 hours light, 23° C. Day 20° C. night temperature, not >60% relative humidity.

At the same time as these haploid inducer plants that were transformed with the Cas9 construct were being identified, we were sowing and growing a population of *Landsberg erecta* (Ler) seed obtained from the *Arabidopsis* Biological Resource Center at Ohio State University (line #CS20). These are wild type seed and the sequence of the GL1 guide RNA target sites in CS20 match that of the guide RNA in our construct. We allowed both populations to flower and made about 2000 controlled crosses, using the wild-type Ler plants as the male pollen-donor, crossing onto the approximately 25 haploid inducers with the Cas9 construct, which was used as the female. We made up to 100 crosses per

female, marking the crossed flowers with a black marker and removing flowers that we did not cross so as to limit the potential of harvesting self-pollinated siliques. In most cases, we emasculated the female flowers prior to pollination by removing the anthers with forceps, again to avoid contamination with self-pollinated seed, but in some cases this was not necessary because the anthers were young or mal-developed.

About 15 days we harvested the siliques which had developed a light brown color. Then we opened the siliques and planted the seeds in the soil. Then put them in the 6° C. (day and night), 8 hours day length, 200 umol/m²s lighting, 60% relative humidity growth chamber for 4 days. Then we transferred them to 16 hours light, 23° C. Day, 20° C. night temperature, not >60% humidity growth chamber for 7-10 days. We observed a high frequency of aborted seed in almost all of the siliques, averaging about 40-50% of the total seeds. This number of aborted embryos is very consistent with the performance of this haploid inducer material in published reports. Without wishing to be constrained by this theory, it has been speculated that the aborted seed is most likely caused by partial or complete genome elimination in the endosperm leading to endosperm imbalance and failure. This is a natural phenomenon in CENH3-type haploid inducer lines during outcross and is likely not connected with the presence of the Cas9 transgene. These aborted embryos do not germinate. Because of the steady and reliable rate of embryo abortion in every outcrossed silique, we ended up using the absence of that phenotype to screen away siliques that were accidental self-pollinations. That way we germinated siliques that had been outcrossed.

In total we recovered approximately 2000 germinated progeny, the majority of which were outcrossed. We identified the edited haploids via a combination of qPCR marker assays and/or phenotypic screening. The markers that we used to detect the edited haploids were as follows.

First, we looked for a "0" score for the ZmCENH3 assay. This indicates that the plant is a haploid because the maternal genome has been lost, and so the ZmCENH3 transgene, which is present in multiple copies of the mother haploid inducer plant, has also been lost. The diploids, in contrast, will be hybrids between the maternal and paternal genome, and will have a "1" or "2" or higher Taqman score for this assay, depending on the copy number of the mother plant. The key is that all diploids will show evidence of this transgene, but paternal haploids, having only the Ler genome, will not and will thus be a "0."

Second, we looked for a "0" score for the Cas9 assay, which indicates that it is non-transgenic. This can also be seen visually by using a fluorescent light and looking for the CFP fluorescent marker.

Third, we looked for a "0" score for one of the GL1 target site assays, which indicates that the plant has been edited. The diploid plants might show a "0," "1" or "2" for those assays, but the haploids either showed a "2" or a "0." The first of the two GL1 guide RNAs apparently had a much higher editing efficiency than the second, because assay 3321 had a high preponderance of "0"s and "1"s in the haploid inducer T0s, but 3322 had mostly "2"s.

Using these assays, we were able to identify unedited haploids (which were "0" for ZmCENH3 and Cas9, but had "2" scores for both GL1 target sites) and also edited haploids (which had a "0" for the ZmCENH3, Cas9 and GL1 (3321) assays). We were also able to identify diploid hybrids that had Cas9 (and often were edited at the GL1 sites) and diploid hybrids that did not have Cas9 (and often had one copy of GL1 edited (from the maternal parent) but not the

other, and thus had a score of “1” for the GL1 assay. We were also able to identify several putative edited haploids because they had a score of “0” for the target site assay (3321), the ZmCENH3 (2298) and the Cas9 (3049). See Table 9 below for an example of progeny Taqman data from parent 5
USR01424136 containing three putative edited haploids (plant 254 in well F2, plant 260 in well D3, and plant 261 in plant E3).

pollination of the fully-edited mother plant and production of null segregant, fully edited (and thus glabrous) progeny. The Taqman assays were able to detect and screen out these false positives, because they directly tested for the presence of not only the Cas9 transgene, but also the ZmCENH3 allele, which would certainly be present in any self-pollinated contaminating seed. We found several examples of self-pollinated seed that all came from one mother plant. The

TABLE 9

Progeny analysis from parent USR01424136.											
		AtGLI-1 cut site 3321		AtGLI-2 cut site 3322		ZmCENH3 2298		Cas9 3049		Putative Haploid	Putative Edited
PLATE 1045	HI parent was single copy Cas9	Raw Copy #	Copy# level	Raw Copy #	Copy# level	Raw Copy #	Copy# level	Raw Copy #	Copy# level		
Well	Plant ID										
E2	USR01424136 X Ler-253	0.06	0	0.87	1	4.30	>2	2.93	>2		X
F2	USR01424136 X Ler-254	0.00	0	0.32	0 or 1	0.00	0	0.00	0	X	X
G2	USR01424136 X Ler-255	1.32	1 or 2	2.06	2	3.16	>2	0.00	0		
H2	USR01424136 X Ler-256	0.02	0	0.99	1	2.51	>2	2.99	>2		X
A3	USR01424136 X Ler-257	0.04	0	0.87	1	2.40	2	2.84	>2		X
B3	USR01424136 X Ler-258	0.03	0	1.64	2	2.99	>2	3.17	>2		X
C3	USR01424136 X Ler-259	0.03	0	1.21	1	5.28	>2	5.28	>2		X
D3	USR01424136 X Ler-260	0.06	0	2.01	2	0.00	0	0.00	0	X	X
E3	USR01424136 X Ler-261	0.00	0	2.01	2	0.01	0	0.00	0	X	X
F3	USR01424136 X Ler-262	2.04	2	2.10	2	0.00	0	0.00	0	X	
G3	USR01424136 X Ler-263	1.36	1 or 2	1.25	1	0.00	0	0.00	0	X	
H3	USR01424136 X Ler-264	1.75	2	1.71	2	0.00	0	0.00	0	X	
A4	USR01424136 X Ler-265	0.00	0	1.67	2	3.06	>2	3.16	>2		X
B4	USR01424136 X Ler-266	1.66	2	2.32	2	0.00	0	0.00	0	X	
C4	USR01424136 X Ler-267	2.09	2	1.94	2	3.99	>2	0.00	0		
D4	USR01424136 X Ler-268	1.47	1 or 2	2.08	2	6.34	>2	1.51	1 or 2		
E4	USR01424136 X Ler-269	1.95	2	1.76	2	3.19	>2	0.00	0		
F4	USR01424136 X Ler-270	1.92	2	2.17	2	4.28	>2	0.02	0		
G4	USR01424136 X Ler-271	2.02	2	1.85	2	4.31	>2	0.00	0		
H4	USR01424136 X Ler-272	0.00	0	1.71	2	1.65	2	1.12	1		X

Simply by germinating seeds and sampling for qPCR Taqman analysis, we were able to identify 8 putative edited haploids. Edited haploids were also identified by phenotypic visual screening, and then confirmed later by Taqman assay. We screened for the edited haploids by looking for trichomeless, or glabrous, plants, which indicated that they did not have any wild-type alleles for the GL1 gene, and by looking for a lack of cyan fluorescent protein (“CFP”) expression in the embryo or seedling root. This indicated that they lacked the Cas9 T-DNA. We observed several of these plants, and submitted them for Taqman assays. For three such plants that we identified phenotypically, we were able to confirm that they were truly edited haploids by the Taqman assays. We were aware of the fact that it is possible that some of these glabrous plants that lack CFP were false positives, either because the CFP was silent or because of self-

pollination notes for that mother indicated that there was highly abundant pollen that may have resulted in some self-pollination. We excluded these progeny from the total analysis.

All of the putative edited haploids identified by Taqman assay were sequenced. We used PCR to amplify the edited alleles, and then subcloned and sequenced at least 8 colonies for each putative edited allele. See Table 10 for the sequence changes we found in the edited haploids at the first guide RNA (assay #3321) target site, as well as the Taqman data from the T0 parents. In total, we found 19 putative edited haploids, and we confirmed that the 3321 target sites had mutations in 11 of the 12 edited haploids that we attempted to sequence. Whether the other 7 would also have mutations will be confirmed upon sequencing. See the sequence alignment for these edits in FIG. 24.

TABLE 10

Taqman and sequence data from 19 edited haploids.												
			AtGLI-1 cut site 3321		AtGLI-2 cut site 3322		ZmCENH3 2298		Cas9 3049		Target site mutation	PA confirm?
Plate	Well	Plant ID	Raw Copy #	Copy# level	Raw Copy #	Copy# level	Raw Copy #	Copy# level	Raw Copy #	Copy# level		
1033	A3	USR01424135 X Ler-425	0.00	0	1.67	2	0.04	0	0.00	0	wild type	Not done
1033	C3	USR01424135 X Ler-427	0.21	0	2.43	>2	0.01	0	0.00	0	insert A	Yes

TABLE 10-continued

Taqman and sequence data from 19 edited haploids.													
Plate	Well	Plant ID	AtGLI-1 cut		AtGLI-2 cut		ZmCENH3		Cas9		Target site mutation	PA confirm?	
			Raw Copy #	Copy# level	Raw Copy #	Copy# level	Raw Copy #	Copy# level	Raw Copy #	Copy# level			
													3321
1033	E4	USR01424135 X Ler-437	0.08	0	2.04	2	0.00	0	0.00	0	insert T	Yes	
1042	E5	USR01424136 X Ler-25	0.16	0	2.95	>2	0.00	0	0.00	0	insert A	Not done	
1042	G10	USR01424136 X Ler-67	0.00	0	2.19	2	0.00	0	0.00	0	delete AG	Not done	
1042	G12	USR01424136 X Ler-83	0.00	0	1.86	2	0.00	0	0.00	0	delete G	Not done	
1043	B11	USR01424136 X Ler-154	0.16	0	1.59	1 or 2	0.01	0	0.00	0	Not done	Not done	
1045	F2	USR01424136 X Ler-254	0.00	0	0.32	0 or 1	0.00	0	0.00	0	delete 8 nt*	Not done	
1045	D3	USR01424136 X Ler-260	0.06	0	2.01	2	0.00	0	0.00	0	insert T	Not done	
1045	E3	USR01424136 X Ler-261	0.00	0	2.01	2	0.01	0	0.00	0	delete TG	Not done	
1046	D11	USR01431609 X Ler-111	0.09	0	1.59	1 or 2	0.02	0	0.01	0	insert A	Not done	
1046	G12	USR01431609 X Ler-122	0.02	0	1.62	2	0.00	0	0.00	0	insert T	Not done	
1046	H12	USR01431609 X Ler-123	0.00	0	0.00	0	0.00	0	0.00	0	delete CTG	Yes	
0583	D12	USR01431603 X Ler-80	0.00	0	1.50	1 or 2	0.00	0	0.00	0	Not done	Not done	
0584	A9	USR01431603 X Ler-137	0.00	0	1.87	2	0.00	0	0.00	0	Not done	Not done	
0584	C11	USR01431603 X Ler-155	0.05	0	2.06	2	0.00	0	0.17	0	Not done	Not done	
0584	G11	USR01431603 X Ler-159	0.09	0	2.15	2	0.00	0	0.00	0	Not done	Not done	
0584	C12	USR01431603 X Ler-163	0.00	0	1.35	1 or 2	0.00	0	0.11	0	Not done	Not done	
0584	F12	USR01431603 X Ler-166	0.00	0	1.65	2	0.00	0	0.00	0	Not done	Not done	
0585	H7	USR01431603 X Ler-212	0.06	0	2.05	2	0.00	0	0.01	0	Not done	Not done	
Female Parent		USR01424135	0.03	0	1.42	1 or 2	4.46	>2	2.98	>2	ΔG, +T chimera	Diploid	
To Plants		USR01424136	0.03	0	1.13	1	3.59	>2	2.76	>2	Not done	Diploid	
		USR01431603	0.14	0	1.25	1	2.48	>2	3.42	>2	Not done	Diploid	
		USR01431609	0.18	0	1.1	1	4.75	>2	5.57	>2	Not done	Diploid	

*delete 16 nt insert CTAAACAT

We further ran leaf samples from three edited haploid plants through ploidy analysis, along with three diploid controls (tissue sampled from the maternal parent plants), which showed that they were true haploids (FIGS. 18-23). This served to reconfirm their status as edited haploids.

maternal Cas9, prior to elimination of the maternal genome. Without wishing to be constrained by this final number, this means that, using this system, as a percentage of total progeny, 9.7%*16.9%=1.64% of all germinated progeny were edited haploids.

TABLE 11

Haploid induction rate and editing rate data from three sets of progeny, each derived from a different SEDHI inducer female parent crossed by Landsberg erecta pollen.							
ID	Parent plant Cas9-05	Parent plant cNpt2-10	Total samples	Haploid number	Haploid rate	Edited Haploid	Edited Haploid rate
USR01431603 X Landsberg erecta	>2	>2	230	36	15.65	7	19.44
USR01431609 X Landsberg erecta	>2	>2	123	14	11.38	3	21.43
USR01431804 X Landsberg erecta	2	1	252	9	3.57	0	0.00

In three parental lines where we were confident that there was no self-pollination contamination, we did not do any phenotypic pre-screening, but instead sampled all germinated progeny for Taqman analysis (Table 11). The three female parents for these progeny were USR01431603, USR01431609, and USR01431604. We found a haploid induction rate of about 9.7% calculated by dividing the number of progeny that lack the ZmCENH3 and Cas9 transgenes (59) by the total number of progeny sampled (605). Of the 59 haploids we found that 10 were edited. That means 16.9% of haploids, on average, were edited by the

The rate of CENH3* type haploid editing or other paternal haploid editing (using a maternal haploid inducer line) might be increased through the use of a promoter that drives the expression of Cas9 and/or the guide RNA to a higher level in the egg cell before fertilization and/or in the zygote cell during or after fertilization. An example of such a promoter would be the promoter for EA1 (EGG APPARATUS1) (GRMZM2G456746), although there are many other examples. One could also express the Cas9 in the context of an egg apparatus—specific enhancer (EASE), which is a 77-bp sequence that stimulates expression of adjoining

genes in the egg cell or the very early zygote (see, e.g., Yang, et al. An Egg Apparatus-Specific Enhancer of *Arabidopsis*, Identified by Enhancer Detection, PLANT PHYSIOLOGY November 2005, 139 (3) 1421-1432; DOI: <https://doi.org/10.1104/pp.105.068262>).

VIII. Simultaneous Haploid Induction and Editing by Directly Modifying a Target Base in Genomic DNA Sequence.

Targeted mutagenesis of DNA sequence can also be achieved through direct conversion of one DNA base to another without requiring double stranded breaks (DSBs). For example, cytidine deaminase APOBEC1, adenine deaminase, and other enhancing components like Uracil DNA glycosylase (UDG) can be fused to Cas9 (A840H) nickase or nuclease-inactivated dead Cas9 (dCas9) to direct editing of DNA sequence without introducing double strand DNA breaks (Komor et al. 2016. Programmable editing of a target base in genomic DNA without double-stranded DNA cleavage. Nature doi:10.1038/nature17946; Gaudelli et al. 2017. Programmable base editing of A:T to G:C in genomic DNA without DNA cleavage. Nature doi:10.1038/nature24644; Komor et al. 2017. Improved base excision repair inhibition and bacteriophage Mu Gam protein yields C:G-to-T:A base editors with higher efficiency and product purity. Science Advances, Vol. 3, no. 8, eaao4774, DOI: 10.1126/sciadv.aao4774). This kind of base editor machinery can also be delivered through haploid induction line to induce base editing in target sequences directly in other varieties. For example, a guide RNA sequence, xZmVLHP-03 (5'-AGGCGTCGAGCAGCGAGGTG-3', SEQ ID NO: 28) is designed to target the cytidine deaminase base editor system to convert ZmVLHP gene exon 2 genomic sequence 5'-AGGCGTCGAGCAGCGAGGTG-3' (SEQ ID NO: 28) into 5'-AGGCGTTGAGCAGCGAGGTG-3' (SEQ ID NO: 29), thus changing the arginine codon CGA into a stop codon (TGA) in the coding sequence and causing premature termination of the protein sequence and functional gene knockout. The C to T mutation is underlined. Similarly, chimeric nCas9- or dCas9-adenine deaminase base editing system can be used to mutate the coding region, splicing junction or promoter sequence of ZmVLHP or other genes to generate variants that have altered gene activity. Both cytidine and adenine deaminase are particularly useful for altering transcript splicing site since canonical splicing junction has 5'-... AG/GT ... 3' sequence (or 5'-... AC/CT ... 3' in the opposite strand).

IX. Simultaneous Haploid Induction and Editing by Allele Replacement with DNA Template

Not only can in vivo haploid induction system be used to introduce protein, RNA or DNA for cleavage or conversion of target sequence, it can also be used to deliver DNA template for homology-dependent repair for precise sequence replacement in the target region in the form of transgenic DNA. The template DNA can be inserted into the inducer line genome carrying genome editing machinery such as CRISPR-Cas9 system, either in the same transgenic

locus or different locus. When both Cas9-sgRNA and template DNA are present in the induced haploid embryos, cleavage of the target sequence will result in repair of the chromosomal break with the homologous transgenic DNA sequence as template. For example, for creating E149L mutation in ZmPYL-D gene (GRMZM2G048733_P02) (see WO16033230, incorporated herein by reference), DNA fragment containing donor sequence (5'-CCTTGGTG TTGCCGTCGGGGACGTCGACGACGAATGACAGGA TGACGAGCGTCCCTGGCCGGCCG TCGATGACCT-3', SEQ ID NO: 30) is used as repair donor. It should be noted that additional homology sequences can be added to flank this core repair donor sequence. One or more copies of this repair donor sequence are inserted into Cas9-sgRNA expression vector 23136 (SEQ ID NO: 31) which expresses guide RNA 5'-GTCGGGGACGTCGACGACGA-3' (SEQ ID NO: 32) to form allele modification vector pBSC23136-AMD. It should be noted that the potential PAM site has been removed from the donor DNA sequences so that the integrated donor sequence will not be cleaved by the Cas9-sgRNA complex expressed from pBSC23136-AMD. pBSC23136-AMD is transformed into haploid inducer line NP2222-HI to generate transgenic editing line. Transgenic editing-haploid induction lines are selfed to produce progeny lines homozygous editing loci. These homozygous lines are used to pollinate target elite maize inbred lines to induce haploid formation and also introduce modified alleles by expressed Cas9-sgRNA using donor DNA present transiently before pollen donor chromosomes are eliminated.

X. Inducing Haploids and Simultaneous Gene Editing in Rice

A HI-rice line is obtained. For example, the rice MATL ortholog, Os03g27610 (SEQ ID NO: 33, is mutated to create a new rice HI line. This line is transformed with a vector comprising a site-directed mutagenesis system for editing the rice genome, for example the CRISPR/Cas9 system.

The rice HI line is crossed with a different rice line, preferably an elite line, to produce at least one progeny haploid embryo. During the cross to produce at least one progeny haploid embryo, the HI parent rice plant also causes the genome editing machinery, e.g., Cas9 plus a guide RNA, to be delivered to the embryo. At that point, the editing machinery operates to edit the genome of the haploid embryo, and thus an edited, haploid progeny plant is obtained.

XI. Taqman Assays and Conditions.

Several assays are mentioned by number or by target name. Provided below is a table of assays mentioned above and the sequences of the relevant primers and probes. Conditions for PCR are standard for all assays and are as follows: Denature at 98° C. for 2 minutes; followed by 35 cycles of (i) denature at 98° C. for 30 seconds, (ii) anneal at 60° C. for 30 seconds, (iii) extension at 72° C. for 1 minute; followed by final extension at 72° C. for 10 minutes with a hold at 4° C. until ready. Assays are carried out at these conditions unless otherwise noted below.

TABLE 11

Assay primers and probes.			
Target Assay No.	Cas9-in corn 2540	Sequence	SEQ ID NO:
Forward Primer	FE09340	TTGTGCTGCTCCACGAACA	39
Reverse Primer	FE09341	GCCAGCCACTACGAGAAGCT	40
Probe	FE09342	CTGCTTCTGCTCGTTGTCTCCGG	41
Target Assay No.	mat1 2827	Sequence	SEQ ID NO:
Forward Primer	FE10299	GCGGATGCTGGCACAGC	42
Reverse Primer	FE10300	GGCATTGCTTCCTTCTCCG	43
Probe	FE10301	CAGGGAGCGAGGTAC	44
Target Assay No.	PMI 1750	Sequence	SEQ ID NO:
Forward Primer	FE07390	CTGGTGGCCAACGTGAAGTT	45
Reverse Primer	FE07391	GCTTCACGGGCTGGGTC	46
Probe	FE07392	AGGCCAAGCCGCCAACCAG	47
Target Assay No.	MATL-WT 2826	Sequence	SEQ ID NO:
Forward Primer	FE10297	GCGGATGCTGGCACAGA	48
Reverse Primer	FE10298	GCATTGCTTCCTTCGCCA	49
Probe	FE10299	CAGGGAGGTACGAACC	50
Target Assay No.	TAV_4A 3252	Sequence	SEQ ID NO:
Forward Primer	FE11306	GCGGCGAAGAAGCGAA	51
Reverse Primer	FE11307	GCGGCGTCTCCAGCTTC	52
Probe	FE11308	CCAGGAACTGCG	53
Target Assay No.	TAV_4B 3253	Sequence	SEQ ID NO:
Forward Primer	FE11309	AAGAAACGCCGCTGAGT	54
Reverse Primer	FE11310	ACCTTGCGGGCGCTT	55
Probe	FE11308	CCAGGAACTGCG	56
Target Assay No.	TAV_4D 3254	Sequence	SEQ ID NO:
Forward Primer	FE11309	AAGAAACGCCGCTGAGT	57
Reverse Primer	FE11311	CCTTGCGCGGCGTC	58
Probe	FE11308	CCAGGAACTGCG	59
Target Assay No.	GW2-01 3065	Sequence	SEQ ID NO:
Forward Primer	FE10799	TGATCCTCGAGGCCAAGCT	60
Reverse Primer	FE10800	AGGTCGAGTCCCTCCA	61
Probe	FE10801	CCTGCTACCCGGGC	62

TABLE 11-continued

Assay primers and probes.			
Target Assay No.	GW2-02 3095	Sequence	SEQ ID NO:
Forward Primer	FE10991	CGCGCCCTGCTACCC	63
Reverse Primer	FE10992	GCGCGTGCTTACCAGGA	64
Probe	FE10993	TCGAGGAGTGCCC	65
Target Assay No.	TaVHLP2- 2A 3332	Sequence	SEQ ID NO:
Forward Primer	FE11312	CACCGATGAGCAGGCG	66
Reverse Primer	FE11313	AGATACACCTTCCGGCCG	67
Probe	FE11314	TTCTCCCCGAAGC	68
Target Assay No.	TaVHLP2- 2D 3333	Sequence	SEQ ID NO:
Forward Primer	FE11312	CACCGATGAGCAGGCG	69
Reverse Primer	FE11313	AGATACACCTTCCGGCCAGT	70
Probe	FE11314	CTCTCCCCGAAGC	71
Target Assay No.	3049	Sequence	SEQ ID NO:
Forward Primer	FE10730	CAAGTTTCTGGACAAGGAGATTCTC	72
Reverse Primer	FE10731	AAGAATTCCTTCTTAATAGCTGGAGA	73
Probe	FE10732	CACGAGCACATTGCTAACCTTGCTGG	74
Target Assay No.	TaVHLP2- 2B 3255	Sequence	SEQ ID NO:
Forward Primer	FE11315	TCACCGATGAGCAGGCA	75
Reverse Primer	FE11316	ATACACCTTCCGGCCAGC	76
Probe	FE11317	TTCTCCCCGAAGC	77
Target Assay No.	3321	Sequence	SEQ ID NO:
Forward Primer	FE11540	GATAGGGCTAAAGAGATGTGGGAA	78
Reverse Primer	FE11541	CTTTGTTACATTAGGGCTCAAATAA	79
Probe	FE11542	TAGACTGAGATGGATG	80
Target Assay No.	3322	Sequence	SEQ ID NO:
Forward Primer	FE11543	AAAACCACCGAGAAGACGA	81
Reverse Primer	FE11544	AGGTGTGGCGGCAGTGA	82
Probe	FE11545	CACCGTCATTGTTC	83
Target Assay No.	Cas9-in <i>Arabidopsis</i> 3049	Sequence	SEQ ID NO:
Forward Primer	FE10730	CAAGTTTCTGGACAAGGAGATTCTC	84
Reverse Primer	FE10731	AAGAATTCCTTCTTAATAGCTGGAGA	85
Probe	FE10732	CACGAGCACATTGCTAACCTTGCTGG	86

TABLE 11-continued

Assay primers and probes.			
Target Assay No.	ZmCENH3 2298	Sequence	SEQ ID NO:
Forward Primer	FE08737	GCGACGCCGAAAGG	87
Reverse Primer	FE08738	TGGCGTGGTTTCGTCTTCTTA	88
Probe	FE08739	AAGAGCGCGTCTGGAGGTGACTCA	89
Target	GL1 3321 target site (PCR)	Sequence	SEQ ID NO:
Forward Primer	3321F	AACCGCATCGTCAGAAAAAC	90
Reverse Primer	3321R	TCAACTTAACCGCCAAATC	91
Annealing Temp.	60 °C		
Target	VLHP2-2A target site (PCR)	Sequence	SEQ ID NO:
Forward Primer	FE4117	CATCCCTTCTCTCCCTCCTG	92
Reverse Primer	FE4118	GCCAGTGTGAGTGTATGAGCA	93
Annealing Temp.	61 °C		
Target	VLHP2-2B target site (PCR)	Sequence	SEQ ID NO:
Forward Primer	FE4120	CATCGTTTCTCCCTCCTCA	94
Reverse Primer	FE4121	ACTGATATGCACGGCGCCA	95
Annealing Temp.	62 °C		
Target	VLHP2-2D target site (PCR)	Sequence	SEQ ID NO:
Forward Primer	FE4121	TGCAGTAGCTTCATTTTCACCG	96
Reverse Primer	FE4122	AGGAATTGATATGTACGCCCGT	97
Annealing Temp.	61 °C		

XII. Use of Codominant Markers for Haploid Testing.

The invention includes methods of testing a haploid plant progeny, e.g. an edited haploid, for the presence of the inducer parent's genomic material. In one embodiment, the method comprises isolating a nucleic acid from the edited haploid progeny, and detecting in the nucleic acid the presence of a plurality of codominant markers. As used herein, codominant markers are markers where the two parents have different alleles at a genetic locus. For example, one parent is a "G" at one locus and the other parent is an "A" at the same locus. The markers will create a distinct haplotype for the first plant, which is the haploid inducer parent, and a distinct haplotype for the second plant which is the source of genomic DNA in the haploid progeny. Haploids may be obtained, for example, by any of the methods described above.

Any number of codominant markers may be used, for example, the plurality of codominant markers include of at least 5, at least 10, at least 15, at least 20, at least 25, at least 30, at least 35, at least 40, at least 45, at least 50, at least 55, at least 60, at least 65, and at least 70 markers. Table 12 provides 74 codominant markers to distinguish between NP2222 and B14 genetics following a cross (marker position identified using brackets), for illustrative purposes, but other may prefer more or less. Similarly, it should be clear that those of ordinary skill in the art may design their own codominant markers as desired. SEQ ID NOs: 99-173 represent the sequence when the male line (inducer line NP2222) is detected. SEQ ID NOs: 474-548 represent the sequence when the female line (inbred line, B14) is detected. The SNP position is indicated by brackets.

TABLE 12

SEQ ID	NO. Sequence (Codominant Marker Position shown in Brackets)
99	GTGGAGAAATGCAGAGCCATGGATCGCATTTCATGGAGTAGCCACAGGGTTC[A] AAGTCAAAATCGCAGATCTCCATCTCGGGGATGATTCAGAGATGCCAG AGAG
100	CGAGAGACCAGATCAACAGCTTCNGGNGGCATNCCTTTTGTGGAANACCTG AAAGATAAAATGTACAAAAAGTAAATTAGTAANACAACAGTAACAGTTTT [T]TTAGAGTATTGTAACAAAATGTAGAAAGATTNNGCACCTTGTGCCATGG GTGTGCTTTGATTGTGGGAACCTAAACTCTGTGTAATTTGGGTTCATGC
101	GTCTTCAAACCTTGCAGTGACGAATTTCAATGTGCGTGATTGGAACCTTTTCG ATGGTGGCCCTTGAACCAGAGTGGATAATCTCTCCAAAATCAGATGTA[G] AGCATCTGGTTGACATACGGAAGACTACTACCTGTTTTCTAGATTTACAATT CATGTGCAGAAATTATGCTATCTGACTTAATCTTCCAAATACTCCTAT
102	AGCGTGCCCATGCCGGAGCCGGTGCCGCCGCCGAGAGAGTGGCAGACCTGG AAGCCTGACAGGCGGGCATAAGATCCAGTCAGCAAAACCAA[A]CAACCTT GCAGAAATGGAAGACGAGACGAGACAAGCAAATAACCAACCAACCTTGCA GGCAGTCGCAGTTCTCGGCTCCTTTTCGCCACCAAG
103	TCGACCGCAGYGTGGAATGCACTGGCAACATYAATGCTATGATCCAAGCTTT CGAATGTGTTTCATGATGTAAGTATATGTATACACTCTCAGCTACTTTC[A]T TCTCCAGGTTCCCTTCATCCAGACATGCATGTTCTAACYGCCGSCSTCGTGAT CCAGGGCTGGGGTGTTCGYGTGCTGGTGGGTGTGCCRCATAAGGAC
104	GCACTCGAGCGACGACTAGGAAGAGGAGGTGATCGATGAGAACGGCGA[A] ATTGTCAAGAGGAAGAAGAAGGGCCTTAAGGAGAAGGTCAAGGAGAAGC T
105	CGCTTCCACACCGGGAGCACATTGGACGCCGCGTCCACAAGAAGGTATACAC AGCATGGTCAAATGTTTGGCTCTGTGGTCTGTGCACTCGTGCAATGCATGCGG ACAGTGCCCATG[C]TTCAGTCATGTTGAGTTGAGTTCTGCTTGC CGGCTG TGATGTTATTTGTTCTTGTTCATCATATCGCAACTGGCAGTGCTGTCCGACG AGCGCGACGCCGCGGCCAGCGGCGG
106	GCGAGCACATACGTCGTGGACACATTGCAAGGCGGATATCATGGCGTTGA [T]GTTACCGGTGCACCTCCACGCTGCGGTGACGCCCGGTTGGTCAGCTCAA
107	GGTGTGACCTTTTSARATCGAGTAGACTGCCATCAGTGCCATGACTTATTGG CAARTTGGAGTCTTTTAGGGTCCACTTGTTTTCTTGAATTCTGCAAG[C]TG AATTTGCTTTTTTTGTTGACTTCAATAMTCCAACCARTTCTCGAAYGACBRCC ATGCTATTTCTCGCCTTGTGCTCTCTGAGATAGAGTGGAGTCATGA
108	GAATATTTCATGGGGCGGTTACAAGATGGCAGGTGAGGACAGGCAGGAAGAT GAACAAGACGACAAGAGCCGACGTAGTCGTCCAAGGATTTCGATCAGGATAT TCCTCCAGATGACAAAGACCACAAAATAGTTGCTTCTTCAGACGTGAACC AATTTTAAACTAATTGCATAAGAGGTGATTAATTAGCAAGACAAAATATT AGCTCTAGTTAGCTCTTGAAGGCATATAATAATATAGCATGATATGAACTAA TTATGGGCTACATATTGAGAATTAATAGGTTTGTCTCGTCAATTAATCACTGT TTACTTGCAAGTAGAGCAAGAGCAAGCGCAGAGAACAGGGGCATAATAAA CTGCTGCGGGTGGTGTACCTAAACCATGATCCAATATTACATCCCTTATGC TGACTCCAGCAGTTTCATCAAACTTTTACCC[A]TAAACAATGTTTTACACTA TAGATTGCACATATTCGTAAATAGAGTTTGAATATGAGTATGAATATGTATA AGCTGTTGGAGATAGTCTTAATACCTAGTTTAATTACTCTAGTATAAACCTC AATCTACATGTATTAAGGTAGATTGGAGTGTAACTTAACTAATTTATATCC CAATCCACCTCAACATACATAACTATAGTCTATCGTACAATATCTAAATGAA GCCTAAAGACAACGTTTAGGATAAGGAACGTGTTAAACAGCCCAAAAGCAT TATGACTCCATGAGTCCAACGCCCAATAGCACGAGACATCGCAACATGCC CGCTCCCCAACCACTGTAGCACCTGATTTGATTGGTACAAAAATTATGATGA CGTCCCATTTGCAGACATCGAGCATATTGCCCAATACCCTTGTTTGAAATTC AAAATTTGAAT
109	CATATGAAAAATCNCTGGTTCGGCTTTCGTTTTTNNGCATACCTAATAACC AACCTCTGCATCTGCTTCAGGTTTCGTACTTTTGGAGCGCAATGTGGGTGCTT GGCGCAGTATGGGATGAAGCACATGCGGTCTTCGCAAAACATCTGCAACGC TGGCATCTTCTTGANGCAGTGTGCAAGGTCGCTGCTCAGGCTTGCACCAGC ATTCCTTCCAACCCCTGGAGTTCTATCCACAAGGGTTTTAGCGCCTAAGAAAT CATAAGGTGAGGCGAAATATTTAGCCGCTCCACCGCAACGAACTGGTTTAC ATTACAGTCCTCAGGGGGTCTAGTCTTGAACANNNNNNNNNNNNNNNN NNNNNNNNNTANTATAGCTGTTCCACCGTNNNTCCAGATTACATAGCTATGC NCAATTTCCGGTGTACATATNATAGTCGGAAGGTTATTTGGCAATTGTATTG GTCGTTGCTGTATATATTCCTATAGTTTGTAGCAGATGTGAGTTGTNAT TCCATAAAAAATGAAG[A]ACGCGTTACTGCTATTTCTATGTAGCCGACTGNT

TABLE 12-continued

SEQ ID NO.	Sequence (Codominant Marker Position shown in Brackets)
	GCTCATGTGAACTTTACCCCATCTTGTGGGAAATGNACTATCCGTGGTG GAATTCNNGCATCGAAACCAATCCCGGGNGATCCCTTATTCAANGTGAANC GTCTGTGNATTTCCATTGAGGNTCGTATTATTTTCANTTGTAGGCTTGTGGC TGGTGGCATCTGTCCACTATGNTNCAAACAAGNNCTATGGGCAGTTTCAT TTGTTTCGTCTCTATATNCTCGACACCTACCCATGAAAAAAT
110	TCNNGNANGNANGTGTATTATTNACCTNCAGATTCCACCTTTGTTCTTCTAT ATTACATGGTACCTTGNAGGTTCTTGCTCTTTGTGATNATCCGTGCCTCTNG GAAAAAGAGGAAACCAATCATTGTTTCAGGTGAATATCNCATCNCNTTCAATT TACAGAACTCCTAAATTCANAGATCTAGTNGTGATATCNGCTATTTTNC CAACTTNAGTGTCAAAGCAACCTAAAAATTTCAAAAAATACAGAGATAGGG CAATCT[G]TCTTCCTTTAAATCAAAGCTGTGGCATTTCCTTTGAAATTAG TAAACATTTATATAAATAGTAAATNTCNTGGAGATCNNNAGGTANTTAA CNTTTTCCCTTCAACTTCCACAGTAATTAACATACCTAGGAANATAGTTTT NGAGTTCTCATGTTAATTGATNTGTTTCATCANAAGAACCATTACNTCNTG CCTAATTATGCATGCCCTTTNATTTTCTTAAATTTCCCTTGATACCATTTT AAGTTGCAAAGATGANTTTTTTTTCTCGTACTGTTTAAATTTTTGTAGC CATAAACTTTCAAAANTAGTTCAGTGTCCCATTTATACATAAATNCTTATGT GTACNTGATGGGTCGNC
111	ATTTATGTGCCTYYAGGCGAACCACTTYCATCYAACCTGGCGAGCCGTTGA GAGTGAACGTGCTMTTCAAGCATATTCAGGCAATGCTGTCTGCGGACAC[C] GGCTGTCTATCGCAGAGACTGGGACTCGTGGTTAACTGCCAGAAGCTGAA GCTACCGGAAGGTTGTGGTAAGCTCCTCTTTCGAANNNTGRTTTTGCT
112	GCACAGAACTCTCCCCGTCTCTTCTGGGGTTTTGGTTACGTGGTGGTAGTA AGCTTGGATTGACATGGATAAAGTTGTTCTAAGCTCCGTGGGTTGCTTGA GATCTTGCTGTTATTGCGTGCCGT[C]CTCACTTTTTTGAATCCGAGGAA TGAATTTGTGCTTACGCTTTTGGTGGATTATTAGCGCGAAAAAAACTC TTTTTTTTTGTCTTTTACTACGAAAAGCATCTTCTGGATTGTGCTATCTTC TTTTACTACGAAAACTCTTGAGTCTAGGAATTTGAATT
113	GGCTATTGTACAACAACAAGGCGCATTGCCATCRCAGTTGCAGTACTASCA ATTGCCWTCGKGATTGCGGTGCGGTGTTGCAGTAGTAGT[G]CTGTAGTA CTATCAAGTGAGTCTGATGCATGAMCAGAGCAGAYGGAGTAGTTAAACGGA GGCGCGGTGAGCTTGCGGTGACGATGTGGTGGTTGTCGT
114	RGACACGGAGGAGCAGTACGACTTCGACCCGCTGGACGACACCAAGACGTG GCCGGAGGACCTGCTSCCGCTCMGSCCGTGGGGAGGCTGGTGTGGAC [C]GGAACGTGGACAACCTCTTCAACGAGAACGAGCAGCTGGCGTTCGSCCG GGGCTGGTGGTGCCAGGGATCTACTACTCGGACGACAAGATGCTGCAGTG
115	ATGAACATATGAAGAACCCTATCACTTTGAGCTAGTCTCAAGTCCAAGCAAAG AKAACAAATCACCAGTAGTAATGGAAAGAGAAGATTAGCAATAACT[A]TT GTTTAGTGGAGCAATAAATATCTTTTTCAGTTTCAAGTMTTAAAGAGAGAAAA TGATAGATGTAAATGGCAAGCACTACTGTACTTGACAACGTTAAAT
116	ATAGTAAAAGGTTGGAATGTTAGTTGAAAACAAGGTGTAAGAAATGTATCA TCTTTTGGACTGGACAACAGACCTGAAGTAGAAGCTTCTGGGTTTCCTCANA NNTGGCCCTCAGCTTNNCNCNAANGNNNNNNNTCCTNTGAAACGAGGAG TTTGTATACAGTNGGCAATGCCAGGATTACCTTGGNGAACNCCATNTCC NCCAGATCAATAGTACCCCTGAAAAATGGCCACTCATTGTACATTTCTTGA GCATGTGGAGGTTCTGATGTCTTCTGGAGGACAT[T]CTTGAATGCGGCT CCAAATCCTAGCCAGACCGGGAGGTGGAACCGCGTNTGTGTCCAAGCAAAG ATCCATGGGATTGCTCGGAGTGAGTCGATACCTCCGCTCGGCTTCTCTTGG ATGGCCTGCTTCTATGTTTATCCTACCATACTCTGTTTCAGGGGTTGCCTGT CAACACAAACATCACAAAGTAAATACTCGCTAAATGTAGGGGTCAAATCGA ATGCTTCAGTAACCATTTTACTTAGGCTTGCAAAAGATAGTTTCTNCAGTAA ATCATGTGCTAGATTAATATTAAGTATCAGGTGAGANTATTTCAAAGTT TGTTTTTGCATCTTATGTTACTGTAGTATCTTAGACACCACTCAGGTGGNAG CCTCNAGTACTGGTCTACNNNCATCTTTTATGTTACTGTAGTATGGCAATC GATGGATCAAGGACTTACAAGCGGAAATACTCGACAAAGCGTGGCTCTT TGAAAGACAATGGACCGATAT
117	ATGCTCTGGGGTCATCTCATGTCTGAATTCTGATCCAATTGAAGGGTTGTAG CTTTAGTGCAATGTTGCTATNTTATCATGCAACCCCTTACCTTTTGTAAAC ATGTCACCAAAACAATTCAATTCATTGGGTTGTTCAAGGATCATGTATTCCC ATGCAATACAACTTGGAGAAGTCANGTATTCCTTTACTATTTTACGTAGCA GTGTTGGCCACGATTATATGTTTAAATATCTGTACTTCAGTTGAGGAAAT ATATTTTCTGAGCGATCAGNTAATCACTTATTTTGGTTCCACTACTTTCCTAT CAGGTCTTGAAGTCTCGGAATATTCAGCTTTTNTTGGGATATTTTGTGAGCNG CATCAAAGAGGCTCCTACATCTGATGANTCAAGTAGCACAGTTACATACTCT GAAGTTGACGGTGATCATAGGAACATAATTTGGACCTTCAACCTGCTGAAA GAGGCCTCAAAGGNCAGACNCTCGATGC[T]GATTTGGTGTGTGGACAGT

TABLE 12-continued

SEQ ID NO. Sequence (Codominant Marker Position shown in Brackets)	
	GGGTTCACATCTCAGATTCNCGGTTACAGCCTCCTGATGCTCCCTANGTT ATTCCTCTGAATGGTCGNGGACAGGNGGAGACAGAGGAANNNNNNNNNN NAAA
118	GCATAGCATACTGCATGCATGGCGCTGTTGCATTGCAGGTTGCAGCTGCAGN NNNNNCNGNNGACCCNGNTGNNNNCGTGACAGACTGNNNNNNNNNNNNNN NNNNNNNCANNTNNNNNTANNNNNNNNNNGTNNNNNNNNNNNNNNNNNT ANATAAATNTANNNNAGTACANTTTTGCNNNNNCCTGCTGCCTGCCTTAGC ACTAGCAGCCAGGCAGCAGGGCTGGCGCCTNTGCTGCGGTGCATGAAAAGG CGACGAGCAAGAAGGGGCGCGTGGAAACTGCCGTCGCGCGCTGCTACC GCCGATGATTGGATCAGTGGAGCGTGGCGG[C/T]TGGCTTTTGGATCCTTT CGGCACGTCAGAAATAAAAAANNTAACAGTTCCTTTATTTTCATCGGTCTCC GACTCTCAAGGCGGCAGNNNCAGCAAGCTCTCGTAGCTNNGTGTAAGCT GTGGAGCACACAGCAGCTGTGAGCCTGTGTTGGTGTGCGAGCAAGAGCAN AGCTGGCAAGCGCAGCANAGGANTTGACACCGGCTTTATACGAATGAGCAG AACANAGGCATCTGTACATGANGACGGGCACGGCACACACAGGCCTTTG GTTGTTGGANGANNGTGTATTGGCCTCCTACTATACTGANACTGNACACCC CTGGACATCCGCATGNCGCCATCTACTACGCATNGATAGCAGCGNTNTGTAA NNNNNTGNANTGGNCTCNNANGAGCCATGCCATGNNANCCAGNANNCC CACAGCAC
119	CACACATAATTACTCCMATCTTCTCAGTGGAGGTACATCAGCAACCGAGGT TATAAAGGAACRCITYGAGGACAACACGATGGTGGCGGTGCATAGATTG[C/G] TCTACGAGGGCTCTGAGGAGGCGTTTCATCAACGGAGGGATGGTYGTGTCCA AGATCGCCACARGAACATCGTCAGAGTTCTGGGCTGCTGTCTGGAACC
120	TGGATCCATCACGGTAGAGCTGACNATGATATGTCCAAANGCTTCTGCTGCT AGTTGTTGGTCGTCCGCAAGAGACTTGACCCNACAGGTTTTTTTCGTTGGGC CTCAAGAACTGTAGGTGCATTTTCCATGGAGGTTATGATGCTTAACCCAC TCTGGTTTCTG[A]TGGCTAAAACATTCTCTCCAGTCGGCTTCTACCATGAG CCCTGGCACTTAGT
121	TGGCTTTATCCGTCGGGATGCCCCGCCATGTGCTCCTTGTCTCGGGACTCGG AGACAGNCTGACAGTCTCGGTTNTTGTGCTAATGGCCNTGGTTAACTTACG TTAATCNTGCGCCTTATTATCATCATCTAGGTGTAATTACTTCCGT[C]TGTG CTAATCACCTCGTGATGTCCTGCAANNNNNNNGGAACCCAGCCCCCNACCT TGCTGATGCCGTGCTTTTCTAGAGAAGTACCCTGCCGCTCAGCTTCGTCCT CGCCTGCTTTGGATTCAACGGNTTTTGGAGNCNNNGGCCA
122	GGTGCCGAAGCCGCTTACAGCGGCCCGCGCTCAGCTA[T]TACGTCAAC AAGTTCAGCCCGCGGTGGCTGCGCGCGCAGACCTCGAGGCCCCCGCC CCCGNCGAGGCACAAGACGGCGCGCCGCCNCCGTGNGNCTCCGGAGGTA CCGGCACCGCAGCTGTCT
123	RCATAAAAYATTGCAAATGCTGAGGAATATAAGCACATAGATACTTAGCAT GTGTCGCCGAGATGGAAATACTTTTACCATTCCCCAC[A]ACTGGATTGCC AGAAAGGTCAATGCCTATGACACCTTGGTCCATYATTTCCAYGGCTAGATTA ACCTGGCCAGGGGAAAYAAATSATTTTGTGCAGATT
124	ACGSYGTAAAGGCTCTACTGCATGTTCTCTGAGGTTGCTCTTGACACCTCAA TTTCTGGAAATCATCCCCATCTRGGGACATAAATGGAGCAAAATTGC[T] GTGAGCTTGAACAAATGGAAGGTCACTGTGATGGTTCTGAATGCCATCAGG CAGATTCTGTGATGTGTTTGCTTCTGATGAAGAAGGAAGTTCAGCGG
125	TCTCAAGCTTATTTAGGTTGACTAAATTTGGAACATCCCCGAGATGCTATT GCCGGCAAGCGCGAGCTCCGTAAGCATGGCAGCTTTGCCAATCGATGC[T] GGTATGGAGCCCTTACGTGGTTACTCTGCAACCGAAGAACARTGAGGTTG AGAATGAGTCAAGCCAGTCTGGCACACTTCATTGAAGTAGTTACCAT
126	TACTAGCACACTTTGCCTCCTGATAGCTTGTAACATTGTGACCTTGTGGAT TTTGGGATTCACCCCAATGTCAAACATACCCAACAGTGACCCGKTATCTTT TTCTTCGTATTTCTATGTCAAGCACTYAAGCATAACTCATACA[C]TTGCCT CTACAGGTGTTTCTGAAACACTAGTGCTCAATATGTCAACTCTGACATGGTC AGTTGTTAGCACTGTGGAAGGACGTGTTCTCTTGCCAGTGAGGTATWTCTC TTGAGCGGTTGAGCAGATTCCAGTTGTTAAGATG
127	GCACAGGTAAGTCTTGATGGCACCTGTCTGAAACAAGTTTCGTGCTAACA AGGGAAATTATCCTTTTTCTGTCGCCAATATCTGTTTTCCATTCAATGATGT ACTGATGTGTTACGGAGTCGTTACCTAGAAATCTCGCTTACTG[C]GACAT ATCACACCGAGCATCATTTTATATCTGCTGTGCGCCTCCTTGCTAATGTAG GCACCTG
128	RYTKCAACMTGCGAGAARTARAAAACTACCTAAGCTTTGCTTTATCAAGTAT AGCAAGGTGAAMGCGAATGCCARGAAGAAGCAGAGTCCRCCGACGRTGAG GTACGCAAGGCCAGAAAAATCGTTCTTCCCCGAGCCAGGTTGCGGT[A] GAAAGCACCAGCTTCTTCTGCCACCAAGCTATAGGTGTGTAGTTGTTGT

TABLE 12-continued

SEQ ID NO.	Sequence (Codominant Marker Position shown in Brackets)
	CCARCTGGACTGTGATTGTRTCGTTTTCCTTGAGATCAACATATATCCTTCCA TACAGCTTTCTGAATGTCGGAAGTGCTGYAGTCCGCATCCAAACA
129	GCGCTCGCCATGAATGCGCGTCGTCTCCRTTGGACCATCAACCATTGCAAT TGATGAGGAGACGAGACAGGAGAAACGGCGAGGGCAGYAGTGTCTGTAG CTGAAAGCTGCCGCAATCGACTGCCATTATTGGACGCC[G]TCCCTTGCCC GACTGAGAATCGGGCGGCCGCTGCACTGCACTRCMCTACCGCCGCCGCC AGGCGCCAGCCRRCTCGTGGTACAAAATTCATCCAGAATTCGCACGAGGC CACCAGCTCGGCTCCGGCCTCGAGGATTTCAA
130	TCATGGAATTGCTTCAAACCCGAGAGCGGCAAGATAATAGACGAACGATCA GCTCCAGCCACCTGTAAGTACAATCACAAYRGTAAGAGCAATGGATCACT YGTGGAGGCTTGTGTTTACAAATAATRGCCAAACA[G]GTTACCTCAGA AATCCTCAAATAATGGCCCCCTGTTGTTGCTGCCAGGTGGAAGAAGACCTT TTCTGATCTGCCCTRATMACCTTAGACATACCATGCCACTAATTTCTTCTTG AGGAGGAGATCAACCAAGCGTTCSACATCTA
131	TAYAAGCAATATTAAGAGATGGAATGTGCGCTCTGTACCACAAGTTAGTA AGTTACTAACTTGATTGTTAAAAACCCATCGAATGTGTGACATGTAGGACC TGGTCCCACATGTCAGACAAAGCCAATGCTATTTGGCATTCTATGTT[G]CA ATGTAGTGCTATAATTTCCACAAAAAACAAGACAGGTCAAATACTGYGT AAACATTGATCCAAAAACAGAGATGCTTGTGATAAAAATCTAGCCAA AAGAATTGGAGCTATTGAGATACAATGRA
132	ACGTTCTTGTGCGACCATTCATTCGCTATCTATTTTGAACGAACGCATTC GCTATCTATGGCCACATCTCCACTTCGTCTTTGCCGAAGCTTCAGTACGCGGT GCGGGTGTCTCCTGGTGTGTATGCTGTTCTTTCTCGGTCCAT[A]TTATCCC CCGTTTGACGCGCCTCTGTCAATCTGCTGTAATTTTATTTGACGTGC ATAAATYATTTCTGGAACGCTCTTGGTACGTCACGTACCTYGAGAA
133	ATCSGGCACCTGATCCWGAATGTGAGWCSGAGTGCAGCTTATACCCGTCCC CAACCGCCCCCATTTGAATCCACCGGTGAATCTATCGCTCCGCCACACGAGG CCCTATATTAACCTCCACAGCATCCATGTGTCGAG[T]CTGTCTGTATACCT GTCACCTACGCTACCGCGTGCCGATCGTTTCGTCTTCCCTCCCTTCGCGGG CCGCGCGCCTATKWTWTTTACTACTSTATTCTRTATCATTAAYKTTTGGTTT CCWTCCCK
134	TTCTCGTAATCTCTATGGAAATTRCCTTTACTAGAGATAAATGGCATCCTTGC AATGTCCCAAATTTCTACCGAAGAACACAAGKAAAAGARAAGAAAAATTG AGCAAGCACAGCACAGGCAGGCAGCATGAATCCACACAGAACCA[G]TGG TTTTTAGCACAGYASGYTGACAGCCAAGGGTTACAGTACATAATCAGACGG GGCAACACGAACGAAGCAGCCGCATCARCAGCAGCCTASAACCCATTAATT GACACACAGCGTATATATATATATRTAGTACTTGTCTTTTAGC
135	AAGTCCAGATGACTTTCAGAAAGAAAATCGAGAACTTCAGACACCTTCTGA CCGCTGAAGCACGAACGACAATGGCCATAC[T]GCACGATGTTGCACGTTT YGTAGAAGTTTGATCTTCGTTAAAAATTATAGCAGTACAAATSATAGCRCA TTCTGTAGGTGTAGGATACAGTAACACAGCAGCAGAACATCAAACCCRTCTC CTGCCAGGCTCTGAGCAACGCCAAAGC
136	CCRTRCAACATTAGSTCGTYYTGTTTCTATTTSTTGACCAAGACATGGTTGYT GTTCTGATGGCAGATTTGAACCTGACATGCTTGCYTTTCTGGGTGCTATTAT GGTGGTGGAGAAAAGGAGAGGAGAAATTAGCTGCAATTCGCA[G]CGA TGGAGAAACCTTGATGTACAAGCTGTGTGATGTGTAATCATTACTCGCTTAC TGAGATAGAATTAYCTTTTGCTAAAAGTTTTCATATTTCTAGATACGTGACCC GGAGAAAGGAAGACGGCAAGGCAGGTGTCCACTAATCTCTGAG
137	NCGCGTGTTGTCGCGATGCWAYKCATGCAGAGGYAGCARARCTAGCTAGY AAGCACGWACKYACGTAGCACATGATAAGAAGGCTGCRTTGAGACAGTAA GACGAAGAATGGCARGCAGAAGAGCACGT[C]A]GCATGCTCCCCGCGGCTT ATAGCTTAGAGGCACTTGAATCCGGTGGGCACKCTTTGCCGCGAGTGGTTGA GGATRAGGCTGAGGTGACSGGYAGGTGAGCTTGATTCCKAGGACCTCTCC CTTGATGGCGGTGCAGAGGCACACGGCG
138	ATGGATGCTTGTGTAGCATTAAACGTGAGAGTTTAATCTSACCACAACAAGC ATGGGTGATCACAGAATAACAAAACAGCCAACTAACCAAMAWKATATT AGCAAATGCAACTTACTTATAACATGGCAACCAACACAAACACATTG[T] GCACACTTATGCTGGCTTAGTCTATCCTGCATTAGACCCAATGCAGTGAATT GCTAGTGCAAGCAACCAATTGCTCCCGTAGTGTTTCTTTTGCTTCTATGAAG GCTATGAATGTCAGGGTTTATTCTAGTGATCTATTATCTAYCA
139	TGGAACCRRTTAATAATTGAAGTTTCTYAGTACAGRTCTGCATAGCATTGA ACTGGACAGCTTGCTG[C]GTTACTCTGTRTAASGAACGATCTACTGCTGAT CTGTACTGTTCCCTTGATTTTYYYYCWCTTTTCTTCTTTGATGGCAAGCAGG ACTGAAGATAAGATGGCTGCCTTRCCATTGGCCACCGCAGAAGCATGTGAT GCTAATGCTG

TABLE 12-continued

SEQ ID NO.	Sequence (Codominant Marker Position shown in Brackets)
140	GGAAATCTTGACAAGGATATGGAGAGAAGGTTTAGCTACGCCCTAAGCAGA GAAGACATCGAGAACGCCATACTCGGAGGACCTTAASCTGAACACATGGCC AAGGGCTCAKAGCTAGGAATTGCTTAGTCGAGGTTGGCACTGAAGAA[A] TAATCGCTCCAGTGATCTGCGTGCTACCATTACGCCATTGTAAGAGCAAATG GATCATGTGTCGAGAAATAATATTCATGAGCAATAATCTCGAACTAGGCTGG AAATGTTGGAWGCCCTCCTGGTGACATGCGYTTGGTTTGTGGCMTC
141	MTAGGAAGGTCAAGGTGCCTTTGGACCCCGGCAGCAACTACACCTTCGACC TGAGTACTACCGCCACGTRCTCGCCACGGGCGGGCTGTTCCAGTCCGACGG CAGCCTCCTGCACGACCCCGTCACCGAGGGCTACGTCGAGAAGGTGG[C]C AAGGCGTCGTGCGCCGACGAGTACTACGCGGACTTCGCCRCCGCGATGGTC AAGATGGGCGCACCGACGTGCTTGTGCGGATCATGGGAGATCAGGCCA ACGTGTGGCATTTTTGTGACTAGGTTCAGGAKTGGGTTGAARATGC
142	GTCAACAAGCACCAGACTTTCAGGACACTCGGTGCTTTTTCGTGGTCTCCA CAGATGGATCCCAGGCAGAYTTCTCTACCTGAAGTGCTGGAGAAGTTTGT GAGGAAGAGCTACACGAGGACGYSGACACATTTCGATGAAGTA[C]TTA AGGCCCGTCGACGGCAGGCACACAGCTGATGTTGGGACAGCATCAGGC RCCCCGGATGAGGTTCCACCGTCAACCGCAGCTGAGACAGAGCAAGGCAT CCTCCAGCCCTCAGGCAGAGGTTTCRCAAGAGAYTTGG
143	ATTACACCWAACTTGTCTTATAAGTTAGAAATCCTACAATATTTATTAGAC TCTGTAGGAGAATCTCTTCTAGATACAAACCCACATCCTTTACTGTAAAAA GGCCCATTTGTAGCTACAATGACTTAACAT[C]TACGGAGGGACCAAATCT CTGGTCTGAAACCCAACTTTCAAGGTTCACTAGGTTCTGCCACTGGCAGAT AAAAAAATCCTCATTAYGATCATAAGTTGACATACTGGACAAAGAAATAC TTGTAAGACGATCCCTTCTTATCTGATGA
144	AGTGTAGAGATACTCACAAGAACAATTTTGAAGTGCCTTGACAGGCCAGGT ACCACGAATAAGTGTTCAACCTTACTCTCATGCCTCAATTTCAAGAACCTT GAACACATTTGCTAAATGATGTAAGAAAAAGTTGCCATCATGGTA[G]CA GATACGAGGTGAGTAGCAACCTGGTGAGGTTAAATGTGTGCCCMAGCCGGT GTGSTTAGTTGAAAAGATAAACRCAGTTTCTGCAGATGTTTGCAGAAAGAG ACACTTTTGTATCTTTCAGAAGATCTAACCTGAAATGAAGCA
145	CCATGCCTACTACAGGCCGGYTCTCGGAGATGTTYGGCTCCACGCCGCC GCCGCAGCGCGAGGACACGCCCTTCCACCCGCGCTCGCCCTACGCCCGCGCC AAGTTCGCCCGCGCACTGGTACACSGTCAACTACCGCGAGGCCATAG[A]CG TATTCGCTGCAACGGCGTGCTYTTCAACACAGTCCCGCGCCCGCGCGA GAACTTCGTACGCGCAAGATCACGCGCGCGTCCGCCCGCATCAAGGTCCG GCTGCAGACCAGGTCCTTCTCGGCAACCTCTCGGCCGCCAGGGA
146	TACATACCTTGCAGCATGTCAAATGCACAAGGATGAAGAGAGAGGGAAAGT TGCAGCAAAGAACTTGTGAGATGGAACCAAAAGCTCATCCACGTACGT GTTCTDTCAAGCTTRCATGCTGCGGCTGGTAACCTGGGTTGAAGCCAA[A] GTAGCCAGAGAAGCAATGCGAGAAAAAGGGGTGATGAAATTTCCAGGGTGT AGTTGGATCACAGTGGGKAACAAACARAGYGTATTTGTTGTACAGGACACA CA
147	ACTCAAATGAGTCAAGAAATGCTAAGGGAAGCCCGTGAAACAATATAAT CGAACTTTTACATTTTATTGATGTGGCCTTYYTAAAAATGACGCCATAAAC WCTATACTGAAAACGGCCTCGGCGCTTAAACCAAGTAGCATTCGTA[G]GTT CATTTCTGATTTGGGACTTCAAACCAATAGTGTGAGACTCACAGGTTTAG AACCGATTGGTGGCTGCCTTCAATCAAGAGCGAAACCAACATGTTATTGCC ACCTTAAATCACAGATTTTGTATTATCGTGTACAATAAAAAAT
148	YCAGATCATCACTGGAATCAACAGCAGTTCTCTGGTTCATTTGTTTCAGATGC TTCKGTGTCACATGCTTCTGAAYGATGGTTTAAGCTTATAACATCATCARCA TTTTTGTAAATATTGCACTGTTCTCGTTGTCCTTTACCACGCTAT[C]CAAA WTTTCTGGCTCCTCAGATACTACAAGATCAGATRTCTGCCGTTCTGAGTAA GAGTGGTTTCAGGACTGACACCCATACCTTGTGCAAGTGCTACATACCGAAG TACTAGWTGCTTGTACGAATTCAATAGGCTCTCTACATCAGC
149	CTTTAGTAGTAGTCYTATAGCTCATAGGTTCTCAGTTCGGTATATRTYGGTG AYATATTTTCATGAACCTASTTTTWAAMCMATTTAAAMGCAMACGCAACA ACGAAGGGAGTGATATTGACCATGGG[C]CTTTCGTAGCTTTTCAATCAAAC GTCACATAAAAAAAGAAACAWCCAAAAGWTCATAATACACAGATTKRAAM SKKAGACAAAAATACGAGTACAAGACATTGGACCTTCAATWWTTTTTTK TGTTTTCTGAGACATGAAAA
150	TTGCACCAACCCATAGTTTAAGAGGCAGTTGCTTTCACACACTTTATTTTGCAT GGCACAGCCAACTGTCCCATCATTTAAACCAAGGCTGCAACTGAGCCACTA CAGAACTGCTAATATTAGATATTCCAGCAAAATAGTCTTGACACTAGAGTGC TA[G]TCCCTAAACACGAAAGACATTTGTGAGGAGACCCGTATTGAGTAC

TABLE 12-continued

SEQ ID NO.	Sequence (Codominant Marker Position shown in Brackets)
	TAGCAGAGTATACTACCGCTAGATATTCCAGTATAATCGTGCAGCAGCTCCG GTCAATGGCAGTGGCACTTGAAGTCTGCAACCTCAGCCAACAGATCCACAT GAGAACCAAC
151	TACACTATACTAATTAGAATGTCTGGAAAGGCTGGGAAGACCACTAAATTTG TCTCATTKTTGGAGGAAATGGTGTCCRAAGGATGTGTTCTTARTCTGATTGCT TATAATACTGTTATTGAGGCTCTTGGTAAGAACAAAGATGGTTGAC[G]AGG CGATTTTTATGCTTTCTAAAATGATTGAGAGTGACTGTGCGGCCAATCAATT CACATATAGCATTATGCTGGATGTTTATCAACAGGGGGACAACCTCCACAGG TTGAATGAGATTCTAGATATTTGTA
152	CCTTCTCCAGTTGGTAATATGCTTCATGTAAGGTACCTGAA[G]GTGGTAA GCRACATGATAGTCAAACCTCCACTCATTATTCGAGGCCTGCGACACTTAGA GACACTCGAGGTGGATGCAGAAARAAGTCGCTGTTCCACTGGATGTTTTCATC TTGAAGAGCCTGTTGCATCTCCGACTTCCGAGCAAGGCT
153	TGCATATGAGAACTACAAGGTTGGCAGGGTCCTTTCCTGTGCCCAGCTGT CATGAAAAGAAAGAGCAATGGAAAGGAAACGCCGGCCAAAAGGTATCAT TAAATAGTATTTACCTGAACATACATTCTKTAGTTAACCTCAGAC[A]GCTC ATTCGTTTTCTGGGTTCTATTTGTCTTTCAGGATCTTCATCAAACGTCTTTG GTACATGTGCTAGGCCTGCCTACCATAAAGATTCTTTATTGATTSTTAGC TGGAAATAATAGKCAGCCATAACAATTCATTGGCTAGAA
154	AAGCACCCGCATGCAGACTTCCAACTATCCAGTGTCCACAAATAGCAGTTT TGCAAACTAAATACATATATGAAATGATGACACAGGCCACCCAAACAAACAA GCCCTAGAGAATCAGACCAACCAACCGAGCAATTCGAGTGCTCCATATCTT CTAGCATTACAACTGACGAACG[T]TGTTGCCCTTCAGCGCCCCAACAT CAATCGTATCAGGAAGATACCTGCAAAAGTTGGTAAACATAGATCAAGTT ACACCACAAGGCTGCCGTGCATGTATGATACATATATGTGAGCTATATACTAT TCAAACCTACATATCTTCTTCTGGCTCATTTTTACAGCATTCCTCGCTATA
155	AGAYGCTACTCTGCTACCATTTTGAGAATCTATTGACAAGAACTTTACAGGA GGRGTCTCATCCCGAGGAGCCTTGCTGATAACTGGAGATGGAGAAGATATG CACATATCCACTCGTGAATGCYTTTTCTTGGACTTATTTCTGTCTT[C]AGC AGTTGTTTTCCCTAGCAACAAAAGGAGATACACATAGCATGAGAAAGAAG AAASAAACTGAAACATYTTATAATGAGAAAAAACATGGGGAGTTGGGGAC AAAGCAACTACCAATGCTTCTACAAATGTTTTGAACCTCCGA
156	TGCTTCTGTGCAATTGCTGCGGGCAGARCATGCTTTGGTGGYGGCTTGCAG TGTGAAACGTTATGAATCTGAGAACCTCGATGATGCTGAGGGACAGGGAAT ACCTGAACAGGGGCCAAGATGACATGAACAGGACGCCGGA[C]AGCACGT GCACAACCTTCTTCTCAAGCTCTGCCTTCATTGATGGCATGGATCAGTACA GAGTTCGGAAGCCAACGACGTGAACARCTGAACATAAGCAAATGTAATGGC CGTTCTAA
157	AGGTCGACAAGGCTGCTAGTAACATGTCTGAGCCGACATCAGGGGAGATGC AGGATGCTGCTTCCAGTCTGACGAAGAGGAGGAAGATGAAGATGTTGATG AAACAGTATTTCGGTCAAGATTGAGATTCGTACACARAATAGCGGCACCG[A] CGACGATGCAAAGTAGACTCACTGCTGGTATTCATCATTATGGTGATTGTAT TGTTTTTAATTAATTGCCAAGCTTGATTTTTGTCAAGGCGACGYTGGAAGG TTGCACAGAATTTTGAAGTGTCTCTGGTTTCATTTTGAAGCACT
158	AGAAAACGAGGCCTTCGTCTCTTAGCGCGCRGGGGCTAGGACGGCGGCA GCGGCAGCCCCACGACCTTGCCGCGAGCGAGGGCGTTGTCGGGCATGTTGT ACTCGTCGCGCAGTGATCGCGCGGGGAGACGACGCTGAT[G]TAGAACTG CTCCCCGAGGTAGTACCGCTCCAGATGTTGGAATGCACGCTCCACATGCCC GCGTTGTGCAACGTCAGCATGATTGCCGTCCATGACGCGGGTACCTTGA TCGTGTGCCGCTCAC
159	TTCTGGTACTTTCGGATCTCGCGCAGMGCCACSGTSCCGGCCTGTACCTGT GGGGCTTCTTCACGCCCGCGGTGTCGGCGCMGACTTCTCGCCGCTGCAA CRARAYAAACCCGCCCGCGTCAGRCRCGTCTTACAACGGGAAC[C]AT CGAGGGGATCGGAACAKCAGATCGGTACGGGTAGGTAGAGACGAACCTTGG TKGCGAGCTGCTTGCGCGGGGCTTGCCTCCGGTGGACTTGCGGGCGGTCTG CTTCGTACGAGCCATCTTCTCTCTCTCTG
160	CTGCGTTAGTTTCAACCTTCTAGCTGCGAGTGAAAGAAACWTGATAACAG CTWSCTGCTAGTTTCTATGRYRGCCATCGAATCTGACATGGCTAYCTCCTGT GMCMACGCAGGTCCGCAAGTACTACAGCCGAGGAAGAGCCA[C]CGGAC GGTGACGGCGGTGATCCACGGCGAGAAGGTGCCGCTGTACGGCGCCGGGGG CGGCCGTGACGCTGTCSACGAGCGCGGGCGGGGSGCGGTGCCCTGACGC
161	AGCACCACGATTACACCTACACATATTCAGACACAAAATGGGCACGCT GGGCATGGACATTTCAATGATAGATGGTATTTAGCTGCCCTACTYTGGCATTT GATTTCGAGCAGCGAAGCTTCAAGAGCTGCAGAATCGCTACAYATATT[T]C TGTGGGTACACATGCTTCTTGTCTCCACTCGACCCAGATCATTGTGATGGA

TABLE 12-continued

SEQ ID NO.	Sequence (Codominant Marker Position shown in Brackets)
	CTACTAACTGSCAGTACAAAAGCAGACATCTATCAAACCTACACAAAAGGC GTGCACAACGAGAYAGYCTGCTTGCGCATCTCCAAGTCTAGTCTAG
162	AWWTTKKWAGYGCARRAATAAACAGGCACTGTWRATTGTACAGAGCAA GCTCTCTGCATTTTTCATGCCGAGCATCTGCAACTGCAAGCAGAGTTTCT A[G]AGGCCAAGGCCGACTGTCTGCCATCTCTCTGCGGAGGCCTGCTAT GTAATGTTTCTCCAGGTTGGTGTGAGCTCCTCCAGTTTCTTGGCTTCTTCCG CCGAGCTCCCGACCTTGTCTCGACCGTGTTAA
163	TCAATTTGTGATTTTCATCCCAATTTCTTTCATATGTGAGGAATGCCGCACRCA CTTCTATGAAATGTGTTCAARGTTTGGAAAACTACTCTAGCCATATATTGC CACAGAAATCAASTGCATGTYTAAT[A]GTTACCGTTACCTGTAGTGTATC AGTCCCCTTCAAATCTGCCCGTGACCTTGCCCTCTGGCTATGGRCCGCGCAT AACAAAGTTAACGAGAGGTTGATGAAGAAGAAAAGGAGTTAGACAATGC TGATCCTTCATTTCTTAAGAT
164	TTGCATACAAACTTGATCMAAATGTTTRCAAMCCATGCTTGAACRRATATAC AAACCCGTTGYGGCAGGAACATYTTAGCTTWGKAMMTYGYWVAGCAAC AATAGCTTCAATAGAGATTGTTGTAAGTGTAGAGTG[T]TGCTGCGATGGA RACTATGCAAAATCACAGGGATGGGAGAAGAGTGACATTGAGAAGGAGGTC CGTGCCGAGATAACCGGCTGCAGTAATCGTGGCATCTGCAACTACACGGGC CAACGTCCCGG
165	ACAAGTCCAAATCAAAGGTGTTGTCCCGCAATGCTTACCCCATGSCYCC CCRAAATAGCATCTGAATCCATTTTTTGTGCGATCTTTTCKCTCTCTTCT GACAGGGAATGCTGCTTGAGTTGGACGAGGAACAGACTGGAAT[G]CAA ATGCACGTGCAAGTTGCAAAATCTTATGTTTGTGTTTGTGKGGCTATCARTRC ACAGTACTCATAGCATGACAAATGGCGTTGCAAGTTTGTGTTTACCSTTT TGGTTTTCC
166	AACCTCGACTCCACTATAASTGCAGGACGACGRCAYGCACAACTGAAACA TCCGGACCAACAGGCAGAGTAGTAGCCAAAGGATGACCCGTCTCATGCAA AAACAGTCAGCTTGATAAACTTATCGAGAAGGCAACA[A]CGGYGCCAA GTACTGGTTGAAATACAGAGCAGACATTGAAGTGCCAAAGAGGGGG
167	GTCTAGAAAAGTCAAAACGTATATACTTGGAACRGGAGTAGATATGACAA AAAGAGCATGCAAGTGATGTCTTACAGAAGAATATATATATTTCATGGCAA GCTAGTACCTAACTTCAGATGGAGTCTTTGCATCGATTGACAGGGC[A]T GCAACCCGGTCTTGAGCTCCTCTGCAGGAGATCATAGACAGCCGTGTGCT CTTRAGCAAGCTCTTCCCTCAAACCTCTTGGACACCACCTCACATTGAAA TGTGTCTCCCCATTGGTCCAGCCACGCCAGCATGGCCCTTGTGG
168	ATTTTACCATACTGCATTCAAAGTTTACAAGATTGGACATTATAGTATAA AACATGAAACTTTTCAAGTGAACAASAACTATAARCAACGTCCGCAAGCGC AATCCACAGCTTGAGGTTACATTGTGAGTGTAGTAGA[A]AAACTAGGTGG TCCCAAATAGCACAAAGATTATCCAAGCTACCCAGTTTCTACGATTATGGG ACRACCATAAGAGCATAGMAGTGCTTCGCATCTGTTGCATTAAAGAAAGTA GTATAGGCATCGGAGGTGCTGAATCTAAGCACACGG
169	TAAACCAGAYTAGATGCCACCTAGKTTTCTGACACAAATCAYGAACAAAAC ACGAGAAGAAAGCYAATCATACCGGATGAYGGATGGACTCAGCGGAGCC CTTATTTGCTACCATTCCTCATGTCTTATGACAGAAATCCATCTATTGC[C]A CTCAACTTCAMTCAGTCTCTGGAACCTGTATCAACAGGGGATGGGAAATGT GTCATGTTCAATGTTTAGCYCATGAAACATAGAAGACCMCATTAGAAGCTA TTATGTGCTTACATTTGATTTTTTTATCCAAGACTCAAGTGTAT
170	RRHYTKRTGGAGGTCGGGGGAAGAAACCTAAGAACGATGTCTACCAAAT TGATTCCTTCCGCCGAGAGATTAGAAGCTAGATCTGTCCAGATTAGTGG ATTGATAGGTTTTGTGAATTTGTCTCTCCAACCTAACTACTTAC[A]GCC AGATTTTACATCAGACCTGATGAAACACTGTTTCTTGACACGAACTGGTG GACGCTGCCCTTGCATCAAGAATCAAGAAATGATTTRCGTTTTATGTYTCT GGTAGCYCCAGACACCTCATACTCTCTCTGTTGCTGTGTSATG
171	AACGGCTAAGAGTCAGGCGATTCTGTTTGTACAGAGACAATCGCAGCACTT GRMTGCTWCGCATGCGSTTCAGAGGGCCTTCAGCTCAGGCTTGACAGATCT AGTTTGGYCGCCACGTCYRCTTTCGGCGAGCACTGYACTCTGTGGC[A]T CTTTAGAAGCCCGATGCTGTGACGATGTGGCCGGTGGTAGATGTTGCGTT GGTGTTTTTTCCCTCGGGTGTCTGTGGT
172	AAACAANACTCAGTGGACTCGCNGAANNNNATAATGAAAGGTGCTCCATC CATACCCATGAGAANGTTCCGATGCTCGCAGTCTCATGTTTCCAGTCA[A] TCTTCTTTTCAATTTCTTCTCCGTATGCACTAATATGCAGATCATGGCGAGAG AAAGGTTCAAGATTGCTCTCTTGATCTCTTCAAACCCGCGCTTTCG
173	AAAATTACTGAAGCAGGTGGGTTGCAGTTGTGTGTTGTTACTGTTTACTG TAWYATGTCAAGCTGTGGCTGCAATTTCTTTGCTGGCAAGCCGAGGCACT

TABLE 12-continued

SEQ ID NO.	Sequence (Codominant Marker Position shown in Brackets)
	GGTGAAGTGTGATAAATACATCATATTCCTGTTGACCTGTGAAGAA[A]CT TGTTCWAGGTRGATTCATTGTACTAGCTCTGTTGTCYAGCATCTCCTTGT GGGAACATTAAACAACCAGCYCTCRMCCCTCAANNNNNNNNNNNNNNNN NNNNNNNNNNNNNNNNNNNNNTAAATAGTCGCATCATGTA
474	GTGGAGAATGCAGAGCCATGGATCGCATTCTGAGTAGCCACAGGGTTC[G] AAGTCAAATCGCAGATCTCCATCTCGGGATGATTTAGAGATGCCAGAG AG
475	CGAGAGACCAGATCAACAGCTTCNGNGGCATNCCTTTTGTGGAANACCTG AAAGATAAAATGTACAAAAGTAAATTAGTAANACAACAGTAACAGTTTT[C] TTAGAGTATTGTACAAAATGTAGAAAGATTNNGCACCTTGTGCCATGGG TGTGCTTTGATTTGTGGGAACCTAAACTCTGTGTAATTTGGGTTTCATGC
476	GTCTTCAAACCTTGCACTGACGAATTTCAATGTGCGTGATTGGAACCTTTTCG ATGGTGGCCCTTGAACCAGAGTGGATAATCTCTCCAAATCAGATGTA[A]A GCATCTGGTTGACATACGGAAGACTACTACCTGTTTTCTAGATTTACAATTC ATTGTAGAAATATGCTATCTGACTTAATCTTCCAAATACCTCTAT
477	AGCGTGGCCATGCCGGAGCCGGTGCCGCCGCGAGAGAGTGGCAGACCTGG AAGCCTGACAGGCGGGCATAAGATCCAGTCAGCAACCAA[G]CAACCTTG CAGAAATGGAAGACGAGACGAGACAAGCAATAACCAACCAACCTTGACG GCAGTCGAGTTCTCGGCCTCCTTTGCAACACG
478	TCGACCGCAGYGTGGAATGCACTGGCAACATYAATGCTATGATCCAAGCTTT CGAATGTGTTTCATGATGTAAGTATATGTATACACTCTCAGCTACTTTC[C]TTC TCCAGGTTCCCTTCATCCAGACATGCATGTTCTAACYGCCGCTCTGTGATCC AGGCTGGGGTGTGCTGCTGGTGGTGTGCCRCATAAGGAC
479	GCAGTCGAGCGACGACTAGGAAGAGGAGTGATCGATGAGAACGGCGA[G] ATTGTCAAGAGGAAGAAGAGGGCCTTAAGGAGAAGGTCAAGGAGAAGCT
480	CGCTTCCACACCGGGAGCACATTGGACGCGCGTCACAAGAAGGTATACAC AGCATGGTCAAATGTTTGCTCTGTGGTCTGTCACCTCGTGCATGCATGCGG ACAGTGGCCATG[A]TTCAGTCATGTTGAGTTGAGTTCTGCTTGCCGGCTGT GATGTTATTTGTTCTTGTTCATCATATCGCAACTGGCAGTGCTGTCCGACGA GCGCGACGCCGCGCCAGCGCGG
481	GCGAGCACATACGTCGTGGACACATTCGAAGCGGATATCATGGCGTTGA[C] GTTACCGGTGCACTCCACGCTGCGGTGACGCCGCCGTTGGTCAGCTCAA
482	GGTTGTACCTTTTSARATCGAGTAGACTGCCATCAGTGCCATGACTTATTGG CAARTTGGAGTCTTTTAGGGTCCACTTGTTTTCTTGAATCTGCAAG[T]TGA ATTTGCTTTTTTGTGACTTCAATAMCCAAACCTTCTCGAAYGACBRCCAT GCYATCTTCGCCTTGTGCTCTCTGAGATAGAGTGGAGTCATGA
483	GAATATTCATGGGGCGGTTACAAGATGGCAGGTGAGGACAGGCAGGAAGAT GAACAAGACGACAAGAGCCGACGTAGTCGTCCAAGGATTCGATCAGGATAT TCCTTCCAAGATGACAAAGACCACAAAATAGTTGCTTCTCAGACGTGAACC AATTTTAAACTAATTCGATAAGAGGTGATTAATTAGCAAGACAAAATATT AGCTCTAGTTAGCTCTTGAAGGCATATAATAATATAGCATGATAGAACTAA TTATGGGCTACATATTAGAATTAATAGGTTTGTCTCGTCAATTAATCACTGT TTACTTGACAGTAGAGCAAGAGCAAGCGCAGAGAACAGGGGCATAATAAA CTGCTGCGGGTGGTGTACCTAAACCATGATCCAATATTACATCCCTTATGC TGACTCCAGCAGTTTCATCAAACTTTACCC[G]TAAACAATGTTTACACTAT AGATTGCACTATTTCGTAATAAGAGTTTGAATATGAGTATGAATATGTATAA GCTGTTGGAGATAGTCTTAATACCTAGTTTAAATTAATCTAGTATAAACCCTCA ATCTACATGTATTAAAGGTAGATTGGAGTGTAACTTAACTAATTTATATCCC AATCCACCTCAACATACATAACTATAGTCTATCGTACAATATCTAATGAAG CCTAAAGACAACGTTTAGGATAAGGAACGTGTTAAACAGCCCCAAAAGCATT ATGACTCCATGAGTCCAACGGCCCAATAGCACGAGACATCGCAACATGCC GCTCCCCAACCACTGTAGCACCTGATTGATTGGTACAAAATTTATGATGAC GTCCCATTTGCAGACATCGAGCATATTGCCCAATACCTTGTGTAATTC AAATTTGAAT
484	CATATGAAAAATCNCTGGTTCGCTTTCGTTTTTNNGCATACCTAATAACC AACCTCTGCATCTGCTTCAGGTTTCGTACTTTGAGGCGCAATGTGGGTCGTT GGCGCAGTATGGGATGAAGCACATGCGGTCCTTCGCAACATCTGCAACGC TGGCATCCTTCCGTGANGCAGTGTGAAGGTGCTGCTCAGGCTTGCAACAGC ATTCCTTCCAACCCCTGGAGTTCTATCCACAAGGTTTGTAGCGCTAAGAA CATAAGGTGAGGCGAATATTTAGCCGCTCCACCGCAACGAAGTGGTTTAC ATTACAGTCTCAGGGGGTCTAGTTCTTGAACANNNNNNNNNNNNNNN NNNNNNNNNTANTATAGCTGTTCCACCGTNNNTCCAGATTACATAGCTATGC NCAATTTCCGGTGTACATATNATAGTCGGAAGTTATTTGGCAATTTGATTG GTCGTTGCTGTATATATTCCTATAGTTTGTAGCAGATGTGAGTTGTNAT TCCATAAAAAATGAAG[G]ACGCGTTACTGCTATTTCTATGTAGCCGACTGNTG

TABLE 12-continued

SEQ ID NO.	Sequence (Codominant Marker Position shown in Brackets)
	CTCATGTGAAACCTTTACCCCATCTTGTGGGAAATGNACTATCCGTGGTGG AATTCNNGCATCGAAAAACAATCCCGGGNGATCCTTTATTCAANGTGAANCG TCTGTGNATTTCATTTGAGGNTCGTATTATTTTCANTTGTAGGCTTGTGGCT GGTGGCATCTGTTCCACTATGNTTNCACAAAGNCTATGGGCAGTTTCATT TGTTTCGTTCTCTATATNCTGCAGCACCTACCCATGGAAAAAT
485	TCNNGNANGNANGTGTATTATTNACCTNCAGATTCCACCTTTGTTCTTCTAT ATTACATGGTACCTTGNAGGTTCTTGCTCTTTGTGATNATCCGTGCCCTCNG GAAAAAGAGGAAACCAATCATTGTTTCAGGTGAATATCNCATCNCNTTCAATT TACAGAACTCCTAAATTCANAGATCTAGTGNGTGATATCNGCTATTTTNC CAACTTNAGTGTCAAAGCAACCTAAAAATTTCAAAAAATACAGAGATAGGG CAATCT[A]TCTTCCTTTAAATCAAAGCTGTGGCATTTCCTTTGAAATTAGTA AACATTTATATAAATAGTAAATNTCNTGGAGATCNCNNAGGTANTTAACN TTTTTCCCTTCAACTTCCACAGTAATTAACATACCTAGGAANATAGTTTNG AGTTCTCATGTTTAATTGATNTGTTTCATCANAAGAACCATTAACNTCNGTCC TAATTATGCATGCCCTTNAATTTTCTCAAATTTCCCTTGATACCATTTCAA GTTGCAAGATGANTTTTTTTTNCCTCGTACTGTTAATATTTTGTAGCCA TAAACTTTCAAAANTAGTTCAGTGTCCCATTTATACATAAATNCTTATGTGT ACNTGATGGGTGCGT
486	ATTTATGTGCCTYYAGGCGAACCACTTYCATCYAACCTGGCGAGCCGTTGA GAGTGAACGTGCTMTTCAAGCATATTCAGGCAATGCTGTCTGCGGACAT[G] CTGTGTATCGCAGAGACTGGGGACTCGTGGTTTAACTGCCAGAAGCTGAAG CTACCCGAAGGMTGTGGTAAAGCTCCTCTTCGAANNNTGRTTTTGCT
487	GCACAGAACTCTCCCTGTCTCTTCTGGGGTTTTGGTTACGTGGTGGTAGTA AGCTTGGATTGTCACATGGATAAAGTTGTTCTAAGCTCCGTGGGTTGCTTGA GATCTTGCTGTTATTGCGTGCCGT[G]CTCACTTTTTTGGCAATCCGAGGAATG AATTTGTGCTTTTACTCGTTTGGTGGATTATTAGCGCGAAAAAAACTCTTT TTTTTTTGTCTTTTACTACGAAAAGCATCTTCTTGGATTTTGTATCTTCTTT TACTACGAAAAACTCTTGAGTCTAGGAATTGAATT
488	GGCTATTGTACAACAACAAGGCGGCTTGGCATCRCAGTTGCAGTACTASCA ATTGCCWTCGKGATTGCGGTGCGGTGTTGCAGTAGTAGT[A]CTGTAGTACT ATCAAGTGAGTCTGATGCATGAMCAGAGCAGAYGGAGTAGTTAAACGGAG GCGCCGGTGAGCTTGCCGTTGACGATGTGGTGGTTGTCGT
489	RGACACGGAGGAGCAGTACGACTTCGACCCGCTGGACGACACCAAGACGTG GCCGGAGGACCTGCTSCCGCTCMGSCCGTGCGGAGGCTGGTGTGGAC[A] GGAACGTGGACAACCTTCTCAACGAGAACGAGCAGCTGGCGTTCCGSCCGG GGCTGTTGGTGCCAGGATCTACTACTCGGACGACAAGATGCTGCAGTG
490	ATGAACATATGAAGAACCCTATCACTTTGAGCTAGTCTCAAGTCCAAGCAAAG AKAACAAATCACCAGTAGTAATGGAAAGAGAAGATTAGCAATAACT[G]TTG TTTAGTGGAGCAATAAATATCTTTTCAGTTTCAAGTMTTAAGAGAGAAAAAT GATAGATGTAAATGGCAAGCACCTACTGTACTTGACAACGTTAAA
491	ATAGTAAAAGGTTGGAATGTTAGTTGAAAACAAGGTGAAGAAATGTATCA TCTTTTGGACTGGACAACAGACCTGAAGTAGAAGCTTCTGGGTTTCTCANA NNTGGCCCTCAGCTTNNCNCNAANGNNNNNNNTCCTNTGAAACGAGGAG TTTGTATACAGTGNGCAATGCCAGGATTACCTTGGNGAACNCCATNTCC NCCAGATCAATAGTACCCCTGAAAAATGGCCACTCATTGTACATTTCTTGA GCATGTGGAGGTTCTGATGTCCTTCTGGAGGACAT[G]CTTGAATGCGGCTC CAATCCTAGCCAGACCGGAGGTGGAACCGCTNTGTGTCCAAGCAAAGA TCCATGGGATTGCTCGGAGTGAGTCGATACCTCCGCTCGGCTTTCTCTTGA TGGCTTCTTCTATGTTTATCCTACCATACTCTGTTTCAGGGGTTGCTGTG AACACAAACATCACAAGTAAATACTCGCTAAATGTAGGGGTCAAATCGAA TGCTTCAGTAACATTTTACTTAGGCTTGCAAAAGATAGTTTCTNCAGTAAA TCATTGTGCTAGATTAAATATTAAGTATCAGGTGAGANCTATTTCAAAGTTT GTTTTTGCATCTTATGTTACTGTAGTATCTCTAGACACCCTCAGGTGGNAGC CTCNACTAGTGGTCTACNNNCATCTTTTATGTTACTGTAGTATGGCAATCG ATGGGATCAAGGACTTACAAGGCGAAATACTCGACAAAGCGTGGCTCTTT GAAGACAATGGACCGATAT
492	ATGCTCTGGGGTCATCTCATGTCTGAATTCTGATCCAATTGAAGGGTTGTAG CTTTAGTGCAATGTTGCTATNTTATCATGCAACCCCTTACCTTTTGTAAAC ATGTCACCAAAACAATTCATTTCATTGGGTTGTTCAAGGATCATGTATTCCC ATGCAATACAACTTGGAGAAGTCANGTATTCCTTTACTATTTTACGTAGCA GTGTTGGCCACGATTATATGTTTAAATATCTGTACTTCAGTTGAGGAAAT ATATTTTCTGAGCGATCAGNTAATCACTTATTTTGGTTCCACTACTTTCCTAT CAGGTCTTCTGAGTCTCGGAATATTCAGCTTTTNTTGGGATATTTTGTGAGCNG CATCAAAGAGGCTCCTACATCTGATGANTCAAGTAGCACAGTTACATACTCT GAAGTTGACGGTGATCATAGGAACATAATTTTGGACCTTCAACCTGCTGAAA GAGGCCTCAAAGGNCAGACNCTCGATGC[C]GATTTGGTGCTGTGGACAGTG

TABLE 12-continued

SEQ ID NO.	Sequence (Codominant Marker Position shown in Brackets)
	GGTTCAACATCTCAGATTCCNCGGTTACAGCCTCCTGATGCTCCCTANGTTA TTCCTCTGAATGGTCGNGGACAGNGGAGACAGAGGAANNNNNNNNNN AAA
493	GCATAGCATACTGCATGCATGGCGCTGTTGCATTGCAGGTTGCAGCTGCAGN NNNNNCNGNGACCCNGNTGNNNNCGTGCAGACTGNNNNNNNNNNNNNN NNNNNNNCANNTNNNNNTANNNNNNNNNGTNNNNNNNNNNNNNNNNNT ANATAAATNTANNNNAGTACANTTTTGCNNNNNCCTGCTGCCTGCCTTAGC ACTAGCAGCCAGGCAGCAGGGCTGGCGCCTNTGCTGCGGTGCATGAAAAGG CGACGAGCAAGAAGGGGCGCTGGAAACTGCCGTCGCGCGCTGCTACC GCCGTATGATTGGATCGAGTGGAGCGTGC GG[C]TGGCTTTTGGATCCTTTT GGCACGCTCAAGAAATAAAAAANNTAACCAGTTCCTTTATTTTCATCGGTCTCCG ACTCTCCAAGGCGGCAGNNNCAGCAAGCTCTCGTAGCTNNGTGTAAGCTG TGGAGCACACAGCAGCTGTGAGCCTGTGTTGGTGTGCGAGCAAGAGCANA GCTGGCAAGCGCAGCANAGGANTTGACACGGCTTTATACGAATGAGCAGA ACANAGGCATCTGTACATGANGACGGGCACGGCACACACACAGGCCTTTGG TTCGTTGGANGANNGTGTATTGGCCTCCTACTATACTGANACTGNACACCCC TGGACATCCGCATGNC CGCATCTACTACGCATNGATAGCAGCGNTNTGTAA NNNNNTGNANTGNC T CNNANGCAGCCATGCCATGNNANCCAGNANNCC CACAGCAC
494	CACACATAATTACTCCMATCTTCTCAGTGGAGGTACATCAGCAACCAGGTT TATAAAGGAACRCITYGAGGACAACACGATGGTGGCGGTGCATAGATTG[C]T CTACGAGGGCTCTGAGGAGGCGTTTCATCAACGAGGGATGGTGTGTCCAA GATCGCCACARGAACATCGTCAGAGTTCGGCTGCTGTCTGGAACC
495	TGGATCCATCACGGTAGAGCTGACNATGATATGTCCAAANGCTTCTGCTGCT AGTTGTTGGTCGTC CGCAAGAGACTTGACCCNACAGGTTTTTTTCGTTGGGC CTCAAGAACTGTAGGTGCATTTTCCATGGAGGTTATGATGCTTAACCCAC TCTGGTTCTG[g]TGGCTAAACATTCTCTCCAGTCGGCTTCTACCATGAGCC CTGGCACTTAGT
496	TGGCTTTATCCGTC CGGATGCCCGCCATGTGCTCCTTGCTCGGGACTCGG AGACAGNCTGACAGTCTGGGTNTTGTGCTAATGGCCNTGGTTAACTTACG TTAATCNTGCGCCTTATTATCATCATCTAGGTGTAATTACTTCCGT[T]TGTGC TAATCAGCTCGTGATTGCCTGCAANNNNNNNGAACCCAGCCCCCNACCTT GCTGATGCCGTGCTTTTCTAGAGAAGTACCCTGCCGCGTCACGCTTCGTCT CGCCTGCTTTGGATTCAACGGNTTTTGGAGNCCNNNGGCCA
497	GGTGCCGGAAGCCGCTACAGCGGCCCGCGCTCAGCTA[C]TACGTACCA AGTTCCAGCCGCGGCTGGCTGCGCCGCGCAGACCCCTCGAGGCCCGCGCCC CCGNCGAGGCACAGAAGCGCGCCGCGCCNCCGTCGNNGTCCGGAGGTAC CGGCACCGCAGCTGTCTGT
498	RCATAAAAYATTGCAAATGCTGAGGAATATAAGCACATAGATACTTAGCAT GTGTCCGCCAGATGGAAATACTTTTACCATTCCCCAC[G]ACTGGATTGCCA GAAAGGTCAATGCCATGACACCTTGGTCCATYATTTCCAYGGCTAGATTAA CCTGGCCAGGGGAAAYAAATSATTTGTGCAGATT C
499	ACGSYGTAAAGGCTCTACTGCATGTTCTTCTGAGGTTGCTCTTGACACCCCTAA TTTCTGGAAATCATCCCCATCTRGGGACATAAATGGAGCAAAATTGC[C]GT GAGCTTGAAACAAATGGAAGTCACTGTGATGGTCTGAATGCCATCAGGCA GATTCTGTGATGTGTTTGTCTCTGATGAAGAAGGGAAGTTCAGCGG
500	TCTCAAGCTTATTTAGGTGACTAAATTTGGAACATCCCCGAGATGCTATT GCCGGCAAGCGCGAGCTCCGTAAGCATGGCAGCTTTGCCAATCGATGC[C]G GTATGGACCCCTTCAGCTGGTTACTCTGCAACCGAAGAACARTGAGGTTGCA GAATGAGTCAAGCCAGTCTGGCACACTTCCATTGAAGTAGTTACCAT
501	TACTAGCACACTTTGCCTCCTGATAGCTTGTAACATTGTGACCTTGTTTGGAT TTTGGGATTCCACCCAATGTCAAACATACCCAAACAGTGACCCGKTATCTTT TTCTTCGTATTTCTATGTCAAGCACTYAAGCATAAATCATACA[T]TTGCGCTCT ACAGGTGTTTCTGAAACACTAGTGCTCAATATGTCAACTCTGACATGGTCAG TTGTTAGCACTGTGGAAGGACGTGTTCTCTTGCCAGTGAGGTATWCTCTT GAGCGGTTGAGCAGCATTCAGTTGTTAAGATG
502	GCACAGGTAACCTTTGCATGGCACCTGTCTGAAACAAGTTTCGTGCTAACA AGGGAAATTATCCTTTTTCTGTGGCACCAATATCTGTTTTCCATTCAATGATGT ACTGATGTGTTACGGAGTCGTTACCTAGAAATCTCGCTTACTG[T]GACATA TCACACCGAGCATCATTTTATATCTGCTGTGYGCCCTCCTTGCTAATGTAGGC ACCTG
503	RYTKAACMTGCGAGAARTARAAAATCACTAACCTTGGCTTTATCAAGTAT AGCAAGGTGAAMGCGAATGCCARGAAGAAGCAGAGTCCRCCGACGTGAG GTACGCAAGGCCCAGAAATCGTTCTTCCCGAGCCAGGTTGCGGT[C]GA

TABLE 12-continued

SEQ ID NO.	Sequence (Codominant Marker Position shown in Brackets)
	AAGCACCAGCTTCTTCTTGCCACCAAAGCTATAGGTGTTGTAGTTGTTGTCC ARCTGGACTGTGATTGTRTCGTTTTCTTGAGATCAACATATATCCTTCCATA CAGCTTTCTGAATGTGCGAAGTGCTGYAGTCCGCATCCAAACA
504	GCGCTCGCCATGAATGCGCGTCGTCTCCRTTGGACCATCAACCATTGCAAT TGATGAGGAGACGAGACAGGAGAAACGGCGAGGGCAGYAGTGTTCGTAG CTGAAAGCTGCCGCGAATCGACTGCCATTATTGGACGCC[A]TCCCTTGCCCG ACTGAGAATCGGGCGGCGCGTGCACCTGCACTRCMCTACCGCCGCGCCGCA GGCGCCAGCCRRCTCTGTTGTTCAAAATTCATCCAGAATTGCGACGAGGCC ACCAGCTCGGCTCCGGCTGCGAGGATTTCAA
505	TCATGGAATTGCTTCAAACCCGAGAGCGGCAAGATAATAGACGAACGATCA GCTCCAGCCACCTGTAAAGTACAATCAGAAAYRGTAGAGCAATGGATCACT YGTGGAGGCTTGTTTACAAATAATRGCCAAACA[A]T]GTTACCTCAGAAA TCCTCAAATAATGGCCCCGTGTTGTTGCTGCCAGGTGCAAGAAGAACCTTTT CTGATCTGCCCTRATMACCTTAGACATACCCATGCCACTAATTTCTTCTTGA GGAGGCAGATCAACCAAGCGTTCSACATCTA
506	TAYAAGCAATATTAAGAGATGGAAATGTCGCCTCTGTACCACAAGTTAGTA AGTTACTAAGTTGATTGTTAAAAACCCATCGAAATGTGTGACATGTAGGACC TGGTCCACATGTCAGACAAAGCCAATGCTATTTTGGCATTGATGT[A]CAA TGTAAGTCTATAATTTCCACAAAAAAGACAGAGTCAAACTACTGYGTA AACATTGATCCAAAAAACAGAGATGCTTTGTGTCATAAAAAATCTAGCCAAA AGAATTGGAGCTATTGAGATACAAATGRA
507	GCTATCTATGGCCACATCTCCACTTCGTCTTTGCCGAAGCTTCAGTACGCGGT ACGTTCTCTTGTGACCATTAATCATTGCTATCTATTTTGAACGAACGCATTTC GCTATCTATGGCCACATCTCCACTTCGTCTTTGCCGAAGCTTCAGTACGCGGT GCGGGTGTCTCCTGGTGTGATGCTGTTCTTTTCTCGGTCCAT[C]TTATCCCC CGTTTGACGCGCCTCTGTCAATCTGCTGTACTGTGAATTTTATTGACGTGCA TAAATYATTCTGGAACGCTCTTGGTACGTCACGTACCTYGAGAA
508	ATCSGGCACCTGATCCWGAATGTGAGWCSGAGTGCACCTATACCCGTCCC CAACCGCCCCCATTTGAATCCACCGGTGAATCTATCGCTCCGCCACACGAGG CCCTATATTAACCTCCACAGCATCCATGTGTCCGAG[C]CTGTCTGTATACCTGT CACTACGCTACCGCGTGCCGATCGTTCTGTTCTCCCTCCCTTCGCGGGCC GCGCGCTATKWTWTTTACTACTSTATTCRTATCATTATAYTKTTTGGTTTCC WTCCCC
509	TTCTCGTAATCTCTATGGAATTRCCTTTACTAGAGATAAATGGCATCCTTGC AATGTCCCAAAATCTCACCGAAGAACACAAGKAAAAGARAAGAAAAATTTG AGCAAGCACAGCACAGGCAGGCAGCATGAATCCACACAGAACCA[A]TGGTT TTTAGCACAGYASGYTGACAGCCAAGGGTTACAGTACATAATCAGACGGGG CAACACGAACGAAGCAGCCGATCARGCAGCCTASAACCCATTAATTGA CACACAGCGTATATATATATATATRTAGTACTTGTCTTTTAGC
510	AAGTCCAGATGACTTTCAGAAAGAAAAATCGAGAACTTCAGACACCTTCTGA CCGCTGAAGCACGAACGACAATGGCCATAC[A]GCACGATGTTGCACGTTTY GTAGAAGTTTGTATCTTCGTTAAAAATTTAGCAGTACAAATSATAGCRCAT TCTGTAGGTGTAGGATACAGTAACACAGCAGCAGAACATCAAAACCCRTCTC CTGCCAGGCTCTGAGCAACGCCAAAGC
511	CCRTRCAACATTAGSTCGTTTYTGTTCTATTTSTTGACCAAGACATGGTTGYT GTTCTGATGGCAGATTTGAACCTGACATGCTTGCTTCTTCTGGGTGCTATTAT GGTGGTGGAGAAAGGAGAGGAGAGAATTAGCTGCAATTTCGCAG[G]CGAT GGAGAACCCTGTCATGTACAAGCTGTGTGATGTGTAATCATTACTCGCTTACT GAGATAGAATTAYCTTTTGCTAAAAGTTTTCATATTCAGATACGTGACCCG GAGAAAGGAAGACGGCAAGGCAGGTGTCCACTAATCTCTGAG
512	NCGCGTGTTCGTGCGGATGCWAYKCATGCAGAGGYAGCARARCTAGCTAGY AAGCACGWACKYACGTAGCACATGATAAGAAGGCTGCRRTTGAGACAGTAA GACGAAGAATGGCARGCAGAAGAGCACGTC[G]GCATGCTCCCCGCGGCTTA TAGCTTAGAGGCACCTTGAATCCGGTGGGCACKCTCTTGCCGAGTGGTTGAG GATRAGGCTGAGGTGACSGGYAGGTTGAGCTTGATTCCKAGGACCTCTCCC TTGATGGCGGTGACAGGCACACGGCG
513	ATGGATGCTTYTGTAGCATTAAACGTGAGAGTTTAATCTSACCACAACAAGC ATGGGTGATCACAAGAAATACAAAACAGCCAACCTAACCAAMAWKATATT AGCAAAATGCAACTTACTTATAACATGGCAACCAACACAAACACATTG[C]G CACACTTATGCTGGCTTAGTCTATCTGCAATTAGACCCAATGCAGTGAATTG CTAGTGAACGAACCAATTGCTCCCGTAGTGTTCCTTTTGCTTCTATGAAGG CTATGAACCTGACGGTTTATTCTAGTGATCTATTATCTAYCA
514	TGGAACCRRTTAATAATTGAAGTTTCTYAGTACAGRTCTGCATAGCATTGA ACTGGACAGCTTGCTG[T]GTTACTCTGTRTAASGAACGATCTACTGCTGATCT GTACTGTTCTTGATTTTTTYYCWTCTTTCTCTTTTGATGGCAAGCAGGAC

TABLE 12-continued

SEQ ID NO.	Sequence (Codominant Marker Position shown in Brackets)
	TGAAGATAAGATGGCTGCCTTRCCATTGGCCACCGCAGAAGCATGTGATGCT AATGCTG
515	GGAAATCTTGACAAGGATATGGAGAGAAGGTTTAGCTACGCCCTAAGCAGA GAAGACATCGAGAACGCCATACTCGGAGGACCTTAASCTGAACACATGGCC AAGGGGCTCAKAGCTAGGAATTGCTTAGTCGAGGTTGGCACTGAAGAA[G]T AATCGCTCCAGTGATCTGCGTGCTACCATTAACGCCATTGTAAGAGCAAAATGG ATCATGTGTGCGAGAAAATAATTATCATGAGCAATAATCTCGAACTAGGCTGGA AATGTTGGAWGCTCCTGGTGACATGCGYTTGTTTGTGGCMTC
516	MTAGGAAGGTCAAGGTGCCTTTGGACCCCGGCAGCAACTACACCTTCGACC TGAGCTACTACCGCCACGTRCTCGCCACGGGCGGGCTGTTCCAGTCCGACGG CAGCCTCCTGCACGACCCCGTCACCGAGAGGCTACGTCGAGAAGGTGG[T]CA AGGCGTCGTGCGCCGACGAGTACTACGCGGACTTCGCCRCGGCGATGGTCA AGATGGGCGGCACCGACGTGCTTGTGCGCGATCATGGGAGATCAGGCCAA CGTGTGGCATTTTTGTGACTAGGTTACGAKTGGGTTGAARATGC
517	GTCAACAAGCACCAGACTTTCAGGACACTCGGTGCTTTTTCGTGGTCTCCA CAGATGGATCCCAGGCAGAYTTCCTTACCTGAAGTGCCTGGAGAACTTTGT GAGGAAGAGCTACACGGAGGACGYSGACACATTCTGCATGAAGTA[T]TTAA GGCCCCGTCGCGAGGACGGCACCACAGCTGATGTTGGGACAGCATCAGGCR CCCCGATGAGGTTCCACCGTCAACCGCAGCTGAGACAGAGCAAGGCACTC CTCCAGCCCCCTCAGGCAGAGGTTCCRCAGAGAYTTGG
518	ATTACACCWAACTTGTCTTATAAGTTAGAAYTCTTACAATATTTATTAGAC TCTGTAGGAGAATCTCTTCTAGATACAAACCCACATCCTTTACTGTAAAAA GGCCCATTTGTTAGCTACAATGACTTAACAT[T]TACGGAGGGACCAAATCTCT GGTCTGAAACCCAACTTTCAAGGTTCACTAGGTTTCTCCACTGGCAGATAA AACAAATCCTCATTTAYGATCATAAGTTGACATACTGGACAAAGAATACTTG TAAGCATATCCCTTCTTATCTGATGA
519	AGTGTAGAGATACTACAAGAACAATTTTGAAGTGCCTTGCGAGGCCAGGT ACCACGAATAAGTGTTCACCTTACTCTCATGCCTCAATTTCAAGAACACCT GAACACATTTGCTAAATGATGTAAGAAAAAGTTGCCATCATGGTA[T]CAG ATACGAGGTGAGTAGCAACCTGGTGAGGTTTAAATGTGTGCCCMAGCGGTG TGSTTAGTTGAAAGATAAACRCAGTTTCTGCAGATGTTTGCGAGAAAGAGA CACTTTTGTATCTTTCAGAAGATCTAACCTGAATGAAGCA
520	CCATGCGCTACTACCGGCCGGYTCTCGGAGATGTTTGGCTCCACGCCGCC GCCGACGCGGAGGACACGCCCTTCCACCCGCGCTCGCCCTACGCCGCCGCC AAGGTGCGCGCGCACTGGTACACSGTCAACTACCGGAGGCCCTACG[G]CGT ATTGCGCTGCAACGGCGTGCTYTTCACCCACGAGTCCC CGCGCCGCGCGAG AATTTCGTACGCGCAAGATCAGCGCGCGCTCGGCCGCATCAAGGTGCGG CTGCAGACCAGGTCCTTCTCGGCAACCTCTCGGCCGCCAGGGA
521	TACATACCTTGCGAGCATGTCAAATGCACAAGGATGAAGAGAGAGGGAAAGT TGCAGCAAAGAACTTGTGAGATGGAACCAAAAGCTCATCCAGGTACGT GTTCTCTDTCAGCTTRCATGCTGCGGCTGGTAACCTGGGTTGAAGCCAA[G]GT AGCCAGAGAAGCAATGCGAGAAAAAGGGTGATGAATTTCCAGGGTGTA GTTGGATCAGAGTGGGKAACAAACARAGYGTATTTGTTGTACAGGCACAC A
522	ACTCAAATTGAGTCAAGAAATGCTAAGGGAAAGCCCGTGAACAATATAAT CGAACTTTTACATTTTATTGATGTGGCTTYYTTAAAAATGACGCCATAAAC WCTATACTGAAAACGGCTCGGCGCTTAAACAGTAGCATTCGTA[A]GTTTC ATTTCTGATTTGGGACTTCAAACCAATAGTGTGTTGAGACTCACAGGTTTAGA ACCGATTGGTGGCTGCCTTCAATCAAGAGCGAAACCAACATGTTATTGCCCA CCTTAAATCACAGATTTTGTATTATCGTGTACAATAAAAAAT
523	YCAGATCATCACTGGAATCAACAGCAGTTCTCTGGTTCATTTGTTTCAGATGC TTCKGTGTCAACTGCTTCTGAAYGATGGTTAAGCTTATAACATCATCARCA TTTTTGTTAATTATGCACTGTCTCGTTGTCTTTTACCACGCTAT[T]CAAAG TTTCTGGCTCCTCAGATACTACAAGATCAGATRTCTGCCGCTTCTGAGTAA AGTGGTTTCAGGACTGACACCCATACCTTGTGCAAGTGCTACATACCGAAGT ACTAGWTGCTTGTACGAATTCATAGGCTCTCTACATCAGC
524	CTTTAGTAGTAGTCYATAGCTCATAGGTTCTCAGTTCGGTATATRTYGGT AYATATTTTCATGAACCTASTTTTWAAMCMATTTAAAAAMGCMACGCAACA ACGAAGGGAGTGATATTGACCATGGG[T]CTTTCGTAGCTTTTCAATCAAACG TCACATAAAAAAGAAACAWCCAAAAGWTATAATACACAGATTKRAAMS KKKAGACAAAAATACGAGTACAAGACATTGGACCTTCAATFWTTTTTTTGT GTTTTCTGAGACATGAAAA
525	TTGCACCACCCATAGTTTAAAGAGCAGTTGCTTTTACACACTTTATTTTGCAT GGCACAGCCAACTGTCCCCATCATTTAAACAGGCTGCAACTGAGCCACTA CAGAACTGCTAATATTAGATATTCCAGCAAATAGTCTTGACACTAGAGTGC

TABLE 12-continued

SEQ ID NO.	Sequence (Codominant Marker Position shown in Brackets)
	TA[A]GTCCCTAAACACGAAAGACATTTGTCTCAGGAGACCCGTATTGAGTACTA GCAGAGTATACTACCGCTAGATATTTCCAGTATAATCGTGCGAGCAGCTCCGGT CAATGGCAGTGGCACTTGAAGTCTGCAACCTCAGCCAACAGATCCACATGA GAACCAAC
526	TACACTATACTAATTAGAATGTCTGGAAGGCTGGGAAGACCCTAAATTTG TCTCATTKTTGGAGGAAATGGTGTCCRAAGGATGTGTCTTARTCTGATTGCT TATAATACTGTTATTGAGGCTCTTGGTAAGAACAAGATGGTTGAC[A]AGGCG ATTTTATGCTTTCTAAAATGATTGAGAGTGACTGTCGGCCCAATCAATTCA CATATAGCATTATGCTGGATGTTTATCAACAGGGGGACAACTCCACAGGTT GAATGAGATTCTAGATATTTGTA
527	CCTTTCTCCAGTTGGTAATATGCTTCATGTAAGGTACCTGAA[A]GTGGTAAG CRACATGATAGTCAAACCTCCCACTCATTATTTCGAGGCCTGCGACACTTAGAG ACACTCGAGGTGGATGCAGAARAAGTCGCTGTTCCACTGGATGTTTTCATCT TGAAGAGCCTGTTGCATCTCCGACTTCCGAGCAAGGCT
528	TGCATATGAGAACTACAAGGTTGGCAGGGTCCTTTCTGTGCCCCGAGCTGT CATGAAAGAAAGCAATGGAAGGGAAACGCCGCCAAAAGGTATCAT TAAATAGTATTTACCTGAACACATTCTKTAGTTAACCCCTCAGAC[T]GCTCAT TCGTTTCTGGGTTCCCTATTTGTCTTTCCAGGATCTTCATCAAACGCTCTTGGT CACATGTGCTAGGCCTGCCTACCACTAAAGATTCTTTATTGATTSTTAGCTG GAAATAATAGKCAGCCCATAAACAATTCATTGGCTAGAA
529	AAGCACCCGCATGCAGACTTCCAACTATCCAGTGTCCACAAATAGCAGTTT TGCAAACTAAATACATATATGAAATGATGACACAGGCCACCCAAACAACAA GCCCTAGAGAATCAGACCAACCAACCGAGCAATTCGAGTGCTCCATATCTT CTAGCATTCAACCTGACGAACG[C]TGTTGCCCTTCAGCGCCCCAACATC AATCGTATCAGGAAGATACCTGCAAAAGTTGGTAAACATAGATCAAGTTA CACCACAAGGCTGCCTGCATGTATGATACATATATGTGAGCTATATACTATT CAAACCTACATATCTTCTCTGGCTCATTTTTTACAGCATTCCTCGCTATA
530	AGAYGCTACTCTGCTACCATTTTGAGAATCTATTGACAAGAACTTTACAGGA GGRGTCTCATCCCCAGGAGCCTTGCTGATAACTGGAGATGGAGAAGATATG CACATATCCACTCGTGAATGCYTTTTCTGGACTTATTCTGTCTCTT[T]AGCA GTTGTTTTCCCTAGCAAACAAAAGGAGATACACATAGCATGAGAAAGAAGA AASAACTGAAACATTYTATAATGAGAAAAAACATGGGGAGTTGGGGACA AAGCAACTCACCAATGCTTCTACAAATGTTTTGAACCTCCGA
531	TGCTTCTGTGCAATTGTCTGCGGCAGARCATGCTTTGGTGGYGGCTTTGCAG TGTGAAACGTTATGAATCTGAGAACCTCGATGCATGCYAGGGACAGGGAAT ACCTGAACAGGGGCCAAGATGACATGAACAGGACGCCGGA[T]AGCACGTGC ACAACCTTCTGTCTCAAGCTCTGCCTTCATTGATGGCATGGATCAGTACAGA GTTCCGAAGCCAACGAGCTGAACARCTGAACATAAGCAAAATGTAATGGCCG TTCCTAA
532	AGGTCGACAAGGCTGCTAGTAACATGTCTGAGCCGACATCAGGGGAGATGC AGGATGCTGCTTTCCAGTCTGACGAAGAGGAGGAAGATGAAGATGTTGATG AAACAGTATTTCGGTCAAGATTGAGATTGTCACARAATAGCGGCACCG[G]C GACGATGCAAAAGTAGACTCACTGCTGGTATTATCATATTATGGTGATTGTATT GTTTTTAATTAAATTGCCAAGCTTGATTTTTGTCAAGCGACGYTGGAAGGT TGCACAGAATTTTGAYAGTGTCTCTGGTTTCATTTTGAAAGCACT
533	AGAAAACGAGGCCTTCGTGCTTTAGCGCGCRGGGGCGTAGGACGGCGGCA GCGGCAGCCCCACGACCTTGCCCGCAGCGGAGGGCGTTGTGCGGCATGTTGT ACTCGTCGCGCAGTGATCGCGCGCGGAGACGACGCTGAT[A]TAGAACTGC TCCCCGAGGTAGTACCGCTCCAGATGTTGGAATGCACGCTCCACATGCCCG CGTTGTCGAACGTCAGCATGATTGCCGTCCATGACCGCGGGTACACCTGGAT CGTGTCGCCGCTCAC
534	TTCTGGTACTTTCGCGATCTCGCGCAGMGCCACSGTSCCGGCTGTACCTGT GGGGCTTCTTCACGCCGCGGTGCTCGGCGCMGACTTCTTCGCCGCTGCAA CRARAYAACCAACCGCCGCGCGTCAGRCCTGCTTACAACGGGAAC[A]ATC GAGGGGATCGGAACAKCAGATCGGTACGGTAGGTAGAGACGAACCTTGGT KGCGAGCTGCTTGC CGGGGCTTGCCCTCCGGTGGACTTGC GGCGGTCTG TTCGTACGAGCCATCTTCTCTCTCCTTG
535	CTGCGTTAGTTTCACCCCTTCTAGCTGCGAGTGAAAGAAACWTGATAACAG CTWSCGTAGTTTCTATGTRYRGCCATCGAATCTGACATGGCTAYCTCCTGT GMCMACGCAAGTCCGCAAGTACTACAGCCGAGGAAGAGCCA[A]CGGACG GTGACGGCGGTGATCCACGGCGAGAAGGTGCCGCTGTACGGCGCCGGGGC GGCCTGACGCTGTCSACGAGCGCGGCGGGGSGCGGTGCCCTGACGC
536	AGCACCACGATTACACCTACACATATTCAGACACAAAATGGGCAGCGAT GGGCATGGACATTTCAATTGATAGATGGTATTTAGCTGCCTACTYTGGCATTT GATTTCGAGCAGCGAAGCTTCAAGAGCTGCAGAATCGCTACAYATATT[C]CT

TABLE 12-continued

SEQ ID NO. Sequence (Codominant Marker Position shown in Brackets)	
	GTGGGTACACATGCTTCTTGTCTCCACTCGACCCAGATCATTGTGATGGAC TACTAAGTGCAGTACAAAAGCAGACATCTATCAAACCTACACAAAAGCGG TGACAAACGAGAYAGYCTGCTTGCGCATCTCCAAGTCTAGTCTAG
537	AWWTTKKWAGYGCARRAATAAAACAAGGCACTGTWRATTGTACAGAGCAA GCTCTCTGCATTTTCAATGCCGCGAGCATCTGCAACTGCAAGCAGAGTTTCT A[C]AGGCCAAGGCCGAGTGTCTGCCATCTCTCCTGCGGAGGCTGTATGT AATGTTTCTCCAGGTGGTGTGAGCTCCTCCAGTTTCTTGGCTTCTTCCGCC GAGTCCCGACCTTGTCTCGACCGTGTAA
538	TCAATTTGTGATTTTATCCACAATTTCTTCATATGTGAGGAATGCCGCACRCA CTTCTATGAAATGTGTTCAARGTTTGGAAAACTACTCTAGCCATATATTGC CACAGAATCAASTGCATGTYTAAT[T]GTTACCGTTACCTGTAGTGTATCA GTCCCCCTTCAAATCTGCCCCTGACCTTGCCCTCTGGCTATGGRCCGCGCATA ACAAAGTTAACGAGAGGTGTATGAAAGAAGAAAGGAGTTAGACAATGCT GATCCTTCATTTCTTAAGAT
539	TTGCATACAAACTTGATCMAAATGTTTTRCAAMCCATGCTTGAACRRATATAC AAACCCGTTGYGGCAGGAACATYTTAGCTTWGKAMMMTYGYWVAGCAAC AATAGCTTCAATAGAGATTGTTGTAAGTGTAGAGTG c TGCTGCGATGGARA CTATGCAAAATCAACAAGGATGGGAGAAGAGTGACATTCAGAAGGAGGTCCG TGCCGAGATAACCGGCTGCAGTAATCGTGGCATCTGCAACTACACGGGCCA ACGTCCCGG
540	ACAAGTCCAAATCAAAGGTGTTGTCCCGCAATGCTTACCCCCCATGSCYCC CCRAAATAGCATCTGAATCCATTTTTTTGTCTCYGATCTTTTTKKCTCCTCTTCT GACAGGGAAATGCTGCTTGAGTTGGACGAGGAACAGACTGGAAT[A]CAAAT GCACGTGCAGTTGCAAATCTTATGTTTRWTTTTTGKGCCTATCARTRCAC AGTACTCATAGCATGACAAATGGCGTTGCAGGTTTTGGTTTTTACCSTTTTTG GTTTC
541	AACCTCGACTCCACTATAASTGCAGGACGCACGRCAYGCAAACTGAAACA TCCGGACCAACAGGCAGAGTAGTAGCCCAAAGGATGACCCGTCTCATGCAA AAACAGTCAGCTTGATAAACTTATCGAGAAGGCAACA[C]CGGYGCCAAGT ACTGGTTGAAATACAGAGCAGACATTGAAGTGCCAAAGAGGGGGG
542	GTCTAGAAAAGTCAAACGTCATATACTTGGAAACRGAGTAGATATGACAA AAAGAGCATGCAAGTGATGTTCTTACAGAAGAATATATATATTTTCATGGCAA GCTAGTACCTAAACTTCAGATGGAGTCTTTGCATCGATTGACAGGGC[G]TGC AACCCGGTCTTGAGCTCCTCCTGCAGGAGATCATAGACAGCCCTGTGCCTCT TRAGCAAGCTCTTCCCCCTCAAACCTCTTGACACCCACCTCACATTGAAATG TGCTCTCCCCATTGGTCCAGCCACGCCAGCATGGCCCTTGTTG
543	ATTTTACCATACTGCATTCAAAGTTTTACAAGATTGGACATTATTAGTATAA AACATGAAAATTTTCAAGTGAACAASAACATAARCAACGTCCGCAAGCGC AATCCACAGCTTGAGGTACATTGTGAGTGTAGTAGA[G]AACTAGGTGGTC CCAAATAGCACAGATTATCCAAGCTACCCAGTTTTCTACGATTATGGGCAC RACCATAAGAGCATAGMAGTGCTTCGCATCCTGTTGCATTAGAAAGTAGT ATAGGCATCGGAGGTGCTGAATCTAAGCACACGG
544	TAAACCAGAYTAGATGCCACCTAGKTTTCTGACACAAATCAYGAACAAAAC ACGAGAAGAAAGCYAATCATACCGGATGAYGGATGGACTCAGCGGAGCC CTTATTTGCTACCATTCCTCATGTCTTATGTCAGAAATCCATCTATTGC[T]AC TCAACTTCAMTCAGTCTCTGGAACCTGTATCAACAGGGGATGGGAAATGT GTCATGTTCAATGTTTAGCYCATGAAACATAGAAGACCMCATTAGAAGCTA TTATGTGCTTACATTTGATTTTTTTATCCAAGACTCAAGTGTAT
545	RRHYTKRTGGAGGTCGGGGGAAGAAACCTAAGAACGATGTCTACCCAAAT TGATTCCTTCCGCCGAGAGATTTAGAAGCTAGATCTGTCAGGATTTAGTGG ATTGATAGGTTTTGTGAATTTGTCATTCTCCAACCTAATCTACTTAC[G]GCCAG ATTTTACATCAGACCTGATGAAACACTGTTTCTTGACACGAACTGGTGGGA CGCTGCCTTTGCATCAAGAATCAAGAAATGATTTTCGTTTTATGTYTCTGGT AGCYCCAGACACCTCACTCTCCTCTGTTGCCTGTSATG
546	AACGGCTAAGAGTCAGGCGATTCTGTTTGTACAGAGACAATCGCAGCACTT GRMTGCTWCGCATGCGSTTCAGAGGGCCTTCAGCTCAGGCTTGACAGATCT AGTTTTGGYCGCCACGCYRCTCTTCGGCGAGCACTGYACTCTGCTGGC[G]TC TTTAGAAGCCCCGATGTGTGACGATGTGGCCGGGTGGTAGATGTTGCGTTG GTGTTTTTTCCCTCGGGTGTCTGTTG
547	AAACAANACTCAGTGGACTCGCNGAAANNNNATAATGAAAGGTGCTCCATC CATACCCATGAGAANGTTCGATGCTCGCAGTCTCATGTTTCCAGTCA[A]T CTTCTTTCATTTCTTCTCCGATGCACTAATATGCAGATCATGGCGCAGAGA AAGGTTCAAGATTGCTCTCTTGATCTCTTCAAACCCGCGCTTTCG

TABLE 12-continued

SEQ ID	NO.	Sequence (Codominant Marker Position shown in Brackets)
548	AAAATTACTGAAGCAGGTGGGTGCAAGTTGTGTTCGTTACTGTTTACTG TAWYATGTCAAGCTGTCGGCTGCAATTTCTTTGCTGGCAAGCCGAGGCACT GGTGAAGTGCTGATAAATACATCATATTCTGTTGACCTGTGAAGAA[G]CTTG TTCWAGGTRGATTCCATTGTACTAGCTCTGTTGCYACGATCTCCTTGTTTG GAACATTAACAACCAGCYCTCRMCCCTCAANNNNNNNNNNNNNNNNNNNN NNNNNNNNNNNNNNNNNNNNNTAAATAGTCGCATCATGTA	

Table 13 provides assay primers and probes usable for detecting the codominant markers in Table 12 by Taqman Assay under the conditions described above. F1 and R1 correspond to forward and reverse primers, respectively.

Probes are indicated by fluorescent reporter (Fluo Color). The detected male line corresponds to inducer line NP2222. The detected female line corresponds to an inbreed line, line B14.

TABLE 13

Assay for	SEQ ID NO:	Primer [[/]] or Probe	Seq	Fluocolor	Allele	Sex
SEQ ID NO: 99 and 474	174	F1	GCCATGGATCGCATTCATG			
	175	R1	CTCTCTGGCATCTCTGAAATC ATC			
	176	FM	CGATTTTGACTTTGAACC	FM	A	Male
	177	TT	CGATTTTGACTTCGAAC	TT	G	Female
SEQ ID NO: 100 and 475	178	R1	ACACCCATGGCACAAGGT			
	179	F1	CGAGAGACCAGATCAACAGC TT			
	180	FM	CAATACTCTAAAAAACTG	FM	A	Male
	181	TT	CAATACTCTAAGAAAACTG	TT	G	Female
SEQ ID NO: 101 and 476	182	R1	CCCTTGAACCAGAGTGGATAA TCTC			
	183	F1	CAGGTAGTAGTCTTCCGTATG TCAAC			
	184	FM	ATCAGATGTAAAGCATC	FM	A	Female
	185	VC	AAATCAGATGTAGAGCATC	VC	G	Male
SEQ ID NO: 102 and 477	186	F1	ACTGCCTGCAAGGTTGGTT			
	187	R1	CGGGCATAAGATCCCAGTCA			
	188	FM	CAACCAACAACCTT	FM	A	Male
	189	TT	AAACCAAGCAACCTT	TT	G	Female
SEQ ID NO: 103 and 478	190	R1	CCCCAGCCCTGGATCAC			
	191	F1	GCTTTCGAATGTGTTTCATGAT GTAAGT			
	192	VC	CCTGGAGAATGAAAGTA	VC	A	Male
	193	1FM	CTGGAGAAGGAAAGTA	FM	C	Female
SEQ ID NO: 104 and 479	194	F1	AGCTTCTCCTTGACCTTCTCCT TAA			
	195	R1	AGGAAGAGGAGGTGATCGAT GAG			
	196	TT	AACGGCGAAATGTCAA	TT	A	Male
	197	FM	CGCGGAGATTGTCAA	FM	G	Female
SEQ ID NO: 105 and 480	198	R1	GCGTCACAAGAAGGTA			
	199	F1	GCAAGCAGAACTCAAC			
	200	TT	TCAACATGACTGAATCA	TT	A	Female
	201	FM	CAACATGACTGAAGCAT	FM	C	Male
SEQ ID NO: 106 and 481	202	R1	CGCAGCGTGGAGTGC			
	203	F1	ACACATTCGAAGGCGGATATC ATG			
	204	FM	CCGGTAACATCAACG	FM	A	Male
	205	TT	CCGGTAACGTCAACG	TT	G	Female
SEQ ID NO: 107 and 482	206	F1	GGAGTCTTTTAGGGTCCACTT GTTT			
	207	R1	AGAGGCACAAAGCGAAGAA T			
	208	TT	TTCTGCAAGTTGAATTT	TT	A	Female
	209	FM	TGCAAGCTGAATTT	FM	G	Male

TABLE 13-continued

Assay for	SEQ ID NO:	Primer [[/]] or Probe	Seq	Fluocolor	Allele	Sex
SEQ ID NO: 108 and 483	210	R1	GAGCAAGCGCAGAGAACAG			
	211	F1	CGGGCATGTTGCGATGTC			
	212	VC	AAAACTTTACCCATAAACAA	VC	A	Male
	213	FM	ACTTTACCCGTAAACAA	FM	G	Female
SEQ ID NO: 109 and 484	214	R1	CCCGGGAATTGTTTTCGATGC			
	215	F1	CCCTATAGTTTGTAGCAGAT			
	216	FM	GTGTAGTT TAACGCGTTCTTCATTT	FM	A	Male
	217	TT	ACGCGTCCTTCATTT	TT	G	Female
SEQ ID NO: 110 and 485	218	F1	GCAACCTAAAAATCTAAAA			
			AATACAGAGATAGGG			
	219	R1	ACTAATTTCAAAGAAAATGCC			
			ACAGCTTT			
SEQ ID NO: 111 and 486	220	VC	ATTTTAAAGGAAGATAGATTG	VC	A	Female
	221	FM	TTTAAAGGAAGACAGATTG	FM	G	Male
	222	F1	GCGAGCCGTTGAGAGTGAA			
	223	R1	CCACGAGTCCCCAGTCTCT			
SEQ ID NO: 112 and 487	224	FM	ACAGCCATGTCGCC	FM	A	Female
	225	TT	ACAGCCGTGTCGCC	TT	G	Male
	226	R1	CCACCAAAACGAGTAAACGA			
			CAAAT			
SEQ ID NO: 113 and 488	227	F1	GGTTGCTTGAGATCTTGCTGT			
			TATT			
	228	TT	AGTGAGCACGGCACG	TT	G	Female
	229	FM	AGTGAGGACGGCACG	FM	C	Male
SEQ ID NO: 114 and 489	230	R1	ACAACAAGGCGGCATTGC			
	231	F1	GGCGCCTCCGTTTAACTACTC			
	232	FM	TTGCAGTAGTAGTACTGTAG	FM	A	Female
	233	TT	TTGCAGTAGTAGTGCTGTAG	TT	G	Male
SEQ ID NO: 115 and 490	234	F1	GCTGCTCGTTCTCGTTGAAGA			
	235	R1	CGCTGGACGACACCAAGA			
	236	FM	TCCACGTTCTCTGTCCAG	FM	A	Female
	237	TT	CACGTTCCGGTCCAG	TT	C	Male
SEQ ID NO: 116 and 491	238	F1	GCTAGTCTCAAGTCCAAGCAA			
			AGA			
	239	R1	AGTAGGTGCTTGCCATTTACA			
			TCT			
SEQ ID NO: 117 and 492	240	TT	AAGATTAGCAATAACTATTGT	TT	A	Male
			T			
	241	FM	AAGATTAGCAATAACTGTTGT	FM	G	Female
			T			
SEQ ID NO: 118 and 493	242	R1	CGGTCTGGCTAGGATTGGA			
	243	F1	GAGCATGTGGAGGTTCCCTGAT			
	244	FM	CATTCAAGAATGTCCTC	FM	A	Male
	245	TT	CATTCAAGCATGTCCTC	TT	C	Female
SEQ ID NO: 119 and 494	246	R1	GAATCTGAGATGTTGAACCCA			
			CTGT			
	247	F1	TGCTGAAAGAGGCCTCAAAG			
			G			
SEQ ID NO: 119 and 494	248	FM	CACCAAATCAGCATCG	FM	A	Male
	249	TT	CACCAAATCGGCATCG	TT	G	Female
	250	R1	TTCTTGACGTGCCGAAAGGAT			
	251	F1	CCGCCGTATGATTGGATCGA			
SEQ ID NO: 119 and 494	252	FM	AAAAAGCCAACCGCACG	FM	A	Male
	253	TT	AAAAAGCCAGCCGCACG	TT	G	Female
	254	F1	GGTACATCAGCAACCGAGGTT			
			TATA			
SEQ ID NO: 119 and 494	255	R1	GTTGATGAACGCCTCCTCAGA			
	256	FM	CTCGTAGAGCAATCT	FM	G	Male
	257	VC	CCTCGTAGACCAATCT	VC	C	Female

TABLE 13-continued

Assay for	SEQ ID NO:	Primer [[/]] or Probe	Seq	Fluocolor	Allele	Sex
SEQ ID NO: 120 and 495	258	R1	GAAGCCGACTGGAGAGAATG TTT			
	259	F1	TTTTCATGGAGGTTATGAT GCTT			
	260	TT	CTCTGGTTTCTGATGGCTA	TT	A	Male
	261	FM	TCTGGTTTCTGGTGGCTA	FM	G	Female
SEQ ID NO: 121 and 496	262	F1	GCGCCTTATTATCATCATCTA GGTGTA			
	263	R1	GAAAAGCACGGCATCAGCAA			
	264	FM	ATTAGCACAAACGGAAGT	FM	A	Female
	265	TT	ATTAGCACAGACGGAAGT	TT	G	Male
SEQ ID NO: 122 and 497	266	R1	CGGCGCCGTCTTGTG			
	267	F1	CCGGAAGCCGCCTACAG			
	268	TT	AAC TTGGTGACGTAATAG	TT	A	Male
	269	FM	AAC TTGGTGACGTAGTAG	FM	G	Female
SEQ ID NO: 123 and 498	270	R1	GACCAAGGTGTCATAGGCATT GA			
	271	F1	GCAATGCTGAGGAATATAA GCACAT			
	272	TT	CCCCCAACTGGAT	TT	A	Male
	273	FM	CCCCACGACTGGAT	FM	G	Female
SEQ ID NO: 124 and 499	274	R1	CCATCACAGTGACCTTCCATT TGT			
	275	F1	TGACACCCCTCAATTGCTGGA AA			
	276	FM	AAGCTCACAGCAATT	FM	A	Male
	277	TT	AAGCTCACGGCAATT	TT	G	Female
SEQ ID NO: 125 and 500	278	R1	GTTGCAGAGTAACCAGCTGA AG			
	279	F1	CGCGAGCTCCGTAAGCAT			
	280	FM	TCCATACCAGCATCG	FM	A	Male
	281	TT	TCCATACCGCATCG	TT	G	Female
SEQ ID NO: 126 and 501	282	R1	CAGAGTTGACATATTGAGCAC TAGTGT			
	283	F1	CCAATGTCAAACATACCCAAA CAGT			
	284	FM	TGTAGAGGCAAAATGTAT	FM	A	Female
	285	VC	CTGTAGAGGCAAGTGTAT	VC	G	Male
SEQ ID NO: 127 and 502	286	F1	GGCACCAATATCTGTTTTCCA TTCAAT			
	287	R1	ACAGCAGATATAAAATGATG CTCGGT			
	288	FM	TGATATGTCACAGTAAGC	FM	A	Female
	289	TT	ATATGTCGAGTAAGC	TT	G	Male
SEQ ID NO: 128 and 503	290	F1	TCCCCGAGCCAGGTT			
	291	R1	CCTATAGCTTTGGTGGCAAGA AGAA			
	292	TT	CTGGTGCTTCTACCG	TT	A	Male
	293	FM	CTGGTGCTTTCGACCG	FM	C	Female
SEQ ID NO: 129 and 504	294	R1	GCCGCCCATTCTCAGT			
	295	F1	CGCGAATCGACTGCCATT			
	296	TT	ACGCCATCCCTTGC	TT	A	Female
	297	FM	CGCGTCCCCTTGC	FM	G	Male
SEQ ID NO: 130 and 505	298	F1	AGCCACCTGTAAGTACAATCA CAAA			
	299	R1	CCTGGGCAGCAACAACAG			
	300	FM	CCAACAACATGTTACC	FM	A	Female
	301	TT	CCAACAACAGGTTACC	TT	C	Male
SEQ ID NO: 131 and 506	302	F1	CATGTCAGACAAAGCCAATG CTATT			
	303	R1	TGGCTAGATTTTTTATGACAA AGCATCTCT			
	304	TT	CATTCATGTTACAATGTAG	TT	A	Female
	305	FM	TCATGTTGCAATGTAG	FM	G	Male

TABLE 13-continued

Assay for	SEQ ID NO:	Primer [[/]] or Probe	Seq	Fluocolor	Allele	Sex
SEQ ID NO: 132 and 507	306	R1	CAGAGGCGCGTCAAACG			
	307	F1	CCTGGTGTGTATGCTGTTCTTT			
			TC			
	308	TT	TCGGTCCATATTATC	TT	A	Male
	309	FM	CGGTCCATCTTATC	FM	C	Female
SEQ ID NO: 133 and 508	310	F1	TTGAATCCACCGGTGAATCTA			
			TCG			
	311	R1	GCGGTAGCGTGAGTGACA			
	312	FM	ATACAGACAGACTCGGAC	FM	A	Male
	313	TT	ATACAGACAGGCTCGGAC	TT	G	Female
SEQ ID NO: 134 and 509	314	R1	GTACTGTAACCCTTGGCTGTC			
			A			
	315	F1	GGCAGGCAGCATGAATCC			
	316	VC	ACAGAACCAATGGTTTT	VC	A	Female
	317	FM	CAGAACCAGTGGTTTT	FM	G	Male
SEQ ID NO: 135 and 510	318	R1	CTGCTGCTGTGTACTGTATC			
			CT			
	319	F1	TGACCGCTGAAGCACGAA			
	320	TT	ATGGCCATACAGCACGA	TT	A	Female
	321	FM	AATGGCCATACTGCACGA	FM	T	Male
SEQ ID NO: 136 and 511	322	R1	CATCACACAGCTTGTACATGC			
			AA			
	323	F	GTGGAGAAAAGGAGAGGAGA			
			GAATT			
	324	TT	CAATTGCGAGACGATGG	TT	A	Male
	325	FM	ATTCGCGAGCGATGG	FM	G	Female
SEQ ID NO: 137 and 512	326	R1	CGGATTCAAGTGCCTCTAAGC			
			TATA			
	327	F1	GAGACAGTAAGACGAAGAAT			
			GGCA			
	328	TT	CACGTGAGCATGCT	TT	A	Male
	329	FM	CACGTGCGCATGCT	FM	G	Female
SEQ ID NO: 138 and 513	330	F1	GCAACTTACTTATAACATGGC			
			AACCAA			
	331	R1	GCAATTCACTGCATTGGGTCT			
			AATG			
	332	FM	ATAAGTGTGCACAATGT	FM	A	Male
	333	TT	ATAAGTGTGCGCAATGT	TT	G	Female
SEQ ID NO: 139 and 514	334	F1	TCTGCATAGCATTGAACTGGA			
			CAG			
	335	R1	AGGAACAGTACAGATCAGCA			
			GTAGA			
	336	FM	ACAGAGTAACACAGCAAG	FM	A	Female
	337	TT	CAGAGTAACGAGCAAG	TT	G	Male
SEQ ID NO: 140 and 515	338	R1	GTAATGGTAGCACGCAGATC			
			ACT			
	339	F1	GCTAGGAATTGCTTAGTCGAG			
			GTT			
	340	TT	ACTGAAGAAATAATCGC	TT	A	Male
	341	FM	CTGAAGAAGTAATCGC	FM	G	Female
SEQ ID NO: 141 and 516	342	R1	GCGAAGTCCGCGTAGTACTC			
	343	F1	CGTCACCAGAGGCTACGT			
	344	FM	ACGCCCTTGACCACCT	FM	A	Female
	345	VC	ACGCCCTTGCCACCT	VC	G	Male
SEQ ID NO: 142 and 517	346	F1	AGAACTTTGTGAGGAAGAGC			
			TACAC			
	347	R1	GCCTGATGCTGTCCCAACAT			
	348	FM	CATGAAGTATTTAAGGCC	FM	A	Female
	349	TT	CATGAAGTACTTAAGGCC	TT	G	Male
SEQ ID NO: 143 and 518	350	F1	CTGTAAAAAGGCCATTGTT			
			AGCT			
	351	R1	TGGGTTTCAGACCAGAGATTT			
			GG			

TABLE 13-continued

Assay for	SEQ ID NO:	Primer [[/]] or Probe	Seq	Fluocolor	Allele	Sex
	352	FM	CCCTCCGTAAATGTTAA	FM	A	Female
	353	TT	CCCTCCGTAGATGTTAA	TT	G	Male
SEQ ID NO: 144 and 519	354	F1	CAAGAACACCTGAACACATTT GCT			
	355	R1	ACCTCACCAGGTTGCTACTCA			
	356	FM	CCATCATGGTATCAGATA	FM	A	Female
	357	TT	CCATCATGGTAGCAGATA	TT	C	Male
SEQ ID NO: 145 and 520	358	F1	CCGCGCACTGGTACAC			
	359	R1	GCGCGTGACGAAGTTCTC			
	360	VC	CCTACGACGTATTTCG	VC	A	Male
	361	FM	TACGGCGTATTTCG	FM	G	Female
SEQ ID NO: 146 and 521	362	F1	TGCTGCGGCTGGTAACTG			
	363	R1	CCCTGGAAATTTTCATCACCCC TTTT			
	364	TT	TTGAAGCCAAAGTAGCCAG	TT	A	Male
	365	FM	AAGCCAAGGTAGCCAG	FM	G	Female
SEQ ID NO: 147 and 522	366	F1	GCCTCGGCGCTTAAACC			
	367	R1	CCTGTGAGTCTCAAACACTAT TGGT			
	368	TT	TAGCATTCGTAAGTTCATT	TT	A	Female
	369	FM	CATTTCGTAGGTTTCATT	FM	G	Male
SEQ ID NO: 148 and 523	370	F1	TTGTTAATTATTGCACTGTTCT CGTTGTC			
	371	R1	TCAGTCCTGAAACCACTCTTA CTCA			
	372	FM	TTACCACGCTATTCAAA	FM	A	Female
	373	TT	TTTACCACGCTATCCAAA	TT	G	Male
SEQ ID NO: 149 and 524	374	F1	GCAACAACGAAGGGAGTGAT ATTGA			
	375	R1	GAAGGTCCAATGTCTTGTA CGTA			
	376	FM	CCATGGGTCTTTTCGT	FM	A	Female
	377	TT	CCATGGGCCTTTTCGT	TT	G	Male
SEQ ID NO: 150 and 525	378	F1	GCTGCTGCACGATTATACTGG AAT			
	379	R1	CTGAGCCACTACAGAACTGC TA			
	380	FM	CTAGAGTGCTAAGTCCCT	FM	A	Female
	381	TT	CTAGAGTGCTAGTCCCT	TT	G	Male
SEQ ID NO: 151 and 526	382	R1	GAAAGGCTGGGAAGACCACT A			
	383	F1	GGCCGACAGTCACTCTCAA			
	384	TT	AAAATCGCCTTGTC AAC	TT	A	Female
	385	FM	AAAATCGCCTCGTCAA	FM	G	Male
SEQ ID NO: 152 and 527	386	R1	CTCCAGTTGGTAATATGCT			
	387	F1	GAGTGGGAGTTTGACTATC			
	388	TT	CTTACCACTTTCAGGTA	TT	A	Female
	389	FM	TTACCACCTTCAGGTA	FM	G	Male
SEQ ID NO: 153 and 528	390	F1	CCTGGAAGACAAATAGGAA CCCA			
	391	R1	GGAAACGCCGGCCAAA			
	392	TT	AACGAATGAGCTGTCTGA	TT	A	Male
	393	FM	AAACGAATGAGCAGTCTG	FM	T	Female
SEQ ID NO: 154 and 529	394	F1	GTGCTCCATATCTTCTAGCAT TCAC			
	395	R1	GGCAGCCTTGTGGTGTAAAC			
	396	TT	AACCTGACGAACGTTGT	TT	A	Male
	397	FM	CTGACGAACGCTGTT	FM	G	Female
SEQ ID NO: 155 and 530	398	F1	TGCACATATCCACTCGTGAAT GC			
	399	R1	ATCTCCTTTTGTGTGCTAGGG AAA			

TABLE 13-continued

Assay for	SEQ ID NO:	Primer [[/]] or Probe	Seq	Fluocolor	Allele	Sex
	400	TT	TATTTCTGTCCTTTAGCAG	TT	A	Female
	401	FM	TTCTGTCTTCAGCAG	FM	G	Male
SEQ ID NO: 156 and 531	402	F1	GGGCCAAGATGACATGAACA			
	403	R1	CAGAGCTTGAGCAGGAAGGT			
			T			
	404	TT	ACGCCGGATAGCAC	TT	A	Female
	405	FM	CCGGACAGCACGT	FM	G	Male
SEQ ID NO: 157 and 532	406	R1	CTGACGAAGAGGAGGAAGAT			
			GAAG			
	407	F1	ACCAGCAGTGAGTCTACTTTG			
			C			
	408	TT	ATCGTCGTCGGTGC	TT	A	Male
	409	FM	ATCGTCGCCGGTG	FM	G	Female
SEQ ID NO: 158 and 533	410	R1	GAGCGTGCATTCCAACATCTG			
	411	F1	GTACTCGTCGCGCAGTGAT			
	412	FM	AGACGACGCTGATATAGA	FM	A	Female
	413	TT	ACGACGCTGATGTAGA	TT	G	Male
SEQ ID NO: 159 and 534	414	R1	ACCAAGGTTCTGTCTCTACCTA			
			CC			
	415	F1	CCTCGCCGCTGCAAC			
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	463	R1	TCAGCTCAGGCTTGACAGA			
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	465	FM	TCTAAAGACGCCAGC	FM	G	Female
SEQ ID NO: 172 and 547	466	R1	GCATATTAGTGCATACGGAGA AGGA			
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	469	FM	CCAGTCAGTCTTCTT	FM	G	Male
SEQ ID NO: 173 and 548	470	F1	CACTGGTGAAGTGCTGATAAA TACATC			
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<211> LENGTH: 99
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;
```

```
<400> SEQUENCE: 104
```

```
gcagtcgagc gacgactagg aagaggaggt gatcgatgag aacggcgaaa ttgtcaagag 60
gaagaagaag ggccttaagg agaaggtaaa ggagaagct 99
```

```
<210> SEQ ID NO 105
<211> LENGTH: 231
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;
```

```
<400> SEQUENCE: 105
```

```
cgcttcacac ccgggagcac attggacgcc gcgtcacaaag aaggtatata cagcatggtc 60
aaattgtttg cctctgttgt cgtgcactcg tgcattgcatg cggacagtgc ccatgcttca 120
gtcatgttga gttgagttct gcttgccggc ctgtgatgtt atttgttctt gttcaatcat 180
atcgcaactg gcagtgtgtt ccgacgagcg cgacgccgcg gccagcggcg g 231
```

```
<210> SEQ ID NO 106
<211> LENGTH: 101
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;
```

```
<400> SEQUENCE: 106
```

```
gcgagcacat acgtogtggc cacattcgaa ggccgatatc atggcgttga tgttacccgt 60
gcactccacg ctgcggtcga cgccgccggtt ggtcagctca a 101
```

```
<210> SEQ ID NO 107
<211> LENGTH: 201
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;
```

```
<400> SEQUENCE: 107
```

```
ggttgtcacc tttssaratc agtagactgc catcagtgcc atgacttatt ggcaarttgg 60
agtcttttag ggtccacttg ttttccttga attctgcaag ctgaatttgc tttttgttg 120
acttcaatam tccaaccart tctcgaaaga cbrccatgcy attcttcgcc ttgtgtcttc 180
tgagatagag tggagtcatt a 201
```

```
<210> SEQ ID NO 108
<211> LENGTH: 889
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;
```

```
<400> SEQUENCE: 108
```

```
gaatattcat ggggcgggta caagatggca ggtgaggaca ggcaggaaga tgaacaagac 60
gacaagagcc gacgtagtgc tccaaggatt cgatcaggat attcctccaa gatgacaaag 120
accacaaaaa tagttgcttc ttccagcgtg aaccaatttt aaaactaatt gcataagagg 180
tgattaatta gcaagacaaa actattagct ctagttagct cttgaaggca tataataata 240
tagcatgata tgaactaatt atgggctaca tattcagaat taatagggtt gtctcgtcaa 300
ttaatcactg tttacttgca cagtagagca agagcaagcg cagagaacag gggcataata 360
```

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```

aactgctgcg ggtggtgtac ctaaacctatg atccaatatt acatcccctt atgctgactc 420
cagcagttca tcaaaacttt acccataaac aatgttttac actatagatt gcactattcg 480
taaaatagag tttgaatatg agtatgaata tgtataagct gttggagata gtcttaatac 540
ctagtttaat tactctagta taaacctcaa tctacatgta ttaaggtaga ttggagtgtg 600
acttaaaacta atttatatcc caatccacct caacatacat aactatagtc tategtacaa 660
tatctaaatg aagcctaaag acaacgttta ggataaggaa cgtgttaaag agcccaaaag 720
cattatgact ccattgagtc aacggcccaa tagcagcaga catcgcaaca tgcccgtccc 780
ccaacaactg tagcacctga ttgtattggt acaaaaatta tgatgacgtc ccatttgtag 840
acatcgagca tattgcccaa tacccttggt tgaaattcaa aatttgaat 889

```

```

<210> SEQ ID NO 109
<211> LENGTH: 824
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (14)..(14)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (36)..(37)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (171)..(171)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (349)..(370)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (374)..(374)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (392)..(394)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (415)..(415)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (435)..(435)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (517)..(517)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (568)..(568)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (608)..(608)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (629)..(629)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (651)..(651)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (666)..(666)

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<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (672)..(672)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (681)..(681)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (696)..(696)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (711)..(711)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (749)..(749)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (752)..(752)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (761)..(762)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (797)..(797)
<223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 109

catatgaaaa atcncctgggt ccgctttcgt tttttnnnga tacctaataa ccaacctctg      60
catctgcttc aggttcgtac ttttgaggcg caatgtgggt cgttggcgca gtatgggatg      120
aagcacatgc ggtccttcgc aaacatctgc aacgctggca tccttcctga ngcagtgctg      180
aaggctcgtg ctcaggcttg caccagcatt ccttccaacc cctggagtgc tatccacaag      240
ggttttagcg cctaagaatc ataaggtgag gcgaaatatt tcagccgctc caccgcaacg      300
aactggttta cattaccagt cctcaggggg gtcctagtgc ttgaaccann nnnnnnnnnn      360
nnnnnnnnnn ctantatagc tgttccaccg tnnntccaga ttacatagct atgencatt      420
tcggtgttac atatnatagt cggaaagtta tttggcaatt gtattggctg ttgctgtata      480
tattccctat agtttgtag cagatgtgta gtttgtnatt ccataaaaaat gaagaacgcg      540
ttactgctat ttctatgtag ccgactgntg ctcatgtgaa actttacccc attcttggtg      600
ggaaatgnac tatcgtgggt ggaattctng catcgaaaac aattcccggg ngatccttta      660
ttcaangtga ancgctctgc natttccatt tgaggntcgt attattttca nttgtaggct      720
tgtggctggt ggcacatggt ccactatgnt tncaaacaag nnctatgggc agtttcattt      780
gtttcgttct ctatatnct gcagcaccta cccatggaaa aaat                        824

<210> SEQ ID NO 110
<211> LENGTH: 643
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (3)..(4)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (6)..(6)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature

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<222> LOCATION: (8)..(8)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (10)..(10)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (12)..(12)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (23)..(23)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (71)..(71)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (92)..(92)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (105)..(105)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (145)..(145)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (151)..(151)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (178)..(178)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (189)..(189)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (198)..(198)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (208)..(208)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (217)..(217)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (337)..(337)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (340)..(340)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (349)..(352)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (358)..(358)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (364)..(364)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:

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<221> NAME/KEY: misc_feature
<222> LOCATION: (407)..(407)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (416)..(416)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (438)..(438)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (448)..(448)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (461)..(461)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (464)..(465)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (488)..(488)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (536)..(536)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (545)..(545)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (589)..(589)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (618)..(618)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (630)..(630)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (641)..(641)
<223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 110

tcnngnangn angtgtttat ttnacctnca gattccacct ttgtttcttc tatatttaca      60
tgggtaccttg naggttcttg ctctttgtga tnatccgtgc ctctnggaaa aagaggaaac      120
caaatcattg ttcaggtgaa tatctcctc nttcaattta cagaactcct aaattcanag      180
atctagtgnng tgatattctg ctatttttnc caacttnagt gtcaaagcaa cctaaaaaatt      240
ctaaaaaata cagagatagg gcaatctgtc ttccttttaa atcaaagctg tggcattttc      300
tttgaaatta gtaaacattt atataaatag taaaatntcn tggagatcnn nnaggtantt      360
aacntttttc ccttcaactt ccacagtaat taaacatacc taggaanata gttttngagt      420
tctcatgttt aattgatntg ttcatcnaa gaaccattac ntcnntgcct aattatgcat      480
gccctttnat ttttctctaa atttcccttg ataccatttc aagttgcaa gatgantttt      540
tttttcttcg tactgtttta tatttttggt agccataaac tttcaaaant agttcagtg      600
cccatttata cataaatnct tatgtgtacn tgatgggtcg nct                          643

<210> SEQ ID NO 111
<211> LENGTH: 201
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<212> TYPE: DNA
<213> ORGANISM: Zea mays ;
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (189)..(191)
<223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 111

atztatgtgc ctyyaggcga accacttyca tcygaacctg gcgagccgtt gagagtgaac      60
gtgctmttca agcatattca ggcaatgctg tcyggcgaca cggctgtcat cgcagagact      120
ggggactcgt ggtttaactg ccagaagctg aagctaccgg aaggmtgtgg gtaagctcct      180
ctttcgaann ntgrttttgc t                                              201

<210> SEQ ID NO 112
<211> LENGTH: 300
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;

<400> SEQUENCE: 112

gcacagaact ctcacctgtc ctttctggg gttttggta cgtggtggtta gtaagcttgg      60
atttgccat ggataaagtt gttctaagct ccgtgggttg cttgagatct tgctgttatt      120
gcgtgccgtc ctcacttttt ttgcaatccg aggaatgaat ttgtcgttta ctcgttttgg      180
tggattatta gcgcgaaaaa aaaactcttt tttttttgtt cttttactac gaaaagcatc      240
ttcttgatt ttgctatctt cttttactac gaaaaactct tgagtctagg aatttgaatt      300

<210> SEQ ID NO 113
<211> LENGTH: 193
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;

<400> SEQUENCE: 113

ggctattgta caacaacaag gcggcattgc catcrcagtt gcagtactas caattgccwt      60
cgkgattgcg gttgcggtgt tgcagtagta gtgctgtagt actatcaagt gagtctgatg      120
catgamcaga gcagayggag tagttaaacg gaggcgcggg tgagcttgcc gttgacgatg      180
tggtggttgt cgt                                              193

<210> SEQ ID NO 114
<211> LENGTH: 201
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;

<400> SEQUENCE: 114

rgacacggag gagcagtacg acttcgaccc gctggacgac accaagacgt ggccggagga      60
cctgctscgg ctcmgscceg tggggaggct ggtgctggac cggaacgtgg acaacttctt      120
caacgagaac gagcagctgg cgttcggscc ggggctggtg gtgccaggga tctactactc      180
ggacgacaag atgctgcagt g                                              201

<210> SEQ ID NO 115
<211> LENGTH: 199
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;

<400> SEQUENCE: 115

atgaactatg aagaacceta tcactttgag ctagtctcaa gtccaagcaa agakaacaat      60
tcaccagtag taatggaaag agaagattag caataactat tgtttagtgg agcaataaat      120

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atctttttca gtttcaagtm ttaagagaga aaatgataga tgtaaatggc aagcacctac 180
tgtacttgac aacgttaaa 199
```

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<210> SEQ ID NO 116
<211> LENGTH: 795
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (102)..(102)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (104)..(105)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (119)..(120)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (122)..(122)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (125)..(125)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (128)..(128)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (130)..(137)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (142)..(142)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (169)..(169)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (193)..(193)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (198)..(198)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (203)..(203)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (207)..(207)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (344)..(344)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (561)..(561)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (605)..(605)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (670)..(670)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
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<222> LOCATION: (677)..(677)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (691)..(693)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (702)..(702)
<223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 116

atagtaaaag gttggaatgt tagttgaaaa caaggtgtaa gaaatgtatc atcttttggg    60
ctggacaaca gacctgaagt agaagcttct gggtttcctc ananntggcc ctcagcttnn    120
cncnnaangn nnnnnnttcc tntgaaacga ggagtttgtc atacagtngg gcaatgccag    180
gattaccctt ggngaacncc atntccncca gatcaatagt caccctgaaa aatggccact    240
cattgtacat ttcttgagc atgtggaggt tcctgatgtc cttctggagg acattcttga    300
atgcggctcc aaatcctagc cagaccggga ggtggaaccg cgtntgtgtc caagcaaaga    360
tccatgggat tgcteggagt gagtcgatac ctccgctcgg cttctctttg gatggcctgc    420
ttctatgtt catcctacca tactctgttt caggggttgc ctgtcaacac aaacatcaca    480
aagtaaatat tcgctaaatg taggggtcaa atcgaatgct tcagtaacca ttttacttag    540
gcttgcaaaa gatagtttct ncagtaaatc attgtgctag attaatatta agtatcaggt    600
caganctatt tcaaaagttt gtttttgcac cttatgttac ttagtatctc ctagacacca    660
ctcaggtggn agcctcnagt actggtctac nnnccatctt tnatgttact gtagtatggc    720
aatcgtatgg atcaaggact tacaaggcgg aaataactcg aaaagcgtgg ctctttgaag    780
acaatggacc gatat                                         795

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<210> SEQ ID NO 117
<211> LENGTH: 627
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (73)..(73)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (170)..(170)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (182)..(182)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (282)..(282)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (348)..(348)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (367)..(367)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (396)..(396)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (486)..(486)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:

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<221> NAME/KEY: misc_feature
<222> LOCATION: (492)..(492)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (543)..(543)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (570)..(570)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (591)..(591)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (599)..(599)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (613)..(624)
<223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 117

atgctctggg gtcattctcat gtcgaattc tgatccaatt gaagggtgt agctttagtg      60
caatgttgct atntttatca tgcaaacccc tttacctttt gtaacatgtc accaaacaat      120
tcatttccat tgggttggtc aaggatcatg tattcccatg caatacaacn tggagaagtc      180
angctattcc tttactatct tacgtagcag tggtggccca cgattatatt gttttaatat      240
ctgtacttca gttgaggaaa ttatatcttc tgagcgatca gntaatcact tatttttggtt      300
ccactacttt cctatcaggt tcttgagtct cggaatatcc agcttttntt gggatatctt      360
gtgagcngca tcaaagaggc tcctacatct gatgantcaa gtagcacagt tacatactct      420
gaagttgacg gtgatcatag gaaactaatt ttggaccttc aacctgctga aagaggcttc      480
aaagncaga cncctgatgc tgatttggtg ctgtggacag tgggttcaac atctcagatt      540
ccnccgttac agcctcctga tgctccctan gttattcctc tgaatggctg nggacaggng      600
gagacagagg aannnnnnnn nnnnaaa                                           627

<210> SEQ ID NO 118
<211> LENGTH: 772
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (52)..(57)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (59)..(59)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (61)..(62)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (68)..(68)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (70)..(70)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (73)..(76)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature

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<222> LOCATION: (88)..(97)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (99)..(109)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (112)..(113)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (115)..(119)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (122)..(131)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (134)..(145)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (147)..(150)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (153)..(153)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (160)..(160)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (163)..(166)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (173)..(173)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (180)..(184)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (233)..(233)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (376)..(377)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (425)..(427)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (446)..(447)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (508)..(508)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (526)..(526)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (531)..(531)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (564)..(564)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:

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<221> NAME/KEY: misc_feature
<222> LOCATION: (581)..(581)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (621)..(621)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (624)..(625)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (651)..(651)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (656)..(656)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (678)..(678)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (696)..(696)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (707)..(707)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (709)..(709)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (715)..(720)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (723)..(723)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (725)..(725)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (729)..(729)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (733)..(734)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (736)..(736)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (751)..(752)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (754)..(754)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (759)..(759)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (761)..(762)
<223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 118

gcatagcata ctgcatacat gccgctgttg cattgcaggt tgcagctgca gnnnnnnncng

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nngacccnngn tgnnnncgtg cagactgnnn nnnnnnnann nnnnnnnnc anntnnnnnt 120
annnnnnnnnn ngtnnnnnnn nnnnnnnnnn tanataaatn tannnnagta canttttgc 180
nnnnccctgct gcctgcccta gcactagcag ccaggcagca gggctggcgc ctntgctgcg 240
gtgcatgaaa aggcgacgag caaagaagg ggcgctggaa aactgcccg cgcgcgctgc 300
taccgccgta tgattggatc gagtggagcg tgcggttggc tttttggatc ctttcggcac 360
gtcaagaaat aaaaanntaa ccagttcctt tatttcacgc gtctccgact ctccaaggcg 420
gcagnnncag caagctctcg tagctnngtg taaagctgtg gagcacacag cagctgtgag 480
cctgtgttgg tgttgcgagc aagagcanag ctggcaagcg cagcanagga nttgcacacg 540
gctttatacg aatgagcaga acanaggcat ctgtacatga ngacgggcac ggcacacaca 600
caggcctttg gttcgttga nganngtga ttggcctcct actatactga nactgnacac 660
ccctggacat ccgcatgncc gcactacta cgcattgata gcagcgtntt gtaannnnnn 720
tgnantggnc tcnnangcag ccattgccatg nnanccagna nccccacgc ac 772

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<210> SEQ ID NO 119
<211> LENGTH: 201
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;

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<400> SEQUENCE: 119

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cacacataat tactccmatc ttctcagtgg aggtacatca gcaaccgagg ttataaaagg 60
aacrcitygag gacaacacga tgggtggcggg gcatagattg gtctacgagg gctctgagga 120
ggcggttcac aacggaggga tggtygtgtc caagatcgcc cacargaaca tcgtcagagt 180
tctgggctgc tgtctggaac c 201

```

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<210> SEQ ID NO 120
<211> LENGTH: 220
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (25)..(25)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (40)..(40)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (84)..(84)
<223> OTHER INFORMATION: n is a, c, g, or t

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<400> SEQUENCE: 120

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tggatccatc acggtagagc tgacnatgat atgtccaaan gcttctgctg ctagttgttg 60
gtcgtccgca agagacttga cccnacaggt ttttttcgtt gggcctcaag aactgtaggt 120
gcattttcca tggagggtta tgatgcttaa cccactctg gtttctgatg gctaaaacat 180
tctctecagt cggcttctac catgagccct ggcacttagt 220

```

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<210> SEQ ID NO 121
<211> LENGTH: 301
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (59)..(59)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:

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<221> NAME/KEY: misc_feature
<222> LOCATION: (76)..(76)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (91)..(91)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (111)..(111)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (181)..(186)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (203)..(203)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (282)..(282)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (292)..(292)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (294)..(296)
<223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 121

tggttttacc cgctccgcatg ccccgccatg tgctccttgt ctccgggactc ggagacagnc      60
tgacagtcctt ggggtnttgt gctaattggcc ntgggttaact tacgttaatc ntgcgcctta      120
ttatcatcat ctagggtgtaa ttacttccgt ctgtgctaata caccctcgtga ttgcctgcaa      180
nnnnnnnggaa ccccgacccc ccnaccttgc tgatgccgtg cttttctaga gaagtaccct      240
gccgcgtcac gcttcgtcct cgctctgttt ggattcaacg gntttttgga gncnnnggcc      300
a                                                                              301

<210> SEQ ID NO 122
<211> LENGTH: 170
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (105)..(105)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (131)..(131)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (138)..(138)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (140)..(140)
<223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 122

ggtgccggaa gccgcctaca gcggcccgcc gctcagctat tacgtcacca agttccagcc      60
ggcggtggct gcgcggcgcc agaccctcga ggcccccgcc cccgncgagg cacaagacgg      120
cgccgcgcc nccgtcgngn ctccggaggt accggcaccc cagctgtcgt      170

<210> SEQ ID NO 123
<211> LENGTH: 190

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<212> TYPE: DNA
<213> ORGANISM: Zea mays ;

<400> SEQUENCE: 123

rcataaaaya ttgcaaatgc tgaggaatat aagcacatag atacttagca tgtgtccgcc      60
agatggaaat acctttacca ttccccaca actggattgc cagaaaggtc aatgcctatg      120
acaccttggt ccatyatttc cayggctaga ttaacctggc caggggaaay aaatsatttt      180
gtgcagattc                                     190

<210> SEQ ID NO 124
<211> LENGTH: 201
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;

<400> SEQUENCE: 124

acgsygtaa ggccttactg catgttctg aggttgctct tgacaccctc aatttctggt      60
aatcatccc catctrugga cataaatgga gcaaaattgc tgtgagcttg aacaaatgga      120
aggctactgt gatggttctg aatgccatca ggcagattct gtgatgtgtt tgettctgat      180
gaagaaggga agttccagcg g                                     201

<210> SEQ ID NO 125
<211> LENGTH: 201
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;

<400> SEQUENCE: 125

tctcaagctt atttaggttg actaaatttg gaacatcccc cgagatgcta ttgccggcaa      60
gcgcgagctc cgtaagcatg gcagctttgc caatcgatgc tggatggac cccttcagct      120
ggttactctg caaccgaaga acartgaggt tcgagaatga gtcaagccag tctggcacac      180
ttcattgaa gtagttacca t                                     201

<210> SEQ ID NO 126
<211> LENGTH: 294
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;

<400> SEQUENCE: 126

tactagcaca ctttgctcc tgatagcttg taccattgtg acctgtttg gatattggga      60
ttccacccaa tgtcaaacat acccaaacag tgaccgkka tctttttctt cgtatttcta      120
tgtcaagcac tyaagcataa ctcatacct tgcccttaca ggtgtttctg aaacactagt      180
gctcaatatg tcaactctga catggtcagt tgtagcact gtggaaggac gtgttcctct      240
tgccagttag gtatwtctct tgagcgggtg agcagcattc cagttgttaa gatg      294

<210> SEQ ID NO 127
<211> LENGTH: 214
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;

<400> SEQUENCE: 127

gcacaggtaa ctcttgcatt gcacctgtcc tgaacaagt ttcgtgctaa caagggaat      60
tattcttttt cgtggcacca atatctgttt tccattcaat gatgtactga tgtgttcacg      120
gagtcggtac ctagaaatct cgcttactgc gacatatcac accgagcatt attttatatc      180
tgctgtgygc cctccttgct aatgtaggca cctg                                     214

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<210> SEQ ID NO 128
<211> LENGTH: 300
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;

<400> SEQUENCE: 128

rytkcaacmt gcgagaarta raaatcact aaccttggtt ttatcaagta tagcaaggtg      60
aamgcgaatg ccargaagaa gcagagtcct cgcacgrtga ggtacgcaag gccagaaaa      120
tcgttctttc ccccgagcca ggttgcggtg gaaagcacca gcttcttctt gccaccaaag      180
ctataggtgt tgtagttggt gtccarctgg actgtgattg trtcgttttc cttgagatca      240
acatatatcc ttccatacag ctttctgaat gtcggaagtg ctgyagtccg catccaaaca      300

<210> SEQ ID NO 129
<211> LENGTH: 287
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;

<400> SEQUENCE: 129

gcgctcgcca tgaatgcgcg tcgtctccrt tggaccatca accatttgca attgatgagg      60
agacgagaca ggagaaaacg cgagggcagy agtgttctgt agctgaaagc tgccgcgaat      120
cgactgccat tattggacgc cgtcccttgc ccgactgaga atcgggcggc cgcgtgcact      180
gcactrcmct accgccgccc gccaggcgcc agccrrctc gtggtacaaa attcatccag      240
aattcgcaag aggccaccag ctcggtccg gcctgcgagg atttcaa      287

<210> SEQ ID NO 130
<211> LENGTH: 288
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;

<400> SEQUENCE: 130

tcatggaatt gcttcaaacc cgagagcggc aagataatag acgaacgac agctccagcc      60
acctgtaagt acaatcacia ayrgtaagag caatggatca ctygtggagg cttgtgttta      120
caaataatrg ccaacaacag gttacctcag aaatcctcaa ataattggccc ctgttgttgc      180
tgcccaggtc gaagaagaac cttttctgat ctgccctrat maccttagac ataccatgc      240
cactaatttc ttcttgagga ggcagatcaa ccaagcggtc sacatcta      288

<210> SEQ ID NO 131
<211> LENGTH: 284
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;

<400> SEQUENCE: 131

tayaagcaat attaagagat ggaaatgtcg cctctgtacc acaagttagt aagttactaa      60
cttgattggt aaaacacccat cgaaatgtgt gacatgtagg acctgggtccc acatgtcaga      120
caaagccaat gctattttgg cattcatggt gcaatgtagt gctataatct ccacaaaaaa      180
wacaagacag gtcaaatact gygtaaacat tcgatccaaa aaacagagat gctttgtcat      240
aaaaaatcta gccaaaagaa ttggagctat tgagatacaa tgra      284

<210> SEQ ID NO 132
<211> LENGTH: 256
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;

<400> SEQUENCE: 132

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acgttccttg tcgaaccatta tcattcgcta tctatTTTTg aacgaacgca ttcgttatct 60
atggccacat ctccacttcg tctttgccga agcttcagta cgcgggtcgg gtgtctctg 120
gtgtgtatgc tgttcttttc tcgggccata ttatcccccg ttgacgcgc ctctgtcaat 180
ctgctgtact gtgaatttta ttgacgtgc ataaatyatt ctggaaacgc tcttggtacg 240
tcacgtacct ygagaa 256

```

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<210> SEQ ID NO 133
<211> LENGTH: 267
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;

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<400> SEQUENCE: 133

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atcsggcacc tgatccwga tgtgagwcs agtgcgacct ataccgctcc ccaaccgccc 60
ccattgaatc caccggtgaa tctatcgctc cgccaccacg aggcctata ttaactccac 120
agcatecatg tgtccgagtc tgtctgtata cctgtcactc acgctaccgc cgtgcgcatc 180
gttctgtctc tccctccctt cgcggggcgc gcgcctatkw twtttactac tstatterta 240
tcattatayt ktttggtttc cwtccck 267

```

```

<210> SEQ ID NO 134
<211> LENGTH: 297
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;

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<400> SEQUENCE: 134

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```

ttctcgtaat ctctatggaa attrcrrta ctagagataa atggcatcct tgcaatgtcc 60
caaattctca ccgaagaaca caagkaaaag araagaaaat attgagcaag cacagcacag 120
gcaggcagca tgaatccaca cagaaccagt ggtttttagc acagyasyt gacagccaag 180
ggttacagta cataatcaga cggggcaaca cgaacgaagc agccgcatca rcagcgcct 240
asaaccatt aattgacaca cagcgtatat atatatatit agtaactgtc ttttagc 297

```

```

<210> SEQ ID NO 135
<211> LENGTH: 231
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;

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```

<400> SEQUENCE: 135

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```

aagtccagat gactttcaga aagaaaatcg agaacttcag acaccttctg accgctgaag 60
cacgaacgac aatggccata ctgcacgatg ttgcacgttt ygtagaagtt tgtatcttcg 120
ttaaaaaata tagcagtaca aatsatagcr cattctgtag gtgtaggata cagtaacaca 180
gcagcagaac atcaaaccrr tctcctgccca ggctctgagc aacgccaaag c 231

```

```

<210> SEQ ID NO 136
<211> LENGTH: 301
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;

```

```

<400> SEQUENCE: 136

```

```

ccrrcaaca ttagstcggt ytgtttctat ttsttgacca agacatgggt gytgttctga 60
tggcagattt gaacctgaca tgcttgcytt ttctgggtgc tattatgggt gtggagaaaa 120
ggagaggaga gaattagctg caattcgagc acgatggaga accttgcatg tacaagctgt 180
gtgatgtgta atcattactc gcttactgag atagaattay cttttgctaa aagttttcat 240

```


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```
attctagata cgtgaccccg agaaaggaag acggcaaggc aggtgtccac taactcctga 300
g 301
```

```
<210> SEQ ID NO 137
<211> LENGTH: 282
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (1)..(1)
<223> OTHER INFORMATION: n is a, c, g, or t
```

```
<400> SEQUENCE: 137
```

```
ncgcgtgttc gtgcggatgc waykcatgca gaggyagcar arctagctag yaagcacgwa 60
ckyacgtagc acatgataag aaggctgcrt tgagacagta agacgaagaa tggcargcag 120
aagagcacgt cagcatgttc cccgcggcct atagcttaga ggcacttgaa tccggtgggc 180
ackctcttgc cgcagtgggt gaggatragg ctgaggtcga csggyaggtt gagcttgatt 240
cckaggacct ctcccttgat ggcggtgcag aggcacacgg cg 282
```

```
<210> SEQ ID NO 138
<211> LENGTH: 301
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;
```

```
<400> SEQUENCE: 138
```

```
atggatgctt ytgtagcatt aaacgtgaga gtttaatctc accacaacaa gcatgggtga 60
tcacaagaaa tacaaaacag ccaactaacc aaamawkata ttagcaaatg caacttactt 120
ataacatggc aacaaaacac aaacacattg tgcacactta tgctggctta gtctatcctg 180
cattagaccc aatgcagtga attgctagtg caagcaacca attgctcccg tagtgtttcc 240
ttttgcttct atgaaggcta tgaactgtca gggtttattc tagtgatcta tttatctayc 300
a 301
```

```
<210> SEQ ID NO 139
<211> LENGTH: 216
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;
```

```
<400> SEQUENCE: 139
```

```
tggaaacrrt taataattga agttttctya gtacagrtct gcatagcatt gaactggaca 60
gcttgctgcg ttactctgtr taasgaacga tctactgctg atctgtactg ttccttgatt 120
tttyyyycwc ttttcttctt ttgatggcaa gcaggactga agataagatg gctgccttrc 180
cattggccac cgcagaagca tgtgatgcta atgctg 216
```

```
<210> SEQ ID NO 140
<211> LENGTH: 301
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;
```

```
<400> SEQUENCE: 140
```

```
ggaaatcttg acaaggatat ggagagaagg ttagctacg ccctaagcag agaagacatc 60
gagaacgcca tactcggagg accttaasct gaacacatgg ccaaggggct cakagctagg 120
aattgcttag tcgaggttgg cactgaagaa ataatoctc cagtgatctg cgtgctacca 180
ttacgccatt gtaagagcaa atggatcatg tgctgagaaa taatattcat gagcaataat 240
ctcgaaactag gctggaaatg ttggawgctt cctggtgaca tgcgytgggt ttgttggcmt 300
```

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c 301

<210> SEQ ID NO 141
 <211> LENGTH: 301
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays ;

<400> SEQUENCE: 141

mtaggaaggt caaggtgctt ttggaccccg gcagcaacta caccttcgac ctgagctact 60
 accgccacgt rctcgccacg ggcgggctgt tccagtcga cggcagcctc ctgcacgacc 120
 ccgtcaccag aggttacgtc gagaaggtgg ccaaggcgtc gtcgcccgac gagtactacg 180
 cggacttcgc crcggcgatg gtcaagatgg gccgcaccga cgtgcttgtc ggcgatcatg 240
 gggagatcag gccaacgtgt ggcatttttg ttgactaggt tcaggaktgg gttgaaratg 300
 c 301

<210> SEQ ID NO 142
 <211> LENGTH: 294
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays ;

<400> SEQUENCE: 142

gtcaacaagc accagacttt ccaggacact cgggtctttt tcgtggtctc cacagatgga 60
 tcccaggcag aytctctcta cctgaagtgc ctggagaact ttgtgaggaa gagctacacg 120
 gaggacgysg acacattctg catgaagtac ttaaggcccc gtcgcaggca ggcaccacca 180
 gctgatgttg ggacagcatc aggcrccccc gatgaggttc caccgtcaac cgcagctgag 240
 acagagcaag gcactcctcc agcccctcag gcagagggtc crcaagagay ttgg 294

<210> SEQ ID NO 143
 <211> LENGTH: 285
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays ;

<400> SEQUENCE: 143

attacaccwa acttgtcctt ataagttaga aytctacaa tatttattag actctgtagg 60
 agaatctctt ctagatacaa acccaccatcc tttactgtaa aaaaggccca ttgttagcta 120
 caatgactta acatctacgg agggaccaaa tctctggtct gaaacccaaa ctttcaaggt 180
 tcactaggtt cgtccactgg cagataaaac aaatcctcat ttaygatcat aagttgacat 240
 actggacaaa gaatacttgt aagacgatcc cttcttatct gatga 285

<210> SEQ ID NO 144
 <211> LENGTH: 297
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays ;

<400> SEQUENCE: 144

agtgtagaga tactcacaag aacaattttg aactgccttg cagagccagg taccacgaat 60
 aagtgttcaa ccttactctc atgcctcaat ttcaagaaca cctgaacaca ttgtctaaat 120
 gatgtaagaa aaaagttgcc atcatggtag cagatacgag gtgagtagca acctggtgag 180
 gtttaaatgt gtgcccmagc ggtgtggtta gttgaaaaga taaacrcagt ttctgcagat 240
 ggtttgcaga aagagacact tttgtatctt tcagaagatc taacctgaaa tgaagca 297

<210> SEQ ID NO 145

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```

<211> LENGTH: 301
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;

<400> SEQUENCE: 145

ccatgcgcta ctaccaggcc ggytcctcgg agatgttygg ctccacgccg ccgcgcgagc    60
gcgaggacac gcccttccac ccgcgctcgc cctacgccgc cgccaaggtc gccgcgcact    120
ggtacacsgt caactaccgc gaggcctacg acgtattcgc ctgcaacggc gtgctyttca    180
accacgagtc cccgcgccgc ggcgagaact tcgtcacgcg caagatcacg cgcgcgcgtcg    240
gccgcatcaa ggtcgggctg cagaccaggg tcttcctcgg caacctctcg gccgccaggg    300
a                                                                    301

<210> SEQ ID NO 146
<211> LENGTH: 255
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;

<400> SEQUENCE: 146

tacatacctt gcagcatgtc aaatgcacaa ggatgaagag agagggaaaag ttgcagcaaa    60
gaaacttggt gagatggaac cacaaagctc atccacgtac gtgttcctdt caagcttrca    120
tgctgcggct ggtaactggg ttgaagccaa agtagccaga gaagcaatgc gagaaaaagg    180
ggtgatgaaa ttccagggt gtagttggat cacagtgggk aacaaacara gygtatttgt    240
tgtacaggac acaca                                                                    255

<210> SEQ ID NO 147
<211> LENGTH: 299
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;

<400> SEQUENCE: 147

actcaaatg agtcaagaaa tgctaaggga aagcccgta aacaatataa tcgaactttt    60
acattttatt gatgtggcct tyttaaaaat gacgccataa accwctatac tgaaaacggc    120
ctcggcgctt aaaccagtag cattcgtagg ttcatttctg atttgggact tcaaaccaat    180
agtgtttgag actcacaggt ttagaaccga ttggtggctg cttcaatca agagcgaaac    240
caacatgtta ttgccacct taaaatcaca gatattgtat tatcgtgtac aataaaaat    299

<210> SEQ ID NO 148
<211> LENGTH: 301
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;

<400> SEQUENCE: 148

ycagatcatc actggaatca acagcagttc tctggttcat ttgttcagat gcttckgtgt    60
caactgcttc tgaaygatgg tttaagctta taacatcatc arcatttttg ttaattattg    120
cactgttctc gttgtccttt accacgctat ccaaawtttc tggctcctca gatactacaa    180
gatcagatrt ctgcccgctc tgagtaagag tgggttcagg actgacaccc atacctgtg    240
caagtgttac ataccgaagt actagwtgct tgtacgaatt caataggctc tctacatcag    300
c                                                                    301

<210> SEQ ID NO 149
<211> LENGTH: 274
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;

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<400> SEQUENCE: 149

ctttagtagc tagtctytata gctcataggt tctcagttcg gtatatrttyg gtgayatatt	60
tcatgaactt asttttwaam cmatttataaa mgcamacgca acaacgaagg gagtgatatt	120
gaccatgggc ctttcgtagc ttttcaatca aacgtcacat aaaaaaagaa acawccaaaa	180
gwtcataata cacagattkr aamskkkaga caaaaatacg agtacaagac attggacctt	240
caatwwtttt tttktgtttt ctgagacatg aaaa	274

<210> SEQ ID NO 150

<211> LENGTH: 319

<212> TYPE: DNA

<213> ORGANISM: Zea mays ;

<400> SEQUENCE: 150

ttgcaccacc catagttaa gaggcagttg ctttcacaca ctttattttg catggcacag	60
cctaactgtcc ccatcattct aaaccaggct gcaactgagc cactacagaa actgctaata	120
ttagatatcc cagcaaatag tcttgacact agagtgttag gtcctaaac acgaaagaca	180
tttgtcagga gaccogtatt gagtactagc agagtatact accgctagat attccagtat	240
aatcgtgcag cagctccggt caatggcagt ggcacttgaa gtctgcaacc tcagccaaca	300
gatccacatg agaaccaac	319

<210> SEQ ID NO 151

<211> LENGTH: 283

<212> TYPE: DNA

<213> ORGANISM: Zea mays ;

<400> SEQUENCE: 151

tacactatac taattagaat gtctggaaag gctgggaaga ccactaaatt tgtctcattk	60
ttggaggaaa tgggtgccra aggatgtgtt cttartctga ttgcttataa tactgttatt	120
gaggctcttg gtaagaacaa gatgggtgac gaggcgattt ttatgctttc taaaatgatt	180
gagagtgtgact gtcggcccaa tcaattcaca tatagcatta tgctggatgt tttatcaaca	240
gggggacaac tccacaggtt gaatgagatt ctagatattt gta	283

<210> SEQ ID NO 152

<211> LENGTH: 193

<212> TYPE: DNA

<213> ORGANISM: Zea mays ;

<400> SEQUENCE: 152

cctttctcca gttggtaata tgcttcatgt aaggtagctg aaggtaggtaa gcracatgat	60
agtcaaaactc ccactcatta ttcgaggcct gcgacactta gagacactcg aggtggatgc	120
agaaraagtc gctgttcac tggtatgttt catcttgaag agcctgttgc atctccgact	180
tccgagcaag gct	193

<210> SEQ ID NO 153

<211> LENGTH: 298

<212> TYPE: DNA

<213> ORGANISM: Zea mays ;

<400> SEQUENCE: 153

tgcatatgag aaactacaag gttggcaggg tcctttcctg tgcccgagct gtcataaaaa	60
gaaagaagca atggaaggga aacgccggcc aaaaggatc attaaatagt atttacctga	120

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actacattct ktagttaacc ctacagacagc tcattcgttt ctgggttcct atttgtcttt	180
ccaggatctt catcaaactg ctttggtcac atgtgctagg cctgcctacc actaaagatt	240
cttttattga ttsttagctg gaaataatag kcagcccata acaattcatt ggctagaa	298

<210> SEQ ID NO 154
 <211> LENGTH: 360
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays ;

<400> SEQUENCE: 154

aagcaccgcg atgcagactt ccaactatc cagtgtccac aaatagcagt ttgcaaact	60
aaatacatat atgaaatgat gacacaggcc acccaaaca caagccctag agaatcagac	120
caaccaaccg agcaattcgc agtgtccat atcttctagc attcacaacc tgacgaacgt	180
tggtgccttc cagcgccccc aacatcaatc gtatcaggaa gatacctgca aaagttggta	240
aaacatagat caagttacac cacaaggctg cctgcatgta tgatacatat atgtgagcta	300
tatactattc aaacttacat atcttcttct ggctcatttt ttacagcatt cctcgctata	360

<210> SEQ ID NO 155
 <211> LENGTH: 299
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays ;

<400> SEQUENCE: 155

agaygtact ctgctaccat ttgagaatc tattgacaag aactttacag gaggrgtctc	60
atccccagga gccttgctga taactggaga tggagaagat atgcacatat ccactcgtga	120
atgcytttct ctggacttat tctgtcctt cagcagttgt ttccctagc aaacaaaagg	180
agatacacat agcatgagaa agaagaaasa aactgaaaca ttttataatg agaaaaaac	240
atggggagtt ggggacaaag caactacca atgcttctac aaatgttttg aacctccga	299

<210> SEQ ID NO 156
 <211> LENGTH: 262
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays ;

<400> SEQUENCE: 156

tgcttctgtc gaattgtgc gggcagarca tgctttggtg gyggctttgc agtgtgaaac	60
gttatgaatc tgagaacctc gatgcatgcy agggacaggg aatacctgaa caggggccaa	120
gatgacatga acaggacgcc ggacagcacg tgcacaacct tcctgctcaa gctctgcctt	180
cattgatggc atggatcagt acagagtctg gaagccaacg acgtgaacar ctgaacataa	240
gcaaatgtaa tggccgttct aa	262

<210> SEQ ID NO 157
 <211> LENGTH: 301
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays ;

<400> SEQUENCE: 157

aggctgacaa ggctgctagt aacatgtctg agccgacatc aggggagatg caggatgctg	60
ctttccagtc tgacgaagag gaggaagatg aagatgttga tgaaacagta ttcggtcaag	120
attcagatcc gtcacaraat agcggcaccg acgacgatgc aaagtagact cactgctggt	180
attcatcatt atggtgattg tattgttttt aattaaattg ccaagcttga tttttgtcaa	240
ggcgacgytg gaaggttgca cagaattttg ayagtgtctc tggtttcatt ttgaaagcac	300

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t 301

<210> SEQ ID NO 158
 <211> LENGTH: 271
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays ;

<400> SEQUENCE: 158

agaaaacgag gccttcgtcg tcttagcgcg crggggcgta ggacggcggc agcggcagcc 60
 ccacgacctt gccgcagcgg agggcggtgt cgggcagtgt gtactcgtcg cgcagtgatc 120
 gcgcggcgga gacgacgctg atgtagaact gctccccgag gtagtaccgc tcccagatgt 180
 tggaatgcac gctccacatg cccgcgttgt cgaacgtcag catgattgcc gtccatgacc 240
 gcgggtacac ctggatcgtg tgccggctca c 271

<210> SEQ ID NO 159
 <211> LENGTH: 286
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays ;

<400> SEQUENCE: 159

ttctgggtact tgccgatctc gcgcagmgcc acsgtscgg gcctgtacct gtggggcttc 60
 ttcacgccgc cggtcgtcgg cgcmgacttc ctgcgcgcct gcaacraray aacaaccgcc 120
 gccgcgtcag crccgtctta caacgggaac catcgagggg atcggaaacac cagatcggta 180
 cgggtaggta gagacgaacc ttggtkcgca gctgcttgcg cggggccttg cctccgggtg 240
 acttgccggc ggtctgcttc gtacgagcca tcttctcttc tccttg 286

<210> SEQ ID NO 160
 <211> LENGTH: 252
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays ;

<400> SEQUENCE: 160

ctgcgttagt ttcacccctt ctagctgcga gtgaaagaaa cwtgataaca gctwsctgct 60
 agtttctatg ryrgccatcg aatctgacat ggctayctcc tgtgmcmacg cagggtccga 120
 ggtactacca gccgaggaag agccaccgga cggtgacggc ggtgatccac ggcgagaagg 180
 tgccgctgta cggcgccggg ggcggcctga cgctgtcsac gagcgcgggc gggggcgcg 240
 tgcccctgac gc 252

<210> SEQ ID NO 161
 <211> LENGTH: 301
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays ;

<400> SEQUENCE: 161

agcaccacga ttcacaccta cacatattca gacacaaaat tgggcagcga tgggcatgga 60
 catttcattg atagatggta tttagctgcc tactytggca tttgattcga gcagcgaagc 120
 ttcaagagct gcagaatcgc tacayatatt tctgtgggta cacatgcttc ttgttctcca 180
 ctgcaccag atcattgtga tggactacta actgscagta caaaagcaga catctatcaa 240
 acttacacaa aaggcgtgca caacgagaya gyctgcttgc gcctctccaa gtctagtcta 300

g 301

<210> SEQ ID NO 162

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<211> LENGTH: 237
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;

<400> SEQUENCE: 162
awwttkkwag ygcarrata aacaaggcac tgtwrattgt acagagcaag ctctctgcat      60
ttttcaatgc cgcagcatct gcaactgcaa gcagagtttc tagaggccaa ggccgactgt      120
cctgccatct ctctcgcgga ggccgtctat gtaatgtttc tccaggttgg tgcgagctc      180
ctccagtttc ttggtttctt ccgcgagct cccgaccttg ttctcgaccg tgtttaa      237

<210> SEQ ID NO 163
<211> LENGTH: 278
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;

<400> SEQUENCE: 163
tcaatttggtg atttcattca caattttctt atatgtgagg aatgccgcac rcattcttat      60
gaaatgtgtt caargtttgg aaaaactact ctagccatat attgccacag aaatcaastg      120
catgtytaat agttaccgtt cacctgtagt gtatcagtc cttcaaactc tgcccgtagc      180
cttgccctct ggctatgggc cgcgcataac aaagttaacg agaggttgat gaaagaagaa      240
aaggagttag acaatgctga tccttcattt cctaagat      278

<210> SEQ ID NO 164
<211> LENGTH: 263
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;

<400> SEQUENCE: 164
ttgcatacaa acttgatcma aatgttttca amccatgctt gaacrratat acaaacccgt      60
tgyggcagga acatyttagc ttwgkammmt ygywvagcaa caatagcttc aatagagatt      120
gttgtaagt tagagtgttg ctgcgatgga ractatgcaa atcacaaggg atgggagaag      180
agtgcattc agaaggaggt ccgtgccgag ataaccggtc gcagtaatcg tggcatctgc      240
aactacacgg gccaacgtcc cgg      263

<210> SEQ ID NO 165
<211> LENGTH: 266
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;

<400> SEQUENCE: 165
acaagtccaa atcaaagggtg ttgtcccgcc aatgcttacc ccccatgscy cccraaata      60
gcatctgaat ccattttttt gtcygatctt tttkkctcct cttctgacag ggaaatgctg      120
cttgagtttg acgaggaaca gactggaatg caaatgcacg tgcaggttgc aaattcttat      180
gttrwttttt gkggcctatc arttcacagt actcatagca tgacaaatgg cgttgacggt      240
tttggttttt accsgttttg gtttcc      266

<210> SEQ ID NO 166
<211> LENGTH: 197
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;

<400> SEQUENCE: 166
aacctcgact ccactataas tgcaggacgc acgrcaygca caactgaaac atccggacca      60
acaggcagag tagtagccca aaggatgacc cgtctcatgc aaaaacagtc agcttgataa      120

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acttatcgag aaggcaacaa cgggcccacaa gtactgggtg aaatacagag cagacattga 180

agtgccaaaag agggggg 197

<210> SEQ ID NO 167

<211> LENGTH: 301

<212> TYPE: DNA

<213> ORGANISM: Zea mays ;

<400> SEQUENCE: 167

gtctagaaaa gtcaaaacgt catatacttg gaacrggagt agatatgaca aaaagagcat 60

gcaagtgatg ttcttacaga agaatatata tatttcattg caagctagta cctaaacttc 120

agatggagtc ttgtcatcga ttgacagggc atgcaaccgc gtcttgagct cctcctgcag 180

gagatcatag acagccctgt gctctttrag caagctcttc cctcaaaact ccttgacac 240

caccctcaca ttgaaatgtg tctccccatt ggtcccagcc acgccagcat ggcccttgtg 300

g 301

<210> SEQ ID NO 168

<211> LENGTH: 291

<212> TYPE: DNA

<213> ORGANISM: Zea mays ;

<400> SEQUENCE: 168

atattaccat actgcattca aagttttaca agattggaca ttattagtat aaaacatgaa 60

acttttcaag tgaacaasaa ctataarcaa cgtccgcaag cgcaatccac agcttgaggt 120

tacattgtga gtgtagtaga aaaactaggt ggtcccaaat agcacaagat tatccaagct 180

acccagtttt ctacgattat gggcacracc ataagagcat agmagtgctt cgcattcctgt 240

tgcattaaga aagtagtata ggcatcggag gtgctgaatc taagcacacg g 291

<210> SEQ ID NO 169

<211> LENGTH: 299

<212> TYPE: DNA

<213> ORGANISM: Zea mays ;

<400> SEQUENCE: 169

taaaccagay tagatgccac ctgktttct gacacaaatc aygaacaaaa cagagaaga 60

aagcyaatca tacacggatg ayggatggac tcagcggagc ccttatttgc taccattcct 120

catgtcttat tgcagaaatc catctattgc cactcaactt camtcagtct ctggaactct 180

gtatcaacag gggatgggaa atgtgtcatg ttcaatgttt agcycatgaa acatagaaga 240

ccmattaga agctattatg tgcttacatt tgattttttt atccaagact caagtgtat 299

<210> SEQ ID NO 170

<211> LENGTH: 300

<212> TYPE: DNA

<213> ORGANISM: Zea mays ;

<400> SEQUENCE: 170

rrhytkrtgg aggtcggggg aagaaaccct aagaacgatg tctcaccaaa ttgattcctt 60

ccgcccgaga gatttagaag ctgatctgt ccagatttag tggattgata ggttttgtga 120

atattgtcatt ctccaactaa tctacttaca gccagatttt acatcagacc tgatgaaaca 180

ctgtttcctt gacacgaac tggatggagc tgcccttgca tcaagaatca agaaattgat 240

ttcgttttta tgtytctggg agcyccagac acctcact ctctctgtt gcctgtsatg 300

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<210> SEQ ID NO 171
<211> LENGTH: 232
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;

<400> SEQUENCE: 171

aacggctaag agtcaggcga ttctgtttgt acagagacaa tcgcagcact tgrmtgctwc      60
gcatgcgstt cagagggcct tcagctcagg cttgacagat ctagttttgg ycgccacgcy      120
rctcttcggc gagcactgya ctctgctggc atcttttagaa gccccgatgc tgtgacgatg      180
tggccgggtg gtagatgttg cgttggtgtt tttccctcg ggtgtctgtg gt                232

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<210> SEQ ID NO 172
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<212> TYPE: DNA
<213> ORGANISM: Zea mays ;
<220> FEATURE:
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<222> LOCATION: (7)..(7)
<223> OTHER INFORMATION: n is a, c, g, or t
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<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(31)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (66)..(66)
<223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 172

aaacaanact cagtggactc gcngaaannn nataatgaaa ggtgctccat ccatacccat      60
gagaangttc cgatgctcgc agtctcatgt ttcccagtcg gtcttctttc atttccttct      120
ccgatgcac taatatgcag atcatggcgc agagaaaggt tcaggattgc tctcttgatc      180
tcttcaaacc cgcgctcttc g                                201

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<210> SEQ ID NO 173
<211> LENGTH: 301
<212> TYPE: DNA
<213> ORGANISM: Zea mays ;
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (239)..(281)
<223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 173

aaaattactg aaggcagggtg ggttgcagtt gtgtgttcgt tactgtttac tgtawyatgt      60
caagctgtcg gctgcaatth ctttgcctggc aagccgcagg cactgggtgaa gtgctgataa      120
atacatcata ttctgttgac ctgtgaagaa acttgttcwa ggtrgattcc attgtactag      180
ctctgttgcy cagcatctcc ttgtttggga acattaacaa ccagcyctcr mcccetcaann      240
nnnnnnnnnn nnnnnnnnnn nnnnnnnnnn nnnnnnnnnn ntaaaatagt cgcacatcatgt      300
a                                                                301

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<210> SEQ ID NO 174
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<212> TYPE: DNA
<213> ORGANISM: Artificial ;Primer
<220> FEATURE:
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<400> SEQUENCE: 174
gccatggatc gcattcatg 19

<210> SEQ ID NO 175
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<212> TYPE: DNA
<213> ORGANISM: Artificial ;Primer
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 175
ctctctggca tctctgaaat catc 24

<210> SEQ ID NO 176
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Probe
<220> FEATURE:
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<400> SEQUENCE: 176
cgattttgac tttgaacc 18

<210> SEQ ID NO 177
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<212> TYPE: DNA
<213> ORGANISM: Artificial ;Probe
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 177
cgattttgac ttcgaac 17

<210> SEQ ID NO 178
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Primer
<220> FEATURE:
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<400> SEQUENCE: 178
acacccatgg cacaaggt 18

<210> SEQ ID NO 179
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<212> TYPE: DNA
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<400> SEQUENCE: 179
cgagagacca gatcaacagc tt 22

<210> SEQ ID NO 180
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<212> TYPE: DNA
<213> ORGANISM: Artificial ;Probe
<220> FEATURE:
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<400> SEQUENCE: 180
caataactcta aaaaaactg 19

<210> SEQ ID NO 181

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<211> LENGTH: 19
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Probe
<220> FEATURE:
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<400> SEQUENCE: 181

caataactcta agaaaactg 19

<210> SEQ ID NO 182
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<212> TYPE: DNA
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cccttgaacc agagtggata atctc 25

<210> SEQ ID NO 183
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caggtagtag tcttcggtat gtcaac 26

<210> SEQ ID NO 184
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<220> FEATURE:
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<400> SEQUENCE: 184

atcagatgta aagcatc 17

<210> SEQ ID NO 185
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<212> TYPE: DNA
<213> ORGANISM: Artificial ;Probe
<220> FEATURE:
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<400> SEQUENCE: 185

aaatcagatg tagagcatc 19

<210> SEQ ID NO 186
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 186

actgcctgca aggttggtt 19

<210> SEQ ID NO 187
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<212> TYPE: DNA
<213> ORGANISM: Artificial ;Primer
<220> FEATURE:
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<400> SEQUENCE: 187

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aaaccaagca acctt	15
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ccccagccct ggatcac	17
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gctttcgaat gtgttcatga tgtaagt	27
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ctggagaagg aaagta	16
 <210> SEQ ID NO 194 <211> LENGTH: 25 <212> TYPE: DNA	

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<213> ORGANISM: Artificial ;Primer
<220> FEATURE:
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<400> SEQUENCE: 194

agcttctcct tgaccttctc cttaa 25

<210> SEQ ID NO 195
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Primer
<220> FEATURE:
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<400> SEQUENCE: 195

aggaagagga ggtgatcgat gag 23

<210> SEQ ID NO 196
<211> LENGTH: 17
<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 196

aacggcgaaa ttgtcaa 17

<210> SEQ ID NO 197
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<220> FEATURE:
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<400> SEQUENCE: 197

cggcgagatt gtcaa 15

<210> SEQ ID NO 198
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<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 198

gcgtcacaag aaggta 16

<210> SEQ ID NO 199
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Primer
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 199

gcaagcagaa ctcaac 16

<210> SEQ ID NO 200
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<212> TYPE: DNA
<213> ORGANISM: Artificial ;Probe
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 200

tcaacatgac tgaatca 17

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<210> SEQ ID NO 201
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<212> TYPE: DNA
<213> ORGANISM: Artificial ;Probe
<220> FEATURE:
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<400> SEQUENCE: 201

caacatgact gaagcat 17

<210> SEQ ID NO 202
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<212> TYPE: DNA
<213> ORGANISM: Artificial ;Primer
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 202

cgcagcgtgg agtgc 15

<210> SEQ ID NO 203
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<212> TYPE: DNA
<213> ORGANISM: Artificial ;Primer
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 203

acacattcga aggcggatat catg 24

<210> SEQ ID NO 204
<211> LENGTH: 15
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Probe
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 204

ccggtaacat caacg 15

<210> SEQ ID NO 205
<211> LENGTH: 15
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Probe
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 205

ccggtaacgt caacg 15

<210> SEQ ID NO 206
<211> LENGTH: 25
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Primer
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 206

ggagtctttt aggtccact tgttt 25

<210> SEQ ID NO 207
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Primer
<220> FEATURE:

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<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 207

agagagcaca aggccaagaa t 21

<210> SEQ ID NO 208
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Probe
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 208

ttctgcaagt tgaattt 17

<210> SEQ ID NO 209
<211> LENGTH: 14
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Probe
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 209

tgcaagctga attt 14

<210> SEQ ID NO 210
<211> LENGTH: 19
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Primer
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 210

gagcaagcgc agagaacag 19

<210> SEQ ID NO 211
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Primer
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 211

cgggcatgtt gcgatgtc 18

<210> SEQ ID NO 212
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Probe
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 212

aaaactttac ccataaacia 20

<210> SEQ ID NO 213
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Probe
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 213

actttaccg taaacia 17

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<210> SEQ ID NO 214
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Primer
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 214

cccggaatt gtttcgatg c 21

<210> SEQ ID NO 215
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<212> TYPE: DNA
<213> ORGANISM: Artificial ;Primer
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 215

ccctatagtt tgtagcaga tgttagtt 29

<210> SEQ ID NO 216
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Probe
<220> FEATURE:
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<400> SEQUENCE: 216

taacgcgttc ttcattt 17

<210> SEQ ID NO 217
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<212> TYPE: DNA
<213> ORGANISM: Artificial ;Probe
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 217

acgcgtcctt cattt 15

<210> SEQ ID NO 218
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<212> TYPE: DNA
<213> ORGANISM: Artificial ;Primer
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 218

gcaacctaaa aattctaaaa aatacagaga taggg 35

<210> SEQ ID NO 219
<211> LENGTH: 29
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Primer
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 219

actaatttca aagaaaatgc cacagcttt 29

<210> SEQ ID NO 220
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Probe
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

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<400> SEQUENCE: 220

attttaaaagg aagatagatt g

21

<210> SEQ ID NO 221

<211> LENGTH: 19

<212> TYPE: DNA

<213> ORGANISM: Artificial ;Probe

<220> FEATURE:

<223> OTHER INFORMATION: Assay component

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gcgagccggt gagagtga

19

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ccacgagtc ccagtcct

19

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acagccatgt cgcc

14

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<220> FEATURE:

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acagccgtgt cgcc

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<400> SEQUENCE: 226

ccacaaaaac gagtaaacga caaat

25

<210> SEQ ID NO 227

<211> LENGTH: 25

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<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 227

ggttgcttga gatcttgetg ttatt 25

<210> SEQ ID NO 228
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<212> TYPE: DNA
<213> ORGANISM: Artificial ;Probe
<220> FEATURE:
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<400> SEQUENCE: 228

agtgagcacg gcacg 15

<210> SEQ ID NO 229
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<212> TYPE: DNA
<213> ORGANISM: Artificial ;Probe
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 229

agtgaggacg gcacg 15

<210> SEQ ID NO 230
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Primer
<220> FEATURE:
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<400> SEQUENCE: 230

acaacaaggc ggcattgc 18

<210> SEQ ID NO 231
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Primer
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 231

ggcgctccg tttaactact c 21

<210> SEQ ID NO 232
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Probe
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 232

ttgcagtagt agtactgtag 20

<210> SEQ ID NO 233
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Probe
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 233

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 ttgcagtagt agtgctgtag 20

<210> SEQ ID NO 234
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 <212> TYPE: DNA
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 <220> FEATURE:
 <223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 234

gctgctcggt ctcgttgaag a 21

<210> SEQ ID NO 235
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 <212> TYPE: DNA
 <213> ORGANISM: Artificial ;Primer
 <220> FEATURE:
 <223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 235

cgctggacga caccaaga 18

<210> SEQ ID NO 236
 <211> LENGTH: 17
 <212> TYPE: DNA
 <213> ORGANISM: Artificial ;Probe
 <220> FEATURE:
 <223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 236

tccacgttcc tgtccag 17

<210> SEQ ID NO 237
 <211> LENGTH: 15
 <212> TYPE: DNA
 <213> ORGANISM: Artificial ;Probe
 <220> FEATURE:
 <223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 237

cacgttccgg tccag 15

<210> SEQ ID NO 238
 <211> LENGTH: 24
 <212> TYPE: DNA
 <213> ORGANISM: Artificial ;Primer
 <220> FEATURE:
 <223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 238

gctagtctca agtccaagca aaga 24

<210> SEQ ID NO 239
 <211> LENGTH: 24
 <212> TYPE: DNA
 <213> ORGANISM: Artificial ;Primer
 <220> FEATURE:
 <223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 239

agtaggtgct tgccatttac atct 24

<210> SEQ ID NO 240
 <211> LENGTH: 22
 <212> TYPE: DNA
 <213> ORGANISM: Artificial ;Probe

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<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 240

aagattagca ataactattg tt 22

<210> SEQ ID NO 241
<211> LENGTH: 22
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Probe
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 241

aagattagca ataactgttg tt 22

<210> SEQ ID NO 242
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Primer
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 242

cggctctggct aggatttgga 20

<210> SEQ ID NO 243
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Primer
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 243

gagcatgtgg aggttcctga t 21

<210> SEQ ID NO 244
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Probe
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 244

cattcaagaa tgtcctc 17

<210> SEQ ID NO 245
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Probe
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 245

cattcaagca tgtcctc 17

<210> SEQ ID NO 246
<211> LENGTH: 25
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Primer
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 246

gaatctgaga tgttgaaccc actgt 25

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<210> SEQ ID NO 247
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Primer
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 247

tgctgaaaga ggcctcaaag g 21

<210> SEQ ID NO 248
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Probe
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 248

caccaaatca gcatcg 16

<210> SEQ ID NO 249
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Probe
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 249

caccaaatcg gcatcg 16

<210> SEQ ID NO 250
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Primer
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 250

ttcttgacgt gccgaaagga t 21

<210> SEQ ID NO 251
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Primer
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 251

ccgccgtatg attggatcga 20

<210> SEQ ID NO 252
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Probe
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 252

aaaaagccaa ccgcacg 17

<210> SEQ ID NO 253
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Probe
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

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<400> SEQUENCE: 253

aaaaagccag cgcacg

17

<210> SEQ ID NO 254

<211> LENGTH: 25

<212> TYPE: DNA

<213> ORGANISM: Artificial ;Primer

<220> FEATURE:

<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 254

ggtacatcag caaccgaggt ttata

25

<210> SEQ ID NO 255

<211> LENGTH: 21

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<213> ORGANISM: Artificial ;Primer

<220> FEATURE:

<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 255

gttgatgaac gcctcctcag a

21

<210> SEQ ID NO 256

<211> LENGTH: 15

<212> TYPE: DNA

<213> ORGANISM: Artificial ;Probe

<220> FEATURE:

<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 256

ctcgtagagc aatct

15

<210> SEQ ID NO 257

<211> LENGTH: 16

<212> TYPE: DNA

<213> ORGANISM: Artificial ;Probe

<220> FEATURE:

<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 257

cctcgtagac caatct

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<210> SEQ ID NO 258

<211> LENGTH: 23

<212> TYPE: DNA

<213> ORGANISM: Artificial ;Primer

<220> FEATURE:

<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 258

gaagccgact ggagagaatg tt

23

<210> SEQ ID NO 259

<211> LENGTH: 25

<212> TYPE: DNA

<213> ORGANISM: Artificial ;Primer

<220> FEATURE:

<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 259

ttttccatgg agggttatga tgctt

25

<210> SEQ ID NO 260

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<211> LENGTH: 19
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Probe
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 260

ctctgggtttc tgatggcta 19

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tgtagaggca aatgtat 17

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ctgtagaggc aagtgtat 18

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cctgggtgtgt atgctgttct tttc 24

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gcggtagcgt gaggtagc 18

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<212> TYPE: DNA

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<212> TYPE: DNA

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gtaatggtag cacgcagatc act

23

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gctaggaatt gcttagtcga gggtt 24

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actgaagaaa taatcgc 17

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ctgaagaagt aatcgc 16

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ccctccgtaa atgttaa 17

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ccctccgtag atgttaa 17

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gcgcgtgacg aagttctc 18

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tacggcgat tcg 13

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tgctgcggct ggtaactg 18

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ttgaagccaa agtagccag 19

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<223> OTHER INFORMATION: Assay component

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gcctcggcgc ttaaacc                                     17

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ttaccacgct attcaaa 17

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tttaccacgc tatccaaa 18

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gaaggtccaa tgtcttgtag tcgta 25

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ccatgggtct ttcgt 15

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ccatgggcct ttcgt 15

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gctgctgcac gattatactg gaat

24

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<211> LENGTH: 23

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ctgagccact acagaaactg cta

23

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18

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<212> TYPE: DNA

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<220> FEATURE:

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gaaaggctgg gaagaccact a

21

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<212> TYPE: DNA

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<220> FEATURE:

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aaaatcgctt tgtcaac

17

<210> SEQ ID NO 385

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ctccagttgg taatatgct 19

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gagtgggagt ttgactatc 19

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ttaccacctt caggta 16

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cctggaaaga caaataggaa ccca 24

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aacgaatgag ctgtctga 18

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aaacgaatga gcagtctg 18

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gtgctccata tcttctagca ttcac 25

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ggcagccttg tgggtgaac 19

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aacctgacga acgttgt 17

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ctgacgaacg ctggt 15

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ttctgtcctt cagcag                        16

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cagagcttga gcaggaaggt t                  21

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acgccggata gcac                          14

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ccggacagca cgt 13

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<400> SEQUENCE: 406

ctgacgaaga ggaggaagat gaag 24

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accagcagtg agtctacttt gc 22

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atcgctgctg gtgc 14

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atcgctgccg gtg 13

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gagcgtgcat tccaacatct g 21

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<400> SEQUENCE: 412
agacgacgct gatataga 18

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<400> SEQUENCE: 413
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<400> SEQUENCE: 414
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<400> SEQUENCE: 415
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cttacaacgg gaacaatc 18

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gcaggtccgc aggtacta 18

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<212> TYPE: DNA
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<210> SEQ ID NO 420
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<400> SEQUENCE: 420

accgtccggt ggct 14

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<400> SEQUENCE: 421

accgtccggt ggc 13

<210> SEQ ID NO 422
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<213> ORGANISM: Artificial ;Primer
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<400> SEQUENCE: 422

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tgggtcgagt ggagaacaag 20

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caggtgaacg gtaactatt                      19

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<212> TYPE: DNA
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gcaacaatag cttcaataga g                  21

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cccttgatgat ttgcatagt                    19

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gtcatgctat gagtactgt 19

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cacgtgcatt tgcatt 15

<210> SEQ ID NO 442
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gatgaccgct ctcatgcaaa 20

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<223> OTHER INFORMATION: Assay component

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<210> SEQ ID NO 445
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tcatggcaag ctagtaccta aac 23

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<212> TYPE: DNA
<213> ORGANISM: Artificial ;Probe
<220> FEATURE:
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<400> SEQUENCE: 449

ttgacagggc gtgc 14

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<212> TYPE: DNA
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<220> FEATURE:
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cacctccgat gcctatacta ctttc 25

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<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 453

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<210> SEQ ID NO 454
<211> LENGTH: 24
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Primer
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 454

cggagccctt attgctacc attc 24

<210> SEQ ID NO 455
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Primer
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 455

ccctgttgat acagagttcc aga 23

<210> SEQ ID NO 456
<211> LENGTH: 19
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Probe
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 456

aaatccatct attgctact 19

<210> SEQ ID NO 457
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Probe
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

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<400> SEQUENCE: 457

catctattgc cactca

16

<210> SEQ ID NO 458

<211> LENGTH: 22

<212> TYPE: DNA

<213> ORGANISM: Artificial ;Primer

<220> FEATURE:

<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 458

ttccgcccga gagatttaga ag

22

<210> SEQ ID NO 459

<211> LENGTH: 20

<212> TYPE: DNA

<213> ORGANISM: Artificial ;Primer

<220> FEATURE:

<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 459

caccagtttc gtgtcaagga

20

<210> SEQ ID NO 460

<211> LENGTH: 19

<212> TYPE: DNA

<213> ORGANISM: Artificial ;Probe

<220> FEATURE:

<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 460

tgtaaaatct ggctgtaag

19

<210> SEQ ID NO 461

<211> LENGTH: 17

<212> TYPE: DNA

<213> ORGANISM: Artificial ;Probe

<220> FEATURE:

<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 461

tgtaaaatct ggccgta

17

<210> SEQ ID NO 462

<211> LENGTH: 19

<212> TYPE: DNA

<213> ORGANISM: Artificial ;Primer

<220> FEATURE:

<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 462

ggccacatcg tcacagcat

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<211> LENGTH: 19

<212> TYPE: DNA

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<220> FEATURE:

<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 463

tcagctcagg cttgacaga

19

<210> SEQ ID NO 464

<211> LENGTH: 17

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<212> TYPE: DNA
<213> ORGANISM: Artificial ;Probe
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 464

cttctaaaga tgccagc 17

<210> SEQ ID NO 465
<211> LENGTH: 15
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Probe
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 465

tctaaagacg ccagc 15

<210> SEQ ID NO 466
<211> LENGTH: 25
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Primer
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 466

gcatattagt gcatacgag aagga 25

<210> SEQ ID NO 467
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Primer
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 467

cgatgctcgc agtctcatgt t 21

<210> SEQ ID NO 468
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Probe
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 468

cccagtcagt cttctt 16

<210> SEQ ID NO 469
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<212> TYPE: DNA
<213> ORGANISM: Artificial ;Probe
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 469

ccagtcagtc ttctt 15

<210> SEQ ID NO 470
<211> LENGTH: 27
<212> TYPE: DNA
<213> ORGANISM: Artificial ;Primer
<220> FEATURE:
<223> OTHER INFORMATION: Assay component

<400> SEQUENCE: 470

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cactggtgaa gtgctgataa atacatc	27
<210> SEQ ID NO 471 <211> LENGTH: 23 <212> TYPE: DNA <213> ORGANISM: Artificial ;Primer <220> FEATURE: <223> OTHER INFORMATION: Assay component <400> SEQUENCE: 471	
atgttcccaa acaaggagat gct	23
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cctgtgaaga aacttggt	18
<210> SEQ ID NO 473 <211> LENGTH: 17 <212> TYPE: DNA <213> ORGANISM: Artificial ;Probe <220> FEATURE: <223> OTHER INFORMATION: Assay component <400> SEQUENCE: 473	
ctgtgaagaa gcttggt	17
<210> SEQ ID NO 474 <211> LENGTH: 104 <212> TYPE: DNA <213> ORGANISM: Zea mays <400> SEQUENCE: 474	
gtggagaaatg cagagccatg gatcgcatc atggagtagc cacagggttc gaagtcaaaa	60
tcgcagatct ccatctcggg gatgatttca gagatgccag agag	104
<210> SEQ ID NO 475 <211> LENGTH: 202 <212> TYPE: DNA <213> ORGANISM: Zea mays <220> FEATURE: <221> NAME/KEY: misc_feature <222> LOCATION: (24)..(24) <223> OTHER INFORMATION: n is a, c, g, or t <220> FEATURE: <221> NAME/KEY: misc_feature <222> LOCATION: (27)..(27) <223> OTHER INFORMATION: n is a, c, g, or t <220> FEATURE: <221> NAME/KEY: misc_feature <222> LOCATION: (33)..(33) <223> OTHER INFORMATION: n is a, c, g, or t <220> FEATURE: <221> NAME/KEY: misc_feature <222> LOCATION: (46)..(46) <223> OTHER INFORMATION: n is a, c, g, or t <220> FEATURE: <221> NAME/KEY: misc_feature <222> LOCATION: (84)..(84) <223> OTHER INFORMATION: n is a, c, g, or t <220> FEATURE: <221> NAME/KEY: misc_feature <222> LOCATION: (135)..(136) <223> OTHER INFORMATION: n is a, c, g, or t	

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<400> SEQUENCE: 475

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cgagagacca gatcaacagc ttcnngnggc atnccttttg tggaanacct gaaagataaa    60
atgtacaaaa agtaaattag taanacaaca gtaacagttt tcttagagta ttgtaacaaa    120
atgtagaaaag atttnngcac cttgtgccat ggggtgtgctt tgatttgtgg gaacttaaac    180
tctgtgtaat ttgggttcat gc                                              202

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<210> SEQ ID NO 476

<211> LENGTH: 201

<212> TYPE: DNA

<213> ORGANISM: Zea mays

<400> SEQUENCE: 476

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gttttcaaac ttgcagtac gaatttcaat gtgcgtgatt ggaacctttt cgatggtggc    60
ccttgaacca gagtggataa tctctccaaa atcagatgta aagcatctgg ttgacatacg    120
gaagactact acctgttttc tagatttaca attcattgtc agaaattatg ctatctgact    180
taatcttcca aatactccta t                                              201

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<210> SEQ ID NO 477

<211> LENGTH: 185

<212> TYPE: DNA

<213> ORGANISM: Zea mays

<400> SEQUENCE: 477

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agcgtgcccc tgccggagcc ggtgccgccg ccgagagagt ggcagacctg gaagcctgac    60
aggcgggcat aagatcccag tcagcaaacc aagcaacctt gcagaaatgg aagacgagac    120
gagacaagca aataaccaac caaccttgca ggcagtcgca gttctcggcc tcctttcgca    180
ccacg                                              185

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<210> SEQ ID NO 478

<211> LENGTH: 201

<212> TYPE: DNA

<213> ORGANISM: Zea mays

<400> SEQUENCE: 478

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tcgaccgcag ygtggaatgc actggcaaca tyaatgctat gatccaagct ttcgaatgtg    60
ttcatgatgt aagtatatgt atacactctc agctacttct cttctccagg ttcccttcat    120
ccagacatgc atgttctaac ygcgcscctc gtgatccagg gctgggggtgt tgcygtgctg    180
gtgggtgtgc crcataagga c                                              201

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<210> SEQ ID NO 479

<211> LENGTH: 99

<212> TYPE: DNA

<213> ORGANISM: Zea mays

<400> SEQUENCE: 479

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gcagtcgagc gacgactagg aagaggaggt gatcgatgag aacggcgaga ttgtcaagag    60
gaagaagaag ggccttaagg agaaggtcaa ggagaagct                                99

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<210> SEQ ID NO 480

<211> LENGTH: 231

<212> TYPE: DNA

<213> ORGANISM: Zea mays

<400> SEQUENCE: 480

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cgcctccaca cggggagcac attggacgcc gcgtcacaag aaggtatata cagcatggtc	60
aaattgtttg cctctgttgt cgtgcactcg tgcctgcatg cggacagtgc ccatgattca	120
gtcatgttga gttgagttct gcttgccggc ctgtgatgtt atttgttctt gttcaatcat	180
atcgcaactg gcagtgtgtt ccgacgagcg cgacgccgcg gccagcggcg g	231

<210> SEQ ID NO 481
 <211> LENGTH: 101
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays

<400> SEQUENCE: 481

gcgagcacat acgtcgtgga cacattcgaa ggccgatatc atggcgttga cgttaccggt	60
gcactccacg ctgcggtcga cgccgccgtt ggtcagctca a	101

<210> SEQ ID NO 482
 <211> LENGTH: 201
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays

<400> SEQUENCE: 482

ggttgtcacc tttaratcg agtagactgc catcagtgcc atgacttatt ggcaarttgg	60
agtcttttag ggtccacttg tttccttga attctgcaag ttgaatttgc tttttgttg	120
acttcaatam tccaaccart tctcgaayga cbrccatgcy attcttcgcc ttgtgtcttc	180
tgagatagag tggagtcag a	201

<210> SEQ ID NO 483
 <211> LENGTH: 889
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays

<400> SEQUENCE: 483

gaatattcat ggggcgggta caagatggca ggtgaggaca ggcagggaaga tgaacaagac	60
gacaagagcc gacgtagtcg tccaaggatt cgatcaggat attcctccaa gatgacaaaag	120
accacaaaaa tagttgcttc ttcagacgtg aaccaatttt aaaactaatt gcataagagg	180
tgattaatta gcaagacaaa actattagct ctagttagct cttgaaggca tataataata	240
tagcatgata tgaactaatt atgggctaca tattcagaat taataggttt gtctcgtcaa	300
ttaatcactg tttacttgca cagtagagca agagcaagcg cagagaacag gggcataata	360
aactgctcgc ggtgggtgac ctaaaccatg atccaatatt acatcccctt atgctgactc	420
cagcagttca tcaaaacttt acccgtaaac aatgttttac actatagatt gcactattcg	480
taaaatagag tttgaatatg agtatgaata tgtataagct gttggagata gtcttaatac	540
ctagttaaat tactctagta taaacctcaa tctacatgta ttaaggtaga ttggagtgta	600
acttaaaacta atttatatcc caatccacct caacatacat aactatagtc tatcgtacaa	660
tatctaaatg aagcctaaag acaacgttta ggataaggaa cgtgttaaac agcccaaaag	720
cattatgact ccatgagtcc aacggcccaa tagcacgaga catcgcaaca tgcccgtccc	780
ccaacaactg tagcacctga tttgatttgt acaaaaatta tgatgacgtc ccatttgtag	840
acatcgagca tattgcctaa tacccttggt tgaaattcaa aatttgaat	889

<210> SEQ ID NO 484
 <211> LENGTH: 824
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays

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<220> FEATURE:
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<222> LOCATION: (14)..(14)
<223> OTHER INFORMATION: n is a, c, g, or t
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<221> NAME/KEY: misc_feature
<222> LOCATION: (36)..(37)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (171)..(171)
<223> OTHER INFORMATION: n is a, c, g, or t
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<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (374)..(374)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (392)..(394)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (415)..(415)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
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<222> LOCATION: (435)..(435)
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<222> LOCATION: (517)..(517)
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<220> FEATURE:
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<222> LOCATION: (568)..(568)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (608)..(608)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (629)..(629)
<223> OTHER INFORMATION: n is a, c, g, or t
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<221> NAME/KEY: misc_feature
<222> LOCATION: (651)..(651)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (666)..(666)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (672)..(672)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (681)..(681)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (696)..(696)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (711)..(711)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (749)..(749)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (752)..(752)

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<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (761)..(762)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (797)..(797)
<223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 484

catatgaaaa atcncgtggtt ccgctttcgt tttttnnnga tacctaataa ccaacctctg      60
catctgcttc aggttcgtac ttttgaggcg caatgtgggt cgttggcgca gtatgggatg      120
aagcacatgc ggtccttcgc aaacatctgc aacgctggca tccttcctga ngcagtgtcg      180
aaggctgctg ctcaggcttg caccagcatt ccttccaacc cctggagtgc tatccacaag      240
ggttttagcg cctaagaatc ataaggtgag gcgaaatatt tcagccgctc caccgcaacg      300
aactggttta cattaccagt cctcaggggg gtcctagtgc ttgaaccann nnnnnnnnnn      360
nnnnnnnnnn ctantatagc tgttcaccg tnnntccaga ttacatagct atgencaatt      420
tcoggtgtac atatnatagt cggaaagtta tttggcaatt gtattggtcg ttgctgtata      480
tattccctat agtttgtag cagatgtgta gtttgtnatt ccataaaaaa gaaggacgcg      540
ttactgctat ttctatgtag ccgactgntg ctcagtgtga actttacccc attcttgttg      600
ggaaatgnac tatcgtggtt ggaattctng catcgaaaac aattcccggg ngatccttta      660
ttcaangtga ancgtctgtc natttccatt tgaggntcgt attattttca nttgtaggct      720
tgtggctggt ggcattctgtt ccactatgnt tncaaacaag nncatggggc agtttcattt      780
gtttcgttct ctatatnctt gcagcaccta cccatggaaa aaat                        824

<210> SEQ ID NO 485
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<212> TYPE: DNA
<213> ORGANISM: Zea mays
<220> FEATURE:
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<222> LOCATION: (3)..(4)
<223> OTHER INFORMATION: n is a, c, g, or t
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<221> NAME/KEY: misc_feature
<222> LOCATION: (6)..(6)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (8)..(8)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (10)..(10)
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<222> LOCATION: (12)..(12)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (23)..(23)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (71)..(71)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
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<222> LOCATION: (92)..(92)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (105)..(105)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (145)..(145)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (151)..(151)
<223> OTHER INFORMATION: n is a, c, g, or t
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<222> LOCATION: (178)..(178)
<223> OTHER INFORMATION: n is a, c, g, or t
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<222> LOCATION: (189)..(189)
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<221> NAME/KEY: misc_feature
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<221> NAME/KEY: misc_feature
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<223> OTHER INFORMATION: n is a, c, g, or t
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<222> LOCATION: (461)..(461)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (464)..(465)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:

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<221> NAME/KEY: misc_feature
<222> LOCATION: (488)..(488)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (536)..(536)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (545)..(545)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (589)..(589)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (618)..(618)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (630)..(630)
<223> OTHER INFORMATION: n is a, c, g, or t
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<221> NAME/KEY: misc_feature
<222> LOCATION: (641)..(641)
<223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 485

tcnngnangn angtgtttat ttnacctnca gattccacct ttgtttcttc tatatttaca      60
tggtagcttg naggttcttg ctctttgtga tnatcgtgc ctctnggaaa aagaggaaac      120
caaatcattg ttcaggtgaa tatctcctc nttcaattta cagaactcct aaattcanag      180
atctagtngn tgatattctg ctatttttnc caacttnagt gtcaaagcaa cctaaaaaatt      240
ctaaaaaata cagagatagg gcaatctatc ttccttttaa atcaaagctg tggcattttc      300
tttgaaatta gtaaacattt atataaatag taaaatntcn tggagatcnn nnaggtantt      360
aacntttttc ccttcaactt ccacagtaat taaacatacc taggaanata gttttngagt      420
tctcatgttt aattgatntg ttcatcanaa gaaccattac ntcnntgcct aattatgcat      480
gcccctttnat ttttctctaa atttcccttg ataccatttc aagttgcaa gatgantttt      540
ttttntctcg tactgtttta tatttttggt agccataaac ttcaaaaant agttcagtg      600
cccatttata cataaatnct tatgtgtach tgatgggtcg nct                        643

<210> SEQ ID NO 486
<211> LENGTH: 201
<212> TYPE: DNA
<213> ORGANISM: Zea mays
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (189)..(191)
<223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 486

atztatgtgc ctyyaggcga accacttyca tcygaacctg gcgagccgtt gagagtgaac      60
gtgctmttca agcatattca ggcaatgctg tcyggcgaca tggctgtcat cgcagagact      120
ggggactcgt ggtttaactg ccagaagctg aagctaccgg aaggmtgtgg gtaagctcct      180
ctttcgaann ntgrttttgc t                                              201

<210> SEQ ID NO 487
<211> LENGTH: 300
<212> TYPE: DNA
<213> ORGANISM: Zea mays

<400> SEQUENCE: 487
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```

gcacagaact ctccoctgtc ctttctggg gttttggta cgtggtgga gtaagcttg 60
atttgccat ggataaagtt gttctaagct ccgtgggttg cttgagatct tgcgttatt 120
gcgtgccgtg ctcaactttt ttgcaatccg aggaatgaat ttgtcgtta ctcgttttg 180
tggattatta gcgcgaaaaa aaaactcttt ttttttgtt cttttactac gaaaagcatc 240
ttcttggatt ttgctatctt cttttactac gaaaaactct tgagtctagg aatttgaatt 300

```

```

<210> SEQ ID NO 488
<211> LENGTH: 193
<212> TYPE: DNA
<213> ORGANISM: Zea mays

```

```

<400> SEQUENCE: 488

```

```

ggctattgta caacaacaag gcggcattgc catcraagtt gcagtactas caattgccwt 60
cgkgattgctg gttgcgggtg tgcagtagta gtactgtagt actatcaagt gagtctgatg 120
catgamcaga gcagayggag tagttaaacg gaggcgcggg tgagcttgcc gttgacgatg 180
tggtggttgt cgt 193

```

```

<210> SEQ ID NO 489
<211> LENGTH: 201
<212> TYPE: DNA
<213> ORGANISM: Zea mays

```

```

<400> SEQUENCE: 489

```

```

rgacacggag gagcagtagc acttcgaccc gctggacgac accaagacgt ggccggagga 60
cctgctscgg ctcmgscggg tggggaggct ggtgctggac aggaacgtgg acaacttctt 120
caacgagaac gagcagctgg cgttcggscg ggggctggtg gtgccaggga tctactactc 180
ggacgacaag atgctgcagt g 201

```

```

<210> SEQ ID NO 490
<211> LENGTH: 199
<212> TYPE: DNA
<213> ORGANISM: Zea mays

```

```

<400> SEQUENCE: 490

```

```

atgaactatg aagaacccta tcactttgag ctagtctcaa gtccaagcaa agakaacaat 60
tcaccagtag taatggaaag agaagattag caataactgt tgtttagtgg agcaataaat 120
atctttttca gtttcaagtm ttaagagaga aaatgataga tgtaaatggc aagcacctac 180
tgtacttgac aacgttaaa 199

```

```

<210> SEQ ID NO 491
<211> LENGTH: 795
<212> TYPE: DNA
<213> ORGANISM: Zea mays
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (102)..(102)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (104)..(105)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (119)..(120)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (122)..(122)

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<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (125)..(125)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (128)..(128)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (130)..(137)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (142)..(142)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (169)..(169)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (193)..(193)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (198)..(198)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (203)..(203)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (207)..(207)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (344)..(344)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (561)..(561)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (605)..(605)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (670)..(670)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (677)..(677)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (691)..(693)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (702)..(702)
<223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 491

atagtaaaag gttggaatgt tagttgaaaa caaggtgtaa gaaatgtatc atcttttga	60
ctggacaaca gacctgaagt agaagcttct gggtttcctc ananntggcc ctcagcttnn	120
cncnnaangn nnnnnntcc tntgaaacga ggagtttgtc atacagtng gcaatgccag	180
gattaccctt ggngaacncc atntcncca gatcaatagt caccctgaaa aatggccact	240
cattgtacat ttcttgagc atgtggaggt tcctgatgtc cttctggagg acatgcttga	300
atgcggctcc aaatcctagc cagaccggga ggtggaaccg cgtntgtgtc caagcaaaga	360

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tccatgggat tgctcggagt gagtcgatac ctccgctcgg cttctctcttg gatggcctgc	420
ttcctatggt catcctacca tactctgttt caggggttgc ctgtcaacac aaacatcaca	480
aagtaaatac tcgctaaatg taggggtcaa atcgaatgct tcagtaacca ttttacttag	540
gcttgcaaaa gatagtttct ncagtaaatc attgtgctag attaatatta agtatcaggt	600
caganctatt tcaaaagttt gtttttgcac cttatgttac tgtagtatct ctagacacca	660
ctcaggtggn agcctcnagt actgggtctac nnnccatctt tnatgttact gtagtatggc	720
aatcgatggg atcaaggact tacaaggcgg aaatactcga caaagcgtgg ctctttgaag	780
acaatggacc gatat	795

```

<210> SEQ ID NO 492
<211> LENGTH: 627
<212> TYPE: DNA
<213> ORGANISM: Zea mays
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (73)..(73)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (170)..(170)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (182)..(182)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (282)..(282)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (348)..(348)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (367)..(367)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (396)..(396)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (486)..(486)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (492)..(492)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (543)..(543)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (570)..(570)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (591)..(591)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (599)..(599)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (613)..(624)
<223> OTHER INFORMATION: n is a, c, g, or t

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<400> SEQUENCE: 492

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atgctctggg gtcattcat gtcgaattc tgatccaatt gaagggtgt agctttagt 60
caatgttgct atntttatca tgcaaacccc ttacctttt gtaacatgac accaaacaat 120
tcatttccat tgggttggtc aaggatcatg tattcccatg caatacaacn tggagaagtc 180
angctattcc ttactatatt tacgtagcag tgttggtccc cgattatatt gttttaatat 240
ctgtacttca gttgaggaaa ttatatattc tgagcgatca gntaatcact tattttggtt 300
ccactacttt cctatcaggt tcttgagtct cggaatatcc agcttttntt gggatatatt 360
gtgagcngca tcaaagaggc tcctacatct gatgantcaa gtagcacagt tacatactct 420
gaagttgacg gtgatcatag gaaactaatt ttggaccttc aacctgctga aagaggcttc 480
aaagncaga cncctgatgc cgatttggtg ctgtggacag tgggttcaac atctcagatt 540
ccnccgttac agcctcctga tgctccctan gttattcctc tgaatggctg nggacaggng 600
gagacagagg aannnnnnnn nnnnaaa 627

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<210> SEQ ID NO 493

<211> LENGTH: 772

<212> TYPE: DNA

<213> ORGANISM: Zea mays

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (52)..(57)

<223> OTHER INFORMATION: n is a, c, g, or t

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (59)..(59)

<223> OTHER INFORMATION: n is a, c, g, or t

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (61)..(62)

<223> OTHER INFORMATION: n is a, c, g, or t

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (68)..(68)

<223> OTHER INFORMATION: n is a, c, g, or t

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (70)..(70)

<223> OTHER INFORMATION: n is a, c, g, or t

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (73)..(76)

<223> OTHER INFORMATION: n is a, c, g, or t

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (88)..(97)

<223> OTHER INFORMATION: n is a, c, g, or t

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (99)..(109)

<223> OTHER INFORMATION: n is a, c, g, or t

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (112)..(113)

<223> OTHER INFORMATION: n is a, c, g, or t

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (115)..(119)

<223> OTHER INFORMATION: n is a, c, g, or t

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (122)..(131)

<223> OTHER INFORMATION: n is a, c, g, or t

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (134)..(145)

<223> OTHER INFORMATION: n is a, c, g, or t

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (134)..(145)

<223> OTHER INFORMATION: n is a, c, g, or t

<220> FEATURE:

<221> NAME/KEY: misc_feature

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<222> LOCATION: (147)..(150)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (153)..(153)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (160)..(160)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (163)..(166)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (173)..(173)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (180)..(184)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (233)..(233)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (376)..(377)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (425)..(427)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (446)..(447)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (508)..(508)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (526)..(526)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (531)..(531)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (564)..(564)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (581)..(581)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (621)..(621)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (624)..(625)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (651)..(651)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (656)..(656)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (678)..(678)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:

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<221> NAME/KEY: misc_feature
<222> LOCATION: (696)..(696)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (707)..(707)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (709)..(709)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (715)..(720)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (723)..(723)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (725)..(725)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (729)..(729)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (733)..(734)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (736)..(736)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (751)..(752)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (754)..(754)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (759)..(759)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (761)..(762)
<223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 493

gcatagcata ctgcatgcat ggcgctgttg cattgcaggt tgcagctgca gnnnnnnncng      60
nngaccncgn tgnnnncgtg cagactgnnn nnnnnnnann nnnnnnnnc anntnnnnnt      120
annnnnnnnn ngtnnnnnnn nnnnnnnnnn tanataaatn tannnnagta canttttgcn      180
nnnncctgct gcctgcccta gcactagcag ccaggcagca ggcgtggcgc ctntgctgcg      240
gtgcatgaaa aggcgacgag caaagaaggg gcgcgtggaa aactgcccg cgcgcgctgc      300
taccgcgcta tgattggatc gaggggagcg tgcgggtggc ttttggatc ctttcggcac      360
gtcaagaaat aaaaanntaa ccagttcctt tatttcacgc gtctccgact ctccaaggcg      420
gcagnnncag caagctctcg tagctnngtg taaagctgtg gagcacacag cagctgtgag      480
cctgtgttgg tggtgcgagc aagagcanag ctggcaagcg cagcanagga nttgcacacg      540
gctttatacg aatgagcaga acanaggcat ctgtacatga ngacgggcac ggcacacaca      600
caggcctttg gttcgttgga nganngtgta ttggcctcct actatactga nactgnacac      660
ccctggacat ccgcatgncc gcactacta cgcattngata gcagcgtntnt gtaannnnnn      720
tgnantggnc tcnnangcag ccatgccatg nnanccagna nccccacagc ac              772

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<210> SEQ ID NO 494
<211> LENGTH: 201
<212> TYPE: DNA
<213> ORGANISM: Zea mays

<400> SEQUENCE: 494

```
cacacataat tactccmatc ttctcagtgg aggtacatca gcaaccgagg ttataaagg      60
aacrcitygag gacaacacga tgggtggcggg gcatagattg ctctacgagg gctctgagga    120
ggcggttcac aacggaggga tggtygtgtc caagatcgcc cacargaaca tcgtcagagt      180
tctggggtgc tgtctggaac c                                                  201
```

<210> SEQ ID NO 495
<211> LENGTH: 220
<212> TYPE: DNA
<213> ORGANISM: Zea mays
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (25)..(25)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (40)..(40)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (84)..(84)
<223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 495

```
tggatccatc acggttagagc tgacnatgat atgtccaaan gcttctgctg ctagtgttg      60
gtcgtccgca agagacttga cccnacaggt ttttttcgtt gggcctcaag aactgtaggt    120
gcattttcca tggagggtta tgatgcttaa cccactctg gtttctgggt gctaaaacat      180
tctctccagt cggttcttac catgagccct ggcacttagt                             220
```

<210> SEQ ID NO 496
<211> LENGTH: 301
<212> TYPE: DNA
<213> ORGANISM: Zea mays
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (59)..(59)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (76)..(76)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (91)..(91)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (111)..(111)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (181)..(186)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (203)..(203)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (282)..(282)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:

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<221> NAME/KEY: misc_feature
<222> LOCATION: (292)..(292)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (294)..(296)
<223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 496

tggttttatc cgctcggatg ccccgccatg tgctccttgt ctctgggactc ggagacagnc      60
tgacagtctc ggggtnttgt gctaattggc ntggttaact tacgttaatc ntgcgcctta      120
ttatcatcat ctagggtgtaa ttacttcctg ttgtgctaac cacctcgtga ttgcctgcaa      180
nnnnnnngaa ccccgacccc ccnaccttgc tgatgccgtg cttttctaga gaagtaccct      240
gccgcgtcac gcttcgtctc cgcttcttt ggattcaacg gntttttgga gncnnnggcc      300
a                                                                           301

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```

<210> SEQ ID NO 497
<211> LENGTH: 170
<212> TYPE: DNA
<213> ORGANISM: Zea mays
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (105)..(105)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (131)..(131)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (138)..(138)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (140)..(140)
<223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 497

ggtgccggaa gccgcctaca gcggcccgcc gctcagctac tacgtcacca agttccagcc      60
ggcgggtggc ggcgcggcgc agaccctcga ggcccccgcc ccgncgagg cacaagacgg      120
cgccgcgcgc nccgtcgn gn ctccggaggt accggcaccc cagctgtcgt      170

```

```

<210> SEQ ID NO 498
<211> LENGTH: 190
<212> TYPE: DNA
<213> ORGANISM: Zea mays

<400> SEQUENCE: 498

rcataaaaya ttgcaaatgc tgaggaatat aagcacatag atacttagca tgtgtccgcc      60
agatggaat acttttacc tccccccacg actggattgc cagaaaggtc aatgcctatg      120
acaccttggt ccatyatttc cayggctaga ttaacctggc caggggaaay aaatsathtt      180
gtgcagattc                                                                           190

```

```

<210> SEQ ID NO 499
<211> LENGTH: 201
<212> TYPE: DNA
<213> ORGANISM: Zea mays

<400> SEQUENCE: 499

acgsygtaa ggcctacttg catgttcctg aggttgcctc tgacaccctc aatttgctgg      60
aaatcatccc catctrggga cataaatgga gcaaaattgc cgtgagcttg aacaaatgga      120

```

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aggtcactgt gatggttctg aatgccatca ggcagattct gtgatgtgtt tgettctgat 180
gaagaaggga agttccagcg g 201

```

```

<210> SEQ ID NO 500
<211> LENGTH: 201
<212> TYPE: DNA
<213> ORGANISM: Zea mays

```

```

<400> SEQUENCE: 500

```

```

tctcaagctt atttaggttg actaaatttg gaacatcccc cgagatgcta ttgccggcaa 60
gcgcgagctc cgtaagcatg gcagctttgc caatcgatgc cggtagggac cccttcagct 120
ggttactctg caaccgaaga acartgaggt tcgagaatga gtcaagccag tctggcacac 180
ttcattgaa gtagttacca t 201

```

```

<210> SEQ ID NO 501
<211> LENGTH: 294
<212> TYPE: DNA
<213> ORGANISM: Zea mays

```

```

<400> SEQUENCE: 501

```

```

tactagcaca ctttgctcc tgatagcttg taccattgtg accttgtttg gatattggga 60
ttccacccaa tgtcaaacat acccaaacag tgacccgka tctttttctt cgtatttcta 120
tgtcaagcac tyaagcataa ctcatcatt tgccctctaca ggtgtttctg aaacactagt 180
gtcaaatatg tcaactctga catggtcagt tgtagcact gtggaaggac gtgttcctct 240
tgccagttag gtatwtctct tgagcgggtg agcagcattc cagttgttaa gatg 294

```

```

<210> SEQ ID NO 502
<211> LENGTH: 214
<212> TYPE: DNA
<213> ORGANISM: Zea mays

```

```

<400> SEQUENCE: 502

```

```

gcacaggtaa ctcttgcatg gcacctgtcc tgaaacaagt ttcgtgctaa caagggaat 60
tatecttttt cgtggcacca atatctgttt tccattcaat gatgtactga tgtgttcacg 120
gagtcgttac ctagaaatct cgcttactgt gacatatcac accgagcattc attttatatc 180
tgctgtgygc cctccttgct aatgtaggca cctg 214

```

```

<210> SEQ ID NO 503
<211> LENGTH: 300
<212> TYPE: DNA
<213> ORGANISM: Zea mays

```

```

<400> SEQUENCE: 503

```

```

rytkcaacmt gcgagaarta raaaatcact aaccttggtt ttatcaagta tagcaagggtg 60
aamcgcaatg ccargaagaa gcagagtccr ccgacgrtga ggtacgcaag gccagaaaa 120
tcgttctttc ccccgagcca ggttgcggtc gaaagcacca gcttcttctt gccaccaaag 180
ctataggtgt tgtagttgtt gtccarctgg actgtgattg trtcgttttc cttgagatca 240
acatatatcc ttccatacag ctttctgaat gtcggaagtg ctgyagtccg catccaaaca 300

```

```

<210> SEQ ID NO 504
<211> LENGTH: 287
<212> TYPE: DNA
<213> ORGANISM: Zea mays

```

-continued

<400> SEQUENCE: 504

gcgctcgcca tgaatgcgcg tcgtctccrt tggaccatca accatttgca attgatgagg	60
agacgagaca ggagaaacgg cgagggcagy agtgttctgt agctgaaagc tgccgcgaat	120
cgactgccat tattggacgc catcccttgc ccgactgaga atcggggcggc cgcgtgcaat	180
gcactrcmct accgccgccc gccaggcgcc agccrrcctc gtggtacaaa attcatccag	240
aattcgcacg aggccaccag ctgggtccg gctgcgagg atttcaa	287

<210> SEQ ID NO 505

<211> LENGTH: 288

<212> TYPE: DNA

<213> ORGANISM: Zea mays

<400> SEQUENCE: 505

tcattggaatt gcttcaaacc cgagagcggc aagataatag acgaacgac agctccagcc	60
acctgtaagt acaatcacaa ayrgtaagag caatggatca ctygtggagg cttgtgttta	120
caaataatrg ccaacaacat gttacctcag aaatcctcaa ataattggccc ctgttgttgc	180
tgcccaggtc gaagaagaac cttttctgat ctgccctrat maccttagac ataccatgc	240
cactaatttc ttcttgagga ggcagatcaa ccaagcgttc sacatcta	288

<210> SEQ ID NO 506

<211> LENGTH: 284

<212> TYPE: DNA

<213> ORGANISM: Zea mays

<400> SEQUENCE: 506

tayaagcaat attaagagat ggaaatgtcg cctctgtacc acaagttagt aagttactaa	60
cttgattgtt aaaacacccat cgaaatgtgt gacatgtagg acctggcccc acatgtcaga	120
caaagccaat gctattttgg cattcatgtt acaatgtagt gctataatct ccacaaaaaa	180
wacaagacag gtcaaatact gygtaaacat tcgatccaaa aaacagagat gctttgtcat	240
aaaaaatcta gccaaaagaa ttggagctat tgagatacaa tgaa	284

<210> SEQ ID NO 507

<211> LENGTH: 256

<212> TYPE: DNA

<213> ORGANISM: Zea mays

<400> SEQUENCE: 507

acgttccttg tcgaccatta tcattcgcta tctatttttg aacgaacgca ttcgctatct	60
atggccacat ctccacttcg tctttgccga agcttcagta cgcgggtcgg gtgtctctg	120
gtgtgtatgc tgttttttc tcggtccatc ttatcccccg tttgacgcgc ctctgtcaat	180
ctgctgtact gtgaatttta tttgacgtgc ataaatyatt ctggaaacgc tcttggtacg	240
tcacgtacct ygagaa	256

<210> SEQ ID NO 508

<211> LENGTH: 267

<212> TYPE: DNA

<213> ORGANISM: Zea mays

<400> SEQUENCE: 508

atcsggcacc tgatccwga tgtgagwscg agtgcgacct ataccgctcc ccaaccgccc	60
ccattgaatc caccggtgaa tctatcgctc cgccaccacg aggccctata ttaactccac	120
agcatccatg tgtccagacc tgtctgtata cctgtcactc acgctaccgc cgtgccgac	180

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```
gttcgttctt tccctccctt cgcgggccgc gcgcctatkw twtttactac tstattorta 240
tcattatayt ktttggtttc cwtccck 267
```

```
<210> SEQ ID NO 509
<211> LENGTH: 297
<212> TYPE: DNA
<213> ORGANISM: Zea mays
```

```
<400> SEQUENCE: 509
```

```
ttctcgtaat ctctatggaa attrccrta ctagagataa atggcatcct tgcaatgtcc 60
caaattctca ccgaagaaca caagkaaaag araagaaaat attgagcaag cacagcacag 120
gcaggcagca tgaatccaca cagaaccaat ggttttttagc acagyasgyt gacagccaag 180
ggttacagta cataatcaga cggggcaaca cgaacgaagc agccgcatca rcagcagcct 240
asaaccatt aattgacaca cagcgtatat atatatatrt agtacttgtc ttttagc 297
```

```
<210> SEQ ID NO 510
<211> LENGTH: 231
<212> TYPE: DNA
<213> ORGANISM: Zea mays
```

```
<400> SEQUENCE: 510
```

```
aagtcagat gactttcaga aagaaaatcg agaacttcag acaccttctg accgctgaag 60
cacgaacgac aatggccata cagcacgatg ttgcacgttt ygtagaagtt tgtatcttcg 120
ttaaaaaata tagcagtaca aatsatagcr cattctgtag gtgtaggata cagtaacaca 180
gcagcagaac atcaaacccr tctcctgcc aagcctctgagc aacgccaaag c 231
```

```
<210> SEQ ID NO 511
<211> LENGTH: 301
<212> TYPE: DNA
<213> ORGANISM: Zea mays
```

```
<400> SEQUENCE: 511
```

```
ccrtrcaaca ttgastcgtt ytgtttctat ttsttgacca agacatgggt gytgttctga 60
tggcagattt gaacctgaca tgettgcyyt ttctgggtgc tattatgggtg gtggagaaaa 120
ggagaggaga gaattagctg caattcgag gcgatggaga accttgcatg tacaagctgt 180
gtgatgtgta atcattactc gcttactgag atagaattay cttttgctaa aagttttcat 240
attctagata cgtgaccgag agaaaggaag acggcaaggc aggtgtccac taactcctga 300
g 301
```

```
<210> SEQ ID NO 512
<211> LENGTH: 282
<212> TYPE: DNA
<213> ORGANISM: Zea mays
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (1)..(1)
<223> OTHER INFORMATION: n is a, c, g, or t
```

```
<400> SEQUENCE: 512
```

```
ncgcgtgttc gtgcggatgc waykcatgca gaggyagcar arctagctag yaagcacgwa 60
ckyacgtagc acatgataag aaggctgcrt tgagacagta agacgaagaa tggcargcag 120
aagagcacgt cggcatgctc cccgcggctt atagcttaga ggcacttgaa tccggtgggc 180
ackctcttgc cgcagtggtt gaggatragg ctgaggtcga csggyaggtt gagcttgatt 240
```


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cckaggacct ctcccttgat ggcgggtgcag aggcacacgg cg 282

<210> SEQ ID NO 513
 <211> LENGTH: 301
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays

<400> SEQUENCE: 513

atggatgctt ytgtagcatt aaacgtgaga gtttaatcts accacaacaa gcatgggtga	60
tcacaagaaa tacaaaacag ccaactaacc aaamawkata ttagcaaatg caacttactt	120
ataacatggc aacaaaacac aaacacattg cgcacaccta tgctggctta gtctatcctg	180
cattagaccc aatgcagtga attgctagtg caagcaacca attgctcccg tagtgtttcc	240
ttttgcttct atgaaggcta tgaactgtca ggggtttattc tagtgatcta tttatctayc	300
a	301

<210> SEQ ID NO 514
 <211> LENGTH: 216
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays

<400> SEQUENCE: 514

tggaaacrrt taataattga agttttctya gtacagrtct gcatagcatt gaactggaca	60
gcttgctgtg ttactctgtr taasgaacga tctactgctg atctgtactg ttccttgatt	120
tttyyyycwc tttttcttct ttgatggcaa gcaggactga agataagatg gctgccttrc	180
cattggccac cgcagaagca tgtgatgcta atgctg	216

<210> SEQ ID NO 515
 <211> LENGTH: 301
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays

<400> SEQUENCE: 515

ggaaatcttg acaaggatat ggagagaagg tttagctacg ccctaagcag agaagacatc	60
gagaacgcca tactcggagg accttaasct gaacacatgg ccaaggggct cakagctagg	120
aattgcttag tcgaggttgg cactgaagaa gtaatcgctc cagtgatctg cgtgctacca	180
ttacgccatt gtaagagcaa atggatcatg tgtcgagaaa taatattcat gagcaataat	240
ctcgaactag gctggaaatg ttggawgct cctggtgaca tgcgytyggt ttgttggcmt	300
c	301

<210> SEQ ID NO 516
 <211> LENGTH: 301
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays

<400> SEQUENCE: 516

mtaggaaggt caaggtgcct ttggaccccg gcagcaacta caccttcgac ctgagctact	60
accgccacgt rctcgccacg ggcgggctgt tccagtcga cggcagcctc ctgcacgacc	120
ccgtcaccag aggetacgtc gagaagggtg tcaaggcgct gtcgcccga gagtactacg	180
cggacttcgc crcggcgatg gtcaagatgg gccgcaccga cgtgcttgct ggcgatcatg	240
gggagatcag gccaacgtgt ggcatttttg ttgactaggt tcaggaktgg gttgaaratg	300
c	301

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<210> SEQ ID NO 517
 <211> LENGTH: 294
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays

<400> SEQUENCE: 517

```
gtcaacaagc accagacttt ccaggacact cggtgctttt tcgtggtctc cacagatgga      60
tcccaggcag aytctctcta cctgaagtgc ctggagaact ttgtgaggaa gagctacacg      120
gaggacgysg acacattctg catgaagtat ttaaggcccc gtgcgaggca ggcaccacca      180
gctgatgttg ggacagcatc aggcrcctcg gatgaggttc caccgtcaac cgcagctgag      240
acagagcaag gcactcctcc agcccctcag gcagaggttc crcaagagay ttgg          294
```

<210> SEQ ID NO 518
 <211> LENGTH: 285
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays

<400> SEQUENCE: 518

```
attacaccwa acttgtcctt ataagttaga aytctacaa tatttattag actctgtagg      60
agaatctctt ctagatacaa acccaccatcc ttactgtaa aaaaggccca ttgttagcta      120
caatgactta acattttacg agggacccaaa tctctggtct gaaacccaaa ctttcaaggt      180
tcactaggtt cgtccactgg cagataaaac aaatcctcat ttaygatcat aagttgacat      240
actggacaaa gaatacttgt aagacgatcc cttcttatct gatga          285
```

<210> SEQ ID NO 519
 <211> LENGTH: 297
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays

<400> SEQUENCE: 519

```
agtgtagaga tactcacaag aacaattttg aactgccttg cagagccagg taccacgaat      60
aagtgttcaa ccttactctc atgcctcaat ttcaagaaca cctgaacaca ttgctaaat      120
gatgtaagaa aaaagttgcc atcatggtat cagatacgag gtgagtagca acctggtgag      180
gtttaaatgt gtgcccmagc ggtgtgstta gttgaaaaga taaacrcagt ttctgcagat      240
ggtttgcaga aagagacact tttgtatctt tcagaagatc taacctgaaa tgaagca          297
```

<210> SEQ ID NO 520
 <211> LENGTH: 301
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays

<400> SEQUENCE: 520

```
ccatgcgcta ctaccaggcc ggytcctcgg agatgttygg ctccacgccg ccgcgcgagc      60
gcgaggacac gcccttccac ccgcgctcgc cctacgccgc cgccaaggtc gccgcgcaact      120
ggtacacsgt caactaccgc gaggcctacg gcgtattcgc ctgcaacggc gtgctyttca      180
accacgagtc ccgcgcgcgc ggcgagaact tcgtcacgcg caagatcacg cgcgcgctcg      240
gccgcatcaa ggtcgggctg cagaccaggg tcttcctcgg caacctctcg gccgccaggg      300
a                                          301
```

<210> SEQ ID NO 521
 <211> LENGTH: 255
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays

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<400> SEQUENCE: 521

tacatacctt gcagcatgtc aaatgcacaa ggatgaagag agagggaaaag ttgcagcaaa	60
gaaacttggt gagatggaac cacaaagctc atccacgtac gtgttcctdt caagcttrca	120
tgctgcggct ggtaactggg ttgaagccaa ggtagccaga gaagcaatgc gagaaaaagg	180
ggtgatgaaa ttccagggt gtagttggat cacagtgggk aacaaacara gygtatttgt	240
tgtacaggac acaca	255

<210> SEQ ID NO 522

<211> LENGTH: 299

<212> TYPE: DNA

<213> ORGANISM: Zea mays

<400> SEQUENCE: 522

actcaaatg agtcaagaaa tgctaaggga aagcccgta aacaatataa tcgaactttt	60
acattttatt gatgtggcct tyttaaaaat gacgccataa accwctatac tgaaaacggc	120
ctcggcgctt aaaccagtag cattcgtaag ttcatttctg atttgggact tcaaaccaat	180
agtgtttgag actcacaggt ttagaaccga ttggtggctg ccttcaatca agagcgaaac	240
caacatgtta ttgccacct taaaatcaca gattttgtat tatcgtgtac aataaaaat	299

<210> SEQ ID NO 523

<211> LENGTH: 301

<212> TYPE: DNA

<213> ORGANISM: Zea mays

<400> SEQUENCE: 523

ycagatcatc actggaatca acagcagttc tctggttcat ttgttcagat gcttckgtgt	60
caactgcttc tgaaygatgg tttaagctta taacatcatc arcatttttg ttaattattg	120
cactgttctc gttgtccttt accacgctat tcaaawtttc tggctcctca gatactacaa	180
gatcagatrt ctgcccgttc tgagtaagag tggtttcagg actgacaccc atacctgtg	240
caagtgttac atacogaagt actagwtgct tgtacgaatt caataggctc tctacatcag	300
c	301

<210> SEQ ID NO 524

<211> LENGTH: 274

<212> TYPE: DNA

<213> ORGANISM: Zea mays

<400> SEQUENCE: 524

cttttagtagc tagtcytata gctcataggt tctcagttcg gtatatrtyg gtgayatatt	60
tcatgaactt asttttwaam cmatttataa mgcamacgca acaacgaagg gagtgatatt	120
gaccatgggt ctttcgtagc ttttcaatca aacgtcacat aaaaaaagaa acawccaaaa	180
gwtcataata cacagattkr aamskkkaga caaaaaatcg agtacaagac attggacctt	240
caatwwtttt tttktgtttt ctgagacatg aaaa	274

<210> SEQ ID NO 525

<211> LENGTH: 319

<212> TYPE: DNA

<213> ORGANISM: Zea mays

<400> SEQUENCE: 525

ttgcaccacc catagttaa gaggcagttg ctttcacaca ctttattttg catggcacag	60
ccaactgtcc ccatcattct aaaccaggct gcaactgagc cactacagaa actgctaata	120

-continued

ttagatattc cagcaaatag tcttgacact agagtgctaa gtcctaaac acgaaagaca	180
tttgtcagga gaccogtatt gactactagc agagtatact accgctagat attccagtat	240
aatcgtgcag cagctccggt caatggcagt ggcacttgaa gtctgcaacc tcagccaaca	300
gatccacatg agaaccaac	319

<210> SEQ ID NO 526
 <211> LENGTH: 283
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays

<400> SEQUENCE: 526

tacactatac taattagaat gtctggaaag gctgggaaga ccactaaatt tgtctcattk	60
ttggaggaaa tgggtccora aggatgtgtt cttartctga ttgcttataa tactgttatt	120
gaggctcttg gtaagaacaa gatggttgac aaggcgattt ttatgctttc taaaatgatt	180
gagagtgact gtcggcccaa tcaattcaca tatagcatta tgctggatgt tttatcaaca	240
gggggacaac tccacaggtt gaatgagatt ctagatattt gta	283

<210> SEQ ID NO 527
 <211> LENGTH: 193
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays

<400> SEQUENCE: 527

cctttctcca gttggtaata tgcttcatgt aaggtacctg aaagtggtaa gcracatgat	60
agtcaaaactc ccactcatta ttogaggcct gcgacactta gagacactcg aggtggatgc	120
agaaraagtc gctgttccac tggatgtttt catcttgaag agcctgttgc atctccgact	180
tccgagcaag gct	193

<210> SEQ ID NO 528
 <211> LENGTH: 298
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays

<400> SEQUENCE: 528

tgcatatgag aaactacaag gttggcaggg tcctttcctg tgcccgagct gtcataaaaa	60
gaaagaagca atggaaggga aacgccggcc aaaaggatc attaaatagt atttacctga	120
actacattct ktagttaacc ctgagactgc tcattcgttt ctgggttccct atttgtcttt	180
ccaggatctt catcaaacgt ctttggtcac atgtgctagg cctgcctacc actaaagatt	240
cttttattga ttsttagctg gaaataatag kcagcccata acaattcatt ggctagaa	298

<210> SEQ ID NO 529
 <211> LENGTH: 360
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays

<400> SEQUENCE: 529

aagcaccgc atgcagactt ccaactatc cagtgtccac aaatagcagt ttgcaaaact	60
aaatacatat atgaaatgat gacacaggcc acccaaaca caagccctag agaatacagac	120
caaccaaccg agcaattcgc agtgctccat atcttctagc attcacaacc tgacgaacgc	180
tgttgccttc cagcgccccc aacatcaatc gtatcaggaa gatacctgca aaagttggta	240
aaacatagat caagttacac cacaaggctg cctgcatgta tgatacatat atgtgagcta	300

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tatactattc aaacttacat atcttcttct ggctcatttt ttacagcatt cctcgtata 360

<210> SEQ ID NO 530
 <211> LENGTH: 299
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays

<400> SEQUENCE: 530

agaygtact ctgctaccat ttgagaatc tattgacaag aactttacag gaggrgtctc	60
atccccagga gccttgctga taactggaga tggagaagat atgcacatat ccactcgtga	120
atgcytttct ctggacttat ttctgtcctt tagcagttgt ttccctagc aaacaaaagg	180
agatacacat agcatgagaa agaagaaasa aactgaaaca tttataatg agaaaaaac	240
atggggagtt ggggacaaag caactacca atgcttctac aaatgttttg aacctccga	299

<210> SEQ ID NO 531
 <211> LENGTH: 262
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays

<400> SEQUENCE: 531

tgcttctgtc gaattgtgc gggcagarca tgctttggtg gyggctttgc agtgtgaaac	60
gttatgaatc tgagaacctc gatgcatgcy agggacaggg aatacctgaa caggggccaa	120
gatgacatga acaggacgcc ggatagcacg tgcacaacct tcctgtcaa gctctgcctt	180
cattgatggc atggatcagt acagagttcg gaagccaacg acgtgaacar ctgaacataa	240
gcaaatgtaa tggcgtttct aa	262

<210> SEQ ID NO 532
 <211> LENGTH: 301
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays

<400> SEQUENCE: 532

aggtcgacaa ggctgctagt aacatgtctg agccgacatc aggggagatg caggatgctg	60
ctttccagtc tgacgaagag gaggaagatg aagatgttga tgaaacagta ttcggtaag	120
attcagatc gtcacaraat agcggcaccg gcgacgatgc aaagtagact cactgctggt	180
attcatcatt atggtgattg tattgttttt aattaaattg ccaagcttga tttttgtcaa	240
ggcgacgytg gaaggttgca cagaattttg ayagtgtctc tggtttcatt ttgaaagcac	300
t	301

<210> SEQ ID NO 533
 <211> LENGTH: 271
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays

<400> SEQUENCE: 533

agaaaacgag gccttcgtcg tcttagcgcg crggggcgta ggacggcggc agcggcagcc	60
ccacgacctt gccgcagcgg agggcggtgt cgggcatgtt gtactcgtcg cgcagtgatc	120
gcgcggcgca gacgacgctg atatagaact gctccccgag gtagtaccgc tcccagatgt	180
tggaatgcac gctccacatg cccgcgttgt cgaacgtcag catgattgcc gtccatgacc	240
gcgggtacac ctggatcgtg tgccggctca c	271

<210> SEQ ID NO 534
 <211> LENGTH: 286

-continued

<212> TYPE: DNA

<213> ORGANISM: Zea mays

<400> SEQUENCE: 534

ttctggtact tgcgcatctc ggcagmgcc acsgtscgg gcctgtacct gtggggcttc	60
ttcacgccgc cggctgtcgg cgcmgacttc ctgcgcgcct gcaacraray aacaaccgcc	120
gccgcgtcag crccgtctta caacgggaac aatcgagggg atcggaacac cagatcggtg	180
cgggtaggta gagaacgaacc ttggtkcgca gctgcttgcg cggggccttg cctccggtgg	240
acttgccggc ggtctgcttc gtacgagcca tcttctcttc tccttg	286

<210> SEQ ID NO 535

<211> LENGTH: 252

<212> TYPE: DNA

<213> ORGANISM: Zea mays

<400> SEQUENCE: 535

ctgcgttagt ttcacccctt ctagctgcga gtgaaagaaa cwtgataaca gctwsctgct	60
agtttctatg ryrgecatcg aatctgacat ggctayctcc tgtgmcmacg cagggtccgca	120
ggtactacca gccgaggaag agccaacgga cggtgacggc ggtgatccac ggcgagaagg	180
tgccgctgta cggcgccggg ggcggcctga cgtgtcsac gagcgcgggc gggggcgcg	240
tgccctgac gc	252

<210> SEQ ID NO 536

<211> LENGTH: 301

<212> TYPE: DNA

<213> ORGANISM: Zea mays

<400> SEQUENCE: 536

agcaccacga ttcacaccta cacatattca gacacaaaat tgggcagcga tgggcatgga	60
catttcattg atagatggta tttagctgcc tactytggca tttgattcga gcagcgaagc	120
ttcaagagct gcagaatcgc tacayatatt cctgtgggta cacatgcttc ttgttctcca	180
ctcgaccag atcattgtga tggactacta actgscagta caaagcaga catctatcaa	240
acttacacaa aaggcgtgca caacgagaya gyctgcttgc gcatctccaa gtctagtcta	300
g	301

<210> SEQ ID NO 537

<211> LENGTH: 237

<212> TYPE: DNA

<213> ORGANISM: Zea mays

<400> SEQUENCE: 537

awwttkkwag ygcarraata aacaaggcac tgtwrattgt acagagcaag ctctctgcat	60
ttttcaatgc cgcagcatct gcaactgcaa gcagagtctc tacaggccaa ggcgactgt	120
cctgccatct ctctgcgga ggctgctat gtaatgttc tccaggttg tgctgagctc	180
ctccagtttc ttggcttctt ccgccgagct cccgaccttg ttctcgaccg tgtttaa	237

<210> SEQ ID NO 538

<211> LENGTH: 278

<212> TYPE: DNA

<213> ORGANISM: Zea mays

<400> SEQUENCE: 538

tcaatttgtg atttcacca caattctctc atatgtgagg aatgccgcac ractttctat	60
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gaaatgtgtt caargtttgg aaaactact ctagccatat attgccacag aatcaastg	120
catgtytaat tgttaccgtt cacctgtagt gtatcagtc cttcaaatac tgcccgtgac	180
cttgccctct ggctatggc cgcgcataac aaagttaacg agaggttgat gaaagaagaa	240
aaggagttag acaatgctga tcttcattt cctaagat	278

<210> SEQ ID NO 539
 <211> LENGTH: 263
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays

<400> SEQUENCE: 539

ttgcatacaa acttgatcma aatgtttrca amccatgctt gaacrratat aaaaaccgt	60
tgyggcagga acatyttagc ttwgkammmt ygywvagca caatagcttc aatagagatt	120
gttgaagtgt tagagtgtgt ctgcatgga ractatgcaa atcacaaggg atgggagaag	180
agtgacattc agaaggaggt cctgcccag ataaccggt gcagtaacg tggcatctgc	240
aactacacgg gccaacgtcc cgg	263

<210> SEQ ID NO 540
 <211> LENGTH: 266
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays

<400> SEQUENCE: 540

acaagtccaa atcaaagggt ttgtccgcc aatgettacc cccatgscy cccraaata	60
gcacttgaat ccattttttt gtcygtctt tttkkctcct cttctgacag ggaaatgtg	120
cttgagttgg acgaggaaca gactggaata caaatgcacg tgcagggtgc aaattcttat	180
gttrwttttt gkggocctatc artcacagt actcatagca tgacaaatgg cgttgaggt	240
tttggttttt accsgttttg gtttcc	266

<210> SEQ ID NO 541
 <211> LENGTH: 197
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays

<400> SEQUENCE: 541

aacctcgact ccactataas tgcaggacgc acgrcaygca caactgaaac atccggacca	60
acaggcagag tagtagccca aaggatgacc cgtctcatgc aaaaacagtc agcttgataa	120
acttatcgag aaggcaacac cgygcccga gtactggttg aaatacagag cagacattga	180
agtgccaaag agggggg	197

<210> SEQ ID NO 542
 <211> LENGTH: 301
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays

<400> SEQUENCE: 542

gtctagaaaa gtcaaacgt catatacttg gaacrggagt agatatgaca aaaagagcat	60
gcaagtgtgt ttcttacaga agaatatata tatttcatgg caagctagta cctaaacttc	120
agatggagtc ttgtcatcga ttgacagggc gtgcaaccgg gtcttgagct cctcctgcag	180
gagatcatag acagccctgt gctctttag caagctcttc cctcaaaact ccttgacac	240
caccctcaca ttgaaatgtg tctccccatt ggtcccagcc acgccagcat ggcccttgtg	300

g

301

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<210> SEQ ID NO 543
<211> LENGTH: 291
<212> TYPE: DNA
<213> ORGANISM: Zea mays

<400> SEQUENCE: 543

at tt t t a c c a t	a c t g c a t t c a	a a g t t t t a c a	a g a t t g g a c a	t t a t t a g t a t	a a a a c a t g a a	60
a c t t t t c a a g	t g a a c a a s a a	c t a t a r c a a	c g t c c g c a a g	c g c a a t c c a c	a g c t t g a g g t	120
t a c a t t g t g a	g t g t a g t a g a	g a a a c t a g g t	g g t c c c a a a t	a g c a c a a g a t	t a t c c a a g c t	180
a c c c a g t t t t	c t a c g a t t a t	g g g c a c r a c c	a t a a g a g c a t	a g m a g t g c t t	c g c a t c c t g t	240
t g c a t t a a g a	a a g t a g t a t a	g g c a t c g g a g	g t g c t g a a t c	t a a g c a c a c g	g	291

<210> SEQ ID NO 544
<211> LENGTH: 299
<212> TYPE: DNA
<213> ORGANISM: Zea mays

<400> SEQUENCE: 544

t a a a c c a g a y	t a g a t g c c a c	c t a g k t t t c t	g a c a c a a a t c	a y g a a c a a a a	c a c g a g a a g a	60
a a g c y a a t c a	t a c a c g g a t g	a y g g a t g g a c	t c a g c g g a g c	c c t t a t t t g c	t a c c a t t c c t	120
c a t g t e t t a t	t g c a g a a a t c	c a t c t a t t g c	t a c t c a a c t t	c a m t c a g t c t	c t g g a a c t c t	180
g t a t c a a c a g	g g g a t g g g a a	a t g t g t c a t g	t t c a a t g t t t	a g c y c a t g a a	a c a t a g a a g a	240
c c m c a t t a g a	a g c t a t t a t g	t g c t t a c a t t	t g a t t t t t t t	a t c c a a g a c t	c a a g t g t a t	299

<210> SEQ ID NO 545
<211> LENGTH: 300
<212> TYPE: DNA
<213> ORGANISM: Zea mays

<400> SEQUENCE: 545

r r h y t k r t g g	a g g t c g g g g g	a a g a a c c c t	a a g a a c g a t g	t c t c a c c a a a	t t g a t t c c t t	60
c c g c c g g a g a	g a t t t a g a a g	c t a g a t c t g t	c c a g a t t t a g	t g g a t t g a t a	g g t t t t g t g a	120
a t t t g t c a t t	c t c c a a c t a a	t c t a c t t a c g	g c c a g a t t t t	a c a t c a g a c c	t g a t g a a a c a	180
c t g t t t c c t t	g a c a c g a a a c	t g g t g g a c g c	t g c c t t t g c a	t c a a g a a t c a	a g a a a t t g a t	240
t t r c g t t t t a	t g t y t c t g g t	a g c y c a g a c	a c c t c a t a c t	c t c c t c t g t t	g c c t g t s a t g	300

<210> SEQ ID NO 546
<211> LENGTH: 232
<212> TYPE: DNA
<213> ORGANISM: Zea mays

<400> SEQUENCE: 546

a a c g g c t a a g	a g t c a g g c g a	t t c t g t t t g t	a c a g a g a c a a	t c g c a g c a c t	t g r m t g e t w c	60
g c a t g c g s t t	c a g a g g g c c t	t c a g c t c a g g	c t t g a c a g a t	c t a g t t t t g g	y c g c c a c g c y	120
r c t c t t c g g c	g a g c a c t g y a	c t c t g c t g g c	g t c t t t a g a a	g c c c c a t g c	t g t g a c g a t g	180
t g g c c g g g t g	g t a g a t g t t g	c g t t g g t g t t	t t t t c c c t c g	g g t g t c t g t g	g t	232

<210> SEQ ID NO 547
<211> LENGTH: 201
<212> TYPE: DNA
<213> ORGANISM: Zea mays
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (7)..(7)

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<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (23)..(23)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(31)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (66)..(66)
<223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 547

aaacaanact cagtggactc gcngaaannn nataatgaaa ggtgctccat ccatacccat      60
gagaangttc cgatgctcgc agtctcatgt ttcccagtc aatctctttc atttccttct      120
ccgtatgcac taatatgcag atcatggcgc agagaaaggt tcaggattgc tctcttgatc      180
tcttcaaacc cgccgtcttc g                                     201

<210> SEQ ID NO 548
<211> LENGTH: 301
<212> TYPE: DNA
<213> ORGANISM: Zea mays
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (239)..(281)
<223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 548

aaaattactg aaggcagggtg ggtgacagtt gtgtgttcgt tactgtttac tgtawyatgt      60
caagctgtcg gctgcaattt cttgtctggc aagccgcagg cactggtgaa gtgctgataa      120
atacatcata ttctgttgac ctgtgaagaa gcttggtcwa ggtrgattcc attgtactag      180
ctctgttgcy cagcatctcc ttgtttggga acattaacaa ccagcyctcr mcccetcaann      240
nnnnnnnnnn nnnnnnnnnn nnnnnnnnnn nnnnnnnnnn ntaaaatagt cgcacatcatgt      300
a                                     301

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What is claimed is:

1. A method of testing an edited haploid maize plant progeny for the absence of a first maize plant's genomic material, wherein the first maize plant is an NP2222 haploid inducer line, the method comprising:

isolating a nucleic acid from the edited haploid maize progeny, and detecting in the nucleic acid the presence of a plurality of codominant markers, wherein the codominant markers have a distinct haplotype for a second maize plant, wherein the second maize plant is a B14 line, and wherein the second maize plant comprises the source of genomic DNA in the edited haploid maize plant progeny,

wherein the plurality of codominant markers includes SEQ ID NOs 474-548, and

wherein the detecting includes the use of at least one set of primers and at least one probe selected from SEQ ID NOs: 174-473,

wherein SEQ ID NOs: 174, 175, and 177 are used to detect SEQ ID NO: 474, SEQ ID NOs: 178, 179, and 181 are used to detect SEQ ID NO: 475, SEQ ID NOs: 182, 183, and 184 are used to detect SEQ ID NO: 476, SEQ ID NOs: 186, 187, and 189 are used to detect SEQ ID NO: 477, SEQ ID NOs: 190, 191, and 193 are used to detect SEQ ID NO: 478, SEQ ID NOs: 194, 195, and

197 are used to detect SEQ ID NO: 479, SEQ ID NOs: 198, 199, and 200 are used to detect SEQ ID NO: 480, SEQ ID NOs: 202, 203, and 205 are used to detect SEQ ID NO: 481, SEQ ID NOs: 206, 207, and 208 are used to detect SEQ ID NO: 482, SEQ ID NOs: 210, 211, and 213 are used to detect SEQ ID NO: 483, SEQ ID NOs: 214, 215, and 217 are used to detect SEQ ID NO: 484, SEQ ID NOs: 218, 219, and 220 are used to detect SEQ ID NO: 485, SEQ ID NOs: 222, 223, and 224 are used to detect SEQ ID NO: 486, SEQ ID NOs: 226, 227, and 228 are used to detect SEQ ID NO: 487, SEQ ID NOs: 230, 231, and 232 are used to detect SEQ ID NO: 488, SEQ ID NOs: 234, 235, and 236 are used to detect SEQ ID NO: 489, SEQ ID NOs: 238, 239, and 241 are used to detect SEQ ID NO: 490, SEQ ID NOs: 242, 243, and 245 are used to detect SEQ ID NO: 491, SEQ ID NOs: 246, 247, and 249 are used to detect SEQ ID NO: 492, SEQ ID NOs: 250, 251, and 253 are used to detect SEQ ID NO: 493, SEQ ID NOs: 254, 255, and 257 are used to detect SEQ ID NO: 494, SEQ ID NOs: 258, 259, and 261 are used to detect SEQ ID NO: 495, SEQ ID NOs: 262, 263, and 264 are used to detect SEQ ID NO: 496, SEQ ID NOs: 266, 267, and 269 are used to detect SEQ ID NO: 497, SEQ ID NOs: 270, 271, and 273 are used to detect SEQ ID NO: 498, SEQ ID NOs: 274, 275, and

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277 are used to detect SEQ ID NO: 499, SEQ ID NOS: 278, 279, and 281 are used to detect SEQ ID NO: 500, SEQ ID NOS: 282, 283, and 284 are used to detect SEQ ID NO: 501, SEQ ID NOS: 286, 287, and 288 are used to detect SEQ ID NO: 502, SEQ ID NOS: 290, 291, and 293 are used to detect SEQ ID NO: 503, SEQ ID NOS: 294, 295, and 296 are used to detect SEQ ID NO: 504, SEQ ID NOS: 298, 299, and 300 are used to detect SEQ ID NO: 505, SEQ ID NOS: 302, 303, and 304 are used to detect SEQ ID NO: 506, SEQ ID NOS: 306, 307, and 309 are used to detect SEQ ID NO: 507, SEQ ID NOS: 310, 311, and 313 are used to detect SEQ ID NO: 508, SEQ ID NOS: 314, 315, and 316 are used to detect SEQ ID NO: 509, SEQ ID NOS: 318, 319, and 320 are used to detect SEQ ID NO: 510, SEQ ID NOS: 322, 323, and 325 are used to detect SEQ ID NO: 511, SEQ ID NOS: 326, 327, and 329 are used to detect SEQ ID NO: 512, SEQ ID NOS: 330, 331, and 333 are used to detect SEQ ID NO: 513, SEQ ID NOS: 334, 335, and 336 are used to detect SEQ ID NO: 514, SEQ ID NOS: 338, 339, and 341 are used to detect SEQ ID NO: 515, SEQ ID NOS: 342, 343, and 344 are used to detect SEQ ID NO: 516, SEQ ID NOS: 346, 347, and 348 are used to detect SEQ ID NO: 517, SEQ ID NOS: 350, 351, and 352 are used to detect SEQ ID NO: 518, SEQ ID NOS: 354, 355, and 356 are used to detect SEQ ID NO: 519, SEQ ID NOS: 358, 359, and 361 are used to detect SEQ ID NO: 520, SEQ ID NOS: 362, 363, and 365 are used to detect SEQ ID NO: 521, SEQ ID NOS: 366, 367, and 368 are used to detect SEQ ID NO: 522, SEQ ID NOS: 370, 371, and 372 are used to detect SEQ ID NO: 523, SEQ ID NOS: 374, 375, and 376 are used to detect SEQ ID NO: 524, SEQ ID NOS: 378, 379, and 38 are used to detect SEQ ID NO: 525, SEQ ID NOS: 382, 383, and 384 are used to detect SEQ ID NO: 526, SEQ ID NOS: 386, 387, and 388 are used to detect SEQ ID NO: 527, SEQ ID NOS: 390, 391, and 393 are used to detect SEQ ID NO: 528, SEQ ID NOS: 394, 395, and 397 are used to detect SEQ ID NO: 529, SEQ ID NOS: 398, 399, and 400 are used to detect SEQ ID NO: 530, SEQ ID NOS: 402, 403, and 404 are used to detect SEQ ID NO: 531, SEQ ID NOS: 406, 407, and 409 are used to detect SEQ ID NO: 532, SEQ ID NOS: 410, 411, and 412 are used to detect SEQ ID NO: 533, SEQ ID NOS: 414, 415, and 416 are used to detect SEQ ID NO: 534, SEQ ID NOS: 418, 419, and 420 are used to detect SEQ ID NO: 535, SEQ ID NOS: 422, 423, and 425 are used to detect SEQ ID NO: 536, SEQ ID NOS: 426, 427, and 429 are used to detect SEQ ID NO: 537, SEQ ID NOS: 430, 431, and 433 are used to detect SEQ ID NO: 538, SEQ ID NOS: 434, 435, and 437 are used to detect SEQ ID NO: 539, SEQ ID NOS: 438, 439, and 440 are used to detect SEQ ID NO: 540, SEQ ID NOS: 442, 443, and 445 are used to detect SEQ ID NO: 541, SEQ ID NOS: 446, 447, and 449 are used to detect SEQ ID NO: 542, SEQ ID NOS: 450, 451, and

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- 453 are used to detect SEQ ID NO: 543, SEQ ID NOS: 454, 455, and 456 are used to detect SEQ ID NO: 544, SEQ ID NOS: 458, 459, and 461 are used to detect SEQ ID NO: 545, SEQ ID NOS: 462, 463, and 465 are used to detect SEQ ID NO: 546, SEQ ID NOS: 466, 467, and 468 are used to detect SEQ ID NO: 547, and SEQ ID NOS: 470, 471, and 473 are used to detect SEQ ID NO: 548.
2. The method of claim 1, wherein the edited haploid plant is first obtained by a method comprising:
 - (a) obtaining the first plant, wherein the first plant is capable of expressing a DNA modification enzyme and optionally at least one guide nucleic acid;
 - (b) obtaining the second plant, wherein the second plant comprises the plant genomic DNA which is to be edited;
 - (c) pollinating the second plant with pollen from the first plant; and
 - (d) selecting at least one haploid progeny produced by the pollination of step (c), wherein the haploid progeny comprises the genome of the second plant but not the first plant, and the genome of the haploid progeny has been modified by the DNA modification enzyme and optionally at least one guide nucleic acid delivered by the first plant.
 3. The method of claim 2, wherein the DNA modification enzyme is a site-directed nuclease selected from the group consisting of meganucleases (MNs), zinc-finger nucleases (ZFNs), transcription-activator like effector nucleases (TALENs), Cas9 nuclease, Cpf1 nuclease, dCas9-FokI, dCpf1-FokI, chimeric Cas9-cytidine deaminase, chimeric Cas9-adenine deaminase, chimeric FEN1-FokI, and Mega-TALs, a nickase Cas9 (nCas9), chimeric dCas9 non-FokI nuclease and dCpf1 non-FokI nuclease.
 4. The method of claim 2, wherein the at least one guide nucleic acid is a guide RNA.
 5. The method of claim 1, wherein the edited haploid progeny is treated with a chromosome doubling agent after testing, thereby creating an edited doubled haploid progeny, optionally wherein the chromosome doubling agent is colchicine, pronamide, dithiopyr, or trifluralin.
 6. The method of claim 2, wherein the optional guide RNA is 18-21 nucleotides long and is at least 90% identical to a sequence selected from the group consisting of SEQ ID NOS: 2, 4, 8, 21, 23, 25, 29, 32, and 33.
 7. The method of claim 1, wherein the first plant expresses a marker gene.
 8. The method of claim 7, wherein the marker gene is selected from the group consisting of GUS, PMI, PAT, GFP, RFP, CFP, B1, C1, R1-nj, and anthocyanin pigments.
 9. The method of claim 1, wherein the first plant acts as the male and the second plant acts as the female in a cross resulting in the haploid progeny.

* * * * *